



Neamen 4e solutions TYU

Electronic Circuit (Islamic University of Technology)

Chapter 1

Exercise Solutions

EX1.1

$$n_i = BT^{3/2} \exp\left(\frac{-E_g}{2kT}\right)$$

$$\text{GaAs: } n_i = (2.1 \times 10^{14})(300)^{3/2} \exp\left(\frac{-1.4}{2(86 \times 10^{-6})(300)}\right) \text{ or } n_i = 1.8 \times 10^6 \text{ cm}^{-3}$$

$$\text{Ge: } n_i = (1.66 \times 10^{15})(300)^{3/2} \exp\left(\frac{-0.66}{2(86 \times 10^{-6})(300)}\right) \text{ or } n_i = 2.40 \times 10^{13} \text{ cm}^{-3}$$

EX1.2

$$(a) (i) \quad n_o = N_d = 2 \times 10^{16} \text{ cm}^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{(1.5 \times 10^{10})^2}{2 \times 10^{16}} = 1.125 \times 10^4 \text{ cm}^{-3}$$

$$(ii) \quad p_o = N_a = 10^{15} \text{ cm}^{-3}$$

$$n_o = \frac{n_i^2}{p_o} = \frac{(1.5 \times 10^{10})^2}{10^{15}} = 2.25 \times 10^5 \text{ cm}^{-3}$$

$$(b) (i) \quad n_o = N_d = 2 \times 10^{16} \text{ cm}^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{(1.8 \times 10^6)^2}{2 \times 10^{16}} = 1.62 \times 10^{-4} \text{ cm}^{-3}$$

$$(ii) \quad p_o = N_a = 10^{15} \text{ cm}^{-3}$$

$$n_o = \frac{n_i^2}{p_o} = \frac{(1.8 \times 10^6)^2}{10^{15}} = 3.24 \times 10^{-3} \text{ cm}^{-3}$$

EX1.3

(a) For n-type;

$$\rho = \frac{1}{e\mu_n N_d} = \frac{1}{(1.6 \times 10^{-19})(6800)(2 \times 10^{16})} = 0.046 \text{ ohm-cm}$$

$$(b) \quad J = \frac{1}{\rho} \cdot E \Rightarrow E = \rho J = (0.046)(175) = 8.04 \text{ V/cm}$$

EX1.4

Diffusion current density due to holes:

$$\begin{aligned} J_p &= -eD_p \frac{dp}{dx} \\ &= -eD_p (10^{16}) \left(\frac{-1}{L_p} \right) \exp\left(\frac{-x}{L_p}\right) \end{aligned}$$

(a) At $x = 0$

$$J_p = \frac{(1.6 \times 10^{-19})(10)(10^{16})}{10^{-3}} = 16 \text{ A/cm}^2$$

(b) At $x = 10^{-3} \text{ cm}$

$$J_p = 16 \exp\left(\frac{-10^{-3}}{10^{-3}}\right) = 5.89 \text{ A/cm}^2$$

EX1.5

(a) $V_{bi} = (0.026) \ln \left[\frac{(10^{16})(10^{17})}{(1.8 \times 10^6)^2} \right] = 1.23 \text{ V}$

(b) $V_{bi} = (0.026) \ln \left[\frac{(10^{16})(10^{17})}{(2.4 \times 10^{13})^2} \right] = 0.374 \text{ V}$

EX1.6

$$C_j = C_{jo} \left(1 + \frac{V_R}{V_{bi}} \right)^{-1/2}$$

and

$$\begin{aligned} V_{bi} &= V_T \ln \left[\frac{N_a N_d}{n_i^2} \right] \\ &= (0.026) \ln \left[\frac{(10^{17})(10^{16})}{(1.5 \times 10^{10})^2} \right] = 0.757 \text{ V} \end{aligned}$$

$$\text{Then } 0.8 = C_{jo} \left(1 + \frac{5}{0.757} \right)^{-1/2} = C_{jo} (7.61)^{-1/2}$$

or

$$C_{jo} = 2.21 \text{ pF}$$

EX1.7

(a) $V_D = V_T \ln \left(\frac{I_D}{I_S} \right)$

(i) $V_D = (0.026) \ln \left(\frac{50 \times 10^{-6}}{2 \times 10^{-14}} \right) = 0.563 \text{ V}$

(ii) $V_D = (0.026) \ln \left(\frac{10^{-3}}{2 \times 10^{-14}} \right) = 0.641 \text{ V}$

(b) (i) $V_D = (0.026) \ln \left(\frac{50 \times 10^{-6}}{2 \times 10^{-12}} \right) = 0.443 \text{ V}$

(ii) $V_D = (0.026) \ln \left(\frac{10^{-3}}{2 \times 10^{-12}} \right) = 0.521 \text{ V}$

EX1.8

$$V_{PS} = I_D R + V_D \text{ and } I_D \cong I_S \exp\left(\frac{V_D}{V_T}\right)$$

$$\text{so } 4 = I_D (4 \times 10^3) + V_D \Rightarrow I_D = \frac{(4 - V_D)}{4 \times 10^3}$$

and

$$I_D = (10^{-12}) \exp\left(\frac{V_D}{0.026}\right)$$

By trial and error, we find $I_D \cong 0.866 \text{ mA}$ and $V_D \cong 0.535 \text{ V}$.

EX1.9

$$(a) \quad I_D = \frac{V_{PS} - V_\gamma}{R} \Rightarrow R = \frac{8 - 0.7}{1.20} = 6.08 \text{ k}\Omega$$

$$(b) \quad I_D = \frac{4 - 0.7}{3.5} = 0.9429 \text{ mA}$$

$$P_D = I_D V_D = (0.9429)(0.7) = 0.66 \text{ mW}$$

EX1.10

PSpice Analysis

EX1.11

$$(a) \quad I_D = \frac{8 - 0.7}{20} = 0.365 \text{ mA}$$

$$r_d = \frac{V_T}{I_D} = \frac{0.026}{0.365} \Rightarrow 71.2 \Omega$$

$$i_d = \frac{0.25 \sin \omega t}{20 + 0.0712} \Rightarrow 12.5 \sin \omega t \text{ (}\mu\text{ A)}$$

$$(b) \quad I_D = \frac{8 - 0.7}{10} = 0.73 \text{ mA}$$

$$r_d = \frac{0.026}{0.73} \Rightarrow 35.6 \Omega$$

$$i_d = \frac{0.25 \sin \omega t}{10 + 0.0356} \Rightarrow 24.9 \sin \omega t \text{ (}\mu\text{ A)}$$

EX1.12

$$\text{For the pn junction diode, } V_D \cong V_T \ln\left(\frac{I_D}{I_S}\right) = (0.026) \ln\left(\frac{1.2 \times 10^{-3}}{4 \times 10^{-15}}\right) \text{ or } V_D = 0.6871 \text{ V}$$

The Schottky diode voltage will be smaller, so $V_D = 0.6871 - 0.265 = 0.4221 \text{ V}$

$$\text{Now } I_D \cong I_S \exp\left(\frac{V_D}{V_T}\right)$$

or

$$I_S = \frac{1.2 \times 10^{-3}}{\exp\left(\frac{0.4221}{0.026}\right)} \Rightarrow I_S = 1.07 \times 10^{-10} \text{ A}$$

EX1.13

$$P = I \cdot V_Z \Rightarrow 10 = I(5.6) \Rightarrow I = 1.79 \text{ mA}$$

$$\text{Also } I = \frac{10 - 5.6}{R} = 1.79 \Rightarrow R = 2.46 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU1.1

(a) $T = 400 \text{ K}$

— Si: $n_i = BT^{3/2} \exp\left(\frac{-E_g}{2kT}\right)$

$$n_i = (5.23 \times 10^{15})(400)^{3/2} \exp\left[\frac{-1.1}{2(86 \times 10^{-6})(400)}\right]$$

or

$$n_i = 4.76 \times 10^{12} \text{ cm}^{-3}$$

— Ge: $n_i = (1.66 \times 10^{15})(400)^{3/2} \exp\left[\frac{-0.66}{2(86 \times 10^{-6})(400)}\right]$

or

$$n_i = 9.06 \times 10^{14} \text{ cm}^{-3}$$

— GaAs:

$$n_i = (2.1 \times 10^{14})(400)^{3/2} \exp\left[\frac{-1.4}{2(86 \times 10^{-6})(400)}\right]$$

or

$$n_i = 2.44 \times 10^9 \text{ cm}^{-3}$$

(b) $T = 250 \text{ K}$

— Si: $n_i = (5.23 \times 10^{15})(250)^{3/2} \exp\left[\frac{-1.1}{2(86 \times 10^{-6})(250)}\right]$

or

$$n_i = 1.61 \times 10^8 \text{ cm}^{-3}$$

— Ge: $n_i = (1.66 \times 10^{15})(250)^{3/2} \exp\left[\frac{-0.66}{2(86 \times 10^{-6})(250)}\right]$

or

$$n_i = 1.42 \times 10^{12} \text{ cm}^{-3}$$

$$\text{GaAs: } n_i = (2.10 \times 10^{14}) (250)^{3/2} \exp \left[\frac{-1.4}{2(86 \times 10^{-6})(250)} \right]$$

or

$$n_i = 6.02 \times 10^3 \text{ cm}^{-3}$$

TYU1.2

$$(a) \quad \sigma = e\mu_p N_a = (1.6 \times 10^{-19})(480)(2 \times 10^{15}) = 0.154 \text{ (ohm-cm)}^{-1}$$

$$\rho = \frac{1}{\sigma} = \frac{1}{0.1536} = 6.51 \Omega \text{-cm}$$

$$(b) \quad \sigma = e\mu_n N_d = (1.6 \times 10^{-19})(1350)(2 \times 10^{17}) = 43.2 \text{ (ohm-cm)}^{-1}$$

$$\rho = \frac{1}{\sigma} = \frac{1}{43.2} = 0.0231 \Omega \text{-cm}$$

TYU1.3

$$(a) \quad J = \sigma E = (0.154)(4) = 0.616 \text{ A/cm}^2$$

$$(b) \quad J = \sigma E = (43.2)(4) = 172.8 \text{ A/cm}^2$$

TYU1.4

$$(a) \quad J_n = eD_n \frac{dn}{dx} = eD_n \frac{\Delta n}{\Delta x} \text{ so } J_n = (1.6 \times 10^{-19})(35) \left(\frac{10^{15} - 10^{16}}{0 - 2.5 \times 10^{-4}} \right)$$

or

$$J_n = 202 \text{ A/cm}^2$$

$$(b) \quad J_p = -eD_p \frac{dp}{dx} = -eD_p \frac{\Delta p}{\Delta x} \text{ so } J_p = -(1.6 \times 10^{-19})(12.5) \left(\frac{10^{14} - 5 \times 10^{15}}{0 - 4 \times 10^{-4}} \right)$$

or

$$J_p = -24.5 \text{ A/cm}^2$$

TYU1.5

$$(a) \quad n_o = N_d = 8 \times 10^{15} \text{ cm}^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{(1.5 \times 10^{10})^2}{8 \times 10^{15}} = 2.81 \times 10^4 \text{ cm}^{-3}$$

$$(b) \quad n = n_o + \delta n = 8 \times 10^{15} + 0.1 \times 10^{15}$$

or

$$n = 8.1 \times 10^{15} \text{ cm}^{-3}$$

$$p = p_o + \delta p = 2.81 \times 10^4 + 10^{14}$$

or

$$p \cong 10^{14} \text{ cm}^{-3}$$

TYU1.6

$$(a) \quad V_{bi} = V_T \ln \left(\frac{N_a N_d}{n_i^2} \right) = (0.026) \ln \left[\frac{(10^{15})(5 \times 10^{16})}{(1.5 \times 10^{10})^2} \right] = 0.679 \text{ V}$$

$$(b) \quad V_{bi} = (0.026) \ln \left[\frac{(10^{15})(5 \times 10^{16})}{(1.8 \times 10^6)^2} \right] = 1.15 \text{ V}$$

$$(c) \quad V_{bi} = (0.026) \ln \left[\frac{(10^{15})(5 \times 10^{16})}{(2.4 \times 10^{13})^2} \right] = 0.296 \text{ V}$$

TYU1.7

$$(a) \quad (i) \quad I_D = I_S \exp \left(\frac{V_D}{V_T} \right) = (10^{-16}) \exp \left(\frac{0.55}{0.026} \right) \Rightarrow 0.154 \mu\text{A}$$

$$(ii) \quad I_D = (10^{-16}) \exp \left(\frac{0.65}{0.026} \right) \Rightarrow 7.20 \mu\text{A}$$

$$(ii) \quad I_D = (10^{-16}) \exp \left(\frac{0.75}{0.026} \right) \Rightarrow 0.337 \text{ mA}$$

$$(b) \quad (i) \quad I_D = -10^{-16} \text{ A}$$

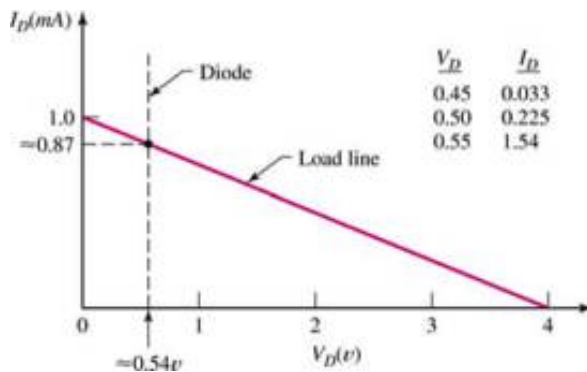
$$(ii) \quad I_D = -10^{-16} \text{ A}$$

TYU1.8

$$\Delta T = 100^\circ\text{C} \text{ so } \Delta V_D \cong 2 \times 100 = 200 \text{ mV}$$

$$\text{Then } V_D = 0.650 - 0.20 = 0.450 \text{ V}$$

TYU1.9



TYU1.10

- (a) $I_D = 0$
(b) $I_D = \frac{2-0.7}{4} = 0.325 \text{ mA}$
(c) $I_D = \frac{5-0.7}{4} = 1.075 \text{ mA}$
(d) $I_D = 0$
(e) $I_D = 0$

TYU1.11

$$P = I_D V_D \Rightarrow 1.05 = I_D (0.7) \text{ so } I_D = 1.5 \text{ mA}$$

$$\text{Now } R = \frac{V_{PS} - V_\gamma}{I_D} = \frac{10 - 0.7}{1.5} \Rightarrow R = 6.2 \text{ k}\Omega$$

TYU1.12

$$g_d = \frac{I_D}{V_T} = \frac{0.8}{0.026} = 30.8 \text{ mS}$$

TYU1.13

$$r_d = \frac{V_T}{I_D} = \frac{0.026}{0.010} = 2.6 \text{ k}\Omega$$

$$r_d = \frac{0.026}{0.10} \Rightarrow 260 \Omega$$

$$r_d = \frac{0.026}{1} \Rightarrow 26 \Omega$$

TYU1.14

$$r_d = \frac{V_T}{I_D} \Rightarrow 50 = \frac{0.026}{I_D} \Rightarrow I_D = \frac{0.026}{50}$$

or

$$I_D = 0.52 \text{ mA}$$

TYU1.15

For the pn junction diode,

$$I_D = \frac{4-0.7}{4} = 0.825 \text{ mA}$$

$$\text{For the Schottky diode, } I_D = \frac{4-0.3}{4} = 0.925 \text{ mA}$$

TYU1.16

$$V_z = V_{zo} + I_z r_z \Rightarrow V_{zo} = V_z - I_z r_z \text{ so } V_{zo} = 5.20 - (10^{-3})(20) = 5.18 \text{ V}$$

$$\text{Then } V_z = 5.18 + (10 \times 10^{-3})(20) \Rightarrow V_z = 5.38 \text{ V}$$

TYU1.17

$$P = I_Z V_Z \Rightarrow I_Z = \frac{P}{V_Z} = \frac{6.5}{3.6} = 1.81 \text{ mA}$$

$$V_{PS} = I_Z R + V_Z = (1.81)(4) + 3.6 = 10.8 \text{ V}$$

Chapter 2

Exercise Solutions

EX2.1

$$i_D(\text{peak}) = \frac{V_S - V_B - V_\gamma}{R} = \frac{12 - 4.5 - 0.6}{0.25} = 27.6 \text{ mA}$$

$$v_R(\text{max}) = V_S + V_B = 12 + 4.5 = 16.5 \text{ V}$$

Conduction cycle:

$$v_I = 12 \sin \omega t_1 = 4.5 + 0.6 = 5.1 \text{ V}$$

or

$$\omega t_1 = \sin^{-1}\left(\frac{5.1}{12}\right) = 25.15^\circ$$

$$\omega t_2 = 180 - 25.15 = 154.85^\circ$$

$$\text{Percent time} = \frac{154.85 - 25.15}{360} \times 100\% = 36.0\%$$

EX2.2

$$(a) \quad v_O = 12 \sin \theta_1 - 1.4 = 0$$

$$\text{or } \sin \theta_1 = \frac{1.4}{12} = 0.1166$$

which yields

$$\theta_1 = 6.7^\circ$$

By symmetry, $\theta_2 = 180 - 6.7 = 173.3^\circ$

Then

$$\% \text{ time} = \frac{173.3 - 6.7}{360} \times 100\% = 46.3\%$$

$$(b) \quad \sin \theta_1 = \frac{1.4}{4} = 0.35$$

which yields

$$\theta_1 = 20.5^\circ$$

By symmetry, $\theta_2 = 180 - 20.5 = 159.5^\circ$

$$\% \text{ time} = \frac{159.5 - 20.5}{360} \times 100\% = 38.6\%$$

Then

EX2.3

$$(a) \quad C = \frac{V_M}{2fRV_r} = \frac{12}{2(60)(2 \times 10^3)(0.4)} \Rightarrow 125 \mu\text{F}$$

$$(b) \quad C = \frac{V_M}{fRV_r} = \frac{12}{(60)(2 \times 10^3)(0.4)} \Rightarrow 250 \mu\text{F}$$

EX2.4

$$V_r = \frac{V_M}{f RC} \Rightarrow R = \frac{V_M}{f C V_r} \quad R = \frac{75}{(60)(50 \times 10^{-6})(4)}$$

Then $R = 6.25 \text{ k}\Omega$

EX2.5

$$10 \leq V_{PS} \leq 14 \text{ V}, \quad V_Z = 5.6 \text{ V}, \quad 20 \leq R_L \leq 100 \Omega$$

$$I_L(\text{max}) = \frac{5.6}{20} = 0.28 \text{ A},$$

$$I_L(\text{min}) = \frac{5.6}{100} = 0.056 \text{ A}$$

$$I_Z(\text{max}) = \frac{[V_{PS}(\text{max}) - V_Z] \cdot I_L(\text{max})}{V_{PS}(\text{min}) - 0.9V_Z - 0.1V_{PS}(\text{max})} - \frac{[V_{PS}(\text{min}) - V_Z] \cdot I_L(\text{min})}{V_{PS}(\text{min}) - 0.9V_Z - 0.1V_{PS}(\text{max})}$$

$$I_L(\text{max}) = \frac{(14 - 5.6)(280) - (10 - 5.6)(56)}{10 - (0.9)(5.6) - (0.1)(14)}$$

or

$$I_L(\text{max}) = 591.5 \text{ mA}$$

$$\text{Power}(\text{min}) = I_Z(\text{max}) \cdot V_Z = (0.5915)(5.6)$$

So $\text{Power}(\text{min}) = 3.31 \text{ W}$

$$R_i = \frac{V_{PS}(\text{max}) - V_Z}{I_Z(\text{max}) + I_L(\text{min})} = \frac{14 - 5.6}{0.5915 + 0.056} \quad \text{or} \quad R_i \cong 13 \Omega$$

EX2.6

$$\text{For } v_{PS} = 13.6 \text{ V}, \quad I_Z = \frac{13.6 - 9}{15.3 + 4} = 0.2383 \text{ A}$$

$$v_{L,\text{max}} = 9 + (4)(0.2383) = 9.9532 \text{ V}$$

$$\text{For } v_{PS} = 11 \text{ V}, \quad I_Z = \frac{11 - 9}{15.3 + 4} = 0.1036 \text{ A}$$

$$v_{L,\text{min}} = 9 + (4)(0.1036) = 9.4144 \text{ V}$$

$$\text{Source Reg} = \frac{\Delta v_L}{\Delta v_{PS}} \times 100\% = \frac{9.9532 - 9.4144}{13.6 - 11} \times 100\%$$

or $\text{Source Reg} = 20.7\%$

$$\text{For } I_L = 0, \quad I_Z = \frac{13.6 - 9}{15.3 + 4} = 0.2383 \text{ A}$$

$$v_{L,\text{no load}} = 9 + (4)(0.2383) = 9.9532 \text{ V}$$

$$\text{For } I_L = 100 \text{ mA}, \quad I_Z = \frac{13.6 - [9 + I_Z(4)]}{15.3} - 0.10$$

which yields

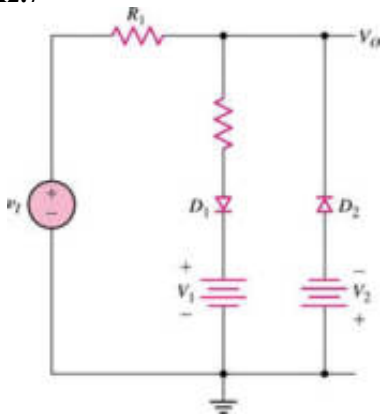
$$I_Z = 0.1591 \text{ A}$$

$$v_{L, \text{full load}} = 9 + (4)(0.1591 \text{ A}) = 9.6363 \text{ V}$$

$$\begin{aligned} \text{Load Reg} &= \frac{v_{L, \text{no load}} - v_{L, \text{full load}}}{v_{L, \text{full load}}} \times 100\% \\ &= \frac{9.9532 - 9.6363}{9.6363} \times 100\% \end{aligned}$$

$$\text{or Load Reg} = 3.29\%$$

EX2.7



For $v_I < 5 \text{ V}$, D_2 on $\Rightarrow V_O = -5 \text{ V}$

Then, $V_2 = 4.3 \text{ V}$.

D_1 turns on when $v_I = 2.5 \text{ V}$,

Then, $V_1 = 1.8 \text{ V}$.

$$\text{For } v_I > 2.5 \text{ V}, \frac{\Delta v_O}{\Delta v_I} = \frac{1}{3} \Rightarrow \frac{R_2}{R_1 + R_2} = \frac{1}{3}$$

For

So that $R_1 = 2R_2$

EX2.8

For $v_O = +2 \text{ V}$, D is on. $\Delta v_I = 10 \text{ V}$, so $\Delta v_O = 10 \text{ V}$.

Output = Square wave between $+2$ and -8 V .

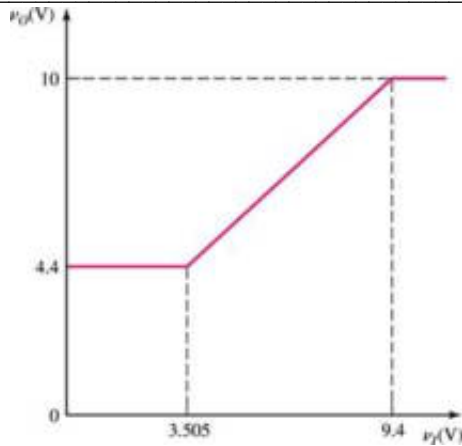
EX2.9

$$v_O = 4.4 \text{ V}, I = \frac{10 - 4.4}{9.5} = 0.5895 \text{ mA}$$

$$\text{Set } I = I_{D1}, \text{ then } v_I = 4.4 - 0.6 - (0.5895)(0.5) = 3.505 \text{ V}$$

Summary: For $0 \leq v_I \leq 3.5 \text{ V}$, $v_O = 4.4 \text{ V}$

For $v_I > 3.5 \text{ V}$, D_2 turns on and when $v_I \geq 9.4 \text{ V}$, $v_O = 10 \text{ V}$



EX2.10

(a) If D_1 is on, $v_O = v_I - V_\gamma - V_B = 5 - 0.7 - 1 = 3.3$ V.

$$\text{Then } I_{D2} = \frac{3.3 - 0.7}{4} = 0.65 \text{ mA}$$

Now $I_{R1} = \frac{5 - 3.3}{1.7} = 1.0$ mA, but $I_{D2} < I_{R1}$ is impossible.

D_1 is cutoff and $I_{D1} = 0$

$$\text{Then } I_{R1} = I_{D2} = \frac{5 - 0.7}{1.7 + 4} = 0.754 \text{ mA}$$

$$v_O = 0.7 + (0.754)(4) = 3.72 \text{ V}$$

(b) $v_I = 10$ V, Both D_1 and D_2 are on.

$$v_O = 10 - 0.7 - 1 \Rightarrow v_O = 8.3 \text{ V}$$

$$I_{D2} = \frac{8.3 - 0.7}{4} = 1.9 \text{ mA}$$

$$I_{R1} = \frac{1.7}{1.7} = 1.0 \text{ mA}$$

$$I_{D1} = 1.9 - 1.0 = 0.9 \text{ mA}$$

EX2.11

D_2 cutoff, $I_{D2} = 0$

$$V_B = -0.7 \text{ V}, I_{D3} = \frac{-0.7 - (-5)}{2} = 2.15 \text{ mA}$$

$$I_{D1} = \frac{5 - 0.7 - (-10)}{R_1 + R_2} = \frac{14.3}{8 + 4} = 1.19 \text{ mA}$$

$$V_A = 5 - (1.19)(8) = -4.53 \text{ V}$$

$V_A < V_B$ so that D_2 is cutoff.

EX2.12

(a) $I_{ph} = \eta e \Phi$

$$I = (0.8)(1.6 \times 10^{-19}) \left[\frac{6.4 \times 10^{-2}}{(2)(1.6 \times 10^{-19})} \right] (0.5)$$

so

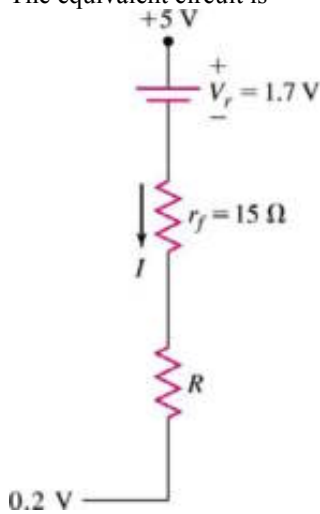
or $I_{ph} = 12.8 \text{ mA}$

(b) We have $v_O = (12.8)(1) = 12.8 \text{ V}$.

The diode must be reverse biased so that $V_{PS} > 12.8 \text{ V}$

EX2.13

The equivalent circuit is



$$I = \frac{5 - 1.7 - 0.2}{r_f + R} = 15 \text{ mA}$$

So

$$\text{Or } r_f + R = \frac{15 - 1.7 - 0.2}{15} = \frac{3.1}{15} = 0.207 \text{ k}\Omega$$

Or

$$\text{Then } R = 207 - 15 \Rightarrow R = 192 \Omega$$

Test Your Understanding Solutions

TYU2.1

(a) $i_D(\text{peak})$ for $V_B = 4 \text{ V}$.

$$i_D(\text{peak}) = 18 = \frac{15 - 0.7 - 4}{R} \Rightarrow R = 572 \Omega$$

(b) $i_D = \frac{15 - 0.7 - 8}{0.572} = 11.0 \text{ mA}$

Then

$$11.0 \leq i_D(\text{peak}) \leq 18 \text{ mA}$$

For $V_B = 4 \text{ V}$,

$$15 \sin \omega t_1 = 4.7 \Rightarrow \omega t_1 = \sin^{-1}\left(\frac{4.7}{15}\right) = 18.26^\circ$$

$$\omega t_2 = 180 - 18.26 = 161.74^\circ$$

$$\text{duty cycle} = \frac{161.74 - 18.26}{360} \times 100\% = 39.9\%$$

For $V_B = 8 \text{ V}$,

$$15 \sin \omega t_1 = 8.7 \Rightarrow \omega t_1 = \sin^{-1}\left(\frac{8.7}{15}\right) = 35.45^\circ$$

$$\omega t_2 = 180 - 35.45 = 144.55^\circ$$

$$\text{duty cycle} = \frac{144.55 - 35.45}{360} \times 100\% = 30.3\%$$

Then

$$30.3 \leq \text{duty cycle} \leq 39.9\%$$

TYU2.2

$$v_I = 120 \sin(2\pi 60t), \quad V_\gamma = 0.7 \text{ V}, \quad \text{and} \quad R = 2.5 \text{ k}\Omega$$

Full-wave rectifier: Turns ratio 1:2 so that

$$v_S = v_I$$

$$V_M = 120 - 0.7 = 119.3 \text{ V}$$

$$V_r = 119.3 - 100 = 19.3 \text{ V}$$

$$\text{So } C = \frac{V_M}{2f R V_r} = \frac{119.3}{2(60)(2.5 \times 10^3)(19.3)} \quad \text{or} \quad C = 20.6 \mu\text{F}$$

TYU2.3

$$v_I = 50 \sin(2\pi 60t), \quad V_\gamma = 0.7 \text{ V}, \quad \text{and} \quad R = 10 \text{ k}\Omega. \quad \text{Full-wave rectifier}$$

$$C = \frac{V_M}{2f R V_r} = \frac{(50 - 1.4)}{2(60)(10 \times 10^3)(2)}$$

$$\text{or } C = 20.3 \mu\text{F}$$

TYU2.4

Using Equation (2.16)

$$\omega\Delta t = \sqrt{\frac{2V_r}{V_M}} = \sqrt{\frac{2(4)}{75}} = 0.327$$

(a) Percent time = $\left(\frac{0.327}{2\pi}\right) \times 100\% = 5.2\%$

$$\omega\Delta t = \sqrt{\frac{2V_r}{V_M}} = \sqrt{\frac{2(19.3)}{119.3}} = 0.569$$

(b) Percent time = $\left(\frac{0.569}{\pi}\right) \times 100\% = 18.1\%$

$$\omega\Delta t = \sqrt{\frac{2V_r}{V_M}} = \sqrt{\frac{2(2)}{48.6}} = 0.287$$

(c) Percent time = $\left(\frac{0.287}{\pi}\right) \times 100\% = 9.14\%$

TYU2.5

(a) $P = I_Z V_Z$

$$1 = I_Z (6.2 + 3I_Z) = 3I_Z^2 + 6.2I_Z$$

$$3I_Z^2 + 6.2I_Z - 1 = 0 \Rightarrow I_Z(\text{max}) = 150 \text{ mA}$$

$$V_Z = 6.2 + 3(0.15) = 6.65 \text{ V}$$

$$R_i = \frac{12 - 6.65}{0.15} \Rightarrow R_i = 35.7 \Omega$$

(b) For $I_Z = (0.1)(150) = 15 \text{ mA}$

$$V_Z = V_O = 6.2 + 3(0.015) = 6.245 \text{ V}$$

$$I_L = I_i - I_Z \Rightarrow I_i = \frac{12 - 6.245}{0.0357} = 161.2 \text{ mA}$$

$$I_L = 161.2 - 15 = 146.2 \text{ mA}$$

$$R_L = \frac{6.245}{0.1462} = 42.7 \Omega$$

$$\begin{aligned} \text{Load Regulation} &= \frac{v_L(\text{no load}) - v_L(\text{full load})}{v_L(\text{no load})} \times 100\% \\ &= \frac{6.65 - 6.245}{6.65} \times 100\% = 6.09\% \end{aligned}$$

TYU2.6

$$I_Z = \frac{V_{PS} - V_Z}{R_i} - I_L$$

For $V_{PS}(\text{min})$ and $I_L(\text{max})$, then $I_Z(\text{min}) = \frac{11 - 9}{20} - 0.1 = 0$ (Minimum Zener current is zero.)

For $V_{PS}(\text{max})$ and $I_L(\text{min})$, then $I_Z(\text{max}) = \frac{13.6 - 9}{20} - 0 \Rightarrow I_Z(\text{max}) = 230 \text{ mA}$

The characteristic of the minimum Zener current being one-tenth of the maximum value is violated. The proper circuit operation is questionable.

TYU2.7

$$I_Z(\min) = \frac{V_{PS}(\min) - V_Z}{R_i} - I_L(\max)$$

$$\text{so } 30 = \frac{10 - 9}{0.0153} - I_L(\max) \quad \text{which yields } I_L(\max) = 35.4 \text{ mA}$$

TYU2.8

For $v_I \leq -3.7 \text{ V}$, D_2 on $\Rightarrow v_O = -3.7 \text{ V}$

D_1 turns on when $v_I = 1.7 \text{ V}$ so

$$-3.7 \leq v_I \leq 1.7 \text{ V} \Rightarrow v_O = v_I$$

For $v_I > 1.7 \text{ V}$,

$$i_1 = \frac{v_I - 1.7}{R_1 + R_2} = \frac{v_I - 1.7}{7}$$

$$v_O = i_1 R_2 + 1.7 = \left(\frac{v_I - 1.7}{7} \right) (2) + 1.7$$

Or

$$v_O = 0.286v_I + 1.21$$

TYU2.9

As v_S goes negative, D turns on and $v_O = +5 \text{ V}$. As v_S goes positive, D turns off. Output is a square wave oscillating between +5 and +35 volts.

TYU2.10

$$V_1 = 3 - 0.7 = 2.3 \text{ V}$$

$$V_2 = 2 - 0.7 = 1.3 \text{ V}$$

For $v_I > 3 \text{ V}$, slope = $\frac{1}{2} \Rightarrow R_1 = R_2$

Put resistor in series with D_2 ,

For $v_I < 2 \text{ V}$, slope = $\frac{1}{3} = \frac{R_3}{R_1 + R_3} \Rightarrow R_1 = 2R_3$

TYU2.11

D_2 and D_3 cutoff so that $I_{D2} = I_{D3} = 0$

$$I_{D1} = \frac{14 - 0.7 - (-5)}{R_1 + R_2 + R_3} = \frac{18.3}{5 + 5 + 5} = 1.22 \text{ mA}$$

$$V_A = 14 - 0.7 - (1.22)(5) = 7.2 \text{ V} \Rightarrow D_2 \text{ cutoff}$$

$$V_B = (1.22)(5) - 5 = 1.1 \text{ V} \Rightarrow D_3 \text{ cutoff}$$

TYU2.12

D_2 cutoff, $I_{D2} = 0$

$$V_B = -0.7 \text{ V}$$

$$I_{D1} = \frac{14 - 0.7 - (-0.7)}{R_1 + R_2} = \frac{14}{8 + 12} = 0.7 \text{ mA}$$

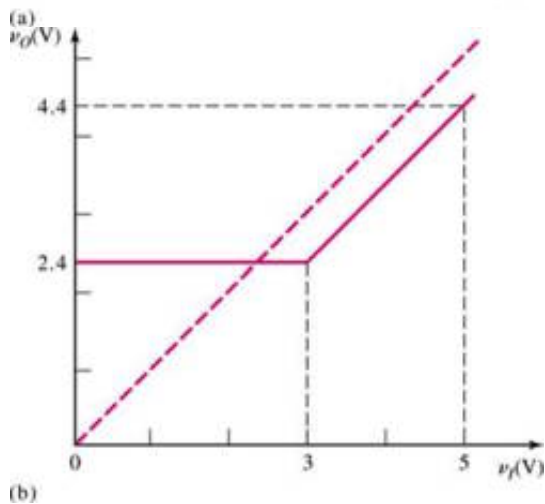
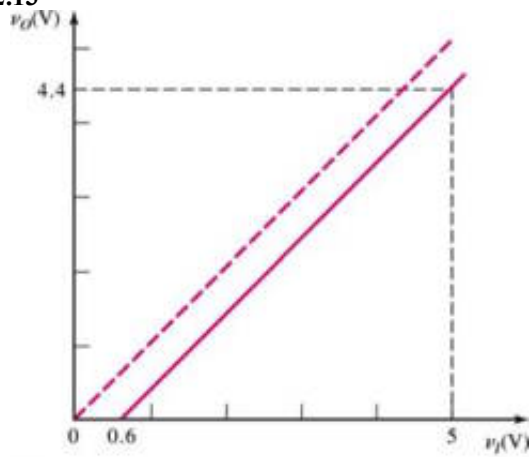
$$V_A = 14 - 0.7 - (0.7)(8) = 7.7 \text{ V} \Rightarrow D_2 \text{ cutoff}$$

$$I_{R3} = \frac{-0.7 - (-5)}{2.5} = 1.72 \text{ mA}$$

$$I_{D1} + I_{D3} = I_{R3}$$

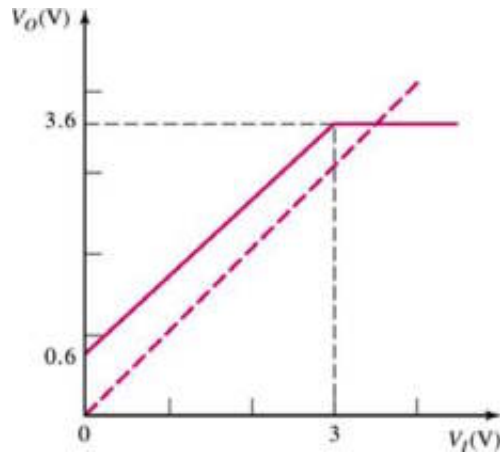
$$I_{D3} = I_{R3} - I_{D1} = 1.72 - 0.7 = 1.02 \text{ mA}$$

TYU2.13



TYU2.14

- (a) $V_O = 0.6\text{ V}$ for all V_I .
(b)



Chapter 3

Exercise Solutions

EX3.1

$$V_{TN} = 1 \text{ V}, V_{GS} = 3 \text{ V}, V_{DS} = 4.5 \text{ V}$$

$$V_{DS} = 4.5 > V_{DS}(\text{sat}) = V_{GS} - V_{TN} = 3 - 1 = 2 \text{ V}$$

Transistor biased in the saturation region

$$I_D = K_n (V_{GS} - V_{TN})^2 \Rightarrow 0.8 = K_n (3 - 1)^2 \Rightarrow K_n = 0.2 \text{ mA/V}^2$$

(a) $V_{GS} = 2 \text{ V}, V_{DS} = 4.5 \text{ V}$

Saturation region:

$$I_D = (0.2)(2 - 1)^2 \Rightarrow I_D = 0.2 \text{ mA}$$

(b) $V_{GS} = 3 \text{ V}, V_{DS} = 1 \text{ V}$

Nonsaturation region:

$$I_D = (0.2) \left[2(3 - 1)(1) - (1)^2 \right] \Rightarrow I_D = 0.6 \text{ mA}$$

EX3.2

$$0.5 = K_p (3 - 1.2)^2 \Rightarrow K_p = 0.154 \text{ mA/V}^2$$

(a) $i_D = K_p (v_{SG} + V_{TP})^2 = 0.154(2 - 1.2)^2 = 0.0986 \text{ mA}$

(b) $i_D = K_p [2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2] = 0.154[2(5 - 1.2)(2) - (2)^2] = 1.72 \text{ mA}$

EX3.3

$$V_{GS} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{DD} = \left(\frac{245}{245 + 355} \right) (2.2) = 0.8983 \text{ V}$$

$$I_D = (25)(0.8983 - 0.35)^2 = 7.52 \mu\text{A}$$

$$V_{DS} = 2.2 - (0.00752)(100) = 1.45 \text{ V}$$

EX3.4

$$I_{DQ} = 0.5 \text{ mA}, V_{SDQ} = 2.0 \text{ V}$$

$$R_D = \frac{3.3 - 2.0}{0.5} = 2.6 \text{ k}\Omega$$

$$I_D = K_p (V_{SGQ} + V_{TP})^2$$

$$0.5 = 0.2(V_{SGQ} - 0.6)^2 \Rightarrow V_{SGQ} = 2.18 \text{ V}$$

$$V_G = 3.3 - 2.18 = 1.12 \text{ V}$$

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{DD} = \frac{1}{R_1} (R_1 \parallel R_2) (3.3)$$

$$1.12 = \frac{1}{R_1} (300)(3.3) \Rightarrow R_1 = 884 \text{ k}\Omega$$

Then

$$\frac{884R_2}{884 + R_2} = 300 \Rightarrow R_2 = 454 \text{ k}\Omega$$

EX3.5

(a) $V_{GSQ} = \left(\frac{30}{30 + 60} \right) (5) = 1.667 \text{ V}$
 $I_{DQ} = (0.5)(1.667 - 0.6)^2 = 0.5689 \text{ mA}$
 $V_{DSQ} = 5 - (0.5689)(4) = 2.724 \text{ V}$

(b) $K_n(+5\%) = 0.525 \text{ mA/V}^2$, $V_{TN}(-5\%) = 0.57 \text{ V}$
 $I_D = 0.525(1.667 - 0.57)^2 = 0.6314 \text{ mA}$
 $V_{DS} = 5 - (0.6314)(4) = 2.474 \text{ V}$
 $K_n(-5\%) = 0.475 \text{ mA/V}^2$, $V_{TN}(+5\%) = 0.63 \text{ V}$
 $I_D = 0.475(1.667 - 0.63)^2 = 0.5105 \text{ mA}$
 $V_{DS} = 5 - (0.5105)(4) = 2.958 \text{ V}$

Then

$$0.5105 \leq I_D \leq 0.6314 \text{ mA}$$

$$2.474 \leq V_{DS} \leq 2.958 \text{ V}$$

EX3.6

(a) $V_G = \left(\frac{345}{345 + 255} \right) (4.4) - 2.2 = 0.330 \text{ V}$
 $2.2 = I_{DQ} R_S + V_{SGQ} + V_G$
 $2.2 = K_p R_S (V_{SGQ} + V_{TP})^2 + V_{SGQ} + 0.330$
 $1.87 = (0.035)(6)(V_{SGQ}^2 - 0.6V_{SGQ} + 0.09) + V_{SGQ}$
 $0.21V_{SGQ}^2 + 0.874V_{SGQ} - 1.8511 = 0 \Rightarrow V_{SGQ} = 1.545 \text{ V}$

We find

$$I_{DQ} = 35(1.545 - 0.3)^2 = 54.22 \mu\text{A}$$

$$V_{SDQ} = 4.4 - (0.05422)(6 + 42) = 1.797 \text{ V}$$

(b)

For $V_{TP} = -0.315 \text{ V}$

We have

$$1.87 = (0.035)(6)(V_{SGQ}^2 - 0.63V_{SGQ} + 0.099225) + V_{SGQ}$$

$$0.21V_{SGQ}^2 + 0.8677V_{SGQ} - 1.849 = 0 \Rightarrow V_{SGQ} = 1.550 \text{ V}$$

Then

$$I_{DQ} = 35(1.550 - 0.315)^2 = 53.36 \mu\text{A}$$

For $V_{TP} = -0.285 \text{ V}$

We have

$$1.87 = (0.035)(6)(V_{SGQ}^2 - 0.57V_{SGQ} + 0.081225) + V_{SGQ}$$

$$0.21V_{SGQ}^2 + 0.8803V_{SGQ} - 1.8529 = 0 \Rightarrow V_{SGQ} = 1.5395 \text{ V}$$

Then

$$I_{DQ} = 35(1.5395 - 0.285)^2 = 55.08 \mu\text{A}$$

Therefore

$$53.36 \leq I_{DQ} \leq 55.08 \mu\text{A}$$

EX3.7

$$4.4 = V_{SD}(\text{sat}) + I_D(6 + 42)$$

$$4.4 = V_{SG} + V_{TP} + (0.035)(48)(V_{SG} + V_{TP})^2$$

$$4.4 = V_{SG} - 0.3 + 1.68(V_{SG}^2 - 0.6V_{SG} + 0.09)$$

$$1.68V_{SG}^2 - 0.008V_{SG} - 4.5488 = 0 \Rightarrow V_{SG} = 1.648 \text{ V}$$

We find

$$I_D = 35(1.648 - 0.3)^2 = 63.59 \mu\text{A}$$

$$V_{SD} = 4.4 - (0.0636)(48) = 1.348 \text{ V}$$

$$\text{Note: } V_{SD}(\text{sat}) = 1.648 - 0.3 = 1.348 \text{ V}$$

EX3.8

$$(a) \quad I_{DQ} = 60 = 30(V_{SGQ} - 0.4)^2 \Rightarrow V_{SGQ} = 1.814 \text{ V}$$

$$I_{DQ} = \frac{3 - 1.814}{R_S} = 0.060 \Rightarrow R_S = 19.77 \text{ k}\Omega$$

$$V_D = 1.814 - 2.5 = -0.686 \text{ V}$$

$$R_D = \frac{-0.686 - (-3)}{0.060} = 38.57 \text{ k}\Omega$$

$$(b) \quad |V_{TP}|(+5\%) = 0.42 \text{ V}, \quad K_p(-5\%) = 28.5 \mu\text{A/V}^2$$

$$3 = I_D R_S + V_{SG} = (0.0285)(19.77)(V_{SG}^2 - 0.84V_{SG} + 0.1764) + V_{SG}$$

which yields

$$V_{SG} = 1.849 \text{ V}$$

$$I_D = (28.5)(1.849 - 0.42)^2 = 58.2 \mu\text{A}$$

$$V_{SD} = 6 - (0.0582)(19.77 + 38.57) = 2.605 \text{ V}$$

$$|V_{TP}|(-5\%) = 0.38 \text{ V}, \quad K_p(+5\%) = 31.5 \mu\text{A/V}^2$$

$$3 = (0.0315)(19.77)(V_{SG}^2 - 0.76V_{SG} + 0.1444) + V_{SG}$$

which yields

$$V_{SG} = 1.780 \text{ V}$$

$$I_D = 31.5(1.780 - 0.38)^2 = 61.72 \mu\text{A}$$

$$V_{SD} = 6 - (0.06172)(19.77 + 38.57) = 2.399 \text{ V}$$

Then

$$58.2 \leq I_D \leq 61.72 \mu\text{A}$$

$$2.399 \leq V_{SD} \leq 2.605 \text{ V}$$

EX3.9

- (a) $V_I = 4$ V, Driver in Non · Sat.

$$K_{nD} [2(V_I - V_{TND})V_O - V_O^2] = K_{nL} [V_{DD} - V_O - V_{TNL}]^2$$

$$5[2(4-1)V_D - V_D^2] = (5 - V_D - 1)^2 = (4 - V_O)^2 = 16 - 8V_O + V_O^2$$

$$6V_D^2 - 38V_O + 16 = 0$$

$$V_D = \frac{38 \pm \sqrt{1444 - 384}}{2(6)}$$

$$V_D = 0.454 \text{ V}$$

- (b) $V_I = 2$ V Driver: Sat

$$K_{nD} [V_I - V_{TND}]^2 = K_{nL} [V_{DD} - V_O - V_{TNL}]^2$$

$$5[2-1]^2 = [5 - V_O - 1]^2$$

$$\sqrt{5} = 4 - V_O \Rightarrow V_O = 1.76 \text{ V}$$

EX3.10

- (a) For $V_I = 5$ V, Load in saturation and driver in nonsaturation.

$$I_{DD} = I_{DL}$$

$$K_{nD} [2(V_I - V_{TND})V_O - V_O^2] = K_{nL} (-V_{TNL})^2$$

$$\frac{K_{nD}}{K_{nL}} [2(5-1)(0.25) - (0.25)^2] = 4 \Rightarrow \frac{K_{nD}}{K_{nL}} = 2.06$$

- (b)

$$I_{DL} = K_{nL} (-V_{TNL})^2 \Rightarrow 0.2 = K_{nL} [-(-2)]^2$$

$$\underline{K_{nL} = 50 \mu A / V^2} \text{ and } \underline{K_{nD} = 103 \mu A / V^2}$$

EX3.11

For M_N

$$I_{DN} = I_{DP}$$

$$K_n (V_{GSN} - V_{TN})^2 = K_p (V_{SGP} + V_{TP})^2$$

$$V_{GSN} = 1 + (5 - 3.25 - 1) = 1.75 \text{ V} = V_I$$

$$V_o = V_{DSN} (\text{sat}) = 1.75 - 1 \Rightarrow V_o = 0.75 \text{ V}$$

For M_P : $V_I = 1.75$ V

$$V_{DD} - V_O = V_{SD} (\text{sat}) = V_{SGP} + V_{TP} = (5 - 3.25) - 1 = 0.75 \text{ V}$$

$$\text{So } V_{ot} = 5 - 0.75 \Rightarrow V_{ot} = 4.25 \text{ V}$$

EX3.12

Transistor in nonsaturation

- (a) $V_{DS} = 0.2$ V, $V_{GS} = 5$ V

$$V_O = V_{DD} - I_D R_D = V_{DD} - K_n R_D [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$$

$$0.2 = 5 - K_n (0.5) [2(5-1)(0.2) - (0.2)^2] \Rightarrow K_n = 6.154 \text{ mA/V}^2$$

$$(b) \quad I_D = (6.154) \left[2(5-1)(0.2) - (0.2)^2 \right] = 9.60 \text{ mA}$$

$$P = I_D V_{DS} = (9.60)(0.2) = 1.92 \text{ mW}$$

EX3.13

a. $V_1 = 5 \text{ V}, \quad V_2 = 0, \quad M_2 \text{ cutoff} \Rightarrow I_{D2} = 0$

$$I_D = K_n \left[2(V_I - V_{TN})V_O - V_O^2 \right] = \frac{5 - V_O}{R_D}$$

$$(0.05)(30) \left[2(5-1)V_O - V_O^2 \right] = 5 - V_O$$

$$1.5V_O^2 - 13V_O + 5 = 0$$

$$V_O = \frac{13 \pm \sqrt{(13)^2 - 4(1.5)(5)}}{2(1.5)} \Rightarrow \underline{V_O = 0.40 \text{ V}}$$

$$I_R = I_{D1} = \frac{5 - 0.40}{30} \Rightarrow \underline{I_R = I_{D1} = 0.153 \text{ mA}}$$

b. $V_1 = V_2 = 5 \text{ V}$

$$\frac{5 - V_O}{R_D} = 2 \left\{ K_n \left[2(V_I - V_{TN})V_O - V_O^2 \right] \right\}$$

$$5 - V_O = 2(0.05)(30) \left[2(5-1)V_O - V_O^2 \right]$$

$$3V_O^2 - 25V_O + 5 = 0$$

$$V_O = \frac{25 \pm \sqrt{(25)^2 - 4(3)(5)}}{2(3)} \Rightarrow \underline{V_O = 0.205 \text{ V}}$$

$$I_R = \frac{5 - 0.205}{30} \Rightarrow \underline{I_R = 0.160 \text{ mA}}$$

$$\underline{I_{D1} = I_{D2} = 0.080 \text{ mA}}$$

EX3.14

$$V_{GS3} = \sqrt{\frac{I_{REF1}}{K_{n3}}} + V_{TN} = \sqrt{\frac{120}{60}} + 0.4 = 1.814 \text{ V}$$

$$V_{GS2} = V_{GS3} = 1.814 \text{ V}$$

$$I_{Q1} = K_{n2} (V_{GS2} - V_{TN})^2 = 30(1.814 - 0.4)^2 = 60 \mu\text{A}$$

$$V_{GS1} = \sqrt{\frac{I_{Q1}}{K_{n1}}} + V_{TN} = \sqrt{\frac{60}{50}} + 0.4 = 1.495 \text{ V}$$

EX3.15

$$0.1 = \left(\frac{0.04}{2} \right) (15) (V_{SGC} - 0.6)^2$$

$$V_{SGC} = 1.177 \text{ V} = V_{SGB}$$

$$0.2 = \left(\frac{0.04}{2} \right) \left(\frac{W}{L} \right)_B (1.177 - 0.6)^2$$

$$\left(\frac{W}{L} \right)_B = 30$$

$$0.2 = \left(\frac{0.04}{2} \right) (25) (V_{SGA} - 0.6)^2$$

$$V_{SGA} = 1.23 \text{ V}$$

EX3.16

$$I_{REF} = K_{n3} (V_{GS3} - V_{TN})^2 = K_{n4} (V_{GS4} - V_{TN})^2$$

$$V_{GS3} = 2 \text{ V} \Rightarrow V_{GS4} = 3 \text{ V}$$

$$(a) \quad (2-1)^2 = \frac{K_{n4}}{K_{n3}} (3-1)^2 \Rightarrow \frac{K_{n4}}{K_{n3}} = \frac{1}{4}$$

$$I_Q = K_{n2} (V_{GS2} - V_{TN})^2$$

$$\text{But } V_{GS2} = V_{GS3} = 2 \text{ V}$$

$$(b) \quad 0.1 = K_{n2} (2-1)^2 \Rightarrow K_{n2} = 0.1 \text{ mA/V}^2$$

$$0.2 = K_{n3} (2-1)^2 \Rightarrow K_{n3} = 0.2 \text{ mA/V}^2$$

$$(c) \quad 0.2 = K_{n4} (3-1)^2 \Rightarrow K_{n4} = 0.05 \text{ mA/V}^2$$

EX3.17

$$V_{S2} = 5 - 5 = 0 \quad R_{S2} = \frac{5}{0.3} = 16.7 \text{ K}$$

$$I_{D2} = K_{n2} (V_{GS2} - V_{TN2})^2$$

$$0.3 = 0.2 (V_{GS2} - 1.2)^2 \Rightarrow V_{GS2} = 2.425 \text{ V} \Rightarrow V_{G2} = V_{GS2} + V_S = 2.425 \text{ V}$$

$$R_{D1} = \frac{5 - 2.425}{0.1} = 25.8 \text{ K}$$

$$V_{S1} = V_{G2} - V_{DSQ1} = 2.425 - 5 = -2.575 \text{ V}$$

$$R_{S1} = \frac{-2.575 - (-5)}{0.1} \Rightarrow R_{S1} = 24.3 \text{ K}$$

$$I_{D1} = K_{n1} (V_{GS1} - V_{TN1})^2$$

$$0.1 = 0.5 (V_{GS1} - 1.2)^2 \Rightarrow V_{GS1} = 1.647 \text{ V}$$

$$V_{G1} = V_{GS1} + V_{S1} = 1.647 + (-2.575) \Rightarrow V_{G1} = -0.928 \text{ V}$$

$$V_{G1} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \frac{1}{R_1} \cdot R_{TN} \cdot (10) - 5$$

$$\begin{aligned}-0.928 &= \frac{1}{R_1}(200)(10) - 5 \Rightarrow R_1 = 491 \text{ K} \\ \frac{491 R_2}{491 + R_2} &= 200 \Rightarrow R_2 = 337 \text{ K}\end{aligned}$$

EX3.18

$$\begin{aligned}V_{S1} &= I_D R_S - 5 = (0.25)(16) - 5 = -1 \text{ V} \\ I_{DQ} &= K_n (V_{GS1} - V_{TN})^2 \Rightarrow 0.25 = 0.5(V_{GS1} - 0.8)^2 \Rightarrow V_{GS1} = 1.507 \text{ V} \\ V_{G1} &= V_{GS1} + V_{S1} = 1.507 - 1 = 0.507 \text{ V} \\ V_{G1} &= \left(\frac{R_3}{R_1 + R_2 + R_3} \right) (5) \Rightarrow 0.507 = \frac{R_3}{500} (5) \Rightarrow R_3 = 50.7 \text{ K} \\ V_{S2} &= V_{S1} + V_{DS1} = -1 + 2.5 = 1.5 \text{ V} \\ V_{G2} &= V_{S2} + V_{GS} = 1.5 + 1.507 = 3.007 \text{ V} \\ V_{G2} &= \left(\frac{R_2 + R_3}{R_1 + R_2 + R_3} \right) (5) \Rightarrow 3.007 = \left(\frac{R_2 + R_3}{500} \right) (5) \\ R_2 + R_3 &= 300.7 \\ R_2 &= 300.7 - 50.7 \Rightarrow R_2 = 250 \text{ K} \\ R_1 &= 500 - 250 - 50.7 \Rightarrow R_1 = 199.3 \text{ K} \\ V_{D2} &= V_{S2} + V_{DS2} = 1.5 + 2.5 = 4 \text{ V} \\ R_D &= \frac{5 - 4}{0.25} \Rightarrow R_D = 4 \text{ K}\end{aligned}$$

EX3.19

$$\begin{aligned}V_{DS}(\text{sat}) &= V_{GS} - V_P = -1.2 - (-4.5) \Rightarrow V_{DS}(\text{sat}) = 3.3 \text{ V} \\ I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 12 \left(1 - \frac{(-1.2)}{(-4.5)} \right)^2 \Rightarrow I_D = 6.45 \text{ mA}\end{aligned}$$

EX3.20

$$\begin{aligned}
 I_D &= 2.5 \text{ mA} \\
 I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\
 2.5 &= 6 \left(1 - \frac{V_{GS}}{(-4)} \right)^2 \Rightarrow V_{GS} = -1.42 \text{ V} \\
 V_S &= I_D R_S - 5 = (2.5)(0.25) - 5 \\
 V_S &= -4.375 \\
 V_{DS} &= 6 \Rightarrow V_D = 6 - 4.375 = 1.625 \\
 R_D &= \frac{5 - 1.625}{2.5} \Rightarrow \underline{R_D = 1.35 \text{ k}\Omega} \\
 \frac{(20)^2}{R_1 + R_2} &= 2 \Rightarrow R_1 + R_2 = 200 \text{ k}\Omega \\
 V_G &= V_{GS} + V_S = -1.42 - 4.375 = -5.795 \\
 V_G &= \left(\frac{R_2}{R_1 + R_2} \right) (20) - 10 \\
 -5.795 &= \left(\frac{R_2}{200} \right) (20) - 10 \Rightarrow \underline{R_2 = 42.05 \text{ k}\Omega} \rightarrow \underline{42 \text{ k}\Omega} \\
 \underline{R_1 = 157.95 \text{ k}\Omega} &\rightarrow \underline{158 \text{ k}\Omega}
 \end{aligned}$$

EX3.21

$$\begin{aligned}
 V_S &= -V_{GS}, \quad I_D = \frac{0 - V_S}{R_S} = \frac{V_{GS}}{R_S} \\
 I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\
 \frac{V_{GS}}{1} &= 6 \left(1 - \frac{V_{GS}}{4} \right)^2 = 6 \left(1 - \frac{V_{GS}}{2} + \frac{V_{GS}^2}{16} \right) \\
 0.375 V_{GS}^2 - 4 V_{GS} + 6 &= 0 \\
 V_{GS} &= \frac{4 \pm \sqrt{16 - 4(0.375)(6)}}{2(0.375)} \\
 \underbrace{V_{GS} = 8.86}_{\text{impossible}} \text{ or } \underline{V_{GS} = 1.806 \text{ V}} \\
 I_D &= \frac{V_{GS}}{R_S} = 1.806 \text{ mA} \\
 V_D &= I_D R_D - 5 = (1.81)(0.4) - 5 = -4.278 \\
 V_{SD} &= V_S - V_0 = -1.81 - (-4.276) \Rightarrow \underline{V_{SD} = 2.47 \text{ V}} \\
 V_{SD}(\text{sat}) &= V_P - V_{GS} = 4 - 1.81 = 2.19 \\
 \text{So } V_{SD} &> V_{SD}(\text{sat})
 \end{aligned}$$

EX3.22

$$R_{ib} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} = 100 \text{ k}\Omega$$

$$I_{DQ} = 5 \text{ mA}, \quad V_S = -I_{DQ} R_S = -(5)(1.2) = -6 \text{ V}$$

$$V_{SDQ} = 12 \text{ V}, \quad V_D = V_S - V_{SDQ} \\ = -6 - 12 = -18 \text{ V}$$

$$R_D = \frac{-18 - (-20)}{5} \Rightarrow R_D = 0.4 \text{ k}\Omega$$

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \Rightarrow 5 = 8 \left(1 - \frac{V_{GS}}{4} \right)^2$$

$$V_{GS} = 0.838 \text{ V}$$

$$V_G = V_{GS} + V_S = 0.838 - 6 = -5.162$$

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) (-20)$$

$$-5.162 = \frac{1}{R_1} (100)(-20) \Rightarrow R_1 = 387 \text{ k}\Omega$$

$$\frac{R_1 R_2}{R_1 + R_2} = 100 \Rightarrow (387) R_2 = 100(387) + 100 R_2$$

$$(387 - 100) R_2 = (100)(387) \Rightarrow R_2 = 135 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU3.1

$$V_{TN} = 1.2 \text{ V}, \quad V_{GS} = 2 \text{ V}$$

(a) $V_{DS}(\text{sat}) = V_{GS} - V_{TN} = 2 - 1.2 = 0.8 \text{ V}$

(i) $V_{DS} = 0.4 \Rightarrow$ Nonsaturation

(ii) $V_{DS} = 1 \Rightarrow$ Saturation

(iii) $V_{DS} = 5 \Rightarrow$ Saturation

$$V_{TN} = -1.2 \text{ V}, \quad V_{GS} = 2 \text{ V}$$

(b) $V_{DS}(\text{sat}) = V_{GS} - V_{TN} = 2 - (-1.2) = 3.2 \text{ V}$

(i) $V_{DS} = 0.4 \Rightarrow$ Nonsaturation

(ii) $V_{DS} = 1 \Rightarrow$ Nonsaturation

(iii) $V_{DS} = 5 \Rightarrow$ Saturation

TYU3.2

$$(a) \quad K_n = \frac{W\mu_n \epsilon_{ox}}{2Lt_{ox}} = \frac{(20 \times 10^{-4})(500)(3.9)(8.85 \times 10^{-14})}{2(0.8 \times 10^{-4})(200 \times 10^{-8})} \Rightarrow 1.08 \text{ mA/V}^2$$

(b)

$$(i) \quad i_D = (1.08)[2(2 - 1.2)(0.4) - (0.4)^2] = 0.518 \text{ mA}$$

$$(ii) \quad i_D = (1.08)(2 - 1.2)^2 = 0.691 \text{ mA}$$

$$(iii) \quad i_D = (1.08)(2 - 1.2)^2 = 0.691 \text{ mA}$$

$$(i) \quad i_D = (1.08)[2(2 + 1.2)(0.4) - (0.4)^2] = 2.59 \text{ mA}$$

$$(ii) \quad i_D = (1.08)[2(2 + 1.2)(1) - (1)^2] = 5.83 \text{ mA}$$

$$(iii) \quad i_D = (1.08)(2 + 1.2)^2 = 11.1 \text{ mA}$$

TYU3.3

$$(a) \quad V_{SD}(\text{sat}) = V_{SG} + V_{TP} = 2 - 1.2 = 0.8 \text{ V}$$

(i) Non Sat (ii) Sat (iii) Sat

$$(b) \quad V_{SD}(\text{sat}) = 2 + 1.2 = 3.2 \text{ V}$$

(i) Non Sat (ii) Non Sat (iii) Sat

TYU3.4

$$(a) \quad K_p = \frac{W\mu_p \epsilon_{ox}}{2Lt_{ox}} = \frac{(10 \times 10^{-4})(300)(3.9)(8.85 \times 10^{-14})}{2(0.8 \times 10^{-4})(200 \times 10^{-8})} \Rightarrow 0.324 \text{ mA/V}^2$$

(b)

$$(i) \quad i_D = (0.324)[2(2 - 1.2)(0.4) - (0.4)^2] = 0.156 \text{ mA}$$

$$(ii) \quad i_D = (0.324)(2 - 1.2)^2 = 0.207 \text{ mA}$$

$$(iii) \quad i_D = (0.324)(2 - 1.2)^2 = 0.207 \text{ mA}$$

$$(i) \quad i_D = (0.324)[2(2 + 1.2)(0.4) - (0.4)^2] = 0.778 \text{ mA}$$

$$(ii) \quad i_D = (0.324)[2(2 + 1.2)(1) - (1)^2] = 1.75 \text{ mA}$$

$$(iii) \quad i_D = (0.324)(2 + 1.2)^2 = 3.32 \text{ mA}$$

TYU3.5

(a), (i) (ii)

$$i_D = (10)(0.5 - 0.25)^2 = 0.625 \mu\text{A}$$

(b)

$$(i) \quad i_D = K_n (\nu_{GS} - V_{TN})^2 (1 + \lambda \nu_{DS})$$

$$i_D = (10)(0.5 - 0.25)^2 [1 + (0.03)(0.5)] = 0.6344 \mu\text{A}$$

$$(ii) \quad i_D = (10)(0.5 - 0.25)^2 [1 + (0.03)(1.2)] = 0.6475 \mu\text{A}$$

(c) For (a),

$$r_o = \infty$$

$$\text{For (b), } r_o = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.03)(0.625)} = 53.3 \text{ M}\Omega$$

TYU3.6

(a) $V_{TN} = V_{TNO} = 0.4 \text{ V}$

(b) $V_{TN} = V_{TNO} + \gamma \left[\sqrt{2\phi_f + v_{SB}} - \sqrt{2\phi_f} \right]$
 $= 0.4 + 0.15 \left[\sqrt{2(0.35) + 0.5} - \sqrt{2(0.35)} \right] \Rightarrow V_{TN} = 0.439 \text{ V}$

(c) $V_{TN} = 0.4 + 0.15 \left[\sqrt{2(0.35) + 1.5} - \sqrt{2(0.35)} \right] \Rightarrow V_{TN} = 0.497 \text{ V}$

TYU3.7

$$V_{DS} = 2.2 - (0.07)R_D = 1.2 \Rightarrow R_D = 14.3 \text{ k}\Omega$$

$$V_{GS} = \sqrt{\frac{I_{DQ}}{K_n}} + V_{TN} = \sqrt{\frac{70}{30}} + 0.25 = 1.778 \text{ V}$$

$$V_{GS} = 1.778 = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{DD} = \left(\frac{R_2}{500} \right) (2.2)$$

We find

$$R_2 = 404 \text{ k}\Omega, \quad R_1 = 96 \text{ k}\Omega$$

TYU3.8

$$I_D = \frac{3.3 - 1.6}{10} = 0.17 \text{ mA}$$

$$0.17 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right) (1.6 - 0.4)^2 \Rightarrow \frac{W}{L} = 2.36$$

TYU3.9

(a) The transition point is

$$V_{It} = \frac{V_{DD} - V_{TNL} + V_{TND} \left(1 + \sqrt{K_{nD} / K_{nL}} \right)}{1 + \sqrt{K_{nD} / K_{nL}}}$$

$$= \frac{5 - 1 + 1 \left(1 + \sqrt{0.05/0.01} \right)}{1 + \sqrt{0.05/0.01}}$$

$$= \frac{7.236}{3.236} \Rightarrow V_{It} = 2.236 \text{ V}$$

$$V_{Ot} = V_{It} - V_{TND} = 2.24 - 1 \Rightarrow V_{Ot} = 1.24 \text{ V}$$

(b) We may write

$$I_D = K_{nD} (V_{GSD} - V_{TND})^2 = (0.05)(2.236 - 1)^2 \Rightarrow I_D = 76.4 \mu\text{A}$$

TYU3.10

- (a) $V_{DS} > [0 - (-1.2)] \Rightarrow$ Saturation

$$I_D = \frac{3.3 - 1.8}{8} = 0.1875 \text{ mA}$$

$$0.1875 = \left(\frac{0.08}{2} \right) \left(\frac{W}{L} \right) [0 - (-1.2)]^2 \Rightarrow \frac{W}{L} = 3.26$$

- (b) $V_{DS} < [0 - (-1.2)] \Rightarrow$ Nonsaturation

$$I_D = \frac{3.3 - 0.8}{8} = 0.3125 \text{ mA}$$

$$0.3125 = \left(\frac{0.08}{2} \right) \left(\frac{W}{L} \right) [2(0 - (-1.2))(0.8) - (0.8)^2] \Rightarrow \frac{W}{L} = 6.10$$

TYU3.11

- (a) Transition point for the load transistor – Driver is in the saturation region.

$$I_{DD} = I_{DL}$$

$$K_{nD} (V_{GSD} - V_{TND})^2 = K_{nL} (V_{GSL} - V_{TNL})^2$$

$$V_{DSL}(\text{sat}) = V_{GSL} - V_{TNL} = -V_{TNL} \Rightarrow V_{DSL} = V_{DD} - V_{Ot} = 2 \text{ V}$$

$$\text{Then } V_{Ot} = 5 - 2 = 3 \text{ V}, \quad \underline{V_{Ot} = 3 \text{ V}}$$

$$\sqrt{\frac{K_{nD}}{K_{nL}}} (V_{It} - 1) = (-V_{TNL})$$

$$\sqrt{\frac{0.08}{0.01}} (V_{It} - 1) = 2 \Rightarrow \underline{V_{It} = 1.89 \text{ V}}$$

- (b) For the driver:

$$V_{Ot} = V_{It} - V_{TND}$$

$$\underline{V_{It} = 1.89 \text{ V}, \quad V_{Ot} = 0.89 \text{ V}}$$

TYU3.12

Transistor biased in nonsaturation

$$I_D = K_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2] = (4) [2(10 - 0.7)(0.2) - (0.2)^2] = 14.72 \text{ mA}$$

$$R_D = \frac{10 - 0.2}{14.72} = 0.666 \text{ k}\Omega$$

TYU3.13

- (a) Transistor biased in the nonsaturation region

$$I_D = \frac{5 - 1.5 - V_{DS}}{R} = 12$$

$$I_D = K_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$$

$$12 = 4 [2(5 - 0.8)V_{DS} - V_{DS}^2]$$

$$4V_{DS}^2 - 33.6V_{DS} + 12 = 0 \Rightarrow \underline{V_{DS} = 0.374 \text{ V}}$$

$$\text{Then } R = \frac{5 - 1.5 - 0.374}{12} \Rightarrow \underline{R = 261 \Omega}$$

TYU3.14

$$I_D = \frac{5 - V_O}{R_D} = K_n \left[2(V_2 - V_{TN})V_O - V_O^2 \right]$$

$$\text{a. } \frac{5 - (0.10)}{25} = K_n \left[2(5 - 1)(0.10) - (0.10)^2 \right] \Rightarrow K_n = 0.248 \text{ mA/V}^2$$

$$\frac{5 - V_0}{25} = 2(0.248) \left[2(5 - 1)V_0 - V_0^2 \right]$$

$$5 - V_0 = 12.4 \left[8V_0 - V_0^2 \right]$$

$$12.4V_0^2 - 100.2V_0 + 5 = 0$$

$$\text{b. } V_0 = \frac{100.2 \pm \sqrt{(100.2)^2 - 4(12.4)(5)}}{2(12.4)} \Rightarrow V_0 = 0.0502 \text{ V}$$

TYU3.15

$$V_{SGC} = \sqrt{\frac{I_{REF2}}{K_{pC}}} - V_{TP} = \sqrt{\frac{40}{40}} + 0.3 \Rightarrow V_{SGC} = V_{SGB} = 1.30 \text{ V}$$

$$I_{Q2} = K_{pB} (V_{SGB} + V_{TP})^2 = 60(1.30 - 0.3)^2 = 60 \mu\text{A}$$

$$V_{SGA} = \sqrt{\frac{I_{Q2}}{K_{pA}}} - V_{TP} = \sqrt{\frac{60}{75}} + 0.3 = 1.19 \text{ V}$$

TYU3.16

$$I_Q = K_{n1} (V_{GS1} - V_{TN})^2$$

$$120 = K_{n1} (1.5 - 0.7)^2 \Rightarrow K_{n1} = 187.5 \mu\text{A/V}^2$$

$$V_{GS2} = V_{GS3} = 2 \text{ V}$$

$$120 = K_{n2} (2 - 0.7)^2 \Rightarrow K_{n2} = 71.0 \mu\text{A/V}^2$$

$$I_{REF} = K_{n3} (V_{GS3} - V_{TN})^2$$

$$80 = K_{n3} (2 - 0.7)^2 \Rightarrow K_{n3} = 47.3 \mu\text{A/V}^2$$

$$V_{GS4} = 5 - 2 = 3 \text{ V}$$

$$80 = K_{n4} (3 - 0.7)^2 \Rightarrow K_{n4} = 15.12 \mu\text{A/V}^2$$

TYU3.17

$$I_{DQ} = K (V_{GS} - V_{TN})^2 \Rightarrow 5 = 50 (V_{GS} - 0.15)^2 \Rightarrow V_{GS} = 0.466 \text{ V}$$

$$V_S = (0.005)(10) = 0.050 \text{ V} \Rightarrow V_{GG} = V_{GS} + V_S = 0.466 + 0.050 \Rightarrow V_{GG} = 0.516 \text{ V}$$

$$V_D = 5 - (0.005)(100) \Rightarrow V_D = 4.5 \text{ V}$$

$$V_{DS} = V_D - V_S = 4.5 - 0.050 \Rightarrow V_{DS} = 4.45 \text{ V}$$

TYU3.18

$$\begin{aligned} I_D &= K \left[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2 \right] \\ &= 100 \left[2(0.7 - 0.2)(0.1) - (0.1)^2 \right] \end{aligned}$$

$$I_D = 9 \mu\text{A}$$

$$R_D = \frac{2.5 - 0.1}{0.009} \Rightarrow \underline{R_D = 267 \text{ k}\Omega}$$

Chapter 4

Exercise Solutions

EX4.1

$$g_m = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)I_{DQ}}$$

$$1.8 = 2\sqrt{\left(\frac{0.1}{2}\right)\left(\frac{W}{L}\right)(0.8)} \Rightarrow \frac{W}{L} = 20.25$$

EX4.2

(a) $K_n = \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right) = \left(\frac{0.1}{2}\right)(50) = 2.5 \text{ mA/V}^2$

$$I_{DQ} = K_n(V_{GSQ} - V_{TN})^2$$

$$0.25 = 2.5(V_{GSQ} - 0.4)^2 \Rightarrow V_{GSQ} = 0.716 \text{ V}$$

$$V_{DSQ} = V_{DD} - I_{DQ}R_D = 3.3 - (0.25)(10) = 0.8 \text{ V}$$

$$V_{DS}(\text{sat}) = V_{GS} - V_{TN} = 0.717 - 0.4 = 0.316 \text{ V} \Rightarrow V_{DS} > V_{DS}(\text{sat})$$

(b)

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(2.5)(0.25)} = 1.58 \text{ mA/V}$$

$$r_o = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.025)(0.25)} = 160 \text{ k}\Omega$$

(c) $A_v = -g_m(r_o \parallel R_D) = -(1.58)(160 \parallel 10) = -14.9$

EX4.3

$$V_{GS} = \left(\frac{R_2}{R_1 + R_2}\right)V_{DD} = \left(\frac{320}{520 + 320}\right)(5) = 1.905 \text{ V}$$

$$I_{DQ} = 0.20(1.905 - 0.8)^2 = 0.244 \text{ mA}$$

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.2)(0.244)} = 0.442 \text{ mA/V}$$

(a) $r_o = \infty$

(b) $A_v = -g_m R_D = -(0.422)(10) = -4.22$

(c) $R_i = R_1 \parallel R_2 = 520 \parallel 320 = 198 \text{ k}\Omega$

(d) $R_O = R_D = 10 \text{ k}\Omega$

EX4.4

At transition point, $I_D = 1 \text{ mA}$

$$I_D = K_n(V_{GS} - V_{TN})^2 = K_n(V_{DS}(\text{sat}))^2$$

$$1 = 0.2(V_{DS}(\text{sat}))^2 \Rightarrow V_{DS}(\text{sat}) = 2.236 \text{ V}$$

$$\text{Want } V_{DSQ} = \frac{5 - 2.236}{2} + 2.236 = 3.62 \text{ V}$$

$$R_D = \frac{5 - 3.62}{0.5} = 2.76 \text{ k}\Omega$$

$$0.5 = 0.2(V_{GSQ} - 0.8)^2 \Rightarrow V_{GSQ} = 2.38 \text{ V}$$

$$V_{GSQ} = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} = \frac{1}{R_1} (R_1 \parallel R_2) V_{DD}$$

$$\text{So } 2.38 = \frac{1}{R_1} (200)(5) \Rightarrow R_1 = 420 \text{ K and } R_2 = 382 \text{ K}$$

$$A_v = -g_m R_D$$

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.2)(0.5)} = 0.6325 \text{ mA/V}$$

$$A_v = -(0.6325)(2.76) \\ = -1.75$$

EX4.5

$$(a) \quad V_G = \left(\frac{35}{35 + 165} \right) (10) - 5 = -3.25 \text{ V}$$

$$V_G = V_{GS} + I_D R_S - 5$$

So

$$5 - 3.25 = V_{GS} + K_n R_S (V_{GS} - V_{TN})^2$$

$$1.75 = V_{GS} + (1)(0.5)(V_{GS}^2 - 1.6V_{GS} + 0.64)$$

or

$$0.5V_{GS}^2 + 0.2V_{GS} - 1.43 = 0 \Rightarrow V_{GS} = 1.503 \text{ V}$$

Then

$$I_{DQ} = (1)(1.503 - 0.8)^2 = 0.4942 \text{ mA}$$

$$V_{DSQ} = 10 - (0.4942)(7 + 0.5) = 6.29 \text{ V}$$

(b)

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1)(0.4942)} = 1.406 \text{ mA/V}$$

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} = \frac{-(1.406)(7)}{1 + (1.406)(0.5)} = -5.78$$

EX4.6

$$(a) \quad K_p = \left(\frac{k'_p}{2} \right) \left(\frac{W}{L} \right) = \left(\frac{0.04}{2} \right) (40) = 0.80 \text{ mA/V}^2$$

$$3 = K_p R_S (V_{SGQ} + V_{TP})^2 + V_{SGQ}$$

$$3 = (0.8)(1.2)(V_{SGQ}^2 - 0.8V_{SGQ} + 0.16) + V_{SGQ}$$

or

$$0.96V_{SGQ}^2 + 0.232V_{SGQ} - 2.846 = 0 \Rightarrow V_{SGQ} = 1.605 \text{ V}$$

$$I_{DQ} = (0.8)(1.605 - 0.4)^2 = 1.162 \text{ mA}$$

$$V_{SDQ} = 6 - (1.162)(1.2 + 2) = 2.283 \text{ V}$$

(b)

$$g_m = 2\sqrt{K_p I_{DQ}} = 2\sqrt{(0.8)(1.162)} = 1.928 \text{ mA/V}$$

$$r_o = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.02)(1.162)} = 43.03 \text{ k}\Omega$$

$$A_v = -g_m(r_o \parallel R_D) = -(1.928)(43.03 \parallel 2) = -3.68$$

EX4.7

$$V_{DSQ} = V_{DD} - I_{DQ} R_S$$

$$5 = 10 - (1.5) R_S \Rightarrow R_S = 3.33 \text{ k}\Omega$$

$$I_{DQ} = K_n (V_{GS} - V_{TN})^2 \Rightarrow 1.5 = (1)(V_{GS} - 0.8)^2$$

$$V_{GS} = 2.025 \text{ V} = V_G - V_S = V_G - 5 \Rightarrow V_G = 7.025 \text{ V} = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} = \frac{R_2}{400} \cdot 10$$

$$\text{So } R_2 = 281 \text{ k}\Omega, \quad R_1 = 119 \text{ k}\Omega$$

$$\text{Neglecting } R_{Si}, \quad A_v = \frac{g_m (R_S \parallel r_o)}{1 + g_m (R_S \parallel r_o)}$$

$$r_o = [\lambda I_{DQ}]^{-1} = [(0.015)(1.5)]^{-1} = 44.4 \text{ k}\Omega$$

$$R_S \parallel r_o = 3.33 \parallel 44.4 = 3.1 \text{ k}\Omega$$

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1)(1.5)} = 2.45 \text{ mA/V}$$

$$A_v = \frac{(2.45)(3.1)}{1 + (2.45)(3.1)} \Rightarrow A_v = 0.884$$

EX4.8

$$(a) \quad V_{SG} = V_{DD} - I_D R_S = 5 - (1.5)(2) = 2 \text{ V}$$

$$I_D = \left(\frac{k'_p}{2} \right) \left(\frac{W}{L} \right) (V_{SG} + V_{TP})^2$$

$$1.5 = \left(\frac{0.04}{2} \right) \left(\frac{W}{L} \right) (2 - 1.2)^2 \Rightarrow \frac{W}{L} = 117$$

$$(b) \quad K_p = \left(\frac{k'_p}{2} \right) \left(\frac{W}{L} \right) = \left(\frac{0.04}{2} \right) (117) = 2.344 \text{ mA/V}^2$$

$$g_m = 2\sqrt{K_p I_D} = 2\sqrt{(2.344)(1.5)} = 3.75 \text{ mA/V}$$

$$A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{(3.75)(2)}{1 + (3.75)(2)} = 0.882$$

$$(c) \quad A_v = \frac{g_m (R_S \parallel R_L)}{1 + g_m (R_S \parallel R_L)} = (0.9)(0.882) = 0.794$$

Then

$$(0.794)[1 + (3.75)(R_S \parallel R_L)] = (3.75)(R_S \parallel R_L) \Rightarrow R_S \parallel R_L = 2 \parallel R_L = 1.028$$

So

$$R_L = 2.12 \text{ k}\Omega$$

EX4.9

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} = \left(\frac{9.3}{70.7 + 9.3} \right) (5) \\ = 0.581 \text{ V}$$

$$I_{DQ} = K_p (V_{SG} - |V_{TP}|)^2 = K_p (V_S - V_G - |V_{TP}|)^2 \\ = \frac{5 - V_S}{R_S}$$

$$\text{Then } (0.4)(5)(V_S - 0.581 - 0.8)^2 = 5 - V_S$$

$$2(V_S - 1.381)^2 = 5 - V_S$$

$$2(V_S^2 - 2.762V_S + 1.907) = 5 - V_S$$

$$2V_S^2 - 4.52V_S - 1.19 = 0$$

$$V_S = \frac{4.52 \pm \sqrt{(4.52)^2 + 4(2)(1.19)}}{2(2)}$$

$$V_S = 2.50 \text{ V} \Rightarrow I_{DQ} = \frac{5 - 2.5}{5} = 0.5 \text{ mA}$$

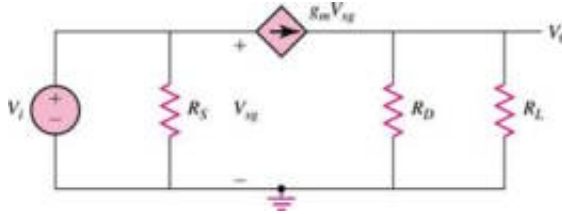
$$g_m = 2\sqrt{K_p I_{DQ}} = 2\sqrt{(0.4)(0.5)} = 0.894 \text{ mA/V}$$

$$A_v = \frac{g_m R_S}{1 + g_m R_S} \cdot \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_{Si}} \\ = \frac{(0.894)(5)}{1 + (0.894)(5)} \cdot \frac{70.7 \parallel 9.3}{70.7 \parallel 9.3 + 0.5} \Rightarrow A_v = 0.770$$

Neglecting R_{Si} , $A_v = 0.817$

$$R_o = R_S \parallel \left(\frac{1}{g_m} \right) = 5 \parallel \frac{1}{0.894} = 5 \parallel 1.12 \Rightarrow R_o = 0.915 \text{ k}\Omega$$

EX4.10



$$V_O = g_m V_{sg} (R_D \parallel R_L) \quad \text{and} \quad V_{sg} = V_i$$

$$A_v = g_m (R_D \parallel R_L)$$

$$I_{DQ} = \frac{5 - V_{SG}}{R_S} = K_p (V_{SG} - |V_{TP}|)^2$$

$$5 - V_{SG} = (1)(4)(V_{SG} - 0.8)^2$$

$$5 - V_{SG} = 4(V_{SG}^2 - 1.6V_{SG} + 0.64)$$

$$4V_{SG}^2 - 5.4V_{SG} - 2.44 = 0$$

$$V_{SG} = \frac{5.4 \pm \sqrt{(5.4)^2 + (4)(4)(2.44)}}{2(4)}$$

$$V_{SG} = 1.71 \text{ V}$$

$$I_{DQ} = \frac{5 - 1.71}{4} = 0.822 \text{ mA}$$

$$g_m = 2\sqrt{K_p I_{DQ}} = 2\sqrt{(1)(0.822)} = 1.81 \text{ mA/V}$$

$$A_v = (1.81)(2\parallel 4) = (1.81)(1.33) \Rightarrow A_v = 2.41$$

$$R_{in} = R_S \parallel \left(\frac{1}{g_m} \right) = 4 \parallel \left(\frac{1}{1.81} \right) = 4 \parallel 0.552 \Rightarrow R_{in} = 0.485 \text{ k}\Omega$$

EX4.11

$$(a) \quad |A_v| = 8 = \sqrt{\frac{(W/L)_D}{(W/L)_L}} = \sqrt{\frac{(W/L)_D}{1.2}} \Rightarrow \left(\frac{W}{L} \right)_D = 76.8$$

$$(b) \quad V_{GSDt} = \frac{(V_{DD} - V_{TNL}) + V_{TND} \left(1 + \sqrt{\frac{K_{nD}}{K_{nL}}} \right)}{1 + \sqrt{\frac{K_{nD}}{K_{nL}}}} = \frac{(3.3 - 0.4) + (0.4)(1 + 8)}{1 + 8}$$

or $V_{GSDt} = 0.7222 \text{ V}$

So

$$V_{GSDQ} = \frac{0.7222 - 0.4}{2} + 0.4 = 0.561 \text{ V}$$

EX4.12

$$A_v = -g_{mD}(r_{oD} \parallel r_{oL})$$

$$r_{oD} = r_{oL} = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.02)(0.1)} = 500 \text{ k}\Omega$$

$$g_{mD} = 2\sqrt{K_{nD} I_{DQ}} = 2\sqrt{(0.25)(0.1)} = 0.3162 \text{ mA/V}$$

Then

$$A_v = -(0.3162)(500 \parallel 500) = -79.1$$

EX4.13

$$A_v = -g_m(r_{on} \parallel r_{op})$$

$$r_{on} = r_{op} = \frac{1}{(0.015)(0.1)} = 666.7 \text{ k}\Omega$$

$$-250 = -g_m(666.7 \parallel 666.7)$$

$$g_m = 0.75 \text{ mA/V} = 2\sqrt{K_n I_{DQ}} = 2\sqrt{K_n(0.1)}$$

$$K_n = 1.406 \text{ mA/V}^2 = \frac{k'_n}{2} \left(\frac{W}{L} \right) = \left(\frac{0.080}{2} \right) \left(\frac{W}{L} \right) \Rightarrow \left(\frac{W}{L} \right)_1 = 35.2$$

EX4.14

(a)

$$R_O = \frac{1}{g_{m1}} \parallel r_{o2} \parallel r_{o1} \approx \frac{1}{g_{m1}}$$

$$\text{So } g_{m1} = \frac{1}{R_O} = \frac{1}{2} = 0.5 \text{ mA/V}$$

$$g_{m1} = 2\sqrt{K_n I_D}$$

$$0.5 = 2\sqrt{(0.2)I_D} \Rightarrow I_D = 0.3125 \text{ mA}$$

(b)

$$A_v = \frac{g_{m1}(r_{o1} \parallel r_{o2})}{1 + g_{m1}(r_{o1} \parallel r_{o2})}$$

$$r_{o1} = r_{o2} = \frac{1}{(0.01)(0.3125)} = 320 \text{ k}\Omega$$

$$A_v = \frac{(0.5)(320 \parallel 320)}{1 + (0.5)(320 \parallel 320)}$$

$$A_v = 0.988$$

EX4.15

(a)

$$A_v = \frac{g_{m1} + \frac{1}{r_{o1}}}{\frac{1}{r_{o2}} + \frac{1}{r_{o1}}} = \frac{2\sqrt{K_n I_D} + \lambda_1 I_D}{\lambda_2 I_D + \lambda_1 I_D}$$

$$120 = \frac{2\sqrt{0.2I_D} + 0.01I_D}{0.01I_D + 0.01I_D}$$

$$2.4I_D - 0.01I_D = 2\sqrt{0.2I_D}$$

$$2.39\sqrt{I_D} = 2\sqrt{0.2} \Rightarrow I_D = 0.140 \text{ mA}$$

$$g_{m1} = 2\sqrt{(0.2)(0.14)} \Rightarrow g_{m1} = 0.335 \text{ mA/V}$$

(b)

$$R_o = r_{o1} \parallel r_{o2}$$

$$r_{o1} = r_{o2} = \frac{1}{(0.01)(0.14)} = 714 \text{ k}\Omega$$

$$R_o = 714 \parallel 714 = 357 \text{ k}\Omega$$

EX4.16

$$R_o = R_{S2} \parallel \frac{1}{g_{m2}}$$

$$g_{m2} = 0.632 \text{ mA/V}, \quad R_{S2} = 8 \text{ k}\Omega$$

$$R_o = 8 \parallel \frac{1}{0.632} = 8 \parallel 1.58 \Rightarrow R_o = 1.32 \text{ k}\Omega$$

EX4.17

$$(a) \quad V_{G1} = \left(\frac{54.6}{54.6 + 150 + 95.4} \right) (5) = \left(\frac{54.6}{300} \right) (5) = 0.91 \text{ V}$$

$$V_{G2} = \left(\frac{54.6 + 150}{300} \right) (5) = 3.41 \text{ V}$$

$$V_{G1} = V_{GS1} + K_{n1} R_S (V_{GS1} - V_{TN})^2 - 5$$

$$5.91 = V_{GS1} + (3)(10)(V_{GS1}^2 - 1.6V_{GS1} + 0.64)$$

or

$$30V_{GS1}^2 - 47V_{GS1} + 13.29 = 0 \Rightarrow V_{GS1} = 1.196 \text{ V}$$

Then

$$I_{DQ} = (3)(1.196 - 0.8)^2 = 0.471 \text{ mA}$$

$$V_{D1} = V_{G2} - V_{GS2} = 3.41 - 1.196 = 2.214 \text{ V}$$

$$V_{S1} = V_{G1} - V_{GS1} = 0.91 - 1.196 = -0.286 \text{ V}$$

Then

$$V_{DSQ1} = 2.214 - (-0.286) = 2.5 \text{ V}$$

$$V_{D2} = 5 - (0.471)(2.5) = 3.8225 \text{ V}$$

$$V_{DSQ2} = 3.8225 - 2.214 = 1.61 \text{ V}$$

(b)

$$g_{m1} = 2\sqrt{K_{n1}I_{DQ}} = 2\sqrt{(3)(0.471)} = 2.377 \text{ mA/V}$$

$$A_v = -g_{m1}R_D = -(2.377)(2.5) = -5.94$$

EX4.18

$$V_S = I_{DQ}R_S = (1.2)(2.7) = 3.24 \text{ V}$$

$$V_D = V_S + V_{DSQ} = 3.24 + 12 = 15.24 \text{ V}$$

$$R_D = \frac{20 - 15.24}{1.2} \Rightarrow R_D = 3.97 \text{ k}\Omega$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$1.2 = 4 \left(1 - \frac{V_{GS}}{V_P} \right)^2 \Rightarrow \frac{V_{GS}}{V_P} = 0.4523$$

$$V_{GS} = (0.4523)(-3) = -1.357 \text{ V}$$

$$V_G = V_S + V_{GS} = 3.24 - 1.357 = 1.883 \text{ V}$$

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) (20) = \left(\frac{R_2}{500} \right) (20) = 1.88 \Rightarrow R_2 = 47 \text{ k}\Omega, \quad R_1 = 453 \text{ k}\Omega$$

$$r_o = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.005)(1.2)} = 167 \text{ k}\Omega$$

$$g_m = \frac{2I_{DSS}}{(-V_P)} \left(1 - \frac{V_{GS}}{V_P} \right) = \frac{2(4)}{(3)} \left(1 - \frac{1.357}{3} \right) = 1.46 \text{ mA/V}$$

$$A_v = -g_m (r_o \parallel R_D \parallel R_L) = -(1.46)(167 \parallel 3.97 \parallel 4) \Rightarrow A_v = -2.87$$

EX4.19

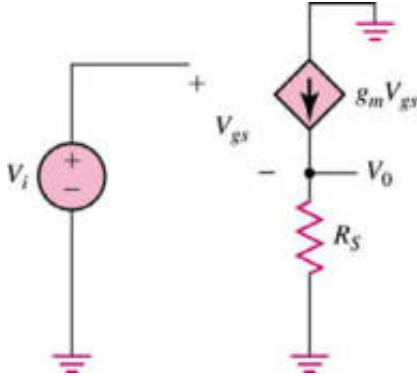
a.

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad 2 = 8 \left(1 - \frac{V_{GS}}{V_P} \right)^2 \Rightarrow \frac{V_{GS}}{V_P} = 0.5$$

$$V_{GS} = (0.5)(-3.5) \Rightarrow V_{GS} = -1.75$$

$$\text{Also } I_{DQ} = \frac{-V_{GS} - (-10)}{R_S} \quad 2 = \frac{1.75 + 10}{R_S} \Rightarrow R_S = 5.88 \text{ k}\Omega$$

b.



$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P} \right) = \frac{2(8)}{3.5} \left(1 - \frac{1.75}{3.5} \right) = 2.29 \text{ mA/V}$$

$$r_o = \frac{1}{(0.01)(2)} = 50 \text{ k}\Omega$$

$$V_i = V_{gs} + g_m R_S V_{gs} \Rightarrow V_{gs} = \frac{V_i}{1 + g_m R_S}$$

$$A_v = \frac{V_o}{V_i} = \frac{g_m (R_S \parallel r_o)}{1 + g_m (R_S \parallel r_o)} = \frac{(2.29)(5.88 \parallel 50)}{1 + (2.29)(5.88 \parallel 50)} \Rightarrow A_v = 0.9234$$

c.

$$A_v = \frac{g_m (R_S \parallel R_L \parallel r_o)}{1 + g_m (R_S \parallel R_L \parallel r_o)} = (0.80)(0.9234) = 0.7387$$

$$0.7387 = \frac{(2.29)(R_S \parallel R_L \parallel r_o)}{1 + (2.29)(R_S \parallel R_L \parallel r_o)} \Rightarrow R_S \parallel R_L \parallel r_o = 1.235 \text{ k}\Omega$$

$$R_S \parallel r_o = 5.261 \text{ k}\Omega$$

$$\text{Then } \frac{(5.261)R_L}{5.261 + R_L} = 1.235 \Rightarrow R_L = 1.61 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU4.1

$$\begin{aligned} \text{(a)} \quad g_m &= 2\sqrt{\left(\frac{k_n'}{2}\right)\left(\frac{W}{L}\right)I_{DQ}} \\ (2.5)^2 &= 4\left[\left(\frac{0.1}{2}\right)\left(\frac{W}{L}\right)(1.2)\right] \Rightarrow \frac{W}{L} = 26.0 \\ \text{(b)} \quad r_o &= \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.015)(1.2)} = 55.6 \text{ k}\Omega \end{aligned}$$

TYU4.2

$$\begin{aligned} \text{(a)} \quad I_{DQ} &= K_n (V_{GSQ} - V_{TN})^2 \\ 0.15 &= 0.5(V_{GSQ} - 0.4)^2 \Rightarrow V_{GSQ} = 0.948 \text{ V} \\ V_{DSQ} &= 3.3 - (0.15)(8) = 2.1 \text{ V} \\ \text{(b)} \quad g_m &= 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.5)(0.15)} = 0.548 \text{ mA/V} \\ r_o &= \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.02)(0.15)} = 333 \text{ k}\Omega \\ A_v &= -g_m (r_o \parallel R_D) = -(0.548)(333 \parallel 8) = -4.28 \end{aligned}$$

TYU4.3

$$\begin{aligned} i_D &= I_{DQ} + i_d = I_{DQ} + g_m v_{gs} \\ i_D &= 0.15 + (0.548)(0.025)\sin \omega t \\ \text{or} \quad i_D &= 0.15 + 0.0137 \sin \omega t \text{ (mA)} \\ \text{Also} \quad v_{DS} &= V_{DSQ} + v_d = 2.1 - (0.0137)(8)\sin \omega t \\ \text{or} \quad v_{DS} &= 2.1 - (0.11)\sin \omega t \text{ (V)} \end{aligned}$$

TYU4.4

$$\begin{aligned} \text{(a)} \quad V_{SDQ} &= V_{DD} - I_{DQ} R_D \\ 3 &= 5 - I_{DQ}(5) \Rightarrow I_{DQ} = 0.4 \text{ mA} \\ I_{DQ} &= K_p (V_{SGQ} + V_{TP})^2 \\ 0.4 &= 0.4(V_{SGQ} - 0.4)^2 \Rightarrow V_{SGQ} = 1.4 \text{ V} \\ \text{(b)} \quad g_m &= 2\sqrt{K_p I_{DQ}} = 2\sqrt{(0.4)(0.4)} = 0.8 \text{ mA/V} \\ A_v &= -g_m R_D = -(0.8)(5) = -4 \end{aligned}$$

TYU4.5

$$\eta = \frac{\gamma}{2\sqrt{2\phi_f + v_{SB}}}$$

$$(a) \quad \eta = \frac{0.40}{2\sqrt{2(0.35)+1}} \Rightarrow \underline{\eta = 0.153}$$

$$\eta = \frac{0.40}{2\sqrt{2(0.35)+3}} \Rightarrow \underline{\eta = 0.104}$$

$$(b) \quad g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.5)(0.75)} = 1.22 \text{ mA/V}$$

$$\text{For (a), } g_{mb} = g_m \eta = (1.22)(0.153) \Rightarrow \underline{g_{mb} = 0.187 \text{ mA/V}}$$

$$\text{For (b), } g_{mb} = (1.22)(0.104) \Rightarrow \underline{g_{mb} = 0.127 \text{ mA/V}}$$

TYU4.6

a. With $R_G \Rightarrow V_{GS} = V_{DS} \Rightarrow$ transistor biased in sat. region

$$I_D = K_n (V_{GS} - V_{TN})^2 = K_n (V_{DS} - V_{TN})^2$$

$$V_{DS} = V_{DD} - I_D R_D = V_{DD} - K_n R_D (V_{DS} - V_{TN})^2$$

$$V_{DS} = 15 - (0.15)(10)(V_{DS} - 1.8)^2$$

$$= 15 - 1.5(V_{DS}^2 - 3.6V_{DS} + 3.24)$$

$$1.5V_{DS}^2 - 4.4V_{DS} - 10.14 = 0$$

$$V_{DS} = \frac{4.4 \pm \sqrt{(4.4)^2 + (4)(1.5)(10.14)}}{2(1.5)} \Rightarrow \underline{V_{DSQ} = 4.45 \text{ V}}$$

$$I_{DQ} = (0.15)(4.45 - 1.8)^2 \Rightarrow \underline{I_{DQ} = 1.05 \text{ mA}}$$

b. Neglecting effect of R_G :

$$A_v = -g_m (R_D \parallel R_L)$$

$$g_m = 2K_n (V_{GS} - V_{TN}) = 2(0.15)(4.45 - 1.8) \Rightarrow g_m = 0.795 \text{ mA/V}$$

$$A_v = -(0.795)(10 \parallel 5) \Rightarrow \underline{A_v = -2.65}$$

c. $R_G \Rightarrow$ establishes $V_{GS} = V_{DS} \Rightarrow$ essentially no effect on small-signal voltage gain.

TYU4.7

a.

$$5 = I_{DQ} R_S + V_{SG} \text{ and } I_{DQ} = K_p (V_{SG} + V_{TP})^2$$

$$0.8 = 0.5(V_{SG} + 0.8)^2 \Rightarrow V_{SG} = 0.465 \text{ V}$$

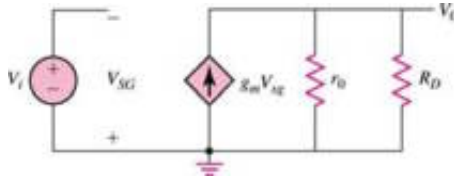
$$5 = (0.8)R_S + 0.465 \Rightarrow \underline{R_S = 5.67 \text{ k}\Omega}$$

$$V_{SDQ} = 10 - I_{DQ} (R_S + R_D)$$

$$3 = 10 - (0.8)(5.67 + R_D)$$

$$R_D = \frac{10 - (0.8)(5.67) - 3}{0.8} \Rightarrow \underline{R_D = 3.08 \text{ k}\Omega}$$

b.



$$V_o = g_m V_{sg} (R_D \parallel r_o) = -g_m V_i (R_D \parallel r_o)$$

$$A_v = \frac{V_o}{V_i} = -g_m (R_D \parallel r_o)$$

$$g_m = 2K_p (V_{SG} + V_{TP}) = 2(0.5)(0.465 + 0.8) = 1.265 \text{ mA/V}$$

$$r_o = \frac{1}{\lambda I_O} = \frac{1}{(0.02)(0.8)} = 62.5 \text{ k}\Omega$$

$$A_v = -(1.265)(3.08 \parallel 62.5) \Rightarrow A_v = -3.71$$

TYU4.8

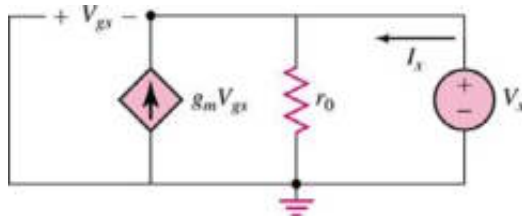
(a)

$$V_o = g_m V_{gs} r_o$$

$$V_i = V_{gs} + V_o \Rightarrow V_{gs} = V_i - V_o$$

$$\text{So } V_o = g_m r_o (V_i - V_o)$$

$$A_v = \frac{V_o}{V_i} = \frac{g_m r_o}{1 + g_m r_o} = \frac{(4)(50)}{1 + (4)(50)} \Rightarrow A_v = 0.995$$



$$I_x + g_m V_{gs} = \frac{V_x}{r_o} \text{ and } V_{gs} = -V_x$$

$$I_x = g_m V_x + \frac{V_x}{r_o} \Rightarrow R_0 = r_o \parallel \frac{1}{g_m} = 50 \parallel \frac{1}{4} \Rightarrow R_0 \cong 0.25 \text{ k}\Omega$$

(b) With $R_S = 4 \text{ k}\Omega \Rightarrow A_v = \frac{g_m (r_o \parallel R_S)}{1 + g_m (r_o \parallel R_S)}$

$$r_o \parallel R_S = 50 \parallel 4 = 3.7 \text{ k}\Omega \Rightarrow A_v = \frac{(4)(3.7)}{1 + (4)(3.7)} \Rightarrow A_v = 0.937$$

TYU4.9

(a)

$$g_m = 2\sqrt{K_n I_{DQ}} \Rightarrow 2 = 2\sqrt{K_n (0.8)} \Rightarrow K_n = 1.25 \text{ mA/V}^2$$

$$K_n = \frac{\mu_n C_{ox}}{2} \cdot \frac{W}{L} \Rightarrow 1.25 = (0.020) \left(\frac{W}{L} \right)$$

$$\text{So } \frac{W}{L} = 62.5$$

$$I_{DQ} = K_n (V_{GS} - V_{TN})^2 \Rightarrow 0.8 = 1.25 (V_{GS} - 2)^2 \Rightarrow \underline{V_{GS} = 2.8 \text{ V}}$$

b.

$$r_o = [\lambda I_{DQ}]^{-1} = [(0.01)(0.8)]^{-1} = 125 \text{ k}\Omega$$

$$A_v = \frac{g_m (r_o \parallel R_L)}{1 + g_m (r_o \parallel R_L)}$$

$$r_o \parallel R_L = 125 \parallel 4 = 3.88 \text{ k}\Omega$$

$$A_v = \frac{(2)(3.88)}{1 + (2)(3.88)} \Rightarrow A_v = 0.886$$

$$R_o = \frac{1}{g_m} \parallel r_o = \frac{1}{2} \parallel 125 \Rightarrow R_o \cong 0.5 \text{ k}\Omega$$

TYU4.10

$$R_{in} = \frac{1}{g_m} = 0.35 \text{ k}\Omega \Rightarrow g_m = 2.86 \text{ mA/V}$$

$$\frac{V_o}{I_i} = R_D \parallel R_L = 2.4 = R_D \parallel 4 \Rightarrow R_D = 6 \text{ k}\Omega$$

$$g_m = 2\sqrt{K_n I_{DQ}}$$

$$2.86 = 2\sqrt{K_n (0.5)} \Rightarrow K_n = 4.09 \text{ mA/V}^2$$

$$I_{DQ} = K_n (V_{GS} - V_{TN})^2$$

$$0.5 = 4.09 (V_{GS} - 1)^2 \Rightarrow V_{GS} = 1.35 \text{ V} \Rightarrow V_S = -1.35 \text{ V}$$

$$V_{DQ} = 5 - (0.5)(6) = 2 \text{ V}$$

$$V_{DS} = V_{DQ} - V_S = 2 - (-1.35) = 3.35 \text{ V}$$

We have $V_{DS} = 3.35 > V_{GS} - V_{TN} = 1.35 - 1 = 0.35 \text{ V} \Rightarrow$ Biased in the saturation region

TYU4.11

$$K_{n1} = \frac{\mu_n C_{ox}}{2} \cdot \left(\frac{W}{L} \right)_1 = (0.020)(80) = 1.6 \text{ mA/V}^2$$

$$K_{n2} = \frac{\mu_n C_{ox}}{2} \cdot \left(\frac{W}{L} \right)_2 = (0.020)(1) = 0.020 \text{ mA/V}^2$$

$$A_v = -\sqrt{\frac{K_{n1}}{K_{n2}}} = -\sqrt{\frac{1.6}{0.020}} \Rightarrow \underline{A_v = -8.94}$$

The transition point is determined from $v_{GSr} - V_{TND} = V_{DD} - V_{TNL} - \sqrt{\frac{K_{n1}}{K_{n2}}} (v_{GSr} - V_{TND})$

$$v_{GSr} - 0.8 = (5 - 0.8) - (8.94)(v_{GSr} - 0.8)$$

$$v_{GSr} = \frac{(5 - 0.8) + (8.94)(0.8) + 0.8}{1 + 8.94}$$

$$v_{GSr} = 1.22 \text{ V}$$

For Q -point in middle of saturation region $V_{GS} = \frac{1.22 - 0.8}{2} + 0.8 \Rightarrow \underline{V_{GS} = 1.01 \text{ V}}$

TYU4.12

(a) $V_{G1} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \left(\frac{135}{135 + 383} \right) (10) - 5 = -2.394 \text{ V}$

$$V_{G1} = V_{GS1} + K_{n1} R_{S1} (V_{GS1} - V_{TN})^2 - 5$$

or

$$5 - 2.394 = V_{GS1} + (1.5)(3.9)(V_{GS1}^2 - 1.2V_{GS1} + 0.36)$$

so

$$5.85V_{GS1}^2 - 6.02V_{GS1} - 0.5 = 0 \Rightarrow V_{GS1} = 1.106 \text{ V}$$

Then

$$I_{DQ1} = (1.5)(1.106 - 0.6)^2 = 0.3845 \text{ mA}$$

$$V_{DSQ1} = 10 - (0.3845)(3.9 + 16.1) = 2.31 \text{ V}$$

$$V_{G2} = 5 - (0.3845)(16.1) = -1.190 \text{ V}$$

$$V_{G2} = V_{GS2} + K_{n2} R_{S2} (V_{GS2} - V_{TN})^2 - 5$$

or

$$5 - 1.19 = V_{GS2} + (2)(8)(V_{GS2}^2 - 1.2V_{GS2} + 0.36)$$

so

$$16V_{GS2}^2 - 18.2V_{GS2} + 1.95 = 0 \Rightarrow V_{GS2} = 1.018 \text{ V}$$

Then

$$I_{DQ2} = (2)(1.018 - 0.6)^2 = 0.349 \text{ mA}$$

$$V_{DSQ2} = 10 - (0.349)(8) = 7.208 \text{ V}$$

(b)

$$g_{m1} = 2\sqrt{K_{n1}I_{DQ1}} = 2\sqrt{(1.5)(0.3845)} = 1.519 \text{ mA/V}$$

$$g_{m2} = 2\sqrt{K_{n2}I_{DQ2}} = 2\sqrt{(2)(0.349)} = 1.671 \text{ mA/V}$$

From Example 4.16

$$\begin{aligned} A_v &= \frac{-g_{m1}g_{m2}R_{D1}(R_{S2} \parallel R_L)}{1 + g_{m2}(R_{S2} \parallel R_L)} \cdot \frac{R_i}{R_i + R_{Si}} \\ &= \frac{-(1.519)(1.671)(16.1)(8 \parallel 4)}{1 + (1.671)(8 \parallel 4)} \cdot \frac{99.8}{99.8 + 4} \end{aligned}$$

or

$$A_v = -19.2$$

(c)

$$R_o = \frac{1}{g_{m2}} \parallel R_{S2} = \frac{1}{1.671} \parallel 8 = 0.5984 \parallel 8 \Rightarrow R_o = 557 \Omega$$

TYU4.13

From Example 6.18

$$g_m = 3.0 \text{ mA/V}, \quad r_o = 41.7 \text{ k}\Omega$$

$$R_1 \parallel R_2 = 420 \parallel 180 = 126 \text{ k}\Omega$$

$$V_{gs} = \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_i} \cdot V_i = \left(\frac{126}{126 + 20} \right) \cdot V_i = 0.863 \cdot V_i$$

$$A_v = \frac{-g_m V_{gs} (r_o \parallel R_D \parallel R_L)}{V_i}$$

$$= -(3.0)(0.863)(41.7 \parallel 2.7 \parallel 4) = -(2.589)(1.55) \Rightarrow A_v = -4.01$$

TYU4.14

a.

$$V_{G1} = \left(\frac{R_2}{R_1 + R_2} \right) (V_{DD})$$

$$V_{G1} = \left(\frac{430}{430 + 70} \right) (20) = 17.2 \text{ V}$$

$$I_{DQ1} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 6 \left(1 - \frac{V_{G1} - V_{S1}}{2} \right)^2$$

$$= 6 \left(1 - \frac{17.2}{2} + \frac{V_{S1}}{2} \right)^2 = 6 \left(\frac{V_{S1}}{2} - 7.6 \right)^2 \quad \text{and} \quad I_{DQ1} = \frac{20 - V_{S1}}{4}$$

$$\text{Then } \frac{20 - V_{S1}}{4} = 6 \left(\frac{V_{S1}}{2} - 7.6 \right)^2$$

$$20 - V_{S1} = 24 \left(\frac{V_{S1}^2}{4} - 7.6V_{S1} + 57.76 \right)$$

$$= 6V_{S1}^2 - 182.4V_{S1} + 1386.24$$

$$6V_{S1}^2 - 181.4V_{S1} + 1366.24 = 0$$

$$V_{S1} = \frac{181.4 \pm \sqrt{(181.4)^2 - 4(6)(1366.24)}}{2(6)}$$

$$V_{S1} = 14.2 \text{ V} \Rightarrow V_{GS1} = 17.2 - 14.2 = 3 \text{ V} > V_P$$

$$\text{So } V_{S1} = 16.0 \Rightarrow V_{GS1} = 17.2 - 16 = 1.2 < V_P - Q$$

$$\text{on } I_{DQ1} = \frac{20 - 16}{4} \Rightarrow I_{DQ1} = 1 \text{ mA}$$

$$V_{SDQ1} = 20 - I_{DQ1} (R_{S1} + R_{D1})$$

$$= 20 - (1)(8) \Rightarrow V_{SDQ1} = 12 \text{ V}$$

$$V_{D1} = I_{DQ1} R_{D1} = (1)(4) = 4 \text{ V} = V_{G2}$$

$$I_{DQ2} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 6 \left(1 - \frac{V_{G2} - V_{S2}}{(-2)} \right)^2$$

$$= 6 \left(1 + \frac{4}{2} - \frac{V_{S2}}{2} \right)^2 = 6 \left(3 - \frac{V_{S2}}{2} \right)^2 \text{ and } I_{DQ2} = \frac{V_{S2}}{R_{S2}} = \frac{V_{S2}}{4}$$

$$\text{Then } \frac{V_{S2}}{4} = 6 \left(3 - \frac{V_{S2}}{2} \right)^2$$

$$V_{S2} = 24 \left(9 - 3V_{S2} + \frac{V_{S2}^2}{4} \right)$$

$$6V_{S2}^2 - 73V_{S2} + 216 = 0$$

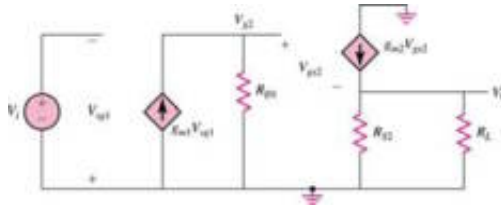
$$V_{S2} = \frac{73 \pm \sqrt{(73)^2 - 4(6)(216)}}{2(6)} \Rightarrow V_{S2} = 7.09 \text{ V or } 5.08 \text{ V}$$

$$\text{For } V_{S2} = 5.08 \text{ V} \Rightarrow V_{GS2} = 4 - 5.08 = -1.08 \text{ transistor biased ON}$$

$$I_{DQ2} = \frac{5.08}{4} \Rightarrow I_{DQ2} = 1.27 \text{ mA}$$

$$V_{DS2} = 20 - V_{S2} = 20 - 5.08 \Rightarrow V_{DS2} = 14.9 \text{ V}$$

b.



$$V_{g2} = g_{m1} V_{gs1} R_{D1} = -g_{m1} V_i R_{D1}$$

$$V_o = g_{m2} V_{gs2} (R_{S2} \parallel R_L)$$

$$V_{g2} = V_{gs2} + V_o \Rightarrow V_{gs2} = \frac{V_{g2}}{1 + g_{m2} (R_{S2} \parallel R_L)}$$

$$A_v = \frac{V_o}{V_i} = \frac{-g_{m1} R_{D1}}{1 + g_{m2} (R_{S2} \parallel R_L)}$$

$$g_{m1} = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P} \right)$$

$$= \frac{2(6)}{2} \left(1 - \frac{1.2}{-2} \right) = 2.4 \text{ mA/V}$$

$$g_{m2} = \frac{2(6)}{2} \left(1 - \frac{1.08}{-2} \right) = 2.76 \text{ mA/V}$$

$$\text{Then } A_v = \frac{-(2.4)(4)}{1 + (2.76)(4 \parallel 2)} = -2.05$$

Chapter 5

Exercise Solutions

EX5.1

$$\begin{aligned}I_E &= (1 + \beta)I_B \\1 + \beta &= \frac{I_E}{I_B} = \frac{1.20}{0.0085} = 141.2 \Rightarrow \beta = 140.2 \\ \alpha &= \frac{\beta}{1 + \beta} = \frac{140.2}{141.2} = 0.9929 \\I_C &= I_E - I_B = 1.20 - 0.0085 = 1.1915 \text{ mA}\end{aligned}$$

EX5.2

$$BV_{CEO} = \frac{BV_{CBO}}{\sqrt[n]{\beta}} = \frac{200}{\sqrt[3]{120}} \text{ or } BV_{CEO} = 40.5 \text{ V}$$

EX5.3

$$\begin{aligned}I_B &= \frac{V_{BB} - V_{BE}(on)}{R_B} = \frac{2 - 0.7}{430} \Rightarrow I_B = 3.02 \mu\text{ A} \\I_C &= \beta I_B = (150)(3.02) \mu\text{ A} \Rightarrow I_C = 0.453 \text{ mA} \\V_{CE} &= V_{CC} - I_C R_C = 3.3 - (0.453)(3.2) = 1.85 \text{ V} \\P &\cong I_C V_{CE} = (0.453)(1.85) = 0.838 \text{ mW}\end{aligned}$$

EX5.4

$$\begin{aligned}I_B &= \frac{V^+ - V_{EB}(on) - V_{BB}}{R_B} = \frac{3.3 - 0.7 - 1.2}{400} \Rightarrow I_B = 3.5 \mu\text{ A} \\I_C &= \beta I_B = (80)(3.5) \mu\text{ A} \Rightarrow I_C = 0.28 \text{ mA} \\V_{EC} &= V^+ - I_C R_C = 3.3 - (0.28)(5.25) = 1.83 \text{ V}\end{aligned}$$

EX5.5

$$\begin{aligned}\text{(a)} \quad I_B &= \frac{V^+ - V_{EB}(on) - V_{BB}}{R_B} = \frac{3.3 - 0.7 - 2}{150} \Rightarrow I_B = 4 \mu\text{ A} \\I_C &= \beta I_B = (110)(4) \mu\text{ A} \Rightarrow I_C = 0.44 \text{ mA} \\V_{EC} &= V^+ - I_C R_C = 3.3 - (0.44)(5) = 1.1 \text{ V} \\ \text{(b)} \quad I_B &= \frac{3.3 - 0.7 - 1}{150} \Rightarrow I_B = 10.7 \mu\text{ A} \\I_C &= \frac{V^+ - V_{EC}(sat)}{R_C} = \frac{3.3 - 0.2}{5} = 0.62 \text{ mA} \\V_{EC} &= 0.2 \text{ V}\end{aligned}$$

EX5.6

For $0 \leq V_I < 0.7 \text{ V}$, Q_n is cutoff, $V_O = 9 \text{ V}$

$$0.2 = 9 - \frac{(100)(V_I - 0.7)(4)}{200} \Rightarrow V_I = 5.1 \text{ V}$$

When Q_n is biased in saturation, we have

So, for $V_I \geq 5.1 \text{ V}$, $V_O = 0.2 \text{ V}$

EX5.7

$$V_{BB} = I_B R_B + V_{BE}(\text{on}) + I_E R_E + V^-$$

$$I_E = (1 + \beta) I_B$$

So

$$I_B = \frac{3.3 - 0.7}{640 + (81)(2.4)} \Rightarrow I_B = 3.116 \mu\text{A}$$

$$I_C = \beta I_B = (80)(3.116) \mu\text{A} \Rightarrow I_C = 0.249 \text{ mA}$$

$$I_E = \left(\frac{1 + \beta}{\beta} \right) I_C = \left(\frac{81}{80} \right) (0.249) = 0.252 \text{ mA}$$

$$V_{CE} = [3.3 - (-3.3)] - (0.249)(10) - (0.252)(2.4) = 3.51 \text{ V}$$

EX5.8

$$I_{EQ} = \frac{V^+ - V_{EB}(\text{on})}{R_E} \Rightarrow R_E = \frac{3 - 0.7}{0.125} = 18.4 \text{ k}\Omega$$

$$V_C = V_{EB}(\text{on}) - V_{ECQ} = 0.7 - 2.2 = -1.5 \text{ V}$$

$$I_{CQ} = \left(\frac{\beta}{1 + \beta} \right) I_{EQ} = \left(\frac{110}{111} \right) (0.125) = 0.1239 \text{ mA}$$

$$R_C = \frac{V_C - V^-}{I_{CQ}} = \frac{-1.5 - (-3)}{0.1239} = 12.1 \text{ k}\Omega$$

EX5.9

$$5 = I_E R_E + V_{EB}(\text{on}) + I_B R_B - 2$$

$$5 + 2 - 0.7 = I_E \left(2 + \frac{180}{41} \right) \quad I_E = 0.9859 \text{ mA}$$

$$I_C = 0.962 \text{ mA}$$

(a)

$$6.3 = I_E \left(2 + \frac{180}{61} \right) \quad I_E = 1.2725 \text{ mA}$$

$$I_C = 1.25 \text{ mA}$$

(b)

$$6.3 = I_E \left(2 + \frac{180}{101} \right) \quad I_E = 1.6657 \text{ mA}$$

$$I_C = 1.64 \text{ mA}$$

(c)

$$6.3 = I_E \left(2 + \frac{180}{151} \right) \quad I_E = 1.97365 \text{ mA}$$

$$I_C = 1.94 \text{ mA}$$

(d)

EX5.10

$$I_E = \frac{V_{BB} - V_{EB}(on)}{R_E} \Rightarrow R_E = \frac{4 - 0.7}{1.0}$$

or $R_E = 3.3 \text{ k}\Omega$

$$I_C = \alpha I_E = (0.992)(1) = 0.992 \text{ mA}$$

$$I_B = I_E - I_C = 1.0 - 0.992 \text{ or } I_B = 8 \mu\text{A}$$

$$V_{CB} = I_C R_C - V_{CC} = (0.992)(1) - 5$$

or $V_{CB} = -4.01 \text{ V}$

EX5.11

(a) $R_1 = \frac{V^+ - V_\gamma - V_{CE}(sat)}{I_{C1}} = \frac{5 - 1.5 - 0.2}{15} \Rightarrow R_1 = 220 \Omega$

$$I_{B1} = \frac{I_{C1}}{50} = \frac{15}{50} = 0.30 \text{ mA}$$

$$R_{B1} = \frac{5 - 0.7}{0.3} = 14.3 \text{ k}\Omega$$

(b)

$$I_{B2} = \frac{I_{C2}}{25} = \frac{2}{25} = 0.08 \text{ A}$$

$$R_{B2} = \frac{12 - 0.7 - 0}{0.08} = 141 \Omega$$

EX5.12

(a) For $V_1 = V_2 = 0$, All currents are zero and $V_O = 5 \text{ V}$.

(b) For $V_1 = 5 \text{ V}$, $V_2 = 0$; $I_{B2} = I_{C2} = 0$,

$$I_{B1} = \frac{5 - 0.7}{0.95} = 4.53 \text{ mA}$$

$$I_{C1} = \frac{5 - 0.2}{0.6} \Rightarrow I_{C1} = I_R = 8 \text{ mA}$$

$$V_O = 0.2 \text{ V}$$

(c) For $V_1 = V_2 = 5 \text{ V}$, $I_{B1} = I_{B2} = 4.53 \text{ mA}$; $I_R = 8 \text{ mA}$, $I_{C1} = I_{C2} = I_R/2 = 4 \text{ mA}$, $V_O = 0.2 \text{ V}$

EX5.13

In active region,

$$v_O = m v_I + b \Rightarrow m = -6.5$$

At $v_I = 0.7 \text{ V}$, $v_O = 5 \text{ V}$

$$5 = -6.5(0.7) + b \Rightarrow b = 9.55$$

Then

$$v_O = -6.5 v_I + 9.55$$

When

$$v_O = 0.2 = -6.5 v_I + 9.55 \Rightarrow v_I = 1.438 \text{ V}$$

Q-point

$$v_{IQ} = \frac{1.438 - 0.7}{2} + 0.7 = 1.069 \text{ V}$$

$$v_{OQ} = \frac{5 - 0.2}{2} + 0.2 = 2.6 \text{ V}$$

Now

$$I_{BQ} = \frac{1.069 - 0.7}{80} \Rightarrow 4.61 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = (120)(4.61) \mu\text{A} \Rightarrow I_{CQ} = 0.5535 \text{ mA}$$

At Q-point

$$v_{OQ} = 5 - I_{CQ} R_C$$

$$2.6 = 5 - (0.5535) R_C \Rightarrow R_C = 4.34 \text{ k}\Omega$$

EX5.14

$$R_C = \frac{V_{CC} - V_{CEQ}}{I_{CQ}} = \frac{2.8 - 1.4}{0.12} = 11.7 \text{ k}\Omega$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.12}{150} \Rightarrow I_{BQ} = 0.80 \mu\text{A}$$

$$R_B = \frac{V_{CC} - V_{BEQ}}{I_{BQ}} = \frac{2.8 - 0.7}{0.80} = 2.625 \text{ M}\Omega$$

EX5.15

(a) $R_{TH} = R_1 \parallel R_2 = 85 \parallel 35 = 24.8 \text{ k}\Omega$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) (V_{CC}) = \left(\frac{35}{35 + 85} \right) (3.3) = 0.9625 \text{ V}$$

(b) $V_{TH} = I_{BQ} R_{TH} + V_{BE}(\text{on}) + (1 + \beta) I_{BQ} R_E$

so

$$I_{BQ} = \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1 + \beta) R_E} = \frac{0.9625 - 0.7}{24.8 + (151)(0.5)} \Rightarrow I_{BQ} = 2.617 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = (150)(0.002617) = 0.3926 \text{ mA}$$

$$I_{EQ} = \left(\frac{1 + \beta}{\beta} \right) I_{CQ} = \left(\frac{151}{150} \right) (0.3926) = 0.3952 \text{ mA}$$

$$V_{CEQ} = 3.3 - (0.3926)(4) - (0.3952)(0.5) = 1.53 \text{ V}$$

(c) $I_{BQ} = \frac{0.9625 - 0.7}{24.8 + (76)(0.5)} \Rightarrow I_{BQ} = 4.18 \mu\text{A}$

Then $I_{CQ} = (75)(0.00418) = 0.3135 \text{ mA}$

$$I_{EQ} = (76)(0.00418) = 0.3177 \text{ mA}$$

$$V_{CEQ} = 3.3 - (0.3135)(4) - (0.3177)(0.5) = 1.89 \text{ V}$$

EX5.16

$$V_{CEQ} \cong V_{CC} - I_{CQ}(R_C + R_E)$$

$$\text{or } 2.5 \cong 5 - I_{CQ}(1 + 0.2)$$

which yields

$$I_{CQ} = 2.08 \text{ mA},$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2.08}{150} = 0.0139 \text{ mA}$$

$$R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(151)(0.2)$$

$$\text{or } R_{TH} = 3.02 \text{ k}\Omega$$

$$\text{Now } V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC}$$

$$\text{so } V_{TH} = \frac{1}{R_1}(3.02)(5)$$

$$\text{We can write } V_{TH} = I_{BQ}R_{TH} + V_{BE}(\text{on}) + (1 + \beta)I_{BQ}R_E$$

$$\text{or } \frac{1}{R_1}(3.02)(5) = (0.0139)(3.02) + 0.7 + (151)(0.0139)(0.2)$$

$$\text{We obtain } R_1 = 13 \text{ k}\Omega \text{ and then } R_2 = 3.93 \text{ k}\Omega$$

EX5.17

$$I_{CQ} = \frac{V^+ - V_o}{R_C} = \frac{5 - 0}{10} = 0.5 \text{ mA}$$

$$I_{EQ} = \left(\frac{1 + \beta}{\beta} \right) I_{CQ} = \left(\frac{151}{150} \right) (0.5) = 0.5033 \text{ mA}$$

$$\begin{aligned} V_{CEQ} &= (V^+ - V^-) - I_{CQ}R_C - I_{EQ}R_E \\ &= 10 - (0.5)(10) - (0.5033)(2) = 3.99 \text{ V} \end{aligned}$$

Now

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.5}{150} \Rightarrow I_{BQ} = 3.33 \mu\text{A}$$

$$R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(151)(2) = 30.2 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \frac{1}{R_1} (R_{TH})(10) - 5 = \frac{1}{R_1} (30.2)(10) - 5$$

Also

$$\begin{aligned} V_{TH} &= I_{BQ}R_{TH} + V_{BE}(\text{on}) + I_{EQ}R_E - 5 \\ &= (0.00333)(30.2) + 0.7 + (0.5033)(2) - 5 = -3.193 \text{ V} \end{aligned}$$

Then

$$\frac{1}{R_1}(30.2)(10) - 5 = -3.193$$

$$\text{or } R_1 = 167 \text{ k}\Omega \text{ and } 167 \parallel R_2 = 30.2 \Rightarrow R_2 = 36.9 \text{ k}\Omega$$

EX5.18

$$V_{EO} = -0.7 \text{ V}, \quad V_{CO} = -0.7 + 1.6 = 0.9 \text{ V}$$

$$R_C = \frac{V^+ - V_{CO}}{I_{CQO}} = \frac{3.3 - 0.9}{0.12} = 20 \text{ k}\Omega$$

$$I_Q = \left(\frac{1 + \beta}{\beta} \right) I_{CQO} = \left(\frac{61}{60} \right) (0.12) = 0.122 \text{ mA}$$

$$I_1 = \left(1 + \frac{2}{\beta} \right) I_Q = \left(1 + \frac{2}{60} \right) (0.122) = 0.126 \text{ mA}$$

$$I_1 = 0.126 = \frac{0 - V_{BE(on)} - (-3.3)}{R_1} = \frac{3.3 - 0.7}{R_1} \Rightarrow R_1 = 20.6 \text{ k}\Omega$$

EX5.19

$$R_{TH} = 50 \parallel 100 = 33.3 \text{ k}\Omega$$

$$V_{TH} = V_{TH} = \left(\frac{50}{50 + 100} \right) (10) - 5 = -1.67 \text{ V}$$

$$I_{B1} = \frac{-1.67 - 0.7 - (-5)}{33.3 + (101)(2)} \Rightarrow 11.2 \mu\text{A}$$

$$I_{C1} = 1.12 \text{ mA}, \quad I_{E1} = 1.13 \text{ mA}$$

$$V_{E1} = I_{E1} R_{E1} - 5 = (1.13)(2) - 5 = -2.74 \text{ V}$$

$$V_{CE1} = 3.25 \text{ V} \Rightarrow V_{C1} = 0.51 \text{ V}$$

$$\text{Now } V_{E2} = 0.51 + 0.7 = 1.21 \text{ V}$$

$$I_{E2} = \frac{5 - 1.21}{2} = 1.90 \text{ mA} \Rightarrow I_{B2} = 18.8 \mu\text{A}$$

$$I_{C2} = 1.88 \text{ mA}$$

$$I_{R1} = I_{C1} - I_{B2} = 1.12 - 0.0188 = 1.10 \text{ mA}$$

$$R_{C1} = \frac{5 - 0.51}{1.10} = 4.08 \text{ k}\Omega$$

$$V_{EC2} = 2.5 \Rightarrow V_{C2} = V_{E2} - V_{EC2} \\ = 1.21 - 2.5 = -1.29 \text{ V}$$

$$R_{C2} = \frac{-1.29 - (-5)}{1.88} = 1.97 \text{ k}\Omega$$

EX5.20

$$\text{We find } \frac{12}{0.05} = 240 \text{ k}\Omega = R_1 + R_2 + R_3$$

$$\text{Then } V_{B1} = (0.5)(2) + 0.7 = 1.7 \text{ V}$$

$$R_3 = \frac{1.7}{0.05} = 34 \text{ k}\Omega$$

$$\text{Also } V_{B2} = (0.5)(2) + 4 + 0.7 = 5.7 \text{ V}$$

$$\Delta V_{R2} = 5.7 - 1.7 = 4 \text{ V}$$

$$\text{so } R_2 = \frac{4}{0.05} = 80 \text{ k}\Omega$$

$$\text{and } R_1 = 240 - 80 - 34 = 126 \text{ k}\Omega$$

$$V_{C2} = 1 + 4 + 4 = 9 \text{ V}$$

$$\text{Then } R_C = \frac{V^+ - V_{C2}}{I_{CQ}} = \frac{12 - 9}{0.5} = 6 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU5.1

$$(a) \quad \alpha = \frac{\beta}{1 + \beta} = \frac{60}{61} = 0.9836$$

$$\alpha = \frac{150}{151} = 0.9934$$

$$(b) \quad \beta = \frac{\alpha}{1 - \alpha} = \frac{0.982}{1 - 0.982} = 54.6$$

$$\beta = \frac{0.9925}{1 - 0.9925} = 132.3$$

TYU5.2

$$I_E = I_C + I_B = 0.620 + 0.005 = 0.625 \text{ mA}$$

$$\beta = \frac{I_C}{I_B} = \frac{0.620}{0.005} = 124$$

$$\alpha = \frac{\beta}{1 + \beta} = \frac{124}{125} = 0.992$$

TYU5.3

$$I_C = \alpha I_E = (0.9915)(1.20) = 1.19 \text{ mA}$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.9915}{1 - 0.9915} = 116.6$$

$$I_B = \frac{I_E}{1 + \beta} = \frac{1.20}{117.6} \Rightarrow I_B = 10.2 \mu\text{A}$$

TYU5.4

$$(a) \quad r_o = \frac{V_A}{I_C} \Rightarrow V_A = (225)(0.8) = 180 \text{ V}$$

$$(b) \quad (i) \quad r_o = \frac{180}{0.08} \Rightarrow 2.25 \text{ M}\Omega$$

$$(ii) \quad r_o = \frac{180}{8} = 22.5 \text{ k}\Omega$$

TYU5.5

$$I_C = I_O \left(1 + \frac{V_{CE}}{V_A} \right)$$

At $V_{CE} = 1 \text{ V}$, $I_C = 1 \text{ mA}$

(a) For $V_A = 75 \text{ V}$, $I_C = 1 = I_O \left(1 + \frac{1}{75} \right) \Rightarrow I_O = 0.9868 \text{ mA}$

Then, at $V_{CE} = 10 \text{ V}$

$$I_C = (0.9868) \left(1 + \frac{10}{75} \right) = 1.12 \text{ mA}$$

(b) For $V_A = 150 \text{ V}$, $I_C = 1 = I_O \left(1 + \frac{1}{150} \right) \Rightarrow I_O = 0.9934 \text{ mA}$

At $V_{CE} = 10 \text{ V}$, $I_C = (0.9934) \left(1 + \frac{10}{150} \right) = 1.06 \text{ mA}$

TYU5.6

$$BV_{CEO} = \frac{BV_{CBO}}{n\sqrt{\beta}} \quad \text{so} \quad BV_{CBO} = \sqrt[3]{100(30)} = 139 \text{ V}$$

TYU5.7

(a) For $V_I = 0.2 \text{ V} < V_{BE}(\text{on}) \Rightarrow I_B = I_C = 0$, $V_O = 5 \text{ V}$ and $P = 0$

(c) For $V_I = 3.6 \text{ V}$, transistor is driven into saturation, so $I_B = \frac{V_I - V_{BE}(\text{on})}{R_B} = \frac{3.6 - 0.7}{0.64} = 4.53 \text{ mA}$

$$I_C = \frac{V^+ - V_{CE}(\text{sat})}{R_C} = \frac{5 - 0.2}{0.440} = 10.9 \text{ mA}$$

and

$$\frac{I_C}{I_B} = \frac{10.9}{4.53} = 2.41 < \beta$$

Note that which shows that the transistor is indeed driven into saturation. Now,

$$P = I_B V_{BE}(\text{on}) + I_C V_{CE}(\text{sat}) \\ = (4.53)(0.7) + (10.9)(0.2) = 5.35 \text{ mW}$$

TYU5.8

For $V_{BC} = 0 \Rightarrow V_O = 0.7 \text{ V}$

Then $I_C = \frac{5 - 0.7}{0.44} = 9.77 \text{ mA}$ and $I_B = \frac{I_C}{\beta} = \frac{9.77}{50} = 0.195 \text{ mA}$

Now $V_I = I_B R_B + V_{BE}(\text{on}) = (0.195)(0.64) + 0.7$ or $V_I = 0.825 \text{ V}$

Also $P = I_B V_{BE}(\text{on}) + I_C V_{CE} \\ = (0.195)(0.7) + (9.77)(0.7) = 6.98 \text{ mW}$

TYU5.9

$$I_C = \frac{3.3 - V_C}{R_C} = \frac{3.3 - 2.27}{4} = 0.2575 \text{ mA}$$

$$I_E = \frac{-V_{BE}(on) - (-3.3)}{R_E} = \frac{3.3 - 0.7}{10} = 0.260 \text{ mA}$$

$$I_B = I_E - I_C = 0.260 - 0.2575 \Rightarrow I_B = 2.5 \mu\text{A}$$

$$\beta = \frac{I_C}{I_B} = \frac{0.2575}{0.0025} = 103$$

$$\alpha = \frac{\beta}{1 + \beta} = \frac{103}{104} = 0.99038$$

TYU5.10

$$I_E = \frac{5 - V_{EB}(on)}{R_E} = \frac{5 - 0.7}{8} = 0.5375 \text{ mA}$$

$$I_C = \left(\frac{\beta}{1 + \beta} \right) I_E = \left(\frac{85}{86} \right) (0.5375) = 0.531 \text{ mA}$$

$$I_B = \frac{I_E}{1 + \beta} = \frac{0.5375}{86} \Rightarrow I_B = 6.25 \mu\text{A}$$

$$V_{EC} = 10 - (0.531)(4) - (0.5375)(8) = 3.575 \text{ V}$$

TYU5.11

$$V_{BB} = I_B R_B + V_{BE}(on) + I_E R_E$$

or $V_{BB} = I_B R_B + V_{BE}(on) + (1 + \beta) I_B R_E$

$$I_B = \frac{V_{BB} - V_{BE}(on)}{R_B + (1 + \beta) R_E} = \frac{2 - 0.7}{10 + (76)(1)}$$

Then $I_B = 15.1 \mu\text{A}$

or $I_B = 15.1 \mu\text{A}$

Also $I_C = (75)(15.1 \mu\text{A}) = 1.13 \text{ mA}$ and $I_E = (76)(15.1 \mu\text{A}) = 1.15 \text{ mA}$

Now $V_{CE} = V_{CC} + V_{BB} - I_C R_C - I_E R_E$

$$= 8 + 2 - (1.13)(2.5) - (1.15)(1) = 6.03 \text{ V}$$

TYU5.12

$$V_E = 5 - V_{CE} = 5 - 2.2 = 2.8 \text{ V}$$

$$I_B = \frac{V_{BB} - V_{BE}(on) - V_E}{R_B} = \frac{5 - 0.7 - 2.8}{10} = 0.15 \text{ mA}$$

$$I_E = (1 + \beta) I_B = (121)(0.15) = 18.15 \text{ mA}$$

$$R_E = \frac{V_E}{I_E} = \frac{2.8}{18.15} = 0.154 \text{ k}\Omega$$

TYU5.13

$$(a) \quad I_E = 1.2 \text{ mA}, \quad I_B = \frac{I_E}{1 + \beta} = \frac{1.2}{91} = 0.01319 \text{ mA}$$

$$V_{BB} = I_E R_E + V_{EB}(\text{on}) + I_B R_B \\ = (1.2)(1) + 0.7 + (0.01319)(50) = 2.56 \text{ V}$$

$$(b) \quad I_C = \left(\frac{\beta}{1 + \beta} \right) I_E = \left(\frac{90}{91} \right) (1.2) = 1.187 \text{ mA}$$

$$V_{EC} = 5 - I_E R_E = 5 - (1.2)(1) = 3.8 \text{ V}$$

TYU5.14

$$(a) \quad \text{For } v_I = 0, \quad i_B = i_C = 0, \quad v_O = 12 \text{ V}, \quad P = 0$$

$$(b) \quad \text{For } v_I = 12 \text{ V}, \quad i_B = \frac{v_I - V_{BE}(\text{on})}{R_B} = \frac{12 - 0.7}{0.24} = 47.1 \text{ mA}$$

$$i_C = \frac{V_{CC} - V_{CE}(\text{sat})}{R_C} = \frac{12 - 0.1}{5} = 2.38 \text{ A}$$

$$v_O = 0.1 \text{ V}$$

and

$$P = i_B V_{BE}(\text{on}) + i_C V_{CE}(\text{sat}) \\ = (0.0471)(0.7) + (2.38)(0.1) = 0.271 \text{ W}$$

TYU5.15

$$(a) \quad I_{CQ} = \frac{5 - V_{CEQ}}{R_C} = \frac{5 - 2.5}{2} = 1.25 \text{ mA}$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{1.25}{120} \Rightarrow I_{BQ} = 10.42 \mu\text{A}$$

$$R_B = \frac{5 - V_{BE}(\text{on})}{I_{BQ}} = \frac{5 - 0.7}{0.01042} = 413 \text{ k}\Omega$$

$$(b) \quad I_{BQ} = 10.42 \mu\text{A}$$

$$\text{For } \beta = 80 \Rightarrow I_{CQ} = 0.8336 \text{ mA}$$

$$\text{For } \beta = 160 \Rightarrow I_{CQ} = 1.667 \text{ mA}$$

Now

$$V_{CEQ} = 5 - I_{CQ}(2)$$

$$\text{For } \beta = 80 \Rightarrow V_{CEQ} = 5 - (0.8336)(2) = 3.33 \text{ V}$$

$$\text{For } \beta = 160 \Rightarrow V_{CEQ} = 5 - (1.667)(2) = 1.67 \text{ V}$$

$$\text{So} \quad 1.67 \leq V_{CEQ} \leq 3.33 \text{ V}$$

TYU5.16

$$I_{BQ} = \frac{5 - 0.7}{800} = 0.005375 \text{ mA}$$

For $\beta = 75$, $I_{CQ} = \beta I_{BQ} = (75)(0.005375)$

Or $I_{CQ} = 0.403 \text{ mA}$

For $\beta = 150$, $I_{CQ} = (150)(0.005375)$

Or $I_{CQ} = 0.806 \text{ mA}$

Largest $I_{CQ} \Rightarrow$ Smallest V_{CEQ}

For $\beta = 150$, $R_C = \frac{5 - 1}{0.806} = 4.96 \text{ k}\Omega$

For $\beta = 75$, $R_C = \frac{5 - 4}{0.403} = 2.48 \text{ k}\Omega$

For a nominal $I_{CQ} = 0.604 \text{ mA}$ and $V_{CEQ} = 2.5 \text{ V}$, $R_C = \frac{5 - 2.5}{0.604} = 4.14 \text{ k}\Omega$

Now for $I_{CQ} = 0.403 \text{ mA}$, $V_{CEQ} = 5 - (0.403)(4.14) = 3.33 \text{ V}$

For $I_{CQ} = 0.806 \text{ mA}$, $V_{CEQ} = 5 - (0.806)(4.14) = 1.66 \text{ V}$

So, for $R_C = 4.14 \text{ k}\Omega$, $1.66 \leq V_{CEQ} \leq 3.33 \text{ V}$

TYU5.17

(a) $R_{TH} = R_1 \parallel R_2 = 440 \parallel 230 = 151 \text{ k}\Omega$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{230}{440 + 230} \right) (5) = 1.716 \text{ V}$$

(b) $I_{BQ} = \frac{V_{TH} - V_{BE(on)}}{R_{TH} + (1 + \beta)R_E} = \frac{1.716 - 0.7}{151 + (151)(1)} \Rightarrow I_{BQ} = 3.364 \mu\text{A}$

$$I_{CQ} = \beta I_{BQ} = 0.5046 \text{ mA}; \quad I_{EQ} = (1 + \beta)I_{BQ} = 0.508 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_{EQ}R_E \\ = 5 - (0.5046)(4) - (0.508)(1) = 2.47 \text{ V}$$

(c) $R_{TH} = 151 \text{ k}\Omega$; $V_{TH} = 1.716 \text{ V}$

$$I_{BQ} = \frac{1.716 - 0.7}{151 + (91)(1)} \Rightarrow I_{BQ} = 4.2 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = 0.378 \text{ mA}; \quad I_{EQ} = (1 + \beta)I_{BQ} = 0.382 \text{ mA}$$

$$V_{CEQ} = 5 - (0.378)(4) - (0.382)(1) = 3.11 \text{ V}$$

TYU5.18

(a) $R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(151)(1) = 15.1 \text{ k}\Omega$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.4}{150} \Rightarrow I_{BQ} = 2.667 \mu\text{A}$$

$$I_{EQ} = \left(\frac{1 + \beta}{\beta} \right) I_{CQ} = \left(\frac{151}{150} \right) (0.4) = 0.4027 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_{EQ}R_E$$

$$2.7 = 5 - (0.4)R_C - (0.4027)(1) \Rightarrow R_C = 4.74 \text{ k}\Omega$$

Now

$$V_{TH} = \frac{1}{R_1}(R_{TH})V_{CC} = I_{BQ}R_{TH} + V_{BE}(on) + I_{EQ}R_E$$

$$\frac{1}{R_1}(15.1)(5) = (0.002667)(15.1) + 0.7 + (0.4027)(1)$$

$$\text{yields } R_1 = 66 \text{ k}\Omega; \quad 66 \parallel R_2 = 15.1 \Rightarrow R_2 = 19.6 \text{ k}\Omega$$

(b) $V_{TH} = 1.143 \text{ V}; \quad R_{TH} = 15.1 \text{ k}\Omega$

$$I_{BQ} = \frac{V_{TH} - V_{BE}(on)}{R_{TH} + (1 + \beta)R_E} = \frac{1.143 - 0.7}{15.1 + (91)(1)} \Rightarrow I_{BQ} = 4.175 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = 0.376 \text{ mA}; \quad I_{EQ} = (1 + \beta)I_{BQ} = 0.380 \text{ mA}$$

$$V_{CEQ} = 5 - (0.376)(4.74) - (0.380)(1) = 2.84 \text{ V}$$

TYU5.19

$$V_{ECQ} = 5 \cong 10 - I_{CQ}(R_C + R_E) = 10 - I_{CQ}(4.5 + 0.5) \Rightarrow I_{CQ} \cong 1 \text{ mA}$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{1}{120} \Rightarrow I_{BQ} = 8.33 \mu\text{A}$$

$$I_{EQ} = \left(\frac{1 + \beta}{\beta} \right) I_{CQ} = \left(\frac{121}{120} \right) (1) = 1.008 \text{ mA}$$

$$R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(121)(0.5) = 6.05 \text{ k}\Omega$$

$$V^+ = I_{EQ}R_E + V_{EB}(on) + I_{BQ}R_{TH} + V_{TH}$$

$$5 = (1.008)(0.5) + 0.7 + (0.00833)(6.05) + V_{TH} \Rightarrow V_{TH} = 3.746 \text{ V}$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \frac{1}{R_1} (R_{TH})(10) - 5$$

So

$$\frac{1}{R_1} (6.05)(10) - 5 = 3.746 \Rightarrow R_1 = 6.92 \text{ k}\Omega$$

$$6.92 \parallel R_2 = 6.05 \Rightarrow R_2 = 48.1 \text{ k}\Omega$$

TYU5.20

(a) $I_{CQ} = \left(\frac{\beta}{1 + \beta} \right) I_Q = \left(\frac{120}{121} \right) (0.25) = 0.2479 \text{ mA}$

$$I_{CQ} = I_S e^{V_{BE}/V_T}$$

or

$$V_{BE} = V_T \ln \left(\frac{I_{CQ}}{I_S} \right) = (0.026) \ln \left(\frac{0.2479 \times 10^{-3}}{3 \times 10^{-14}} \right) = 0.5937 \text{ V}$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.2479}{120} \Rightarrow I_{BQ} = 2.066 \mu\text{A}$$

$$V_B = -(0.002066)(75) = -0.155 \text{ V}$$

$$V_E = V_B - V_{BE} = -0.155 - 0.5937 = -0.7487 \text{ V}$$

$$V_C = V^+ - I_{CQ}R_C = 2.5 - (0.2479)(4) = 1.508 \text{ V}$$

$$V_{CEQ} = V_C - V_E = 1.508 - (-0.7487) = 2.26 \text{ V}$$

$$(b) \quad I_{CQ} = \left(\frac{60}{61}\right)(0.25) = 0.2459 \text{ mA}$$

$$V_{BE} = (0.026) \ln \left(\frac{0.2459 \times 10^{-3}}{3 \times 10^{-14}} \right) = 0.5935 \text{ V}$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.2459}{60} \Rightarrow I_{BQ} = 4.098 \mu \text{ A}$$

$$V_B = -(0.004098)(75) = -0.307 \text{ V}$$

$$V_E = -0.307 - 0.5935 = -0.901 \text{ V}$$

$$V_C = 2.5 - (0.2459)(4) = 1.516 \text{ V}$$

$$V_{CEQ} = 1.516 - (-0.901) = 2.42 \text{ V}$$

Chapter 6

Exercise Solutions

EX6.1

$$\begin{aligned} \text{(a)} \quad I_{BQ} &= \frac{V_{BB} - V_{BE}(on)}{R_B} = \frac{0.85 - 0.7}{180} \Rightarrow I_{BQ} = 0.833 \mu\text{A} \\ I_{CQ} &= \beta I_{BQ} = (120)(0.000833) = 0.10 \text{ mA} \\ V_{CEQ} &= V_{CC} - I_{CQ} R_C = 3.3 - (0.1)(15) = 1.8 \text{ V} \\ \text{(b)} \quad g_m &= \frac{I_{CQ}}{V_T} = \frac{0.1}{0.026} = 3.846 \text{ mA/V} \\ r_\pi &= \frac{\beta V_T}{I_{CQ}} = \frac{(120)(0.026)}{0.1} = 31.2 \text{ k}\Omega \\ \text{(c)} \quad A_v &= -g_m R_C \left(\frac{r_\pi}{r_\pi + R_B} \right) = -(3.846)(15) \left(\frac{31.2}{31.2 + 180} \right) = -8.52 \end{aligned}$$

EX6.2

$$\begin{aligned} \text{(a)} \quad I_{BQ} &= \frac{V_{BB} - V_{BE}(on)}{R_B} = \frac{1.025 - 0.7}{100} = 0.00325 \text{ mA} \\ I_{CQ} &= \beta I_{BQ} = (150)(0.00325) = 0.4875 \text{ mA} \\ g_m &= \frac{I_{CQ}}{V_T} = \frac{0.4875}{0.026} = 18.75 \text{ mA/V} \\ r_\pi &= \frac{\beta V_T}{I_{CQ}} = \frac{(150)(0.026)}{0.4875} = 8 \text{ k}\Omega \\ r_o &= \frac{V_A}{I_{CQ}} = \frac{150}{0.4875} = 308 \text{ k}\Omega \\ \text{(b)} \quad A_v &= -g_m (r_o \parallel R_C) \left(\frac{r_\pi}{r_\pi + R_B} \right) = -(18.75)(308 \parallel 6) \left(\frac{8}{8 + 100} \right) = -8.17 \end{aligned}$$

EX6.3

$$\begin{aligned} \text{(a)} \quad I_{BQ} &= \frac{V_{BB} - V_{EB}(on)}{R_B} = \frac{1.145 - 0.7}{50} \\ \text{or } I_{BQ} &= 0.0089 \text{ mA} \\ \text{Then } I_{CQ} &= \beta I_{BQ} = (90)(0.0089) = 0.801 \text{ mA} \\ \text{Now} \end{aligned}$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.801}{0.026} = 30.8 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(90)(0.026)}{0.801} = 2.92 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{120}{0.801} = 150 \text{ k}\Omega$$

(b) We have $V_o = g_m V_\pi (r_o \parallel R_C)$

$$V_\pi = -\left(\frac{r_\pi}{r_\pi + R_B}\right)V_s$$

and
so

$$\begin{aligned} A_v = \frac{V_o}{V_s} &= -g_m \left(\frac{r_\pi}{r_\pi + R_B}\right)(r_o \parallel R_C) \\ &= -(30.8) \left(\frac{2.92}{2.92 + 50}\right)(150 \parallel 2.5) \end{aligned}$$

which yields $A_v = -4.18$

EX6.4

Using Figure 6.23

(a) For $I_{CQ} = 0.2 \text{ mA}$, $7.8 < h_{ie} < 15 \text{ k}\Omega$, $60 < h_{fe} < 125$, $6.2 \times 10^{-4} < h_{re} < 50 \times 10^{-4}$,
 $5 < h_{oe} < 13 \text{ }\mu\text{mhos}$

(b) For $I_{CQ} = 5 \text{ mA}$, $0.7 < h_{ie} < 1.1 \text{ k}\Omega$, $140 < h_{fe} < 210$, $1.05 \times 10^{-4} < h_{re} < 1.6 \times 10^{-4}$,
 $22 < h_{oe} < 35 \text{ }\mu\text{mhos}$

EX6.5

$$R_{TH} = R_1 \parallel R_2 = 250 \parallel 75 = 57.7 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{CC} = \left(\frac{75}{75 + 250}\right)(5)$$

or

$$V_{TH} = 1.154 \text{ V}$$

$$I_{BQ} = \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1 + \beta)R_E} = \frac{1.154 - 0.7}{57.7 + (121)(0.6)}$$

or

$$I_{BQ} = 3.48 \text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = (120)(3.48 \text{ }\mu\text{A}) = 0.418 \text{ mA}$$

(a)

Now

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.418}{0.026} = 16.08 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(120)(0.026)}{0.418} = 7.46 \text{ k}\Omega$$

We have

$$V_o = -g_m V_\pi R_C$$

We find

$$R_{ib} = r_\pi + (1 + \beta) R_E = 7.46 + (121)(0.6)$$

or

$$R_{ib} = 80.1 \text{ k}\Omega$$

Also

$$R_1 \parallel R_2 = 250 \parallel 75 = 57.7 \text{ k}\Omega$$

$$R_1 \parallel R_2 \parallel R_{ib} = 57.7 \parallel 80.1 = 33.54 \text{ k}\Omega$$

We find

$$V'_s = \left(\frac{R_1 \parallel R_2 \parallel R_{ib}}{R_1 \parallel R_2 \parallel R_{ib} + R_s} \right) \cdot V_s = \left(\frac{33.54}{33.54 + 0.5} \right) \cdot V_s$$

or

$$V'_s = (0.985) V_s$$

Now

$$V'_s = V_\pi \left[1 + \left(\frac{1 + \beta}{r_\pi} \right) R_E \right] = V_\pi \left[1 + \left(\frac{121}{7.46} \right) (0.6) \right]$$

or

$$V_\pi = (0.0932) V'_s = (0.0932)(0.985) V_s$$

So

$$A_v = \frac{V_o}{V_s} = -(16.08)(0.0932)(0.985)(5.6)$$

or

$$A_v = -8.27$$

EX6.6

(a) $R_{TH} = R_1 \parallel R_2 = 14.4 \parallel 110 = 12.73 \text{ k}\Omega$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \left(\frac{110}{110 + 14.4} \right) (12) = 10.61 \text{ V}$$

$$12 = (101) I_{BQ} (0.3) + 0.7 + I_{BQ} (12.73) + 10.61$$

so

$$I_{BQ} = 0.0160 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 1.60 \text{ mA}; \quad I_{EQ} = (1 + \beta) I_{BQ} = 1.62 \text{ mA}$$

$$V_{ECQ} = 12 - (1.6)(4) - (1.62)(0.3) = 5.11 \text{ V}$$

(b) $g_m = \frac{I_{CQ}}{V_T} = \frac{1.60}{0.026} = 61.54 \text{ mA/V}$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{1.60} = 1.625 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \infty$$

(c) $A_v = \frac{-\beta(R_C \parallel R_L)}{r_\pi + (1 + \beta)R_E} = \frac{-(100)(4 \parallel 10)}{1.625 + (101)(0.3)} = -8.95$

EX6.7

(a) $R_i = R_S + R_B \parallel r_\pi$

$$I_{BQ}R_B + 0.7 + (1 + \beta)I_{BQ}R_E + V^- = 0$$

$$5 - 0.7 = I_{BQ}[100 + (121)(4)] \Rightarrow I_{BQ} = 0.007363 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 0.8836 \text{ mA}$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.8836}{0.026} = 33.98 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(120)(0.026)}{0.8836} = 3.53 \text{ k}\Omega$$

$$R_i = 0.5 + 100 \parallel 3.53 = 3.91 \text{ k}\Omega$$

(b) $r_o = \frac{V_A}{I_{CQ}} = \frac{80}{0.8836} = 90.5 \text{ k}\Omega$

$$V_\pi = \left(\frac{R_B \parallel r_\pi}{R_B \parallel r_\pi + R_S} \right) \cdot v_s = \left(\frac{100 \parallel 3.53}{100 \parallel 3.53 + 0.5} \right) \cdot v_s = (0.872)v_s$$

$$A_v = -g_m(R_C \parallel r_o) \frac{V_\pi}{v_s} = -(33.98)(4 \parallel 90.5)(0.872) = -114$$

EX6.8

(a)

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.25}{0.026} = 9.615 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{100}{0.25} = 400 \text{ k}\Omega$$

$$A_v = -g_m(r_o \parallel r_c) = -(9.615)(400 \parallel 100) = -769$$

(b)

$$A_v = -g_m(r_o \parallel r_c \parallel r_L) = -(9.615)(400 \parallel 100 \parallel 100)$$

$$A_v = -427$$

EX6.9

$$I_{BQ} = \frac{5 - 0.7}{10 + (126)(5)} = 0.00672 \text{ mA}$$

$$I_{CQ} = 0.84 \text{ mA}, I_{EQ} = 0.847 \text{ mA}$$

$$V_{CEQ} = 10 - (0.84)(2.3) - (0.847)(5)$$

or

$$V_{CEQ} = 3.83 \text{ V}$$

dc load line

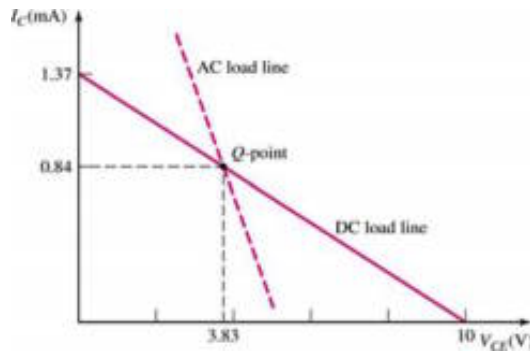
$$V_{CE} \cong (V^+ - V^-) - I_C(R_C + R_E)$$

or

$$V_{CE} = 10 - I_C(7.3)$$

ac load line (neglecting r_o)

$$v_{ce} = -i_c (R_C \parallel R_L) = -i_c (2.3 \parallel 5) = -i_c (1.58)$$



EX6.10

$$(b) I_{BQ} R_B + 0.7 + (1 + \beta) R_E - 5 = 0$$

$$\Rightarrow I_{BQ} = 0.007363 \text{ mA}; I_{CQ} = \beta I_{BQ} = 0.884 \text{ mA}; I_{EQ} = 0.8909 \text{ mA}$$

$$V_{CEQ} = 10 - (0.8836)(4) - (0.8909)(4) = 2.90 \text{ V}$$

$$(c) \Delta V_{CE} = -\Delta I_C (R_C \parallel r_o) = -\Delta I_C (4 \parallel 90.5) = -\Delta I_C (3.831)$$

$$\text{For } |\Delta V_{CE}| = 2.9 - 0.5 = 2.4 \text{ V}$$

$$\text{Then } |\Delta I_C| = \frac{2.4}{3.831} = 0.626 \text{ mA}; \Delta v_{ce} = 4.8 \text{ V, peak-to-peak}$$

EX6.11

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC}$$

$$(a) R_{TH} = (0.1)(1 + \beta) R_E = (0.1)(121)(1)$$

so

$$R_{TH} = 12.1 \text{ k}\Omega, V_{TH} = \frac{1}{R_1} (12.1)(12)$$

We can write

$$V_{CC} = (1 + \beta) I_{BQ} R_E + V_{EB}(\text{on}) + I_{BQ} R_{TH} + V_{TH}$$

We have

$$I_{CQ} = 1.6 \text{ mA}, I_{BQ} = \frac{1.6}{120} = 0.0133 \text{ mA}$$

Then

$$I_{BQ} = 0.0133 = \frac{12 - 0.7 - \frac{1}{R_1} (12.1)(12)}{12.1 + (121)(1)}$$

which yields

$$R_1 = 15.24 \text{ k}\Omega$$

Since $R_{TH} = R_1 \parallel R_2 = 12.1 \text{ k}\Omega$, we find $R_2 = 58.7 \text{ k}\Omega$

$$\text{Also } V_{ECQ} = 12 - (1.6)(4) - (1.61)(1) = 3.99 \text{ V}$$

(b) ac load line $\Delta v_{ec} = -i_c (R_C \parallel R_L)$
 Want $\Delta i_c = I_{CQ} - 0.1 = 1.6 - 0.1 = 1.5 \text{ mA}$
 Also $\Delta v_{ec} = 3.99 - 0.5 = 3.49 \text{ V}$
 Now $\frac{\Delta v_{ec}}{\Delta i_c} = \frac{3.49}{1.5} = 2.327 \text{ k}\Omega = R_C \parallel R_L$
 So $4 \parallel R_L = 2.327 \text{ k}\Omega$ which yields $R_L = 5.56 \text{ k}\Omega$

EX6.12

(a) $R_{TH} = R_1 \parallel R_2 = 1.3 \parallel 4.2 = 0.9927 \text{ k}\Omega$
 $V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{4.2}{1.3 + 4.2} \right) (12) = 9.1636 \text{ V}$
 $I_{BQ} = \frac{9.1636 - 0.7}{0.9927 + (81)(0.03)} = 2.473 \text{ mA}$
 $I_{EQ} = (1 + \beta) I_{BQ} = 0.20 \text{ A}, I_{CQ} = \beta I_{BQ} = 0.1978 \text{ A}$
 $V_{CEQ} = 12 - (0.20)(30) = 6.0 \text{ V}$
 (b) $g_m = \frac{I_{CQ}}{V_T} = \frac{0.1978}{0.026} = 7.608 \text{ A/V}$
 $r_\pi = 10.52 \text{ }\Omega, r_o = 379.2 \text{ }\Omega$
 $A_v = \frac{(1 + \beta)(r_o \parallel R_E)}{r_\pi + (1 + \beta)(r_o \parallel R_E)} = \frac{(81)(379.2 \parallel 30)}{10.52 + (81)(379.2 \parallel 30)} = 0.9953$
 (c) $R_{ib} = r_\pi + (1 + \beta)(r_o \parallel R_E) = 10.52 + (81)(379.2 \parallel 30)$
 or
 $R_{ib} = 2.26 \text{ k}\Omega$

EX6.13

$$R_o = R_E \parallel r_o \parallel \frac{r_\pi}{1 + \beta} = 30 \parallel 379.2 \parallel \frac{10.52}{81} = 0.129 \text{ }\Omega$$

EX6.14

$I_{CQ} = 1.25 \text{ mA}$ and $\beta = 100$, we find
 (a) For $I_{EQ} = 1.26 \text{ mA}$ and $I_{BQ} = 0.0125 \text{ mA}$
 Now $V_{CEQ} = 10 - I_{EQ} R_E$
 Or
 $4 = 10 - (1.26) R_E$
 which yields
 $R_E = 4.76 \text{ k}\Omega$
 Then
 $R_{TH} = (0.1)(1 + \beta) R_E = (0.1)(101)(4.76)$
 or
 $R_{TH} = 48.1 \text{ k}\Omega$

We have

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \frac{1}{R_1} \cdot R_{TH} (10) - 5$$

or

$$V_{TH} = \frac{1}{R_1} (481) - 5$$

$$I_{BQ} = \frac{V_{TH} - 0.7 - (-5)}{R_{TH} + (1 + \beta) R_E}$$

We can write

Or

$$0.0125 = \frac{\frac{1}{R_1} (481) - 5 - 0.7 + 5}{48.1 + (101)(4.76)}$$

which yields

$$R_1 = 65.8 \text{ k}\Omega$$

Since $R_1 \parallel R_2 = 48.1 \text{ k}\Omega$, we obtain

$$R_2 = 178.8 \text{ k}\Omega$$

(b)

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{1.25} = 2.08 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{125}{1.25} = 100 \text{ k}\Omega$$

We may note that

$$g_m V_\pi = g_m (I_b r_\pi) = \beta I_b$$

Also

$$\begin{aligned} R_{ib} &= r_\pi + (1 + \beta)(R_E \parallel R_L \parallel r_o) \\ &= 2.08 + (101)(4.76 \parallel 1 \parallel 100) \end{aligned}$$

or

$$R_{ib} = 84.9 \text{ k}\Omega$$

Now

$$I_o = \left(\frac{R_E \parallel r_o}{R_E \parallel r_o + R_L} \right) (1 + \beta) I_b$$

where

$$I_b = \left(\frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_{ib}} \right) \cdot I_s$$

We can then write

$$A_I = \frac{I_o}{I_s} = \left(\frac{R_E \parallel r_o}{R_E \parallel r_o + R_L} \right) (1 + \beta) \left(\frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_{ib}} \right)$$

We have

$$R_E \parallel r_o = 4.76 \parallel 100 = 4.54 \text{ k}\Omega$$

so

$$A_I = \left(\frac{4.54}{4.54 + 1} \right) (101) \left(\frac{48.1}{48.1 + 84.9} \right)$$

or

$$A_I = 29.9$$

(c)

$$R_o = R_E \parallel r_o \parallel \frac{r_\pi}{1 + \beta} = 4.76 \parallel 100 \parallel \frac{2.08}{101}$$

or

$$R_o = 20.5 \, \Omega$$

EX6.15

(a)

$$R_{TH} = R_1 \parallel R_2 = 70 \parallel 6 = 5.53 \, k\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5 = \left(\frac{6}{70 + 6} \right) (10) - 5$$

or

$$V_{TH} = -4.2105 \, V$$

We find

$$I_{BQ1} = \frac{-4.2105 - 0.7 - (-5)}{5.53 + (126)(0.2)} \Rightarrow 2.91 \, \mu A$$

and

$$I_{CQ1} = \beta I_{BQ1} = (125)(2.91 \, \mu A) = 0.364 \, mA$$

$$I_{EQ1} = (1 + \beta) I_{BQ1} = 0.368 \, mA$$

At the collector of Q_1 ,

$$\frac{5 - V_{C1}}{R_{C1}} = I_{CQ1} + \frac{V_{C1} - 0.7 - (-5)}{(1 + \beta)(R_{E2})}$$

or

$$\frac{5 - V_{C1}}{5} = 0.364 + \frac{V_{C1} - 0.7 - (-5)}{(126)(1.5)}$$

which yields

$$V_{C1} = 2.99 \, V$$

also

$$V_{E1} = I_{EQ1} R_{E1} - 5 = (0.368)(0.2) - 5$$

or

$$V_{E1} = -4.93 \, V$$

Then

$$V_{CEQ1} = V_{C1} - V_{E1} = 2.99 - (-4.93) = 7.92 \, V$$

We find

$$I_{EQ2} = \frac{V_{C1} - 0.7 - (-5)}{1.5} = 4.86 \, mA$$

and

$$I_{CQ2} = \left(\frac{\beta}{1 + \beta} \right) \cdot I_{EQ1} = \left(\frac{125}{126} \right) (4.86) = 4.82 \, mA$$

We find

$$V_{E2} = V_{C1} - 0.7 = 2.99 - 0.7 = 2.29 \, V$$

and

$$V_{CEQ2} = 5 - V_{E2} = 5 - 2.29 = 2.71 \, V$$

(b)

The small-signal transistor parameters are:

$$r_{\pi 1} = \frac{\beta V_T}{I_{CQ1}} = \frac{(125)(0.026)}{0.364} = 8.93 \text{ k}\Omega$$

$$g_{m1} = \frac{I_{CQ1}}{V_T} = \frac{0.364}{0.026} = 14.0 \text{ mA/V}$$

$$r_{\pi 2} = \frac{\beta V_T}{I_{CQ2}} = \frac{(125)(0.026)}{4.82} = 0.674 \text{ k}\Omega$$

$$g_{m2} = \frac{I_{CQ2}}{V_T} = \frac{4.82}{0.026} = 185 \text{ mA/V}$$

We find $R_{ib1} = r_{\pi 1} + (1 + \beta) R_{E1} = 8.93 + (126)(0.2)$

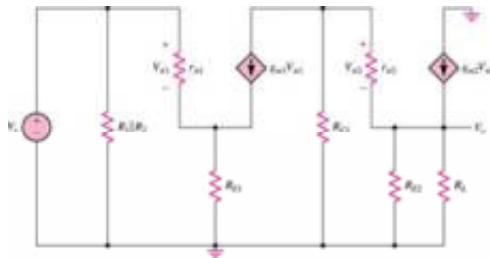
Or

$$R_{ib1} = 34.1 \text{ k}\Omega$$

and

$$\begin{aligned} R_{ib} &= r_{\pi 2} + (1 + \beta)(R_{E2} \parallel R_L) \\ &= 0.674 + (126)(1.5 \parallel 10) = 165 \text{ k}\Omega \end{aligned}$$

The small-signal equivalent circuit is:



We can write

$$V_o = (1 + \beta) I_{b2} (R_{E2} \parallel R_L)$$

where

$$I_{b2} = \left(\frac{R_{C1}}{R_{C1} + R_{ib2}} \right) (-g_{m1} V_{\pi 1})$$

$$V_{\pi 1} = \frac{V_s}{R_{ib1}} \cdot r_{\pi 1}$$

Then

$$A_v = \frac{V_o}{V_i} = (1 + \beta)(R_{E2} \parallel R_L) \left(\frac{R_{C1}}{R_{C1} + R_{ib2}} \right) \left(\frac{-g_{m1} r_{\pi 1}}{R_{ib1}} \right)$$

so

$$A_v = -(126)(1.5 \parallel 10) \left(\frac{5}{5 + 165} \right) \left(\frac{125}{34.1} \right)$$

or

$$A_v = -17.7$$

(c)

$$R_i = R_1 \parallel R_2 \parallel R_{ib1} = 70 \parallel 6 \parallel 34.1 = 4.76 \text{ k}\Omega$$

$$R_o = R_{E2} \parallel \left(\frac{r_{\pi 2} + R_{C1}}{1 + \beta} \right) = 1.5 \parallel \left(\frac{0.676 + 5}{126} \right)$$

and
or

$$R_o = 43.7 \Omega$$

Test Your Understanding Solutions

TYU6.1

$$i_b = \frac{v_s}{R_B + r_{\pi}} = \frac{v_s}{180 + 31.2} = \frac{v_s}{211.2}$$

$$i_b = \frac{0.065 \sin \omega t}{211.2} \Rightarrow i_b = 0.308 \sin \omega t \text{ (}\mu\text{ A)}$$

$$i_B = I_{BQ} + i_b = 0.833 + 0.308 \sin \omega t \text{ (}\mu\text{ A)}$$

$$v_{be} = \left(\frac{r_{\pi}}{r_{\pi} + R_B} \right) \cdot v_s = \left(\frac{31.2}{31.2 + 180} \right) (0.065 \sin \omega t) = 0.00960 \sin \omega t \text{ (V)}$$

$$v_{BE} = V_{BE}(on) + v_{be} = 0.7 + 0.00969 \sin \omega t \text{ (V)}$$

$$|v_{ce}| = |A_v| \cdot v_s = (8.52)(0.065 \sin \omega t) = 0.554 \sin \omega t \text{ (V)}$$

$$v_{CE} = V_{CEQ} + v_{ce} = 1.8 - 0.554 \sin \omega t \text{ (V)}$$

TYU6.2

$$(a) \quad I_{BQ} = \frac{V^+ - V_{EB}(on) - V_{BB}}{R_B} = \frac{3.3 - 0.7 - 2.455}{80} \Rightarrow I_{BQ} = 1.8125 \mu\text{ A}$$

$$I_{CQ} = \beta I_{BQ} = 0.20 \text{ mA}$$

$$V_{ECQ} = 3.3 - (0.2)(7) = 1.9 \text{ V}$$

$$(b) \quad g_m = \frac{I_{CQ}}{V_T} = \frac{0.2}{0.026} = 7.692 \text{ mA/V}$$

$$r_{\pi} = \frac{\beta V_T}{I_{CQ}} = \frac{(110)(0.026)}{0.20} = 14.3 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{80}{0.20} = 400 \text{ k}\Omega$$

$$(c) \quad A_v = -g_m (R_C \parallel r_o) \left(\frac{r_{\pi}}{r_{\pi} + R_B} \right) = -(7.692)(7 \parallel 400) \left(\frac{14.3}{14.3 + 80} \right) = -8.02$$

$$(d) \quad R_i = R_B + r_{\pi} = 80 + 14.3 = 94.3 \text{ k}\Omega$$

$$R_o = R_C \parallel r_o = 7 \parallel 400 = 6.88 \text{ k}\Omega$$

TYU6.3

$$R_{TH} = R_1 \parallel R_2 = 100 \parallel 25 = 20 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{25}{100 + 25} \right) (5) = 1.0 \text{ V}$$

$$I_{BQ} = \frac{V_{TH} - V_{BE(on)}}{R_{TH} + (1 + \beta)R_E} = \frac{1.0 - 0.7}{20 + (121)(0.25)} = 0.00597 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 0.7164 \text{ mA}$$

Now

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.7164}{0.026} = 27.55 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(120)(0.026)}{0.7164} = 4.355 \text{ k}\Omega$$

We find

$$R_{TH} \parallel [r_\pi + (1 + \beta)R_E] = 20 \parallel [4.355 + (121)(0.25)] = 20 \parallel 34.605 = 12.67 \text{ k}\Omega$$

Then

$$A_v = \frac{-\beta R_C}{r_\pi + (1 + \beta)R_E} \cdot \left(\frac{12.67}{12.67 + 0.25} \right) = \frac{-(120)(4)}{4.355 + (121)(0.25)} \cdot (0.9807)$$

or

$$A_v = -13.6$$

TYU6.4

$$A_v \cong -\frac{R_C}{R_E}$$

As a first approximation,

Resulting gain is always smaller than this value. The effect of R_S is very small.

$$\frac{R_C}{R_E} = 10$$

Set

$$\text{Now } 5 \cong I_C (R_C + R_E) + V_{CEQ}$$

$$\text{or } 5 = (0.5)(R_C + R_E) + 2.5$$

$$\text{which yields } R_C + R_E = 5 \text{ k}\Omega$$

$$\text{We have } R_C = 10R_E$$

$$\text{so } R_E = 0.454 \text{ k}\Omega \text{ and } R_C = 4.54 \text{ k}\Omega$$

We have

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.5}{100} = 0.005 \text{ mA}$$

and

$$R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(101)(0.454) \quad \text{or} \quad R_{TH} = 4.59 \text{ k}\Omega$$

Also

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC}$$

or

$$V_{TH} = \frac{1}{R_1} (4.59)(5) = \frac{23}{R_1}$$

Also $V_{TH} = I_{BQ}R_{TH} + V_{BE}(on) + (1 + \beta)I_{BQ}R_E$
 so that $\frac{23}{R_1} = (0.005)(4.59) + 0.7 + (101)(0.005)(0.454)$
 which yields $R_1 = 24.1 \text{ k}\Omega$
 Since $R_1 \parallel R_2 = 4.59 \text{ k}\Omega$, then $R_2 = 5.67 \text{ k}\Omega$

TYU6.5

As a first approximation

$$A_v \cong -\frac{R_C}{R_E}$$

Set

$$\frac{R_C}{R_E} = 9$$

Now

$$V_{CC} \cong I_{CQ}(R_C + R_E) + V_{ECQ}$$

$$7.5 = (0.6)(9R_E + R_E) + 3.75$$

which yields

$$R_E = 0.625 \text{ k}\Omega \quad \text{and} \quad R_C = 5.62 \text{ k}\Omega$$

We have

$$R_{TH} = (0.1)(1 + \beta)R_E = (0.1)(101)(0.625) \quad \text{or} \quad R_{TH} = 6.31 \text{ k}\Omega$$

Also

$$V_{TH} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC} = \frac{1}{R_1}(6.31)(7.5)$$

We have

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.6}{100} = 0.006 \text{ mA}$$

and

$$V_{CC} = (1 + \beta)I_{BQ}R_E + V_{EB}(on) + I_{BQ}R_{TH} + V_{TH}$$

$$7.5 = (101)(0.006)(0.625) + 0.7 + (0.006)(6.31) + \frac{1}{R_1}(6.31)(7.5)$$

which yields

$$R_1 = 7.41 \text{ k}\Omega$$

Since $R_{TH} = R_1 \parallel R_2 = 6.31 \text{ k}\Omega$, then $R_2 = 42.5 \text{ k}\Omega$

TYU6.6

We have

$$A_v = \frac{-\beta R_C}{r_\pi + (1 + \beta)R_E} = -(0.95)\left(\frac{R_C}{R_E}\right)$$

or

$$A_v = -(0.95)\left(\frac{2}{0.4}\right) = -4.75$$

Assume, from Example 6.5 that, $r_\pi = 1.2 \text{ k}\Omega$

Then

$$\frac{-\beta(2)}{1.2 + (1 + \beta)(0.4)} = -4.75$$

or

$$\beta = 76$$

TYU6.7

Dc analysis: by symmetry, $V_{TH} = 0$

$$R_{TH} = R_1 \parallel R_2 = 20 \parallel 20 = 10 \text{ k}\Omega$$

We can write

$$I_{BQ} = \frac{0 - 0.7 - (-5)}{10 + (126)(5)} = 0.00672 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = (125)(0.00672) = 0.84 \text{ mA}$$

Small-signal transistor parameters:

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(125)(0.026)}{0.84} = 3.87 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.84}{0.026} = 32.3 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{200}{0.84} = 238 \text{ k}\Omega$$

(a) We can write

$$V_o = -g_m V_\pi (r_o \parallel R_C \parallel R_L) \text{ and } V_\pi = V_s$$

so

$$A_v = \frac{V_o}{V_s} = -g_m (r_o \parallel R_C \parallel R_L) = -(32.3)(238 \parallel 2.3 \parallel 5)$$

or $A_v = -50.5$

(b)

$$R_o = r_o \parallel R_C = 238 \parallel 2.3 = 2.28 \text{ k}\Omega$$

TYU6.8

We find

$$I_{CQ} = 0.418 \text{ mA},$$

$$V_{CEQ} = 5 - (0.418)(5.6) - \left(\frac{121}{120}\right)(0.418)(0.6)$$

or $V_{CEQ} = 2.41 \text{ V}$

So

$$\Delta v_{CE} = (2.41 - 0.5) \times 2, \text{ or } \Delta v_{CE} = 3.82 \text{ V}$$

TYU6.9

For $I_{CQ} \cong I_{EQ}$,

$$V_{CEQ} = 10 - I_{CQ}(4 + 4) = 10 - I_{CQ}(8)$$

$$\Delta I_C = I_{CQ} - 0.1$$

$$\Delta V_{CE} = V_{CEQ} - 0.7$$

$$\Delta V_{CE} = \Delta I_C(4) = (I_{CQ} - 0.1)(4) = V_{CEQ} - 0.7$$

So

$$V_{CEQ} = (I_{CQ} - 0.1)(4) + 0.7$$

Then

$$(I_{CQ} - 0.1)(4) + 0.7 = 10 - I_{CQ}(8) \Rightarrow I_{CQ} = 0.8083 \text{ mA}$$

$$V_{CEQ} = 10 - (0.8083)(8) \Rightarrow V_{CEQ} = 3.533 \text{ V}$$

Then peak-to-peak values are

$$\Delta V_{CE} = (3.533 - 0.7)(2) = 5.67 \text{ V}$$

$$\Delta I_C = (0.8083 - 0.1)(2) = 1.42 \text{ mA}$$

TYU6.10

We can write

$$I_{BQ} = \frac{0 - 0.7 - (-10)}{100 + (131)(10)} \Rightarrow 6.60 \mu A$$

$$I_{CQ} = (130)(6.60 \mu A) = 0.857 \text{ mA}$$

Assume nominal small-signal parameters of:

$$h_{ie} = 4 \text{ k}\Omega, h_{fe} = 134$$

$$h_{re} = 0, h_{oe} = 12 \mu S \Rightarrow \frac{1}{h_{oe}} = 83.3 \text{ k}\Omega$$

We find

$$\begin{aligned} R_{ib} &= h_{ie} + (1 + h_{fe}) \left(R_E \parallel R_L \parallel \frac{1}{h_{oe}} \right) \\ &= 4 + (135)(10 \parallel 10 \parallel 83.3) = 641 \text{ k}\Omega \end{aligned}$$

To find the voltage gain:

$$V'_s = \frac{R_B \parallel R_{ib}}{R_B \parallel R_{ib} + R_s} \cdot V_s = \frac{100 \parallel 641}{100 \parallel 641 + 10} \cdot V_s = (0.896)V_s$$

Also

$$\frac{V_o}{V'_s} = \frac{(1 + h_{fe})R'}{h_{ie} + (1 + h_{fe})R'}$$

where

$$R' = R_E \parallel R_L \parallel \frac{1}{h_{oe}} = 10 \parallel 10 \parallel 83.3 = 4.72 \text{ k}\Omega$$

Then

$$A_v = \frac{V_o}{V_s} = \frac{(0.896)(135)(4.72)}{4 + (135)(4.72)} = 0.891$$

To find the current gain

$$A_i = \frac{I_o}{I_i} = \left(\frac{R_E \parallel \frac{1}{h_{oe}}}{R_E \parallel \frac{1}{h_{oe}} + R_L} \right) \left(1 + h_{fe} \left(\frac{R_B}{R_B + R_{ib}} \right) \right)$$

$$= \left(\frac{10 \parallel 83.3}{10 \parallel 83.3 + 10} \right) (135) \left(\frac{100}{100 + 641} \right)$$

or

$$A_i = 8.59$$

To find the output resistance:

$$R_o = R_E \parallel \frac{1}{h_{oe}} \parallel \frac{h_{ie} + R_S \parallel R_B}{1 + h_{fe}}$$

$$= 10 \parallel 83.3 \parallel \frac{4 + 10 \parallel 100}{135} \Rightarrow R_o = 96.0 \Omega$$

TYU6.11

$$R_{TH} = R_1 \parallel R_2 = 50 \parallel 50 = 25 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{1}{2} \right) (5) = 2.5 \text{ V}$$

Now

$$I_{BQ} = \frac{V_{CC} - V_{EB}(on) - V_{TH}}{R_{TH} + (1 + \beta) R_E}$$

$$= \frac{5 - 0.7 - 2.5}{25 + (101)(2)} = 0.00793 \text{ mA}$$

and

$$I_{CQ} = (100)(0.00793) = 0.793 \text{ mA}$$

Small-signal transistor parameters:

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.793}{0.026} = 30.5 \text{ mA/V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{0.793} = 3.28 \text{ k}\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{125}{0.793} = 158 \text{ k}\Omega$$

(a) Define

$$R' = R_E \parallel R_L \parallel r_o = 2 \parallel 0.5 \parallel 158 \cong 0.40 \text{ k}\Omega$$

$$A_v = \frac{(1 + \beta) R'}{r_\pi + (1 + \beta) R'} = \frac{(101)(0.4)}{3.28 + (101)(0.4)}$$

or $A_v = 0.925$

(b)

$$R_{ib} = r_\pi + (1 + \beta) R' = 3.28 + (101)(0.4)$$

$$R_{ib} = 43.7 \text{ k}\Omega$$

$$R_o = R_E \parallel r_o \parallel \frac{r_\pi}{1 + \beta} = 2 \parallel 158 \parallel \frac{3.28}{101}$$

or

$$R_o = 32.0 \, \Omega$$

TYU6.12

(a) $I_{BQ}R_B + V_{BE}(on) + (1 + \beta)I_{BQ}R_E + V^- = 0$

$$I_{BQ} = \frac{3.3 - 0.7}{100 + (121)(15)} = 0.001358 \, \text{mA}$$

$$I_{EQ} = 0.1643 \, \text{mA}; \quad I_{CQ} = 0.1629 \, \text{mA}$$

$$V_{CEQ} = 6.6 - (0.1643)(15) = 4.14 \, \text{V}$$

(b) $A_v = \frac{(1 + \beta)(R_E \parallel R_L)}{r_\pi + (1 + \beta)(R_E \parallel R_L)} \cdot \left(\frac{R_i}{R_i + R_s} \right)$

We find

$$g_m = \frac{0.1629}{0.026} = 6.265 \, \text{mA/V}$$

$$r_\pi = \frac{(120)(0.026)}{0.1629} = 19.15 \, \text{k}\Omega$$

Now

$$R_{ib} = r_\pi + (1 + \beta)(R_E \parallel R_L) = 19.15 + (121)(15 \parallel 2) = 232.7 \, \text{k}\Omega$$

$$R_i = R_{ib} \parallel R_B = 232.7 \parallel 100 = 69.94 \, \text{k}\Omega$$

Then

$$A_v = \frac{(121)(15 \parallel 2)}{19.15 + (121)(15 \parallel 2)} \cdot \left(\frac{69.94}{69.94 + 2} \right) = 0.892$$

Also

$$A_i = (1 + \beta) \left(\frac{R_B}{R_B + R_{ib}} \right) \left(\frac{R_E}{R_E + R_L} \right) = (121) \left(\frac{100}{100 + 232.7} \right) \left(\frac{15}{15 + 2} \right)$$

$$A_i = 32.1$$

(c) We found

$$R_{ib} = 232.7 \, \text{k}\Omega$$

Now

$$R_o = \left(\frac{r_\pi + R_B \parallel R_s}{1 + \beta} \right) \parallel R_E = \left(\frac{19.15 + 100 \parallel 2}{121} \right) \parallel 15 = 0.1745 \parallel 15$$

$$R_o = 172 \, \Omega$$

TYU6.13

(a) dc analysis:

$$I_{EQ} = \frac{V_{EE} - V_{EB}(on)}{R_E} = \frac{10 - 0.7}{10} = 0.93 \, \text{mA}$$

$$I_{CQ} = \left(\frac{\beta}{1 + \beta} \right) I_{EQ} = \left(\frac{100}{101} \right) (0.93) = 0.921 \, \text{mA}$$

$$\begin{aligned} V_{ECQ} &= V_{EE} - I_{EQ}R_E - I_{CQ}R_C - (-V_{CC}) \\ &= 10 - (0.93)(10) - (0.921)(5) - (-10) \end{aligned}$$

or

$$V_{ECQ} = 6.1 \text{ V}$$

(b) Small-signal transistor parameters:

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{0.921} = 2.82 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.921}{0.026} = 35.42 \text{ mA/V}$$

Small-signal current gain:

$$I_o = g_m V_\pi, \text{ and } V_\pi = V_s$$

Also

$$I_i = \frac{V_s}{R_E \parallel r_\pi} + g_m V_\pi = V_s \left(\frac{1}{R_E \parallel r_\pi} + g_m \right)$$

Then

$$\begin{aligned} A_i = \frac{I_o}{I_i} &= \frac{g_m V_\pi}{V_\pi \left(\frac{1}{R_E \parallel r_\pi} + g_m \right)} = \frac{g_m (R_E \parallel r_\pi)}{1 + g_m (R_E \parallel r_\pi)} \\ &= \frac{(35.42)(10 \parallel 2.82)}{1 + (35.42)(10 \parallel 2.82)} \end{aligned}$$

or $A_i = 0.987$

(c) Small-signal voltage gain:

$$V_o = g_m V_\pi R_C = g_m V_s R_C$$

$$A_v = \frac{V_o}{V_s} = g_m R_C = (35.42)(5)$$

or $A_v = 177$

TYU6.14

(a) $I_{BQ}R_B + V_{BE}(on) + (1 + \beta)I_{BQ}R_E = V_{EE}$

$$I_{BQ} = \frac{3.3 - 0.7}{100 + (121)(12)} = 0.001675 \text{ mA}$$

$$I_{CQ} = 0.201 \text{ mA}$$

$$g_m = \frac{0.201}{0.026} = 7.73 \text{ mA/V}; \quad r_\pi = \frac{(120)(0.026)}{0.201} = 15.52 \text{ k}\Omega; \quad r_o = \infty$$

(b)
$$A_i = \left(\frac{R_E}{R_E + \frac{r_\pi}{1 + \beta}} \right) \left(\frac{\beta}{1 + \beta} \right) \left(\frac{R_C}{R_C + R_L} \right) = \left(\frac{12}{12 + \frac{15.52}{121}} \right) \left(\frac{120}{121} \right) \left(\frac{12}{12 + 6} \right)$$

$$A_i = 0.654$$

Now

$$A_v = g_m \left(\frac{R_C \parallel R_L}{R_S} \right) \left[\frac{r_\pi}{1 + \beta} \parallel R_E \parallel R_S \right] = (7.73) \left(\frac{12 \parallel 6}{0.5} \right) \left[\frac{15.52}{121} \parallel 12 \parallel 0.5 \right]$$

$$A_v = (7.73)(8)(0.1012) = 6.26$$

$$(c) \quad R_i = R_E \parallel \frac{r_\pi}{1 + \beta} = 12 \parallel \frac{15.52}{121} \Rightarrow R_i = 127 \, \Omega$$

$$R_o = R_C = 12 \, k\Omega$$

TYU6.15

dc analysis

$$5 = I_{BQ} R_B + V_{BE} (on) + I_{EQ} R_E$$

$$I_{BQ} = \frac{5 - 0.7}{R_B + (101) R_E} = \frac{4.3}{R_B + (101) R_E}$$

$$I_{CQ} = \frac{(100)(4.3)}{R_B + (101) R_E}$$

Also

$$5 = I_{CQ} R_C + V_{CEQ} + I_{EQ} R_E - 5$$

or

$$V_{CEQ} = 10 - I_{CQ} \left[R_C + \left(\frac{101}{100} \right) R_E \right]$$

ac analysis :

$$V_o = -g_m V_\pi (R_C \parallel R_L)$$

and

$$V_s = -V_\pi - \frac{V_\pi}{r_\pi} \cdot R_B = -V_\pi \left(1 + \frac{R_B}{r_\pi} \right)$$

or

$$V_\pi = - \left(\frac{r_\pi}{r_\pi + R_B} \right) \cdot V_s$$

Then

$$A_v = \frac{V_o}{V_s} = \frac{\beta}{r_\pi + R_B} (R_C \parallel R_L)$$

where

$$\beta = g_m r_\pi$$

For

$$I_{CQ} = 1 \, mA,$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{1} = 2.6 \, k\Omega$$

Then

$$A_v = 20 = \frac{(100)(2 \parallel 2)}{2.6 + R_B}$$

which yields

$$R_B = 2.4 \, k\Omega$$

Then from

$$I_{CQ} = 1 = \frac{(100)(4.3)}{2.4 + (101)R_E}$$

we find

$$R_E = 4.23 \text{ k}\Omega$$

TYU6.16

(a) dc analysis

For $I_{EQ2} = 1 \text{ mA}$,

$$I_{CQ2} = \left(\frac{100}{101}\right)(1) = 0.990 \text{ mA}$$

$$I_{EQ1} = \frac{I_{EQ2}}{1 + \beta} = \frac{1}{101} = 0.0099 \text{ mA}$$

$$I_{BQ1} = \frac{I_{EQ1}}{1 + \beta} = \frac{0.0099}{101} = 0.000098 \text{ mA}$$

$$I_{CQ1} = (100)(0.000098) = 0.0098 \text{ mA}$$

$$V_{B1} = -I_{BQ1}R_B = -(0.000098)(10)$$

$$V_{B1} = -0.00098 \approx 0$$

$$V_{E1} = -0.7 \text{ V}$$

$$V_{E2} = -1.4 \text{ V}$$

$$I_1 = I_{CQ1} + I_{CQ2} = 0.0098 + 0.990 \approx 1 \text{ mA}$$

$$V_O = 5 - (1)(4) = 1 \text{ V}$$

$$V_{CEQ2} = 1 - (-1.4) = 2.4 \text{ V}$$

$$V_{CEQ1} = 1 - (-0.7) = 1.7 \text{ V}$$

(b) small-signal transistor parameters:

$$r_{\pi 1} = \frac{\beta V_T}{I_{CQ1}} = \frac{(100)(0.026)}{0.0098} = 265 \text{ k}\Omega$$

$$g_{m1} = \frac{I_{CQ1}}{V_T} = \frac{0.0098}{0.026} = 0.377 \text{ mA/V}$$

$$r_{\pi 2} = \frac{\beta V_T}{I_{CQ2}} = \frac{(100)(0.026)}{0.990} = 2.63 \text{ k}\Omega$$

$$g_{m2} = \frac{I_{CQ2}}{V_T} = \frac{0.990}{0.026} = 38.1 \text{ mA/V}$$

$$r_{o1} = r_{o2} = \infty$$

(c) small-signal voltage gain

$$V_o = -(g_{m1}V_{\pi 1} + g_{m2}V_{\pi 2})R_C$$

$$V_s = V_{\pi 1} + V_{\pi 2}$$

$$V_{\pi 2} = \left(\frac{V_{\pi 1}}{r_{\pi 1}} + g_{m1}V_{\pi 1}\right) \cdot r_{\pi 2} = \left(\frac{1 + \beta}{r_{\pi 1}}\right) \cdot V_{\pi 1}r_{\pi 2}$$

$$V_o = -\left[g_{m1}V_{\pi 1} + g_{m2}\left(\frac{1 + \beta}{r_{\pi 1}}\right) \cdot r_{\pi 2}V_{\pi 1}\right] \cdot R_C$$

$$V_s = V_{\pi 1} + \left(\frac{1 + \beta}{r_{\pi 1}} \right) \cdot r_{\pi 2} V_{\pi 1}$$

$$= V_{\pi 1} \left[1 + (1 + \beta) \left(\frac{r_{\pi 2}}{r_{\pi 1}} \right) \right]$$

$$V_{\pi 1} = \frac{V_s}{1 + (1 + \beta) \left(\frac{r_{\pi 2}}{r_{\pi 1}} \right)}$$

Now

$$A_v = \frac{V_o}{V_s} = - \frac{\left[g_{m1} + g_{m2} (1 + \beta) \left(\frac{r_{\pi 2}}{r_{\pi 1}} \right) \right] \cdot R_C}{1 + (1 + \beta) \left(\frac{r_{\pi 2}}{r_{\pi 1}} \right)}$$

$$A_v = - \frac{\left[0.377 + (38.1)(101) \left(\frac{2.63}{265} \right) \right] (4)}{1 + (101) \left(\frac{2.63}{265} \right)}$$

or

$$A_v = -77.0$$

(d)

$$R_i = r_{\pi 1} + (1 + \beta) r_{\pi 2} = 265 + (101)(2.63)$$

or

$$R_i = 531 \text{ k}\Omega$$

TYU6.17

$$(a) \quad R_1 + R_2 + R_3 = \frac{9}{0.1} = 90 \text{ k}\Omega$$

$$R_E = \frac{0.7}{1} = 0.7 \text{ k}\Omega$$

$$V_{B1} = 0.7 + 0.7 = 1.4 \text{ V} \Rightarrow \left(\frac{R_3}{90} \right) (9) = 1.4 \Rightarrow R_3 = 14 \text{ k}\Omega$$

$$V_{B2} = 0.7 + 2.5 + 0.7 = 3.9 \text{ V} \Rightarrow \left(\frac{R_2 + R_3}{90} \right) (9) = 3.9 \Rightarrow R_2 = 25 \text{ k}\Omega$$

Then $R_1 = 51 \text{ k}\Omega$

$$V_{C2} = 0.7 + 2.5 + 2.5 = 5.7 \text{ V}$$

So

$$R_C = \frac{9 - 5.7}{1} = 3.3 \text{ k}\Omega$$

$$(b) \quad g_m = \frac{1}{0.026} = 38.46 \text{ mA/V}; \quad r_{\pi} = \frac{(100)(0.026)}{1} = 2.6 \text{ k}\Omega$$

$$(c) \quad A_v = -g_{m1} g_{m2} \left(\frac{r_{\pi 2}}{1 + \beta_2} \right) (R_C \parallel R_L) = -(38.46)^2 \left(\frac{2.6}{101} \right) (3.3 \parallel 10) = -94.5$$

TYU6.18

(a) dc analysis

$$R_{TH} = R_1 \parallel R_2 = 125 \parallel 30 = 24.2 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{30}{125 + 30} \right) (12)$$

or

$$V_{TH} = 2.32 \text{ V}$$

Now

$$V_{TH} = I_{BQ} R_{TH} + V_{BE}(\text{on}) + (1 + \beta) I_{BQ} R_E$$

$$I_{BQ} = \frac{2.32 - 0.7}{24.2 + (81)(0.5)} = 0.0250 \text{ mA}$$

$$I_{CQ} = (80)(0.025) = 2.00 \text{ mA}$$

$$\begin{aligned} V_{CEQ} &= V_{CC} - I_{CQ} \left[R_C + \left(\frac{1 + \beta}{\beta} \right) R_E \right] \\ &= 12 - (2) \left[2 + \left(\frac{81}{80} \right) (0.5) \right] \end{aligned}$$

or

$$V_{CEQ} = 6.99 \text{ V}$$

Power dissipated in R_C :

$$P_{RC} = I_{CQ}^2 R_C = (2.0)^2 (2) = 8.0 \text{ mW}$$

Power dissipated in R_L :

$$I_{LQ} = 0 \Rightarrow P_{RL} = 0$$

Power dissipated in transistor:

$$\begin{aligned} P_Q &= I_{BQ} V_{BEQ} + I_{CQ} V_{CEQ} \\ &= (0.025)(0.7) + (2.0)(6.99) = 14.0 \text{ mW} \end{aligned}$$

(b) With

$$v_s = 18 \cos \omega t \text{ (mV)}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(80)(0.026)}{2.0} = 1.04 \text{ k}\Omega$$

We can write

$$v_{ce} = \frac{\beta}{r_\pi} (R_C \parallel R_L) V_P \cos \omega t$$

Power dissipated in R_L :

$$\begin{aligned} \bar{P}_{RL} &= \frac{|v_{ce}(\text{rms})|^2}{R_L} = \frac{1}{2} \cdot \frac{1}{R_L} \cdot \left[\frac{\beta}{r_\pi} (R_C \parallel R_L) V_P \right]^2 \\ &= \frac{1}{2} \cdot \frac{1}{2 \times 10^3} \cdot \left[\frac{80}{1.04} (2 \parallel 2) (0.018) \right]^2 \end{aligned}$$

or

$$\bar{P}_{RL} = 0.479 \text{ mW}$$

Power dissipated in R_C :

Since $R_C = R_L = 2 \text{ k}\Omega$, we find

$$\bar{P}_{RC} = 8.0 + 0.479 = 8.48 \text{ mW}$$

$$\begin{aligned}\bar{P}_Q &\cong I_{CQ}V_{CEQ} - \left(\frac{\beta}{r_\pi}\right)^2 \left(\frac{V_P}{\sqrt{2}}\right)^2 (R_C \parallel R_L) \\ &= (2 \times 10^{-3})(6.99) - \left(\frac{80}{1.04 \times 10^3}\right)^2 \left(\frac{0.018}{\sqrt{2}}\right)^2 (2 \times 10^3 \parallel 2 \times 10^3)\end{aligned}$$

or

$$\bar{P}_Q = 13.0 \text{ mW}$$

TYU6.19

(a) dc analysis

$$R_{TH} = R_1 \parallel R_2 = 53.8 \parallel 10 = 8.43 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{CC} = \left(\frac{10}{53.8 + 10}\right)(5)$$

or

$$V_{TH} = 0.7837 \text{ V}$$

Now

$$I_{BQ} = \frac{0.7837 - 0.7}{8.43} = 0.00993 \text{ mA}$$

$$I_{CQ} = (100)(0.00993) = 0.993 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C$$

$$2.5 = 5 - (0.993)R_C$$

which yields

$$R_C = 2.52 \text{ k}\Omega$$

(b)

Power dissipated in R_C :

$$P_{RC} = I_{CQ}^2 R_C = (0.993)^2 (2.52)$$

or

$$P_{RC} = 2.48 \text{ mW}$$

Power dissipated in transistor:

$$P_Q \cong I_{CQ}V_{CEQ} = (0.993)(2.5)$$

or

$$P_Q = 2.48 \text{ mW}$$

(c) ac analysis

Maximum ac collector current:

$$i_c = (0.993)\cos\omega t \text{ (mA)}$$

Power dissipated in R_C :

$$\bar{P}_{RC} = \frac{1}{2}(0.993)^2 R_C = \frac{1}{2}(0.993)^2 (2.52)$$

or

$$\bar{P}_{RC} = 1.24 \text{ mW}$$

Now

$$\text{Fraction} = \frac{\bar{P}_{RC}}{P_{RC} + P_Q} = \frac{1.24}{2.48 + 2.48} = 0.25$$

Chapter 7

Exercise Solutions

EX7.1

$$(a) (i) f_L = \frac{1}{2\pi\tau_s} \Rightarrow \tau_s = \frac{1}{2\pi f_L} = \frac{1}{2\pi(50)} \Rightarrow 3.183 \text{ ms}$$

$$\tau_s = (R_s + R_p)C_s$$

$$3.183 \times 10^{-3} = (2 + 8) \times 10^3 (C_s) \Rightarrow C_s = 0.318 \mu\text{F}$$

$$(ii) |T| = \left(\frac{R_p}{R_p + R_s} \right) \left[\frac{f/f_L}{\sqrt{1 + (f/f_L)^2}} \right] = \left(\frac{8}{8 + 2} \right) \left[\frac{f/f_L}{\sqrt{1 + (f/f_L)^2}} \right]$$

$$\text{For } f = 20, \frac{f}{f_L} = \frac{20}{50} = 0.4; |T| = (0.8) \left[\frac{0.4}{\sqrt{1 + (0.4)^2}} \right] = 0.297$$

$$\text{For } f = 50, \frac{f}{f_L} = \frac{50}{50} = 1; |T| = (0.8) \frac{(1)}{\sqrt{2}} = 0.566$$

$$\text{For } f = 100, \frac{f}{f_L} = \frac{100}{50} = 2; |T| = (0.8) \frac{(2)}{\sqrt{1 + (2)^2}} = 0.716$$

(b)

$$(i) \tau_p = (R_s \| R_p)C_p = (4.7 \| 25) \times 10^3 \times (120 \times 10^{-12}) = 4.747 \times 10^{-7} \text{ s}$$

$$f_H = \frac{1}{2\pi\tau_p} = \frac{1}{2\pi(4.747 \times 10^{-7})} \Rightarrow 335 \text{ kHz}$$

$$(ii) |T| = \left(\frac{R_p}{R_p + R_s} \right) \cdot \frac{1}{\sqrt{1 + (f/f_H)^2}} = \frac{(0.84175)}{\sqrt{1 + (f/f_H)^2}}$$

$$\text{For } f = 0.2f_H; |T| = \frac{(0.84175)}{\sqrt{1 + (0.2)^2}} = 0.825$$

$$\text{For } f = f_H; |T| = \frac{(0.84175)}{\sqrt{2}} = 0.595$$

$$\text{For } f = 8f_H; |T| = \frac{(0.84175)}{\sqrt{1 + (8)^2}} = 0.104$$

EX7.2

$$\begin{aligned}
 \text{(a)} \quad A_v(\text{mid}) &= \frac{R_p}{R_p + R_s} \\
 -2 &= 20 \log_{10} \left(\frac{R_p}{R_p + R_s} \right) \Rightarrow \frac{R_p}{R_p + R_s} = 0.7943 \\
 \frac{7.5}{7.5 + R_s} &= 0.7943 \Rightarrow R_s = 1.942 \text{ k}\Omega \\
 f_L &= \frac{1}{2\pi\tau_s} \Rightarrow \tau_s = \frac{1}{2\pi f_L} = \frac{1}{2\pi(200)} = 0.7958 \times 10^{-3} \text{ s} \\
 \tau_s &= (R_s + R_p)C_s \\
 0.7958 \times 10^{-3} &= (1.942 + 7.5) \times 10^3 (C_s) \Rightarrow C_s = 0.0843 \mu\text{F} \\
 \tau_p &= (R_s \parallel R_p)C_p = (1.942 \parallel 7.5) \times 10^3 \times (80 \times 10^{-12}) = 1.234 \times 10^{-7} \text{ s} \\
 f_H &= \frac{1}{2\pi\tau_p} = \frac{1}{2\pi(1.234 \times 10^{-7})} \Rightarrow 1.29 \text{ MHz} \\
 \text{(b)} \quad \tau_s &= 0.796 \text{ ms} \\
 \tau_p &= 0.123 \mu\text{s}
 \end{aligned}$$

EX7.3

$$\begin{aligned}
 \text{(a)} \quad R_{TH} &= R_1 \parallel R_2 = 110 \parallel 42 = 30.39 \text{ k}\Omega \\
 V_{TH} &= \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_{CC} = \left(\frac{42}{110 + 42} \right) (3) = 0.8289 \text{ V} \\
 I_{BQ} &= \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1 + \beta)R_E} = \frac{0.8289 - 0.7}{30.39 + (151)(0.5)} = 0.001217 \text{ mA} \\
 I_{CQ} &= \beta I_{BQ} = 0.1826 \text{ mA} \\
 \tau_s &= (R_1 \parallel R_2 \parallel R_{ib})C_C \\
 \text{where } R_{ib} &= r_\pi + (1 + \beta)R_E \\
 \text{Now} \quad r_\pi &= \frac{(150)(0.026)}{0.1826} = 21.36 \text{ k}\Omega \\
 \text{Then } R_{ib} &= 21.36 + (151)(0.5) = 96.86 \text{ k}\Omega \\
 \tau_s &= (110 \parallel 42 \parallel 96.86) \times 10^3 \times (0.47 \times 10^{-6}) \Rightarrow 10.87 \text{ ms} \\
 \text{(b)} \quad f_L &= \frac{1}{2\pi\tau_s} = \frac{1}{2\pi(10.87 \times 10^{-3})} = 14.6 \text{ Hz} \\
 A_v &= \frac{-\beta R_C}{r_\pi + (1 + \beta)R_E} = \frac{-(150)(7)}{21.36 + (151)(0.5)} = -10.84
 \end{aligned}$$

EX7.4

$$\begin{aligned} \text{(a)} \quad I_{DQ} &= K_n (V_{GSQ} - V_{TN})^2 \\ 250 &= 100(V_{GSQ} - 0.4)^2 \Rightarrow V_{GSQ} = 1.981 \text{ V} \\ R_S &= \frac{-V_{GSQ} - (-3)}{I_{DQ}} = \frac{3 - 1.981}{0.25} = 4.08 \text{ k}\Omega \\ V_D &= -V_{GSQ} + V_{DSQ} = -1.981 + 1.7 = -0.281 \text{ V} \\ R_D &= \frac{3 - V_D}{I_{DQ}} = \frac{3 - (-0.281)}{0.25} = 13.1 \text{ k}\Omega \\ \text{(b)} \quad \tau_s &= (R_D + R_L)C_C = (13.1 + 20) \times 10^3 \times (0.7 \times 10^{-6}) \Rightarrow \tau_s = 23.17 \text{ ms} \\ f_L &= \frac{1}{2\pi\tau_s} = \frac{1}{2\pi(23.17 \times 10^{-3})} = 6.87 \text{ Hz} \end{aligned}$$

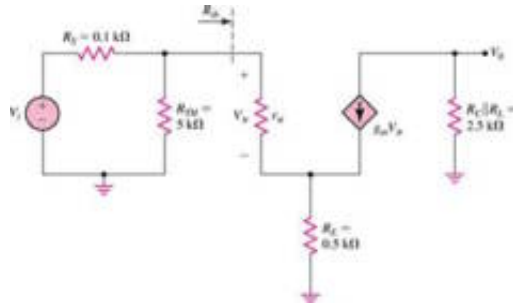
EX7.5

$$\begin{aligned} \tau_s &= (R_L + R_o)C_{C2} \\ f &= \frac{1}{2\pi\tau_s} \Rightarrow C_{C2} = \frac{1}{2\pi(R_L + R_o)} \\ R_o &= R_E \parallel r_o \left\| \left(\frac{r_\pi + (R_S \parallel R_B)}{1 + \beta} \right) \right\} \\ \text{From Example 7-5, } R_o &= 35.5 \Omega \\ C_{C2} &= \frac{1}{2\pi(10)[10 \times 10^3 + 35.5]} \\ C_{C2} &= 1.59 \mu\text{F} \end{aligned}$$

EX7.6

$$\begin{aligned} \text{(a)} \quad R_{TH} &= 5 \text{ K} \\ V_{TH} &= -3.7527 \\ I_{BQ} &= \frac{-3.7527 - 0.7 - (-5)}{5 + (101)(0.5)} = \frac{0.54726}{55.5} \\ &= 0.00986 \\ I_{CQ} &= 0.986 \text{ mA} \\ g_m &= 37.925 \quad r_\pi = 2.637 \text{ K} \\ 0.986 &= \frac{5 - V_o}{5} - \frac{V_o}{5} = 1 - V_o(0.4) \\ V_o &= 0.035 \end{aligned}$$

(b)



$$V_o = -g_m V_\pi (R_C \parallel R_L)$$

$$R_{ib} = r_\pi + (1 + \beta)R_E = 2.64 + (101)(0.5) = 53.14 \text{ k}\Omega$$

$$V_\pi = \frac{V_b}{1 + \left(\frac{1 + \beta}{r_\pi}\right)R_E} = \frac{V_b}{1 + \left(\frac{101}{2.637}\right)(0.5)} = \frac{V_b}{14.885}$$

$$V_b = \frac{R_{TH} \parallel R_{ib}}{R_{TH} \parallel R_{ib} + R_S} \cdot V_i = \frac{5 \parallel 53.14}{5 \parallel 53.14 + 0.1} \cdot V_i = (0.9786)V_i$$

$$A_v = \frac{-(37.925)(2.5)}{14.885} (0.9786) \Rightarrow A_v = -6.23$$

(c)

EX7.7

a.

$$I_{BQ} = \frac{0 - 0.7 - (-10)}{0.5 + (101)(4)} = 0.0230 \text{ mA}$$

$$I_{CQ} = 2.30 \text{ mA}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{2.30} = 1.13 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{2.30}{0.026} = 88.46 \text{ mA/V}$$

$$\tau_B = \frac{R_E (R_S + r_\pi) C_E}{R_S + r_\pi + (1 + \beta)R_E}$$

$$= \frac{(4 \times 10^3)(0.5 + 1.13)C_E}{0.5 + 1.13 + (101)(4)}$$

$$\tau_B = \frac{1}{2\pi f_B} = \frac{1}{2\pi(200)} \Rightarrow \tau_B = 0.7958 \text{ ms}$$

$$\tau_B = 16.07 C_E \Rightarrow C_E = \frac{0.796 \times 10^{-3}}{16.07} \Rightarrow C_E = 49.5 \mu\text{F}$$

b.

$$\tau_A = R_E C_E = (4 \times 10^3)(49.5 \times 10^{-6}) \Rightarrow \tau_A = 0.198 \text{ s}$$

$$f_A = \frac{1}{2\pi \tau_A} = \frac{1}{2\pi(0.198)} \Rightarrow f_A = 0.804 \text{ Hz}$$

EX7.8

$$(a) \quad f_{\beta} = \frac{1}{2\pi r_{\pi}(C_{\pi} + C_{\mu})}$$

$$r_{\pi} = \frac{(120)(0.026)}{0.2} = 15.6 \text{ k}\Omega$$

$$C_{\pi} + C_{\mu} = \frac{1}{2\pi r_{\pi} f_{\beta}} = \frac{1}{2\pi(15.6 \times 10^3)(90 \times 10^6)} \Rightarrow 0.113 \text{ pF}$$

$$C_{\pi} = 0.113 - 0.02 = 0.093 \text{ pF}$$

$$(b) \quad |h_{fe}| = \frac{\beta_o}{\sqrt{1 + \left(\frac{f}{f_{\beta}}\right)^2}}$$

$$(i) \quad \text{For } f = 50 \text{ MHz, } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{50}{90}\right)^2}} = 105$$

$$(ii) \quad \text{For } f = 125 \text{ MHz, } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{125}{90}\right)^2}} = 70.1$$

$$(iii) \quad \text{For } f = 500 \text{ MHz, } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{500}{90}\right)^2}} = 21.3$$

EX7.9

$$r_{\pi} = \frac{(150)(0.026)}{0.15} = 26 \text{ k}\Omega$$

$$f_{\beta} = \frac{1}{2\pi r_{\pi}(C_{\pi} + C_{\mu})} = \frac{1}{2\pi(26 \times 10^3)(0.8 + 0.012) \times 10^{-12}}$$

or

$$f_{\beta} = 7.54 \text{ MHz}$$

$$f_T = \beta_o f_{\beta} = (150)(7.54) \Rightarrow f_T = 1.13 \text{ GHz}$$

EX7.10

$$(a) \quad R_{TH} = R_1 \| R_2 = 200 \| 220 = 104.8 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{CC} = \left(\frac{220}{200 + 220}\right)(5) = 2.619 \text{ V}$$

$$I_{BQ} = \frac{V_{TH} - V_{BE(on)}}{R_{TH} + (1 + \beta)R_E} = \frac{2.619 - 0.7}{104.8 + (101)(1)} = 0.009325 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 0.9325 \text{ mA}$$

Now

$$r_{\pi} = \frac{(100)(0.026)}{0.9325} = 2.788 \text{ k}\Omega \quad ; \quad g_m = \frac{0.9325}{0.026} = 35.87 \text{ mA/V}$$

$$I_b = \left(\frac{r_s \| R_1 \| R_2}{r_s \| R_1 \| R_2 + r_\pi} \right) \cdot I_s = \left(\frac{100 \| 104.8}{100 \| 104.8 + 2.788} \right) \cdot I_s = 0.9483 I_s$$

$$I_o = -\beta I_b \left(\frac{R_C}{R_C + R_L} \right) = -(100)(0.9483)I_s \left(\frac{2.2}{2.2 + 4.7} \right)$$

or

$$A_i = \frac{I_o}{I_s} = -30.24$$

(b) (i) For $C_\mu = 0 \Rightarrow C_M = 0$

$$\begin{aligned} \text{(ii) For } C_\mu = 0.08 \text{ pF} \Rightarrow C_M &= C_\mu [1 + g_m (R_C \| R_L)] \\ C_M &= (0.08) [1 + (35.87)(2.2 \| 4.7)] = 4.38 \text{ pF} \end{aligned}$$

(c) $R_{eq} = r_s \| R_{TH} \| r_\pi = 100 \| 104.8 \| 2.788 = 2.644 \text{ k}\Omega$

$$\text{(i) } f_{3dB} = \frac{1}{2\pi R_{eq} C_\pi} = \frac{1}{2\pi (2.644 \times 10^3)(10^{-12})} \Rightarrow 60.2 \text{ MHz}$$

$$\text{(ii) } f_{3dB} = \frac{1}{2\pi (2.644 \times 10^3)(1 + 4.38) \times 10^{-12}} \Rightarrow 11.2 \text{ MHz}$$

EX7.11

$$g_m = 2\pi f_T (C_{gs} + C_{gd}) = 2\pi (3 \times 10^9) (60 + 8) \times 10^{-15} \Rightarrow 1.282 \text{ mA/V}$$

$$g_m = 2\sqrt{K_n I_{DQ}} \Rightarrow I_{DQ} = \frac{1}{K_n} \left(\frac{g_m}{2} \right)^2 = \frac{1}{1.2} \left(\frac{1.282}{2} \right)^2$$

or

$$I_{DQ} = 0.342 \text{ mA}$$

EX7.12

$$\text{(a) } V_G = \left(\frac{166}{166 + 234} \right) (10) = 4.15 \text{ V}$$

$$V_G = V_{GS} + K_n R_s (V_{GS} - V_{TN})^2$$

$$4.15 = V_{GS} + (0.8)(0.5)(V_{GS}^2 - 4V_{GS} + 4)$$

or

$$0.4V_{GS}^2 - 0.6V_{GS} - 2.55 = 0$$

which yields

$$V_{GS} = 3.384 \text{ V}$$

$$I_D = (0.8)(3.384 - 2)^2 = 1.532 \text{ mA}$$

Now

$$g_m = 2\sqrt{K_n I_D} = 2\sqrt{(0.8)(1.532)} = 2.214 \text{ mA/V}$$

$$R_{TH} = R_1 \| R_2 = 166 \| 234 = 97.11 \text{ k}\Omega$$

So

$$A_v = -g_m (R_D \| R_L) \left(\frac{R_{TH}}{R_{TH} + R_i} \right) = -(2.214)(4 \| 20) \left(\frac{97.11}{97.11 + 10} \right)$$

or

$$A_v = -6.69$$

$$(b) C_M = C_{gd} [1 + g_m (R_D \parallel R_L)] = 20 [1 + (2.214)(4 \parallel 20)] = 167.6 \text{ fF}$$

$$(c) f_{3-dB} = \frac{g_m}{2\pi(C_{gs} + C_M)} = \frac{2.214 \times 10^{-3}}{2\pi(100 + 167.6) \times 10^{-15}}$$

or

$$f_{3-dB} = 1.32 \text{ GHz}$$

EX7.13

dc analysis

$$V_{TH} = 0, \quad R_{TH} = 10 \text{ k}\Omega$$

$$I_{BQ} = \frac{0 - 0.7 - (-5)}{10 + (126)(5)} = 0.00672 \text{ mA}$$

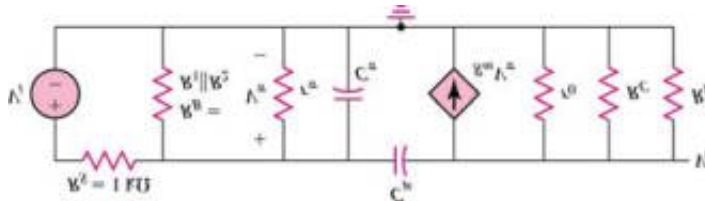
$$I_{CQ} = 0.840 \text{ mA}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(125)(0.026)}{0.840} = 3.87 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.840}{0.026} = 32.3 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_{CQ}} = \frac{200}{0.84} = 238 \text{ k}\Omega$$

High-frequency equivalent circuit



a.

Miller Capacitance

$$C_M = C_\mu (1 + g_m R'_L)$$

$$R'_L = r_o \parallel R_C \parallel R_L = 238 \parallel 2.3 \parallel 5 = 1.565 \text{ k}\Omega$$

$$C_M = (3) [1 + (32.3)(1.565)] \Rightarrow C_M = 155 \text{ pF}$$

b.

$$R_{eq} = R_S \parallel R_B \parallel r_\pi = R_S \parallel R_1 \parallel R_2 \parallel r_\pi$$

$$= 1 \parallel 20 \parallel 20 \parallel 3.87 = 0.736 \text{ k}\Omega$$

$$\tau_p = R_{eq} (C_\pi + C_M) = (0.736 \times 10^3) (24 + 155) \times 10^{-12}$$

$$= 1.314 \times 10^{-7} \text{ s}$$

$$f_H = \frac{1}{2\pi(1.314 \times 10^{-7})} \Rightarrow f_H = 1.21 \text{ MHz}$$

c.

$$(A_v)_M = -g_m R'_L \left[\frac{R_B \parallel r_\pi}{R_B \parallel r_\pi + R_S} \right] = -(32.3)(1.565) \left[\frac{10 \parallel 3.87}{10 \parallel 3.87 + 1} \right]$$

or

$$(A_v)_M = -37.2$$

EX7.14

The dc analysis

$$I_{BQ} = \frac{10 - 0.7}{100 + (101)(10)} = 0.00838 \text{ mA}$$

$$I_{CQ} = 0.838 \text{ mA}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{0.838} = 3.10 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = 32.22 \text{ mA/V}$$

For the input

$$\tau_{p\pi} = \left[\left(\frac{r_\pi}{1 + \beta} \right) \parallel R_E \parallel R_S \right] C_\pi = \left[\left(\frac{3.10}{101} \right) \parallel 10 \parallel 1 \right] \times 10^3 \times 24 \times 10^{-12}$$

$$= 7.13 \times 10^{-10} \text{ s}$$

$$f_{H\pi} = \frac{1}{2\pi\tau_{p\pi}} = \frac{1}{2\pi(7.13 \times 10^{-10})} \Rightarrow f_{H\pi} = 223 \text{ MHz}$$

For the output

$$\tau_{p\mu} = (R_C \parallel R_L) C_\mu = (10 \parallel 1) \times 10^3 \times 3 \times 10^{-12}$$

$$= 2.73 \times 10^{-9} \text{ s}$$

$$f_{H\mu} = \frac{1}{2\pi\tau_{p\mu}} = \frac{1}{2\pi(2.73 \times 10^{-9})} \Rightarrow f_{H\mu} = 58.4 \text{ MHz}$$

$$(A_v)_M = g_m (R_C \parallel R_L) \left[\frac{R_E \parallel \left(\frac{r_\pi}{1 + \beta} \right)}{R_E \parallel \left(\frac{r_\pi}{1 + \beta} \right) + R_S} \right]$$

$$= (32.22)(10 \parallel 1) \left[\frac{10 \parallel \left(\frac{3.1}{101} \right)}{10 \parallel \left(\frac{3.1}{101} \right) + 1} \right] \Rightarrow (A_v)_M = 0.870$$

EX7.15

$$V_{B1} = \left(\frac{R_3}{R_1 + R_2 + R_3} \right) (12) = \left(\frac{7.92}{58.8 + 33.3 + 7.92} \right) (12) = 0.9502 \text{ V}$$

Neglecting base currents

$$I_C = \frac{0.9502 - 0.7}{0.5} = 0.50 \text{ mA}$$

$$r_\pi = \frac{\beta V_T}{I_C} = \frac{(100)(0.026)}{0.5} = 5.2 \text{ K}$$

$$g_m = \frac{I_C}{V_T} = \frac{0.5}{0.026} = 19.23 \text{ mA/V}$$

From Eq (7.119(a)),

$$\tau_{p\pi} = (R_S \parallel R_{B1} \parallel r_\pi) (C_{\pi 1} + C_{M1})$$

$$R_{B1} = R_2 \parallel R_3 = 33.3 \parallel 7.92 = 6.398 \text{ k}\Omega$$

$$C_{M1} = 2C_{\mu 1} = 6 \text{ pF}$$

Then

$$\tau_{p\pi} = (1 \parallel 6.398 \parallel 5.2) \times 10^3 \times (24 + 6) \times 10^{-12} \Rightarrow \tau_{p\pi} = 22.24 \text{ ns}$$

$$f_{H\pi} = \frac{1}{2\pi\tau_{p\pi}} = \frac{1}{2\pi(22.24 \times 10^{-9})} \Rightarrow f_{H\pi} = 7.15 \text{ MHz}$$

From Eq (7.120(a)),

$$\tau_{p\mu} = (R_C \parallel R_L) C_{\mu 2} = (7.5 \parallel 2) \times 10^3 \times 3 \times 10^{-12} \Rightarrow \tau_{p\mu} = 4.737 \text{ ns}$$

$$f_{H\mu} = \frac{1}{2\pi\tau_{p\mu}} = \frac{1}{2\pi(4.737 \times 10^{-9})} \Rightarrow f_{H\mu} = 33.6 \text{ MHz}$$

From Eq. (7.125),

$$|A_v|_M = g_{m2} (R_C \parallel R_L) \left[\frac{R_{B1} \parallel r_{\pi 1}}{R_{B1} \parallel r_{\pi 1} + R_S} \right] = (19.23) (7.5 \parallel 2) \left[\frac{6.40 \parallel 5.2}{6.40 \parallel 5.2 + 1} \right]$$

$$|A_v|_M = 22.5$$

Test Your Understanding Solutions

TYU7.1

a.

$$V_0 = -(g_m V_\pi) R_L$$

$$V_\pi = \frac{r_\pi}{r_\pi + \frac{1}{sC_C} + R_S} \times V_i$$

$$T(s) = \frac{V_0(s)}{V_i(s)} = \frac{-g_m r_\pi R_L}{r_\pi + R_S + (1/sC_C)}$$

$$= \frac{-g_m r_\pi R_L (sC_C)}{1 + s(r_\pi + R_S)C_C}$$

$$g_m r_\pi = \beta$$

$$T(s) = \frac{-\beta R_L}{r_\pi + R_S} \times \left(\frac{s(r_\pi + R_S)C_C}{1 + s(r_\pi + R_S)C_C} \right)$$

Then $\tau = (r_\pi + R_S)C_C$

b.

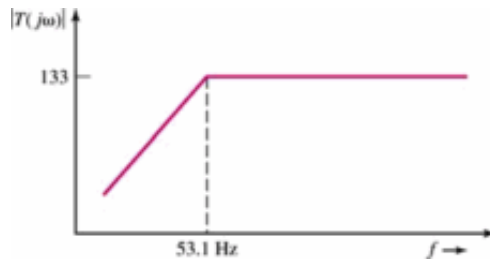
$$f_{3-dB} = \frac{1}{2\pi(r_\pi + R_S)C_C}$$

$$f_{3-dB} = \frac{1}{2\pi[2 \times 10^3 + 1 \times 10^3][10^{-6}]} \Rightarrow f_{3-dB} = 53.1 \text{ Hz}$$

$$|T(j\omega)|_{\max} = \frac{r_\pi g_m R_L}{r_\pi + R_S} = \frac{(2)(50)(4)}{2 + 1}$$

$$|T(j\omega)|_{\max} = 133$$

c.



TYU7.2

(a) $\tau = R_L C_L = (10 \times 10^3)(2 \times 10^{-12}) \Rightarrow 0.02 \mu\text{s}$

(b) $f_{3-dB} = \frac{1}{2\pi\tau} = \frac{1}{2\pi(0.02 \times 10^{-6})} \Rightarrow 7.96 \text{ MHz}$

$$|A_v|_{\max} = \frac{r_\pi}{r_\pi + R_S} \cdot (g_m R_L) = \left(\frac{2.4}{2.4 + 0.1} \right) (50)(10) = 480$$

TYU7.3

$$\begin{aligned} \text{(a)} \quad \tau_S &= (R_S + r_\pi)C_C = (0.1 + 2.4) \times 10^3 \times (5 \times 10^{-6}) \Rightarrow 12.5 \text{ ms} \\ \tau_P &= R_L C_L = (10 \times 10^3)(4 \times 10^{-12}) \Rightarrow 0.04 \mu\text{s} \\ \text{(b)} \quad A_v &= -\left(\frac{r_\pi}{r_\pi + R_S}\right)(g_m R_L) = -\left(\frac{2.4}{2.4 + 0.1}\right)(50)(10) = -480 \\ \text{(c)} \quad f_L &= \frac{1}{2\pi\tau_S} = \frac{1}{2\pi(12.5 \times 10^{-3})} = 12.7 \text{ Hz} \\ f_H &= \frac{1}{2\pi\tau_P} = \frac{1}{2\pi(0.04 \times 10^{-6})} \Rightarrow 3.98 \text{ MHz} \end{aligned}$$

TYU7.4 Computer Analysis

TYU7.5 Computer Analysis

TYU7.6

$$\begin{aligned} r_\pi &= \frac{(120)(0.026)}{0.12} = 26 \text{ k}\Omega \\ f_\beta &= \frac{1}{2\pi r_\pi (C_\pi + C_\mu)} \Rightarrow (C_\pi + C_\mu) = \frac{1}{2\pi r_\pi f_\beta} \\ \text{or} \\ C_\pi + C_\mu &= \frac{1}{2\pi(26 \times 10^3)(15 \times 10^6)} \Rightarrow 0.408 \text{ pF} \\ \text{Then} \\ C_\pi &= 0.408 - 0.08 = 0.328 \text{ pF} \end{aligned}$$

TYU7.7

$$\begin{aligned} |h_{fe}| &= \frac{\beta_o}{\sqrt{1 + \left(\frac{f}{f_\beta}\right)^2}} ; \quad \phi = -\tan^{-1}\left(\frac{f}{f_\beta}\right) \\ f_\beta &= 167 \text{ MHz} ; \quad \beta_o = 120 \end{aligned}$$

Then

$$\begin{aligned} \text{(a)} \quad \text{For } f &= 150 \text{ MHz; } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{150}{167}\right)^2}} = 89.3 \\ \phi &= -\tan^{-1}\left(\frac{150}{167}\right) = -41.9^\circ \\ \text{For } f &= 500 \text{ MHz; } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{500}{167}\right)^2}} = 38.0 \end{aligned}$$

$$\phi = -\tan^{-1}\left(\frac{500}{167}\right) = -71.5^\circ$$

$$\text{For } f = 4 \text{ GHz; } |h_{fe}| = \frac{120}{\sqrt{1 + \left(\frac{4000}{167}\right)^2}} = 5.0$$

$$\phi = -\tan^{-1}\left(\frac{4000}{167}\right) = -87.6^\circ$$

TYU7.8

$$(a) \quad f_\beta = \frac{f_T}{\beta_o} = \frac{10^9}{150} \Rightarrow f_\beta = 6.67 \text{ MHz}$$

$$f_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)} \Rightarrow (C_\pi + C_\mu) = \frac{1}{2\pi r_\pi f_\beta}$$

or

$$(C_\pi + C_\mu) = \frac{1}{2\pi (12 \times 10^3) (6.667 \times 10^6)} \Rightarrow 1.989 \text{ pF}$$

Then

$$C_\pi = 1.989 - 0.15 = 1.84 \text{ pF}$$

$$(b) \quad r_\pi = \frac{\beta V_T}{I_{CQ}} \Rightarrow I_{CQ} = \frac{\beta V_T}{r_\pi} = \frac{(150)(0.026)}{12} = 0.325 \text{ mA}$$

TYU7.9

(a)

$$g_m = 2K_n (V_{GS} - V_{TN}) = 2(0.4)(3 - 1) \Rightarrow g_m = 1.6 \text{ mA/V}$$

$$g'_m = 80\% \text{ of } g_m = 1.28 \text{ mA/V}$$

$$g'_m = \frac{g_m}{1 + g_m r_S}$$

$$1 + g_m r_S = \frac{g_m}{g'_m}$$

$$r_S = \frac{1}{g_m} \left(\frac{g_m}{g'_m} - 1 \right) = \frac{1}{1.6} \left(\frac{1.6}{1.28} - 1 \right)$$

$$r_S = 0.156 \text{ k}\Omega \Rightarrow r_S = 156 \text{ ohms}$$

(b)

$$g_m = 2K_n (V_{GS} - V_{TN}) = 2(0.4)(5 - 1) \Rightarrow g_m = 3.2 \text{ mA/V}$$

$$g'_m = \frac{g_m}{1 + g_m r_S} = \frac{3.2}{1 + (3.2)(0.156)} = 2.134$$

$$\frac{\Delta g_m}{g_m} = \frac{3.2 - 2.134}{3.2} \Rightarrow \text{A } 33.3\% \text{ reduction}$$

TYU7.10

$$g_m = 2\sqrt{\left(\frac{0.1}{2}\right)(15)(0.1)} = 0.5477 \text{ mA/V}$$

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gsp} + C_{gdp})} \Rightarrow (C_{gs} + C_{gsp} + C_{gdp}) = \frac{g_m}{2\pi f_T}$$

$$= \frac{(0.5477 \times 10^{-3})}{2\pi(1.2 \times 10^9)} \Rightarrow 72.64 \text{ fF}$$

Then

$$C_{gs} = 72.64 - 3 - 3 = 66.6 \text{ fF}$$

TYU7.11

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gsp} + C_{gdp})} \Rightarrow (C_{gs} + C_{gsp} + C_{gdp}) = \frac{g_m}{2\pi f_T}$$

$$= \frac{1.2 \times 10^{-3}}{2\pi(2.5 \times 10^9)} \Rightarrow 76.39 \text{ fF}$$

$$C_{gsp} + C_{gdp} = 76.39 - 60 = 16.39$$

or

$$C_{gsp} = C_{gdp} = 8.2 \text{ fF}$$

TYU7.12

dc analysis

$$V_G = \left(\frac{50}{50+150}\right)(10) - 5 = -2.5$$

$$V_S = V_G - V_{GS}, \quad I_D = \frac{V_S - (-5)}{R_S}$$

$$K_n(V_{GS} - V_{TN})^2 = \frac{V_G - V_{GS} + 5}{R_S}$$

$$(1)(2)[V_{GS}^2 - 1.6V_{GS} + 0.64] = -2.5 - V_{GS} + 5$$

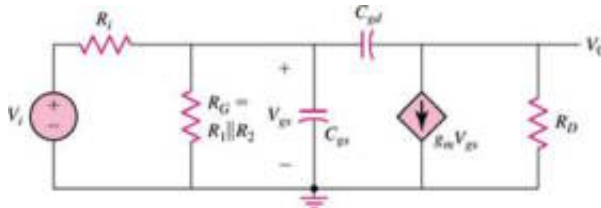
$$2V_{GS}^2 - 2.2V_{GS} - 1.22 = 0$$

$$V_{GS} = \frac{2.2 \pm \sqrt{(2.2)^2 + 4(2)(1.22)}}{2(2)} \Rightarrow V_{GS} = 1.505 \text{ V}$$

$$g_m = 2K_n(V_{GS} - V_{TN}) = 2(1)(1.505 - 0.8)$$

$$= 1.41 \text{ mA/V}$$

Equivalent circuit



(a) $C_M = C_{gd} (1 + g_m R_D) = (0.2) [1 + (1.42)(5)] \Rightarrow \underline{C_M = 1.61 \text{ pF}}$

(b)
$$\tau_P = (R_i \parallel R_G)(C_{gs} + C_M)$$
$$= (20 \parallel 50 \parallel 150) \times 10^3 \times (2 + 1.61) \times 10^{-12} = 4.71 \times 10^{-8} \text{ s}$$

$$f_H = \frac{1}{2\pi\tau_P} = \frac{1}{2\pi(4.71 \times 10^{-8})} \Rightarrow f_H = 3.38 \text{ MHz}$$

$$(A_v)_M = -g_m R_D \left(\frac{R_G}{R_G + R_S} \right)$$

c.
$$(A_v)_M = -(1.41)(5) \left(\frac{37.5}{37.5 + 20} \right) \Rightarrow \underline{(A_v)_M = -4.60}$$

TYU7.13 Computer Analysis

Chapter 8

Exercise Solutions

EX8.1

(a) $P_T = I_C V_{CE}$; At $V_{CEQ} = \frac{1}{2} V_{CC} = 12 \text{ V}$

$$25 = I_{CQ}(12) \Rightarrow I_{C,\max} = 2I_{CQ} = 2\left(\frac{25}{12}\right) = 4.17 \text{ A}$$

$$R_L = \frac{24}{4.167} = 5.76 \Omega$$

$$P_{Q,\max} = 25 \text{ W}$$

(b) $25 = I_{CQ}\left(\frac{1}{2} V_{CC}\right) = I_{CQ}(6) \Rightarrow I_{CQ} = 4.17 \text{ A}, \Rightarrow I_{C,\max} = 5 \text{ A}$

$$R_L = \frac{12}{5} = 2.4 \Omega$$

At $I_{CQ} = 2.5 \text{ A}, V_{CEQ} = 6 \text{ V}$

$$P_{Q,\max} = (2.5)(6) = 15 \text{ W}$$

EX8.2

$$P_Q = (2)(8) = 16 \text{ W}$$

(a) $T_{dev} = 25 + (16)(3 + 1 + 4) = 153^\circ \text{C}$

(b) $T_{case} = 25 + (16)(1 + 4) = 105^\circ \text{C}$

(c) $T_{snk} = 25 + (16)(4) = 89^\circ \text{C}$

EX8.3

$$\theta_{dev-case} = \frac{T_{J,\max} - T_{amb}}{P_{D,\text{rated}}} = \frac{200 - 25}{50} = 3.5^\circ \text{C/W}$$

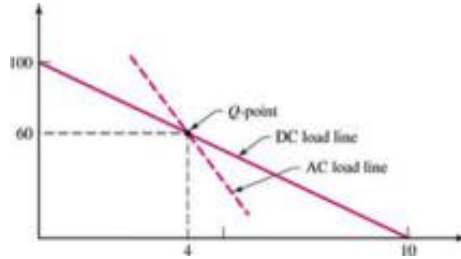
$$P_{D,\max} = \frac{T_{J,\max} - T_{amb}}{\theta_{dev-case} + \theta_{case-snk} + \theta_{snk-amb}}$$
$$= \frac{200 - 25}{3.5 + 0.5 + 2} \Rightarrow P_{D,\max} = 29.2 \text{ W}$$

$$T_{case} = T_{amb} + P_{D,\max} (\theta_{case-snk} + \theta_{snk-amb})$$
$$= 25 + (29.2)(0.5 + 2) \Rightarrow T_{case} = 98^\circ \text{C}$$

EX8.4

a. $I_{DQ} = \frac{10-4}{0.1} \Rightarrow I_{DQ} = 60 \text{ mA}$

b.



$$v_{ds} = -\left(\frac{9}{10}\right)(60)(0.050) = -2.7 \text{ V} \Rightarrow v_{DS}(\text{min}) = 4 - 2.7 = 1.3 \text{ V}$$

So maximum swing is determined by drain-to-source voltage.

$$V_{PP} = 2 \times (2.5) = 5.0 \text{ V}$$

c.

$$\overline{P_L} = \frac{1}{2} \cdot \frac{V_P^2}{R_L} = \frac{1}{2} \cdot \frac{(2.5)^2}{0.1} \Rightarrow \overline{P_L} = 31.25 \text{ mW}$$

$$\overline{P_S} = V_{DD} \cdot I_{DQ} = (10)(60) = 600 \text{ mW}$$

$$\eta = \frac{\overline{P_L}}{\overline{P_S}} = \frac{31.25}{600} \Rightarrow \eta = 5.2\%$$

EX8.5 Computer Analysis

EX8.6 No Exercise Problem

EX8.7

(a) For $v_O = 8 \text{ V}$, $i_L = \frac{8}{25} = 0.32 \text{ A}$

$$I_{DQ} = (0.2)(0.32) \Rightarrow 64 \text{ mA}$$

$$I_{DQ} = K(V_{GS} - V_{TN})^2$$

$$64 = 250(V_{GS} - 1.2)^2 \Rightarrow V_{GS} = \frac{V_{BB}}{2} = 1.706 \text{ V}$$

$$\text{Then } V_{BB} = 3.412 \text{ V}$$

(b) $v_O = v_I + \frac{V_{BB}}{2} - v_{GSn} \Rightarrow v_I = v_O - \frac{V_{BB}}{2} + v_{GSn}$

$$\frac{dv_I}{dv_O} = 1 + \frac{dv_{GSn}}{dv_O}$$

We have

$$\frac{dv_{GSn}}{dv_O} = \frac{dv_{GSn}}{di_{dn}} \cdot \frac{di_{dn}}{dv_O}$$

$$v_{GSn} = \sqrt{\frac{i_{dn}}{K}} + V_{TN} \Rightarrow \frac{dv_{GSn}}{di_{dn}} = \frac{1}{2} \cdot \frac{1}{\sqrt{K}} \cdot \frac{1}{\sqrt{i_{dn}}}$$

(i) For $v_O \approx 0$ (very small), then

$$i_{dn} = i_L + i_{dp} \Rightarrow -\Delta i_{dp} \cong \Delta i_{dn}$$

$$\text{so } \Delta i_{dn} \cong \frac{1}{2} \Delta i_L$$

$$\text{then } \frac{\Delta i_{dn}}{\Delta v_O} = \frac{1}{2} \cdot \frac{\Delta i_L}{\Delta v_O} = \frac{1}{2} \cdot \frac{1}{R_L} = \frac{1}{50} = 0.02$$

For $v_O \approx 0$, $i_{dn} = 0.064$ A

$$\frac{dv_{GSn}}{di_{dn}} = \frac{1}{2} \cdot \frac{1}{\sqrt{0.25}} \cdot \frac{1}{\sqrt{0.064}} = 3.953$$

Then

$$\frac{dv_I}{dv_O} = 1 + (3.953)(0.02) = 1.079$$

Or

$$\frac{dv_O}{dv_I} = 0.927$$

(ii) For $v_O \cong 8$ V, $i_{dn} = i_L$

$$\frac{di_{dn}}{dv_O} = \frac{di_L}{dv_O} = \frac{1}{R_L} = 0.04$$

then

$$\frac{dv_{GSn}}{di_{dn}} = \frac{1}{2} \cdot \frac{1}{\sqrt{0.25}} \cdot \frac{1}{\sqrt{0.32}} = 1.768$$

$$\frac{dv_I}{dv_O} = 1 + (1.768)(0.04) = 1.0707$$

Or

$$\frac{dv_O}{dv_I} = 0.934$$

EX8.8

a.

$$R_b = r_\pi + (1 + \beta) R'_E \text{ and } R'_E = a^2 R_L = (10)^2 (8) = 800 \Omega$$

$$R_i = 1.5 \text{ k}\Omega = R_{TH} \parallel R_b$$

$$I_Q = \frac{V_{CC}}{a^2 R_L} = \frac{18}{(10)^2 (8)} = 22.5 \text{ mA}$$

$$r_\pi = \frac{(100)(0.026)}{22.5} = 0.116 \text{ k}\Omega$$

$$R_b = 0.116 + (101)(0.8) = 80.9 \text{ k}\Omega$$

$$1.5 = R_{TH} \parallel 80.9 = \frac{R_{TH}(80.9)}{R_{TH} + (80.9)} \Rightarrow (80.9 - 1.5) R_{TH} = (1.5)(80.9) \Rightarrow R_{TH} = 1.53 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC}$$

$$I_{BQ} = \frac{I_Q}{\beta} = \frac{22.5}{100} = 0.225 \text{ mA}$$

$$I_{BQ} = \frac{V_{TH} - 0.7}{R_{TH}} \Rightarrow \frac{1}{R_1}(1.53)(18) = (0.225)(1.53) + 0.7 \Rightarrow \underline{R_1 = 26.4 \text{ k}\Omega}$$

$$\frac{26.4R_2}{26.4 + R_2} = 1.53$$

$$(26.4 - 1.53)R_2 = (1.53)(26.4) \Rightarrow \underline{R_2 = 1.62 \text{ k}\Omega}$$

b.

$$v_E = 0.9V_{CC} = (0.9)(18) = 16.2 \text{ V}$$

$$i_E = 0.9I_{CQ} = (0.9)(22.5) = 20.25 \text{ mA}$$

$$v_o = \frac{v_E}{a} = \frac{16.2}{10} \Rightarrow \underline{V_P = 1.62 \text{ V}}$$

$$i_o = ai_E = (10)(20.25) \Rightarrow \underline{I_P = 203 \text{ mA}}$$

$$\overline{P_L} = \frac{1}{2}(1.62)(0.203) \Rightarrow \underline{\overline{P_L} = 0.164 \text{ W}}$$

EX8.9

(a) For $I_Q = 1 \text{ mA}$, $v_{BE} = (0.026)\ln\left(\frac{10^{-3}}{2 \times 10^{-14}}\right) = 0.6405 \text{ V}$

Then $V_{BB} = 1.281 \text{ V}$

For D_1, D_2 ; $I_{Bias} = (1.2 \times 10^{-14})\exp\left(\frac{0.6405}{0.026}\right) = 0.60 \text{ mA}$

(b) $v_O = 1.2 \text{ V}$, $i_L = \frac{1.2}{1} = 1.2 \text{ mA}$

1st approximation:

$$i_{Cn} = 1.6 \text{ mA}, i_{Bn} = 0.016 \text{ mA}$$

$$v_{BE_n} = (0.026)\ln\left(\frac{1.6 \times 10^{-3}}{2 \times 10^{-14}}\right) = 0.65274 \text{ V}$$

$$I_D = 0.60 - 0.016 = 0.584 \text{ mA}$$

$$V_{BB} = 2(0.026)\ln\left(\frac{0.584 \times 10^{-3}}{1.2 \times 10^{-14}}\right) = 1.27963 \text{ V}$$

then $v_{EB_p} = 1.27963 - 0.65274 = 0.62689 \text{ V}$

$$i_{C_p} = (2 \times 10^{-14})\exp\left(\frac{0.62689}{0.026}\right) = 0.59206 \text{ mA}$$

2nd approximation:

$$i_{E_n} = 1.2 + 0.59206 = 1.792 \text{ mA}; i_{C_n} = 1.7743 \text{ mA}$$

After 4 iterations:

$$i_{C_n} = 1.73 \text{ mA}; i_{C_p} = 0.547 \text{ mA}$$

$$v_{BE_n} = 0.6547 \text{ V}; v_{EB_p} = 0.6248 \text{ V}$$

$$I_D = 0.5827 \text{ mA}$$

(c) $v_O = 3 \text{ V}$; $i_L = \frac{3}{1} = 3 \text{ mA}$

1st approximation;

$$i_{C_n} = 3.3 \text{ mA}, i_{B_n} = 0.033 \text{ mA}$$

$$v_{BE_n} = (0.026) \ln \left(\frac{3.3 \times 10^{-3}}{2 \times 10^{-14}} \right) = 0.67156 \text{ V}$$

$$I_D = 0.6 - 0.033 = 0.567 \text{ mA}$$

$$V_{BB} = 2(0.026) \ln \left(\frac{0.567 \times 10^{-3}}{1.2 \times 10^{-14}} \right) = 1.2781 \text{ V}$$

$$v_{EB_p} = 1.2781 - 0.67156 = 0.6065 \text{ V}$$

$$i_{C_p} = (2 \times 10^{-14}) \exp \left(\frac{0.6065}{0.026} \right) = 0.2706 \text{ mA}$$

Then $i_{E_n} = 3 + 0.2706 = 3.2706 \text{ mA}$; $i_{C_n} = 3.2382 \text{ mA}$

After 4 iterations:

$$i_{C_n} = 3.24 \text{ mA}; i_{C_p} = 0.276 \text{ mA}$$

$$v_{BE_n} = 0.671 \text{ V}, v_{EB_p} = 0.607 \text{ V}$$

$$I_D = 0.5676 \text{ mA}$$

EX8.10 No Exercise EX8.10

EX8.11

a.

$$v_I = 0 = v_0, v_{B3} = 0.7 \text{ V}$$

$$I_{R1} = \frac{12 - 0.7}{R_1} = \frac{11.3}{0.25} \Rightarrow I_{R1} = 45.2 \text{ mA}$$

If transistors are matched, then

$$i_{E1} = i_{E3}$$

$$i_{R1} = i_{E1} + i_{B3} = i_{E1} + \frac{i_{E3}}{1 + \beta}$$

$$i_{R1} = i_{E1} \left(1 + \frac{1}{1 + \beta} \right) = i_{E1} \left(1 + \frac{1}{41} \right)$$

$$i_{E1} = \frac{45.2}{1.024} \Rightarrow i_{E1} = i_{E2} = 44.1 \text{ mA}$$

$$i_{B1} = i_{B2} = \frac{i_{E1}}{1 + \beta} = \frac{44.1}{41} \Rightarrow i_{B1} = i_{B2} = 1.08 \text{ mA}$$

b.

$$\text{For } v_I = 5 \text{ V} \Rightarrow v_0 = 5 \text{ V}$$

$$i_0 = \frac{5}{8} \Rightarrow i_0 = 0.625 \text{ A}$$

$$i_{E3} \cong 0.625 \text{ A}, i_{B3} = \frac{0.625}{41} \Rightarrow i_{B3} = 15.2 \text{ mA}$$

$$v_{B3} = 5.7 \text{ V} \Rightarrow i_{R1} = \frac{12 - 5.7}{0.25} = 25.2 \text{ mA}$$

$$i_{E1} = 25.2 - 15.2 \Rightarrow i_{E1} = 10.0 \text{ mA} \Rightarrow i_{B1} = \frac{10}{41} = 0.244 \text{ mA}$$

$$v_{B4} = 5 - 0.7 = 4.3 \text{ V}$$

$$I_{R2} = \frac{4.3 - (-12)}{0.25} = 65.2 \text{ mA} \equiv i_{E2}$$

$$i_{B2} = \frac{65.2}{41} = 1.59 \text{ mA}$$

$$i_I = i_{B2} - i_{B1} = 1.59 - 0.244 \Rightarrow i_I = 1.35 \text{ mA}$$

$$A_I = \frac{i_0}{i_I} = \frac{625}{1.35} \Rightarrow A_I = 463$$

c.

From Equation (8.55)

$$A_I = \frac{(1 + \beta)R}{2R_L} = \frac{(41)(250)}{2(8)} = 641$$

Test Your Understanding Solutions

TYU8.1

For $V_{DS} = 0$, $I_D(\text{max}) = \frac{24}{20} = 1.2 \text{ A} = I_{D(\text{max})}$

For $I_D = 0 \Rightarrow V_{DS}(\text{max}) = 24 \text{ V}$

Maximum power when

$$V_{DS} = \frac{V_{DS}(\text{max})}{2} = 12 \text{ V and}$$

$$I_D = \frac{I_D(\text{max})}{2} = 0.6 \text{ A} \Rightarrow P_D(\text{max}) = (12)(0.6) = 7.2 \text{ Watts}$$

TYU8.2

(a) $R_E = \frac{V_{CC} - (-V_{CC})}{I_{C,\text{max}}} = \frac{12 - (-12)}{0.25} = 96 \Omega$

(b) For $I_{CQ} = 0.125 \text{ A}$, $V_{CEQ} = 12 \text{ V}$

$$P_{Q,\text{max}} = (0.125)(12) = 1.5 \text{ W}$$

TYU8.3

(a) $\Delta T = P \cdot \theta = (6)(1.8) = 10.8^\circ \text{ C}$

(b) $P = \frac{\Delta T}{\theta} = \frac{100}{2.5} = 40 \text{ W}$

TYU8.4

- (a) $V_{CEQ} = 6 = V_{CC} - I_{CQ} R_L = 12 - I_{CQ}(1) \Rightarrow I_{CQ} = 6 \text{ mA}$
 $P_Q = I_{CQ} V_{CEQ} = (6)(6) = 36 \text{ mW}$
- (b) (i) $\bar{P}_L = \frac{1}{2} \cdot \frac{V_P^2}{R_L} = \frac{1}{2} \cdot \frac{(4.5)^2}{1} = 10.1 \text{ mW}$
- (ii) $\eta = \frac{10.1}{I_{CQ} V_{CC}} \times 100\% = \frac{10.1}{(6)(12)} \times 100\% = 14.1\%$
- (iii) $\bar{P}_Q = 36 - 10.1 = 25.9 \text{ mW}$

TYU8.5

- a. $\bar{P}_L = \frac{1}{2} \cdot \frac{V_P^2}{R_L} \Rightarrow V_P = \sqrt{2R_L \bar{P}_L} = \sqrt{2(8)(25)} \Rightarrow V_P = 20 \text{ V} \Rightarrow V_{CC} = \frac{20}{0.8} \Rightarrow V_{CC} = 25 \text{ V}$
- b. $I_P = \frac{V_P}{R_L} = \frac{20}{8} \Rightarrow I_P = 2.5 \text{ A}$
- $\bar{P}_Q = \frac{V_{CC} V_P}{\pi R_L} - \frac{V_P^2}{4R_L}$
- c. $\bar{P}_Q = \frac{(25)(20)}{\pi(8)} - \frac{(20)^2}{4(8)} = 19.9 - 12.5 \Rightarrow \bar{P}_Q = 7.4 \text{ W}$
- d. $\eta = \frac{\pi V_P}{4V_{CC}} = \frac{\pi}{4} \cdot \frac{20}{25} \Rightarrow \eta = 62.8\%$

TYU8.6

- a. $\bar{P}_L = \frac{1}{2} \cdot \frac{V_P^2}{R_L} = \frac{(4)^2}{2(0.1)} \Rightarrow \bar{P}_L = 80 \text{ mW}$
- b. $I_P = \frac{V_P}{R_L} = \frac{4}{0.1} \Rightarrow I_P = 40 \text{ mA}$
- $\bar{P}_Q = \frac{V_{CC} V_P}{\pi R_L} - \frac{V_P^2}{4R_L}$
- c. $\bar{P}_Q = \frac{(5)(4)}{\pi(0.1)} - \frac{(4)^2}{4(0.1)} = 63.7 - 40 \Rightarrow \bar{P}_Q = 23.7 \text{ mW}$
- d. $\eta = \frac{\pi V_P}{4V_{CC}} = \frac{\pi}{4} \cdot \frac{4}{5} \Rightarrow \eta = 62.8\%$

TYU8.7

a.

$$I_{CQ} \cong \frac{1}{2} \cdot \left(\frac{2V_{CC}}{R_L} \right) = \frac{V_{CC}}{R_L} = \frac{12}{1.5} = 8 \text{ mA}$$

$$R_{TH} = R_1 \parallel R_2$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \frac{1}{R_1} \cdot R_{TH} \cdot V_{CC}$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{8}{75} = 0.107 \text{ mA} = \frac{V_{TH} - V_{BE}}{R_{TH} + (1 + \beta) R_E}$$

$$\text{Let } R_{TH} = (1 + \beta) R_E = (76)(0.1) = 7.6 \text{ k}\Omega$$

$$0.107 = \frac{\frac{1}{R_1} \cdot (7.6)(12) - 0.7}{7.6 + 7.6}$$

$$\frac{1}{R_1} \cdot (91.2) = 2.33 \Rightarrow R_1 = 39.1 \text{ k}\Omega$$

$$\frac{39.1 R_2}{39.1 + R_2} = 7.6 \Rightarrow (39.1 - 7.6) R_2 = (7.6)(39.1) \Rightarrow R_2 = 9.43 \text{ k}\Omega$$

b.

$$\overline{P}_L = \frac{1}{2} \cdot (0.9 I_{CQ})^2 R_L = \frac{1}{2} [(0.9)(8)]^2 (1.5) \Rightarrow \overline{P}_L = 38.9 \text{ mW}$$

$$\overline{P}_S = V_{CC} I_{CQ} = (12)(8) = 96 \text{ mW}$$

$$\overline{P}_Q = \overline{P}_S - \overline{P}_L = 96 - 38.9 \Rightarrow \overline{P}_Q = 57.1 \text{ mW}$$

$$\eta = \frac{\overline{P}_L}{\overline{P}_S} = \frac{38.9}{96} \Rightarrow \eta = 40.5\%$$

TYU8.8

$$(a) \quad v_{BE_n} = (0.026) \ln \left(\frac{10^{-3}}{5 \times 10^{-16}} \right) = 0.73643 \text{ V}$$

$$v_{EB_p} = (0.026) \ln \left(\frac{10^{-3}}{8 \times 10^{-16}} \right) = 0.72421 \text{ V}$$

$$V_{BB} = v_{BE_n} + v_{EB_p} = 1.4606 \text{ V}$$

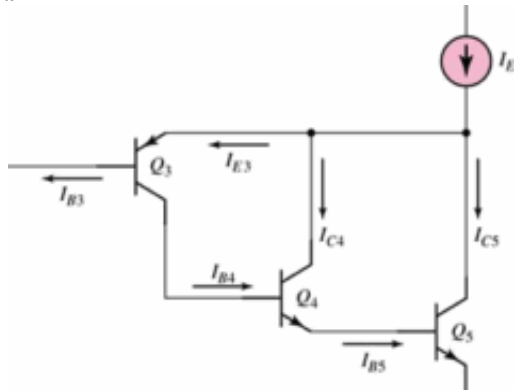
(b) See above

$$(c) \quad v_I = v_{BE_n} - \frac{V_{BB}}{2} = 0.73643 - \frac{1.4606}{2}$$

or

$$v_I = 6.1 \text{ mV}$$

TYU8.9



$$\begin{aligned}
 I_E &= I_{E3} + I_{C4} + I_{C5} \\
 &= I_{E3} + I_{C4} + \beta_5 I_{B5} \\
 &= I_{E3} + \beta_4 I_{B4} + \beta_5 (1 + \beta_4) I_{B4} \\
 I_E &= (1 + \beta_3) I_{B3} + \beta_4 \beta_3 I_{B3} + \beta_5 (1 + \beta_4) \beta_3 I_{B3}
 \end{aligned}$$

If β_4 and β_5 are large, then $I_E \cong \beta_3 \beta_4 \beta_5 I_{B3}$

So that composite current gain is $\beta \cong \beta_3 \beta_4 \beta_5$

Chapter 9

Exercise Solutions

EX9.1

$$(a) \quad i_1 = \frac{v_I}{R_1} = \frac{25 \times 10^{-3}}{R_1} = 10 \times 10^{-6} \Rightarrow R_1 = 2.5 \text{ k}\Omega$$

$$A_v = -\frac{R_2}{R_1} \Rightarrow -25 = \frac{-R_2}{2.5} \Rightarrow R_2 = 62.5 \text{ k}\Omega$$

$$(b) \quad |v_O| = |A_v| \cdot v_I = (25)(25 \times 10^{-3}) = 0.625 \text{ V}$$

$$-0.625 \leq v_O \leq 0.625 \text{ V}$$

EX9.2

$$A_v = -\frac{R_2}{R_1} \left(1 + \frac{R_3}{R_4} + \frac{R_3}{R_2} \right)$$

$$A_v = -75, \text{ Let } R_1 = 20 \text{ k}\Omega$$

$$A_v = -75 = -\frac{R_2}{R_1} \left(1 + \frac{R_3}{R_4} \right) - \frac{R_3}{R_1}$$

$$\text{Let } \frac{R_2}{R_1} = \frac{R_3}{R_1} = 8$$

$$\text{Then } R_2 = R_3 = 160 \text{ k}\Omega$$

$$75 = 8 \left(1 + \frac{R_3}{R_4} \right) + 8$$

$$\text{or } \frac{R_3}{R_4} = 7.375$$

$$\text{So } R_4 = \frac{160}{7.375} = 21.7 \text{ k}\Omega$$

EX9.3

$$(a) \quad A_v = -\frac{R_2}{R_1} \cdot \frac{1}{\left[1 + \frac{1}{A_{od}} \left(1 + \frac{R_2}{R_1} \right) \right]}$$

$$R_1 = 25 \text{ k}\Omega, \quad A_v = -15.0, \quad A_{od} = 10^4$$

Then

$$-15.0 = -\frac{R_2}{R_1} \cdot \frac{1}{\left[1 + \frac{1}{10^4} \left(1 + \frac{R_2}{R_1} \right) \right]}$$

$$\text{which yields } \frac{R_2}{R_1} = 15.024 \Rightarrow R_2 = 375.6 \text{ k}\Omega$$

(b) (i) $A_{od} = 10^5$

$$\text{Then } A_v = -(15.024) \cdot \frac{1}{\left[1 + \frac{1}{10^5}(16.024)\right]} = -15.0216$$

(ii) $A_{od} = 10^3$

$$A_v = -(15.024) \cdot \frac{1}{\left[1 + \frac{1}{10^3}(16.024)\right]} = -14.787$$

EX9.4

(a) $v_O = -3(v_{I1} + 2v_{I2} + 0.3v_{I3} + 4v_{I4})$

$$v_O = -3v_{I1} - 6v_{I2} - 0.9v_{I3} - 12v_{I4}$$

Then $\frac{R_F}{R_1} = 3$, $\frac{R_F}{R_2} = 6$, $\frac{R_F}{R_3} = 0.9$, $\frac{R_F}{R_4} = 12$

R_3 will be the maximum resistance.

Let $R_3 = 400 \text{ k}\Omega \Rightarrow R_F = 360 \text{ k}\Omega$, $R_1 = 120 \text{ k}\Omega$, $R_2 = 60 \text{ k}\Omega$, $R_4 = 30 \text{ k}\Omega$

(b) (i) $v_O = -3(0.1) - 6(-0.2) - 0.9(-1) - 12(0.05) = +1.2 \text{ V}$

(ii) $v_O = -3(-0.2) - 6(0.3) - 0.9(1.5) - 12(-0.1) = -1.35 \text{ V}$

EX9.5

We may note that $\frac{R_2}{R_1} = \frac{3}{1.5} = 2$ and $\frac{R_F}{R_1} = \frac{20}{10} = 2$ so that $\frac{R_3}{R_2} = \frac{R_F}{R_1}$
Then

$$i_L = \frac{-v_L}{R_2} = \frac{-(-3)}{1.5 \text{ k}\Omega} \Rightarrow i_L = 2 \text{ mA}$$

$$v_L = i_L Z_L = (2 \times 10^{-3})(200) = 0.4 \text{ V}$$

$$i_4 = \frac{v_L}{R_2} = \frac{0.4}{1.5 \text{ k}\Omega} = 0.267 \text{ mA}$$

$$i_3 = i_4 + i_L = 0.267 + 2 = 2.267 \text{ mA}$$

$$v_O = i_3 R_3 + v_L = (2.267 \times 10^{-3})(3 \times 10^3) - 0.4 \Rightarrow v_O = 7.2 \text{ V}$$

EX9.6

(a) $A_d = \frac{R_2}{R_1} = 50$

For $v_{I2} = 50 \text{ mV}$ and $v_{I1} = -50 \text{ mV}$

$$v_O = 50(0.05 - (-0.05)) = 5 \text{ V}$$

$$|i_{R2}| \cong \frac{5 - 0.05}{R_2} = 50 \mu\text{A}$$

Set $R_2 = R_4 = 100 \text{ k}\Omega$

$$R_1 = R_3 = 2 \text{ k}\Omega$$

(b) $i_{R3} = \frac{0.05}{R_3 + R_4} = \frac{0.05}{100 + 2} \Rightarrow 0.49 \mu\text{A}$

EX9.7

We have the general relation that

$$v_0 = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{[R_4/R_3]}{1 + [R_4/R_3]}\right) v_{I2} - \frac{R_2}{R_1} v_{I1}$$

$$R_1 = R_3 = 10 \text{ k}\Omega, \quad R_2 = 20 \text{ k}\Omega, \quad R_4 = 21 \text{ k}\Omega$$

$$v_0 = \left(1 + \frac{20}{10}\right) \left(\frac{[21/10]}{1 + [21/10]}\right) v_{I2} - \left(\frac{20}{10}\right) v_{I1}$$

$$v_0 = 2.0323 v_{I2} - 2.0 v_{I1}$$

$$v_{I1} = 1, \quad v_{I2} = -1$$

$$v_0 = -2.0323 - 2.0 \Rightarrow v_0 = -4.032 \text{ V}$$

a.

$$v_{I1} = v_{I2} = 1 \text{ V}$$

b.

$$v_0 = 2.0323 - 2.0 \Rightarrow v_0 = 0.0323 \text{ V}$$

c.

$$v_{cm} = v_{I1} = v_{I2} \text{ so common-mode gain}$$

$$A_{cm} = \frac{v_0}{v_{cm}} = 0.0323$$

d.

$$CMRR_{dB} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right)$$

$$A_d = \frac{2.0323}{2} - (2.0) \left(-\frac{1}{2} \right) = 2.016$$

$$CMRR_{dB} = 20 \log_{10} \left(\frac{2.016}{0.0323} \right) = 35.9 \text{ dB}$$

EX9.8

$$(a) \quad A_d = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right)$$

$$A_d(\max) = \frac{90}{30} \left(1 + \frac{2(50)}{2} \right) = 153$$

$$A_d(\min) = \frac{90}{30} \left(1 + \frac{2(50)}{2 + 100} \right) = 5.94$$

$$5.94 \leq A_d \leq 153$$

$$(b) \quad i_1 = \frac{v_{I1} - v_{I2}}{R_1} = \frac{0.025 - (-0.025)}{2} \Rightarrow i_1 = 25 \mu\text{A}$$

EX9.9

$$(a) \quad R_1 C_2 = (10^4)(0.1 \times 10^{-6}) \Rightarrow 1 \text{ ms}$$

$$(i) \quad 0 < t < 1$$

$$v_O = 0 - \frac{1}{1} \cdot t \Big|_0^1 = -1 \text{ V}$$

$$(ii) \quad 1 < t < 2$$

$$v_O = -1 - \frac{(-1)}{1} \cdot t \Big|_1^2 = -1 + 1(2 - 1) = 0$$

(iii) $2 < t < 3$

$$v_O = 0 - \frac{1}{1} \cdot t' \Big|_2^3 = -1 \text{ V}$$

(iv) $3 < t < 4$

$$v_O = -1 - \frac{(-1)}{1} \cdot t' \Big|_3^4 = 0$$

(b) $R_1 C_2 = (10^4)(10^{-6}) \Rightarrow 10 \text{ ms}$

(i) $0 < t < 1$

$$v_O = 0 - \frac{(1)}{10} \cdot t' \Big|_0^1 = -0.1 \text{ V}$$

(ii) $1 < t < 2$

$$v_O = -0.1 - \frac{(-1)}{10} \cdot t' \Big|_1^2 = 0$$

(iii) $2 < t < 3$

$$v_O = 0 - \frac{(1)}{10} \cdot t' \Big|_2^3 = -0.1 \text{ V}$$

(iv) $3 < t < 4$

$$v_O = -0.1 - \frac{(-1)}{10} \cdot t' \Big|_3^4 = 0$$

EX9.10

(a) $R_N = R_1 \parallel R_2 = 40 \parallel 20 = 13.33 \text{ k}\Omega$

$$R_P = R_A \parallel R_B \parallel R_C = 50 \parallel 50 \parallel 100 = 20 \text{ k}\Omega$$

$$v_O = -\frac{R_F}{R_1} v_{I1} - \frac{R_F}{R_2} v_{I2} + \left(1 + \frac{R_F}{R_N}\right) \left[\frac{R_P}{R_A} v_{I3} + \frac{R_P}{R_B} v_{I4} \right]$$

$$v_O = -\frac{80}{40} v_{I1} - \frac{80}{20} v_{I2} + \left(1 + \frac{80}{13.33}\right) \left[\frac{20}{50} v_{I3} + \frac{20}{50} v_{I4} \right]$$

$$v_O = -2v_{I1} - 4v_{I2} + 2.8v_{I3} + 2.8v_{I4}$$

(b) (i) $v_O = -2(0.1) - 4(0.15) + 2.8(0.2) + 2.8(0.3) = 0.6 \text{ V}$

(ii) $v_O = -2(-0.2) - 4(0.25) + 2.8(-0.1) + 2.8(0.2) = -0.32 \text{ V}$

EX9.11 Computer Analysis

Test Your Understanding Solutions

TYU9.1

- (a) $A_v = -\frac{R_2}{R_1} \Rightarrow -12 = -\frac{240}{R_1} \Rightarrow R_1 = 20 \text{ k}\Omega$
- (b) (i) $i_1 = \frac{v_I}{R_1} = \frac{-0.15}{20} \Rightarrow i_1 = -7.5 \mu\text{A}$
- (ii) $i_1 = \frac{0.25}{20} \Rightarrow i_1 = 12.5 \mu\text{A}$

TYU9.2

(a)

$$A_v = \frac{-R_2}{R_1 + R_s}$$

$$A_v(\text{min}) = \frac{-100}{19 + 1.3} = -4.926$$

$$A_v(\text{max}) = \frac{-100}{19 + 0.7} = -5.076$$

so $4.926 \leq |A_v| \leq 5.076$

(b)

$$i_1(\text{max}) = \frac{0.1}{19 + 0.7} = 5.076 \mu\text{A}$$

$$i_1(\text{min}) = \frac{0.1}{19 + 1.3} = 4.926 \mu\text{A}$$

so $4.926 \leq i_1 \leq 5.076 \mu\text{A}$

(c) Maximum current specification is violated.

TYU9.3

$$A_v = -\frac{R_2}{R_1} \cdot \frac{1}{\left[1 + \frac{1}{A_{od}} \left(1 + \frac{R_2}{R_1}\right)\right]} = -\frac{200}{20} \cdot \frac{1}{\left[1 + \frac{1}{10^4} \left(1 + \frac{200}{20}\right)\right]}$$

or

$$A_v = -9.989$$

(a) $v_O = (-9.989)(50) \Rightarrow v_O = -0.49945 \text{ V}$

$$v_1 = -\frac{v_O}{A_{od}} = -\frac{(-0.49945)}{10^4} \Rightarrow v_1 = 49.945 \mu\text{V}$$

(b) $v_I = \frac{v_O}{A_v} = \frac{+5}{-9.989} = -0.50055 \text{ V}$

$$v_1 = -\frac{v_O}{A_{od}} = -\frac{+5}{10^4} \Rightarrow v_1 = -0.5 \text{ mV}$$

(c) $v_O = -A_{od}v_1 = -(10^4)(0.2) \Rightarrow v_O = -2 \text{ V}$

$$v_I = \frac{v_O}{A_v} = \frac{-2}{-9.989} = 0.20022 \text{ V}$$

TYU9.4

$$\begin{aligned} v_O &= -\frac{R_F}{R_1}v_{I1} - \frac{R_F}{R_2}v_{I2} - \frac{R_F}{R_3}v_{I3} \\ &= -\frac{200}{20}v_{I1} - \frac{200}{40}v_{I2} - \frac{200}{50}v_{I3} \\ &= -10v_{I1} - 5v_{I2} - 4v_{I3} \end{aligned}$$

(a) $v_O = -10(-0.25) - 5(0.30) - 4(-0.50) \Rightarrow v_O = 3 \text{ mV}$

(b) $v_O = -10(10) - 5(-40) - 4(25) = 0$

TYU9.5

$$|v_O| = \frac{v_{I1} + v_{I2} + v_{I3}}{3} = \frac{R_F}{R}(v_{I1} + v_{I2} + v_{I3})$$

$$\frac{R_F}{R} = \frac{1}{3} \Rightarrow R_1 = R_2 = R_3 \equiv R = 1 \text{ M}\Omega$$

Then $R_F = \frac{1}{3} \text{ M}\Omega = 333 \text{ k}\Omega$

TYU9.6

(a) $A_v = 10 = \left(1 + \frac{R_2}{R_1}\right) \Rightarrow \frac{R_2}{R_1} = 9$

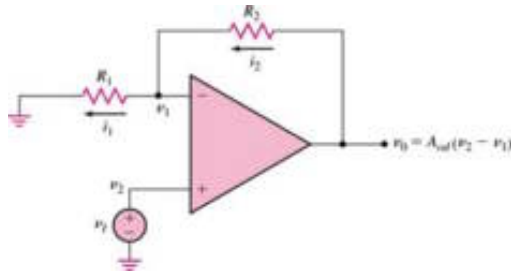
Set $R_2 = 180 \text{ k}\Omega$, $R_1 = 20 \text{ k}\Omega$

(b) $A_v = 5 = \left(1 + \frac{R_2}{R_1}\right) \Rightarrow \frac{R_2}{R_1} = 4$

For $v_O = 5 \text{ V}$, $v_1 = 1 \text{ V}$

$$|i_{R2}| = 100 \mu\text{A} = \frac{5-1}{R_2} \Rightarrow R_2 = 40 \text{ k}\Omega, \text{ then } R_1 = 10 \text{ k}\Omega$$

TYU9.7



$$v_0 = A_{od}(v_2 - v_1) = A_{od}(v_1 - v_1)$$

$$\frac{v_0}{A_{od}} - v_1 = -v_1 \text{ or } v_1 = v_1 - \frac{v_0}{A_{od}}$$

$$i_1 = \frac{v_1}{R_1} = i_2 \text{ and } i_2 = \frac{v_0 - v_1}{R_2}$$

$$\begin{aligned}\text{Then } v_1 \left(\frac{1}{R_1} \right) &= \frac{v_0 - v_1}{R_2} \\ v_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) &= \frac{v_0}{R_2} \\ v_0 \left(1 + \frac{R_2}{R_1} \right) v_1 &= \left(1 + \frac{R_2}{R_1} \right) \left(v_1 - \frac{v_0}{A_{od}} \right) \\ \text{So } A_v = \frac{v_0}{v_1} &= \frac{\left(1 + \frac{R_2}{R_1} \right)}{1 + \frac{1}{A_{od}} \left(1 + \frac{R_2}{R_1} \right)}\end{aligned}$$

TYU9.8

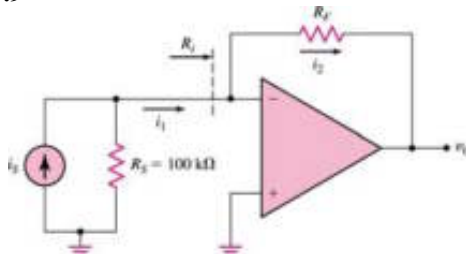
$$\begin{aligned}\text{For } v_{I2} = 0, v_2 &= \left(\frac{R_b}{R_b + R_a} \right) v_{I1} \text{ and} \\ v_0(v_{I1}) &= \left(1 + \frac{R_2}{R_1} \right) \left(\frac{R_b}{R_b + R_a} \right) v_{I1} \\ &= \left(1 + \frac{70}{5} \right) \left(\frac{50}{50 + 25} \right) v_{I1} \\ &= 10v_{I1}\end{aligned}$$

$$\begin{aligned}\text{For } v_{I1} = 0, v_2 &= \left(\frac{R_a}{R_b + R_a} \right) v_{I2} \\ v_0(v_{I2}) &= \left(1 + \frac{R_2}{R_1} \right) \left(\frac{R_a}{R_b + R_a} \right) v_{I2} \\ &= \left(1 + \frac{70}{5} \right) \left(\frac{25}{25 + 50} \right) v_{I2} \\ &= 5v_{I2}\end{aligned}$$

Then

$$\begin{aligned}v_0 &= v_0(v_{I1}) + v_0(v_{I2}) \\ v_0 &= 10v_{I1} + 5v_{I2}\end{aligned}$$

TYU9.9



$$R_s \gg R_f \text{ so } i_1 = i_2 = i_s = 100 \mu\text{A}$$

$$v_0 = -i_s R_f$$

$$\text{We want } -10 = -(100 \times 10^{-6}) R_f \Rightarrow R_f = 100 \text{ k}\Omega$$

TYU9.10

We want $i_L = 1 \text{ mA}$ when $v_I = -5 \text{ V}$

$$i_L = \frac{-V_I}{R_2} \Rightarrow R_2 = \frac{-v_I}{i_2} = \frac{-(-5)}{10^{-3}} \Rightarrow R_2 = 5 \text{ k}\Omega$$

$$v_L = i_L Z_L = (10^{-3})(500) \Rightarrow v_L = 0.5 \text{ V}$$

$$i_4 = \frac{v_L}{R_2} = \frac{0.5}{5 \text{ k}\Omega} \Rightarrow i_4 = 0.1 \text{ mA}$$

$$i_3 = i_4 + i_L = 0.1 + 1 = 1.1 \text{ mA}$$

If op-amp is biased at $\pm 10 \text{ V}$, output must be limited to $\approx 8 \text{ V}$.

So $v_0 = i_3 R_3 + v_L$

$$8 = (1.1 \times 10^{-3}) R_3 + 0.5 \Rightarrow R_3 = 6.82 \text{ k}\Omega$$

Let $R_3 = 7.0 \text{ k}\Omega$

$$\frac{R_3}{R_2} = \frac{R_F}{R_1} = \frac{7}{5} = 1.4$$

Then we want

Can choose $R_1 = 10 \text{ k}\Omega$ and $R_F = 14 \text{ k}\Omega$

TYU9.11

$$(a) \quad v_{O1} = v_{I1} + i_1 R'_2 = v_{I1} + \left(\frac{v_{I1} - v_{I2}}{R_1} \right) R'_2$$

$$v_{O1} = \left(1 + \frac{R'_2}{R_1} \right) v_{I1} - \frac{R'_2}{R_1} v_{I2}$$

$$v_{O2} = \left(1 + \frac{R_2}{R_1} \right) v_{I2} - \frac{R_2}{R_1} v_{I1}$$

Now

$$v_O = \frac{R_4}{R_3} (v_{O2} - v_{O1})$$

We can write $v_{I1} = v_{cm} - \frac{v_d}{2}$ and $v_{I2} = v_{cm} + \frac{v_d}{2}$

Then

$$v_{O1} = \left(1 + \frac{R'_2}{R_1} \right) \left(v_{cm} - \frac{v_d}{2} \right) - \frac{R'_2}{R_1} \left(v_{cm} + \frac{v_d}{2} \right) = v_{cm} - \left(1 + \frac{2R'_2}{R_1} \right) \left(\frac{v_d}{2} \right)$$

$$v_{O2} = \left(1 + \frac{R_2}{R_1} \right) \left(v_{cm} + \frac{v_d}{2} \right) - \frac{R_2}{R_1} \left(v_{cm} - \frac{v_d}{2} \right) = v_{cm} + \left(1 + \frac{2R_2}{R_1} \right) \left(\frac{v_d}{2} \right)$$

Then

$$v_O = \frac{R_4}{R_3} \left[\left(1 + \frac{2R_2}{R_1} \right) \left(\frac{v_d}{2} \right) + v_{cm} + \left(1 + \frac{2R'_2}{R_1} \right) \left(\frac{v_d}{2} \right) - v_{cm} \right]$$

or

$$v_O = \frac{R_4}{R_3} \left[1 + \frac{R_2 + R'_2}{R_1} \right] \cdot v_d$$

So

$$A_{cm} = 0$$

(b) For $A_d(\max)$, let $R_1 = 2 \text{ k}\Omega$, $R'_2 = 50 \text{ k}\Omega + 5\% = 52.5 \text{ k}\Omega$

$$\text{Then } A_d(\max) = \frac{90}{30} \left[1 + \frac{50 + 52.5}{2} \right] = 156.75$$

For $A_d(\min)$, let $R_1 = 102 \text{ k}\Omega$, $R'_2 = 50 \text{ k}\Omega - 5\% = 47.5 \text{ k}\Omega$

$$\text{Then } A_d(\min) = \frac{90}{30} \left[1 + \frac{50 + 47.5}{102} \right] = 5.87$$

(c) $\text{CMRR} = \infty$

TYU9.12

$$i_1 = \frac{v_{I1} - v_{I2}}{R_1} \Rightarrow R_1(\text{fixed}) = \frac{[2 - (-2)] \times 10^{-3}}{2 \times 10^{-6}} \Rightarrow 2 \text{ k}\Omega$$

$$A_d(\max) = (2.5) \left(1 + \frac{2R_2}{2} \right) = 500 \Rightarrow R_2 = 199 \text{ k}\Omega$$

$$A_d(\min) = (2.5) \left(1 + \frac{2(199)}{2 + R_1(\text{var})} \right) = 5 \Rightarrow R_1(\text{var}) = 396 \text{ k}\Omega$$

TYU9.13

$$\text{End of 1st pulse: } v_o = \frac{-1}{\tau} \times t \Big|_0^{10 \mu\text{s}} = \frac{-10 \times 10^{-6}}{\tau}$$

$$\text{After 10 pulses: } v_o = -5 = \frac{-(10)(10 \times 10^{-6})}{\tau}$$

$$\text{So } \tau = \frac{100 \times 10^{-6}}{5} \Rightarrow \tau = 20 \mu\text{s}$$

$$\tau = 20 \times 10^{-6} = R_1 C_2$$

$$\text{For example, } C_2 = 0.01 \times 10^{-6} = 0.01 \mu\text{F} \Rightarrow R_1 = 2 \text{ k}\Omega$$

TYU9.14

(a)

$$v_A = \left(\frac{R - \Delta R}{R - \Delta R + R + \Delta R} \right) \cdot V^+ = \left(\frac{R - \Delta R}{2R} \right) \cdot V^+$$

$$v_B = \left(\frac{R + \Delta R}{R + \Delta R + R - \Delta R} \right) \cdot V^+ = \left(\frac{R + \Delta R}{2R} \right) \cdot V^+$$

$$v_{O1} = v_A - v_B = \left[\left(\frac{R - \Delta R}{2R} \right) - \left(\frac{R + \Delta R}{2R} \right) \right] \cdot V^+$$

so

$$v_{O1} = \left(\frac{-\Delta R}{R} \right) \cdot V^+ = \left(\frac{-5}{20 \times 10^3} \right) \Delta R = (-2.5 \times 10^{-4}) \Delta R$$

$$\text{We have } v_{O1} = (-2.5 \times 10^{-4})(-100) \Rightarrow v_{O1} = 25 \text{ mV}$$

(b)

For the instrumentation amplifier

$$v_O = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right) (v_{O1})$$

$$3 = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right) (0.025)$$

$$\text{or } 120 = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right) ; \quad \text{For example, set } \frac{R_4}{R_3} = 10 \quad \text{and} \quad \frac{R_2}{R_1} = 5.5$$

TYU9.15

(a)

$$v_A = \left(\frac{R}{R+R} \right) \cdot V^+ = \frac{1}{2}(3) = 1.5 \text{ V}$$

$$v_B = \left(\frac{R}{R+R(1+\delta)} \right) \cdot V^+ = \left(\frac{1}{2+\delta} \right) (3)$$

$$v_{O1} = v_A - v_B = 1.5 - \left(\frac{1}{2+\delta} \right) (3) = \frac{(2+\delta)(1.5) - 3}{2+\delta} = \frac{1.5\delta}{2+\delta} \cong 0.75\delta \text{ V}$$

(b)

For an instrumentation amplifier

$$v_O = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right) (v_{O1})$$

For $\delta = 0.025$, want $v_O = 3 \text{ V}$

$$3 = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right) (0.75)(0.025)$$

or

$$160 = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_1} \right)$$

$$\text{For example, set } \frac{R_4}{R_3} = 10 \quad \text{and} \quad \frac{R_2}{R_1} = 7.5$$

Chapter 10

Exercise Solutions

EX10.1

$$I_{REF} = \frac{V^+ - V_{BE(on)} - V^-}{R_1} = \frac{3 - 0.7 - (-3)}{47} = 0.1128 \text{ mA}$$

$$I_O = \frac{I_{REF}}{\left(1 + \frac{2}{\beta}\right)} = \frac{0.112766}{\left(1 + \frac{2}{120}\right)} = 0.1109 \text{ mA}$$

$$I_{B1} = I_{B2} = \frac{I_O}{\beta} \Rightarrow 0.9243 \mu\text{A}$$

EX10.2

$$I_{REF} = \frac{V^+ - V_{BE(on)} - V^-}{R_1} = \frac{5 - 0.7 - (-5)}{12}$$

$$I_{REF} = 0.775 \text{ mA}$$

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}} = \frac{0.775}{1 + \frac{2}{75}} = 0.7549 \text{ mA}$$

$$\Delta I_O = (0.02)(0.7549) = 0.0151 \text{ mA and } \Delta I_O = \frac{1}{r_0} \Delta V_{CE2} \Rightarrow r_0 = \frac{\Delta V_{CE2}}{\Delta I_O}$$

$$r_0 = \frac{4}{0.0151} = 265 \text{ k}\Omega = \frac{V_A}{I_O} \Rightarrow V_A = (265)(0.7549) \Rightarrow \underline{V_A \cong 200 \text{ V}}$$

EX10.3

$$I_{REF} = \frac{3 - 0.6 - 0.7 - (-3)}{30} = 0.15667 \text{ mA}$$

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta(1 + \beta_3)}} = \frac{0.15667}{1 + \frac{2}{(120)(81)}} = 0.15663 \text{ mA}$$

$$I_{C1} = I_{C2} = I_O$$

$$I_{B1} = I_{B2} = \frac{I_O}{\beta} \Rightarrow 1.3053 \mu\text{A}$$

$$I_{E3} = I_{B1} + I_{B2} = 2.6106 \mu\text{A}$$

$$I_{B3} = \frac{I_{E3}}{1 + \beta_3} = 0.03223 \mu\text{A}$$

EX10.4

$$R_1 = \frac{V^+ - V_{BE1} - V^-}{I_{REF}} = \frac{3 - 0.6 - (-3)}{0.10} = 54 \text{ k}\Omega$$

$$R_E = \frac{V_T}{I_O} \ln\left(\frac{I_{REF}}{I_O}\right) = \frac{0.026}{0.02} \ln\left(\frac{100}{20}\right) = 2.09 \text{ k}\Omega$$

$$V_{BE2} = V_{BE1} - I_O R_E = 0.6 - (0.02)(2.09) = 0.558 \text{ V}$$

EX10.5

$$(a) \quad V_{BE1} = V_T \ln\left(\frac{I_{REF}}{I_{S1}}\right) = (0.026) \ln\left(\frac{120 \times 10^{-6}}{2 \times 10^{-16}}\right) = 0.7051 \text{ V}$$

$$(b) \quad V_{BE2} = V_T \ln\left(\frac{I_O}{I_{S2}}\right) = (0.026) \ln\left(\frac{50 \times 10^{-6}}{2 \times 10^{-16}}\right) = 0.6824 \text{ V}$$

$$R_E = \frac{V_T}{I_O} \ln\left(\frac{I_{REF}}{I_O}\right) = \frac{0.026}{0.05} \ln\left(\frac{120}{50}\right) \Rightarrow 455 \Omega$$

$$(c) \quad I_O R_E = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

$$I_O(0.7) = (0.026) \ln\left(\frac{0.120}{I_O}\right)$$

By trial and error, $I_O = 40.4 \mu\text{A}$

Now,

$$V_{BE2} = V_{BE1} - I_O R_E = 0.7051 - (0.0404)(0.7) = 0.6768 \text{ V}$$

EX10.6

$$I_0 R_E = V_T \ln\left(\frac{I_{REF}}{I_0}\right)$$

$$R_E = \frac{0.026}{0.025} \ln\left(\frac{0.70}{0.025}\right) \Rightarrow R_E = 3.465 \text{ k}\Omega$$

$$g_{m2} = \frac{I_0}{V_T} = \frac{0.025}{0.026} \Rightarrow g_{m2} = 0.9615 \text{ mA/V}$$

$$r_{\pi 2} = \frac{\beta V_T}{I_0} = \frac{(150)(0.026)}{0.025} = 156 \text{ k}\Omega$$

$$r_{02} = \frac{V_A}{I_0} = \frac{100}{0.025} = 4000 \text{ k}\Omega$$

$$R'_E = R_E \parallel r_{\pi 2} = 3.47 \parallel 156 = 3.39 \text{ k}\Omega$$

$$R_0 = r_{02} (1 + g_{m2} R'_E) = 4000 [1 + (0.962)(3.39)]$$

$$R_0 = 17.04 \text{ M}\Omega$$

$$dI_0 = \frac{1}{R_0} \cdot dV_{C2} = \frac{3}{17,040} \Rightarrow dI_0 = 0.176 \mu\text{A}$$

EX10.7

$$I_{REF} = I_R + I_{BR} + I_{B1} + \dots + I_{BN}$$

$$I_R = I_{01} = I_{02} = \dots = I_{0N} \text{ and } I_{BR} = I_{B1} = I_{B2} = \dots = I_{BN} = \frac{I_{01}}{\beta}$$

$$I_{REF} = I_{01} + (N+1) \left(\frac{I_{01}}{\beta} \right) = I_{01} \left(1 + \frac{N+1}{\beta} \right)$$

$$\text{So } I_{01} = I_{02} = \dots = I_{0N} = \frac{I_{REF}}{1 + \frac{N+1}{\beta}}$$

$$\frac{I_{01}}{I_{REF}} = 0.90 = \frac{1}{1 + \frac{N+1}{50}}$$

$$1 + \frac{N+1}{50} = \frac{1}{0.9}$$

$$N+1 = \left(\frac{1}{0.9} - 1 \right) (50)$$

$$N = \left(\frac{1}{0.9} - 1 \right) (50) - 1$$

$$N = 4.55 \Rightarrow \underline{N = 4}$$

EX10.8

$$V_{DS2}(sat) = 0.4 = V_{GS2} - 0.4 \Rightarrow V_{GS2} = 0.8 \text{ V}$$

$$I_O = \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right)_2 (V_{GS2} - V_{TN})^2$$

$$0.1 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right)_2 (0.8 - 0.4)^2 \Rightarrow \left(\frac{W}{L} \right)_2 = 12.5$$

$$I_{REF} = 0.5 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right)_1 (0.8 - 0.4)^2 \Rightarrow \left(\frac{W}{L} \right)_1 = 62.5$$

$$V_{GS3} = (V^+ - V^-) - V_{GS1} = 1.8 - (-1.8) - 0.8 = 2.8 \text{ V}$$

$$I_{REF} = 0.5 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right)_3 (2.8 - 0.4)^2 \Rightarrow \left(\frac{W}{L} \right)_3 = 1.74$$

EX10.9

$$I_{REF} = K_n (V_{GS} - V_{TN})^2$$

$$0.020 = 0.080 (V_{GS} - 1)^2$$

a. $\underline{V_{GS} = 1.5 \text{ V}}$ all transistors

b.

$$V_{G4} = V_{GS3} + V_{GS1} + V^- = 1.5 + 1.5 - 5 = -2 \text{ V}$$

$$V_{S4} = V_{G4} - V_{GS4} = -2 - 1.5 = -3.5 \text{ V}$$

$$V_{D4}(\min) = V_{S4} + V_{DS4}(sat) \text{ and } V_{DS4}(sat) = V_{GS4} - V_{TN} = 1.5 - 1 = 0.5 \text{ V}$$

$$\text{So } V_{D4}(\min) = -3.5 + 0.5 \Rightarrow \underline{V_{D4}(\min) = -3.0 \text{ V}}$$

c.

$$R_0 = r_{04} + r_{02}(1 + g_m r_{04})$$

$$r_{02} = r_{04} = \frac{1}{\lambda I_0} = \frac{1}{(0.02)(0.020)} = 2500 \text{ k}\Omega$$

$$g_m = 2K_n(V_{GS} - V_{TN}) = 2(0.080)(1.5 - 1) \Rightarrow g_m = 0.080 \text{ mA/V}$$

$$R_0 = 2500 + 2500(1 + (0.080)(2500)) \Rightarrow R_0 = 505 \text{ M}\Omega$$

EX10.10

For Q_2 : $v_{DS}(\min) = |V_P| = 2 \text{ V} \Rightarrow V_S(\min) = v_{DS}(\min) - 5 = 2 - 5 \Rightarrow V_S(\min) = -3 \text{ V}$

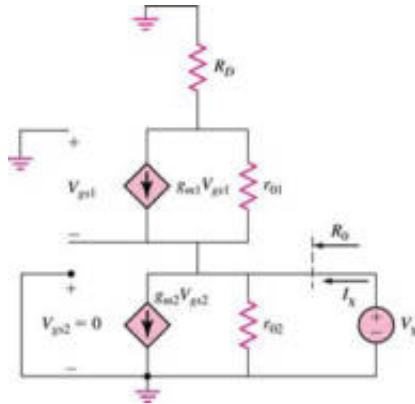
$$I_0 = I_{DSS2}(1 + \lambda v_{DS2}) = 0.5(1 + (0.15)(2)) \Rightarrow I_0 = 0.65 \text{ mA}$$

$$I_0 = I_{DSS1} \left(1 - \frac{v_{GS1}}{V_{P1}} \right)^2$$

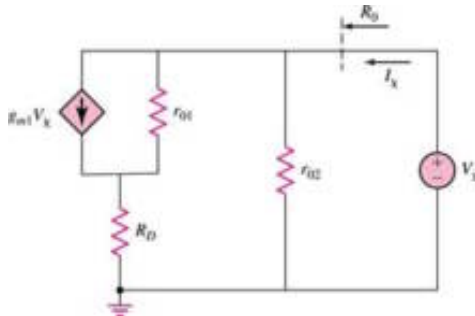
$$0.65 = 0.80 \left(1 - \frac{v_{GS1}}{-2} \right)^2$$

$$\frac{v_{GS1}}{-2} = 0.0986 \Rightarrow v_{GS1} = -0.197 \text{ V}$$

$$v_{GS1} = V_I - V_S - 0.197 = V_I - (-3) \Rightarrow V_I(\min) = -3.2 \text{ V}$$



$$V_{gs2} = 0, V_{gs1} = -V_X$$



$$I_X = \frac{V_X}{r_{02}} + \frac{V_X - V_I}{r_{01}} + g_{m1}V_X \quad (1)$$

$$\frac{V_1}{R_D} + \frac{V_1 - V_X}{r_{o1}} = g_{m1} V_X \quad (2)$$

$$V_1 = \frac{V_X \left(\frac{1}{r_{o1}} + g_{m1} \right)}{\frac{1}{R_D} + \frac{1}{r_{o1}}}$$

$$\frac{I_X}{V_X} = \frac{1}{R_0} = \frac{1}{r_{o2}} + \frac{1}{r_{o1}} + g_{m1} - \frac{\frac{1}{r_{o1}} \left(\frac{1}{r_{o1}} + g_{m1} \right)}{\frac{1}{R_D} + \frac{1}{r_{o1}}}$$

$$= \frac{1}{r_{o2}} + \left(\frac{1}{r_{o1}} + g_{m1} \right) \left[1 - \frac{\frac{1}{r_{o1}}}{\frac{1}{R_D} + \frac{1}{r_{o1}}} \right]$$

$$= \frac{1}{r_{o2}} + \left(\frac{1}{r_{o1}} + g_{m1} \right) \left(\frac{\frac{1}{R_D}}{\frac{1}{R_D} + \frac{1}{r_{o1}}} \right)$$

$$\text{For } R_D \ll r_{o1} \Rightarrow \frac{1}{R_0} \cong \frac{1}{r_{o2}} + \left(\frac{1}{r_{o1}} + g_{m1} \right)$$

For Q_1 :

$$g_{m1} = \frac{2I_{DSS1}}{|V_P|} \left(1 - \frac{V_{GS1}}{V_P} \right) = \frac{2(0.8)}{2} \left(1 - \frac{-0.197}{-2} \right)$$

$$g_{m1} = 0.721 \text{ mA/V}$$

$$r_0 = \frac{1}{\lambda I_0} = \frac{1}{(0.15)(0.65)} = 10.3 \text{ k}\Omega$$

$$\frac{1}{R_0} = \frac{1}{10.3} + \frac{1}{10.3} + 0.721 = 0.915 \Rightarrow \underline{R_0 = 1.09 \text{ k}\Omega}$$

EX10.11

$$I_{REF} = I_S \exp \left(\frac{V_{EB2}}{V_T} \right)$$

$$\text{a. } V_{EB2} = V_T \ln \left(\frac{I_{REF}}{I_S} \right) = (0.026) \ln \left(\frac{0.5 \times 10^{-3}}{10^{-12}} \right) \Rightarrow \underline{V_{EB2} = 0.521 \text{ V}}$$

$$\text{b. } R_1 = \frac{5 - 0.521}{0.5} \Rightarrow \underline{R_1 = 8.96 \text{ k}\Omega}$$

(c) Combining Equations (10.79), (10.80), and (10.81), we find

$$I_{S0} \left[\exp \left(\frac{V_I}{V_T} \right) \right] \left(1 + \frac{V_{CE0}}{V_{AN}} \right) = I_{REF} \times \frac{\left(1 + \frac{V_{EC2}}{V_{AP}} \right)}{\left(1 + \frac{V_{EB2}}{V_{AP}} \right)}$$

$$10^{-12} \left[\exp \left(\frac{V_I}{V_T} \right) \right] \left(1 + \frac{2.5}{100} \right) = (0.5 \times 10^{-3}) \frac{\left(1 + \frac{2.5}{100} \right)}{\left(1 + \frac{0.521}{100} \right)}$$

$$1.025 \times 10^{-12} \exp \left(\frac{V_I}{V_T} \right) = 5.098 \times 10^{-4} \exp \left(\frac{V_I}{V_T} \right) = 4.974 \times 10^8 \Rightarrow \underline{V_I = 0.521 \text{ V}}$$

$$A_v = \frac{-\left(\frac{1}{V_T} \right)}{\frac{1}{V_{AN}} + \frac{1}{V_{AP}}} = \frac{-\frac{1}{0.026}}{\frac{1}{100} + \frac{1}{100}} = \frac{-38.46}{0.01 + 0.01} \Rightarrow \underline{A_v = -1923}$$

d.

EX10.12

(a) $A_v = -g_{mo} (r_{on} \parallel r_{op})$

$$r_{on} = \frac{V_{AN}}{I_{Co}} = \frac{100}{0.25} = 400 \text{ k}\Omega$$

$$r_{op} = \frac{V_{AP}}{I_{Co}} = \frac{60}{0.25} = 240 \text{ k}\Omega$$

$$g_{mo} = \frac{0.25}{0.026} = 9.615 \text{ mA/V}$$

$$A_v = -(9.615)(400 \parallel 240) = -1442$$

(b) $A_v = -(0.6)(1442) = -865 = -g_{mo} (r_{on} \parallel r_{op} \parallel R_L)$

$$-865 = -(9.615)(150 \parallel R_L) \Rightarrow R_L = 225 \text{ k}\Omega$$

EX10.13

(a) $I_{REF} = I_O = K_n (V_{IQ} - V_{TN})^2$

$$0.20 = 0.10 (V_{IQ} - 0.5)^2 \Rightarrow V_{IQ} = 1.914 \text{ V}$$

(b) $A_v = -g_{mo} (r_{on} \parallel r_{op})$

$$g_{mo} = 2\sqrt{K_n I_Q} = 2\sqrt{(0.1)(0.2)} = 0.2828 \text{ mA/V}$$

$$r_{on} = r_{op} = \frac{1}{\lambda I_Q} = \frac{1}{(0.015)(0.2)} = 333 \text{ k}\Omega$$

$$A_v = -(0.2828)(333 \parallel 333) = -47.1$$

(c) $A_v = -(0.5)(47.1) = -23.55 = -g_{mo} (r_{on} \parallel r_{op} \parallel R_L)$

$$23.55 = (0.2828)(333 \parallel 333 \parallel R_L) \Rightarrow R_L = 166.5 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU10.1

$$I_{REF} = \left(1 + \frac{2}{\beta}\right) I_O = \left(1 + \frac{2}{120}\right) (0.20) = 0.2033 \text{ mA}$$

$$R_1 = \frac{2.5 - 0.7 - (-2.5)}{0.2033} = 21.15 \text{ k}\Omega$$

TYU10.2

Neglecting base currents

$$V_{BE1} = (0.026) \ln \left(\frac{150 \times 10^{-6}}{8 \times 10^{-15}} \right) = 0.6150 \text{ V}$$

$$I_O = (5 \times 10^{-15}) \exp \left(\frac{0.6150}{0.026} \right) \Rightarrow 93.75 \mu\text{A}$$

TYU10.3

$$I_0 = I_{REF} \cdot \frac{1}{\left(1 + \frac{2}{\beta(1+\beta)}\right)} = \frac{0.50}{\left(1 + \frac{2}{50(51)}\right)} \Rightarrow \underline{I_0 = 0.4996 \text{ mA}}$$

$$I_{B3} = \frac{I_0}{\beta} \Rightarrow \underline{I_{B3} = 9.99 \mu\text{A}}$$

$$I_{E3} = \left(\frac{1+\beta}{\beta}\right) I_{C3} = \underline{I_{E3} = 0.5096 \text{ mA}}$$

$$I_{C2} = \frac{I_{E3}}{\left(1 + \frac{2}{\beta}\right)} = \frac{0.5096}{\left(1 + \frac{2}{50}\right)} \Rightarrow \underline{I_{C2} = 0.490 \text{ mA} = I_{C1}}$$

$$I_{B1} = I_{B2} = \frac{I_{C2}}{\beta} \Rightarrow \underline{I_{B1} = I_{B2} = 9.80 \mu\text{A}}$$

TYU10.4

For circuit - Figure 10.2(b)

(a) $I_O \cong I_{REF} = 1 \text{ mA}$

(b) $R_o = \frac{V_A}{I_O} = \frac{50}{1} = 50 \text{ k}\Omega$

(c) $dI_O = \frac{\Delta V_{C2}}{R_o} = \frac{3}{50} = 0.06 \text{ mA}$

$$\frac{dI_O}{I_O} = \frac{0.06}{1} \Rightarrow 6\%$$

From circuit - Figure 10.9

$$(a) \quad I_O R_E = V_T \ln \left(\frac{I_{REF}}{I_O} \right)$$

$$I_O(2) = (0.026) \ln \left(\frac{1}{I_O} \right)$$

By trial and error, $I_O = 41.4 \mu A$

$$(b) \quad r_{o2} = \frac{50}{0.0414} \Rightarrow 1.208 \text{ M}\Omega$$

$$g_{m2} = \frac{0.0414}{0.026} = 1.5923 \text{ mA/V} ; \quad r_{\pi 2} = \frac{(200)(0.026)}{0.0414} = 125.6 \text{ k}\Omega$$

$$R_E \| r_{\pi 2} = 2 \| 125.6 = 1.969 \text{ k}\Omega$$

$$R_o = (1.208) [1 + (1.5923)(1.969)] \Rightarrow 5 \text{ M}\Omega$$

$$(c) \quad dI_O = \frac{3}{5} = 0.6 \mu A$$

$$\frac{dI_O}{I_O} = \frac{0.6}{41.4} \Rightarrow 1.45\%$$

TYU10.5

$$(a) \quad I_{REF} = \left(\frac{k'_{n1}}{2} \right) \left(\frac{W}{L} \right)_1 (V_{GS1} - V_{TN1})^2 = \left(\frac{k'_{n3}}{2} \right) \left(\frac{W}{L} \right)_3 (V_{GS3} - V_{TN3})^2$$

$$V_{GS3} = V^+ - V_{GS1} = 2.5 - V_{GS1}$$

Then

$$\left(\frac{100}{2} \right) (12.5)(V_{GS1} - 0.38)^2 = \left(\frac{95}{2} \right) (1.18)(2.5 - V_{GS1} - 0.42)^2$$

$$\text{We find } 25(V_{GS1} - 0.38) = (7.4867)(2.08 - V_{GS1})$$

$$\text{Or } V_{GS1} = V_{GS2} = 0.7718 \text{ V}$$

$$I_{REF} = \left(\frac{100}{2} \right) (12.5)(0.7718 - 0.38)^2 = 95.93 \mu A$$

$$I_O = \left(\frac{k'_{n2}}{2} \right) \left(\frac{W}{L} \right)_2 (V_{GS2} - V_{TN2})^2 = \left(\frac{105}{2} \right) (7.5)(0.7718 - 0.40)^2$$

or

$$I_O = 54.43 \mu A$$

$$(b) \quad \frac{\Delta I_{REF}}{I_{REF}} = \frac{95.93 - 100}{100} \times 100\% = -4.07\%$$

$$\frac{\Delta I_O}{I_O} = \frac{54.43 - 60}{60} \times 100\% = -9.28\%$$

TYU10.6

$$(a) \quad I_{REF} = \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right)_1 (V_{GS1} - V_{TN})^2 = \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right)_3 (V_{GS3} - V_{TN})^2$$

$$V_{GS3} = (V^+ - V^-) - V_{GS1} = 6 - V_{GS1}$$

then

$$\sqrt{12}(V_{GS1} - 0.5) = \sqrt{3}(6 - V_{GS1} - 0.5) \Rightarrow V_{GS1} = 2.167 \text{ V}$$

$$I_{REF} = \left(\frac{80}{2}\right)(12)(2.167 - 0.5)^2 \Rightarrow 1.33 \text{ mA}$$

$$\begin{aligned} \text{(b)} \quad I_O &= \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_2 (V_{GS2} - V_{TN})^2 [1 + \lambda V_{DS}] \\ &= \left(\frac{0.08}{2}\right)(6)(2.167 - 0.5)^2 [1 + (0.02)(2)] = 0.6936 \text{ mA} \end{aligned}$$

$$\text{(c)} \quad I_O = \left(\frac{0.08}{2}\right)(6)(2.167 - 0.5)^2 [1 + (0.02)(4)] = 0.7203 \text{ mA}$$

TYU10.7

$$I_{REF} = K_{n1}(V_{GS1} - V_{TN})^2 = K_{n3}(V_{GS3} - V_{TN})^2$$

$$\text{We have} \quad V_{GS3} = (V^+ - V^-) - V_{GS1} = 6 - V_{GS1}$$

Then

$$\sqrt{0.35}(V_{GS1} - 0.7) = \sqrt{0.10}(6 - V_{GS1} - 0.7)$$

which yields

$$V_{GS1} = 2.302 \text{ V}$$

then

$$I_{REF} = 0.35(2.302 - 0.7)^2 = 0.8986 \text{ mA}$$

$$I_O = 3(0.30)(2.302 - 0.7)^2 = 2.31 \text{ mA}$$

TYU10.8

All transistors are identical

$$\Rightarrow I_0 = I_{REF} = 250 \text{ } \mu\text{A}$$

$$I_{REF} = K_n (V_{GS} - V_{TN})^2$$

$$0.25 = 0.20(V_{GS} - 1)^2 \Rightarrow V_{GS} = 2.12 \text{ V}$$

TYU10.9

$$\text{a.} \quad V_{EB2} = (0.026) \ln \left(\frac{0.1 \times 10^{-3}}{5 \times 10^{-14}} \right) \Rightarrow V_{EB2} = 0.557 \text{ V}$$

$$\text{b.} \quad R_1 = \frac{5 - 0.557}{0.1} \Rightarrow R_1 = 44.4 \text{ k}\Omega$$

c.

$$I_{S0} \left[\exp \left(\frac{V_I}{V_T} \right) \right] \left(1 + \frac{V_{CE0}}{V_{AN}} \right) = I_{REF} \times \left(\frac{1 + \frac{V_{EC2}}{V_{AP}}}{1 + \frac{V_{EB2}}{V_{AP}}} \right)$$

$$5 \times 10^{-14} \left[\exp \left(\frac{V_I}{V_T} \right) \right] \left(1 + \frac{2.5}{100} \right) = (0.1 \times 10^{-3}) \left(\frac{1 + \frac{2.5}{100}}{1 + \frac{0.557}{100}} \right)$$

$$(5.125 \times 10^{-14}) \exp \left(\frac{V_I}{V_T} \right) = 1.019 \times 10^{-4}$$

$$\exp \left(\frac{V_I}{V_T} \right) = 1.988 \times 10^9 \Rightarrow \underline{V_I = 0.557 \text{ V}}$$

$$A_v = \frac{-\frac{1}{\frac{0.026}{1} + \frac{1}{100}}}{\frac{1}{100} + \frac{1}{100}} \Rightarrow \underline{A_v = -1923}$$

d.

TYU10.10

(a) $I_{REF} = K_p (V_{SG} + V_{TP})^2$
 $0.15 = 0.12(V_{SG} - 0.7)^2 \Rightarrow V_{SG} = 1.818 \text{ V}$

(b) $V_O = \frac{[1 + \lambda_p(V^+ - V_{SG})]}{\lambda_n + \lambda_p} - \frac{K_n(V_I - V_{TN})^2}{I_{REF}(\lambda_n + \lambda_p)}$
 Set $V_O = 2.5 \text{ V}$
 $2.5 = \frac{[1 + (0.02)(5 - 1.818)]}{0.04} - \frac{0.12(V_I - 0.7)^2}{0.15(0.04)} \Rightarrow V_I = 1.798 \text{ V}$

(c) $A_v = \frac{-2K_n(V_I - V_{TN})}{I_{REF}(\lambda_n + \lambda_p)} = \frac{-2(0.12)(1.798 - 0.7)}{(0.15)(0.04)} = -43.9$

TYU10.11

(a) $I_{REF} = 80 = 50(V_{SG} - 0.7)^2 \Rightarrow V_{SG} = 1.965 \text{ V}$

(b) $V_O = \frac{[1 + \lambda_p(V^+ - V_{SG})]}{\lambda_n + \lambda} - \frac{K_n(V_I - V_{TN})^2}{I_{REF}(\lambda_n + \lambda_p)}$
 $2.5 = \frac{[1 + (0.02)(5 - 1.965)]}{0.04} - \frac{(0.05)(V_I - 0.7)^2}{(0.08)(0.04)} \Rightarrow V_I = 1.940 \text{ V}$

(c) $A_v = \frac{-2K_n(V_I - V_{TN})}{I_{REF}(\lambda_n + \lambda_p)} = \frac{-2(0.05)(1.940 - 0.7)}{(0.08)(0.04)} = -38.74$

TYU10.12

a.

$$g_m = \frac{I_{C0}}{V_T} = \frac{0.5}{0.026} \Rightarrow \underline{g_m = 19.2 \text{ mA/V}}$$

$$r_0 = \frac{V_{AN}}{I_{CQ}} = \frac{120}{0.5} \Rightarrow \underline{r_0 = 240 \text{ k}\Omega}$$

$$r_{02} = \frac{V_{AP}}{I_{CQ}} = \frac{80}{0.5} \Rightarrow \underline{r_{02} = 160 \text{ k}\Omega}$$

b. $A_v = -g_m (r_0 \parallel r_{02} \parallel R_L) = -(19.2)[240 \parallel 160 \parallel 50] \Rightarrow \underline{A_v = -631}$

TYU10.13

$$I_C = 1\text{mA}, \quad g_m = \frac{1}{0.026} = 38.46 \text{ mA/V}$$

$$r_{\pi 1} = r_{\pi 2} = \frac{(100)(0.026)}{1} = 2.6 \text{ K}$$

$$r_{O1} = r_{O2} = \frac{80}{1} = 80 \text{ K}$$

$$r_O = \frac{120}{1} = 120 \text{ K}$$

$$R_{O1} = 2.6 \parallel \frac{1}{38.46} \parallel 80 = 0.0257 \text{ K}$$

For $R_1 = 9.3 \text{ K}$

$$R' = R_1 \parallel (R_{O1} + R_E) = 9.3 \parallel (0.0257 + 1) = 0.924 \text{ K}$$

$$R_E'' = 1 \parallel [2.6 + 0.924] = 0.779 \text{ K}$$

$$R_{O2} = 80 \parallel [1 + (38.46)(0.779)] = 2476.7 \text{ K}$$

$$A_v = -g_m (r_O \parallel R_{O2}) = -(38.46)(120 \parallel 2476.7) = -(38.46)(114.5)$$

$$A_v = -4404$$

For $R_L = 100 \text{ K}$

$$A_v = -38.46 [114.5 \parallel 100] = -2053$$

For $R_L = 10 \text{ K}$

$$A_v = -38.46 [114.5 \parallel 10] = -354$$

TYU10.14

M_1 and M_2 identical $\Rightarrow I_o = I_{REF}$

a.

$$I_o = K_n (V_I - V_{TN})^2$$

$$0.25 = 0.2 (V_I - 1)^2$$

$$V_I = 2.12 \text{ V}$$

$$g_m = 2K_n (V_I - V_{TN}) = 2(0.2)(2.12 - 1) \Rightarrow \underline{g_m = 0.447 \text{ mA/V}}$$

$$r_{on} = \frac{1}{\lambda_n I_o} = \frac{1}{(0.01)(0.25)} \Rightarrow \underline{r_{on} = 400 \text{ k}\Omega}$$

$$r_{op} = \frac{1}{\lambda_p I_o} = \frac{1}{(0.02)(0.25)} \Rightarrow \underline{r_{op} = 200 \text{ k}\Omega}$$

$$A_v = -g_m (r_o \parallel r_{o2} \parallel R_L)$$

b.
$$A_v = -(0.447)[400 \parallel 200 \parallel 100] \Rightarrow \underline{A_v = -25.5}$$

Chapter 11

Exercise Solutions

EX11.1

- (a) $v_E = 0 - 0.7 = -0.7 \text{ V}$
 $v_{C1} = v_{C2} = 5 - (0.15)(20) = 2 \text{ V}$
 $v_{CE1} = v_{CE2} = 2 - (-0.7) = 2.7 \text{ V}$
- (b) $v_E = -1 - 0.7 = -1.7 \text{ V}$
 $v_{C1} = v_{C2} = 2 \text{ V}$
 $v_{CE1} = v_{CE2} = 3.7 \text{ V}$
- (c) $v_E = +1 - 0.7 = +0.3 \text{ V}$
 $v_{C1} = v_{C2} = 2 \text{ V}$
 $v_{CE1} = v_{CE2} = 2 - 0.3 = 1.7 \text{ V}$

EX11.2

- (a) $\frac{i_{C1}}{I_Q} = 0.25 = \frac{1}{1 + \exp\left(\frac{-v_d}{V_T}\right)}$
- We find, $\exp\left(\frac{-v_d}{V_T}\right) = 3 \Rightarrow -v_d = (0.026)\ln(3)$
- Or $v_d = -28.56 \text{ mV}$
- (b) $\frac{i_{C2}}{I_Q} = \frac{1}{1 + \exp\left(\frac{+v_d}{V_T}\right)} = 0.9$
- We find, $\exp\left(\frac{v_d}{V_T}\right) = 0.1111 \Rightarrow v_d = (0.026)\ln(0.1111)$
- Or $v_d = -57.13 \text{ mV}$

EX11.3

- (a) $\text{CMRR}_{dB} = 75 \text{ dB} \Rightarrow \text{CMRR} = 5623.4$
- $$5623.4 = \frac{1}{2} \left[1 + \frac{(101)(0.8)R_o}{(0.026)(100)} \right] \Rightarrow R_o = 362 \text{ k}\Omega$$
- (b) $\text{CMRR}_{dB} = 95 \text{ dB} \Rightarrow \text{CMRR} = 56,234$
- $$56,234 = \frac{1}{2} \left[1 + \frac{(101)(0.8)R_o}{(0.026)(100)} \right] \Rightarrow R_o = 3.62 \text{ M}\Omega$$

EX11.4

$$(a) \quad v_{C1} = -g_m \cdot \frac{v_d}{2} \cdot R_{C1}$$

$$\frac{v_{C1}}{v_d} = -150 = -\frac{g_m R_{C1}}{2}$$

$$\text{We find } g_m = \frac{0.1}{0.026} = 3.846 \text{ mA/V}$$

$$\text{Then } R_{C1} = \frac{2(150)}{3.846} = 78.0 \text{ k}\Omega$$

$$v_{C2} = +g_m \cdot \frac{v_d}{2} \cdot R_{C2}$$

$$\frac{v_{C2}}{v_d} = 100 = \frac{g_m R_{C2}}{2}$$

$$\text{Then } R_{C2} = \frac{2(100)}{3.846} = 52.0 \text{ k}\Omega$$

$$(b) \quad \text{For } V_{CB} = 0, \quad v_{C1} = v_{C2} = 1.5 \text{ V for } v_{cm} = 1.5 \text{ V}$$

$$\text{Then } 1.5 = V^+ - I_{CQ} R_C = V^+ - (0.1)(78) \Rightarrow V^+ = +9.3 \text{ V}$$

$$\text{So } V^+ = -V^- = 9.3 \text{ V}$$

EX11.5

$$(a) \quad v_o = A_d v_d + A_{cm} v_{cm}$$

$$v_d = v_1 - v_2 = -20 \mu\text{V}$$

$$v_{cm} = \frac{v_1 + v_2}{2} = 0$$

$$\text{Then } v_o = (150)(-20) \Rightarrow v_o = -3 \text{ mV}$$

$$(b) \quad v_d = v_1 - v_2 = -20 \mu\text{V}$$

$$v_{cm} = \frac{v_1 + v_2}{2} = 200 \mu\text{V}$$

$$\text{Now } \text{CMRR}_{dB} = 50 \text{ dB} \Rightarrow \text{CMRR} = 316.2 = \left| \frac{A_d}{A_{cm}} \right| = \frac{150}{A_{cm}} \Rightarrow A_{cm} = 0.474$$

$$\text{Then } v_o = A_d v_d + A_{cm} v_{cm} = (150)(-20) + (0.474)(200)$$

$$\text{Or } v_o = -2.905 \text{ mV}$$

EX11.6

$$R_{id} = 2[r_\pi + (1 + \beta)R_E]$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{(100)(0.026)}{0.25} = 10.4 \text{ K}$$

$$R_{id} = 2[10.4 + (101)(0.5)] = 122 \text{ K}$$

EX11.7

$$A_d = \frac{g_m R_C}{2(1 + g_m R_E)}$$

$$10 = \frac{(9.62)(10)}{2[1 + (9.62)R_E]}$$

$$1 + (9.62)R_E = 4.81 \Rightarrow R_E = 0.396 \text{ K}$$

$$R_{id} = 2[r_\pi + (1 + \beta)R_E] = 2[10.4 + (101)(0.396)]$$

$$R_{id} = 100.8 \text{ K}$$

EX11.8

$$I_1 = \frac{10 - V_{GS4}}{R_1} = K_{n3} (V_{GS4} - V_{TN})^2$$

$$10 - V_{GS4} = (0.1)(80)(V_{GS4} - 0.8)^2$$

$$10 - V_{GS4} = 8(V_{GS4}^2 - 1.6V_{GS4} + 0.64)$$

$$8V_{GS4}^2 - 11.8V_{GS4} - 4.88 = 0$$

$$V_{GS4} = \frac{11.8 \pm \sqrt{(11.8)^2 + 4(8)(4.88)}}{2(8)} = 1.81 \text{ V}$$

$$I_1 = I_Q = \frac{10 - 1.81}{80} = 0.102 \text{ mA}$$

$$I_{D1} = I_{D2} = \frac{0.102}{2} = 0.0512 \text{ mA}$$

$$= K_{n1} (V_{GS1} - V_{TN})^2$$

$$0.0512 = 0.050(V_{GS1} - 0.8)^2 \Rightarrow V_{GS1} = 1.81 \text{ V}$$

$$v_{01} = v_{02} = 5 - (0.0512)(40) = 2.95 \text{ V}$$

$$\begin{aligned} \text{Max } v_{cm} : V_{DS1}(\text{sat}) &= V_{GS1} - V_{TN} \\ &= 1.81 - 0.8 = 1.01 \text{ V} \end{aligned}$$

$$\begin{aligned} v_{cm}(\text{max}) &= v_{01} - V_{DS1}(\text{sat}) + V_{GS1} \\ &= 2.95 - 1.01 + 1.81 \\ v_{cm}(\text{max}) &= 3.75 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Min } v_{cm} : V_{DS4}(\text{sat}) &= V_{GS4} - V_{TN} \\ &= 1.81 - 0.8 = 1.01 \text{ V} \end{aligned}$$

$$\begin{aligned} v_{cm}(\text{min}) &= V_{GS1} + V_{DS4}(\text{sat}) - 5 \\ &= 1.81 + 1.01 - 5 \\ v_{cm}(\text{min}) &= -2.18 \text{ V} \end{aligned}$$

$$-2.18 \leq v_{cm} \leq 3.75 \text{ V}$$

EX11.9

$$(a) \quad A_d = \sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)\left(\frac{I_Q}{2}\right)} \cdot R_D$$

$$15 = \sqrt{\left(\frac{0.1}{2}\right)\left(\frac{W}{L}\right)\left(\frac{0.2}{2}\right)} \cdot (15) \Rightarrow \left(\frac{W}{L}\right) = 200$$

$$(b) \quad g_f(\text{max}) = \sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)\left(\frac{I_Q}{2}\right)} = \sqrt{\left(\frac{0.1}{2}\right)(200)\left(\frac{0.2}{2}\right)} = 1 \text{ mA/V}$$

EX11.10

$$(a) \quad r_{o2} = \frac{V_{A2}}{I_{CQ}} = \frac{150}{0.2} = 750 \text{ k}\Omega$$

$$r_{o4} = \frac{V_{A4}}{I_{CQ}} = \frac{90}{0.2} = 450 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.2}{0.026} = 7.692 \text{ mA/V}$$

$$A_d = g_m(r_{o2} \parallel r_{o4}) = (7.692)(750 \parallel 450) = 2163$$

$$(b) \quad A_d = g_m(r_{o2} \parallel r_{o4} \parallel R_L) = (7.692)(750 \parallel 450 \parallel 250) = 1018$$

$$(c) \quad R_{id} = 2r_\pi, \quad r_\pi = \frac{(120)(0.026)}{0.2} = 15.6 \text{ k}\Omega$$

$$R_{id} = 31.2 \text{ k}\Omega$$

$$(d) \quad R_o = r_{o2} \parallel r_{o4} = 750 \parallel 450 = 281 \text{ k}\Omega$$

EX11.11

We have $\left(\frac{W}{L}\right)_4 = 10, \left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_6 = 0.33$

$$I_{REF} = \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_4 (V_{GS4} - V_{TN})^2 = \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_5 (V_{GS5} - V_{TN})^2$$

$$(10)(V_{GS4} - 0.5)^2 = (0.33)(V_{GS5} - 0.5)^2$$

$$V_{GS5} = \frac{6 - V_{GS4}}{2}$$

Then $(5.505)(V_{GS4} - 0.5) = (V_{GS5} - 0.5) = \frac{6 - V_{GS4}}{2} - 0.5$

Which yields $V_{GS4} = 0.8747 \text{ V}$

Then $I_{REF} = I_Q = \left(\frac{80}{2}\right)(10)(0.8747 - 0.5)^2 = 56.16 \mu\text{A}$

$$A_d = 2\sqrt{2\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_n\left(\frac{1}{I_Q}\right)} \cdot \frac{1}{\lambda_n + \lambda_p} = 2\sqrt{2\left(\frac{80}{2}\right)(10)\left(\frac{1}{56.16}\right)} \cdot \frac{1}{0.02 + 0.02}$$

$$A_d = 188.7$$

EX11.12

$$A_d = g_m (r_{o2} \parallel R_o)$$

$$400 = g_m (500 \parallel 101000) \Rightarrow g_m = 0.8 \text{ mA/V}$$

$$g_m = 2\sqrt{K_n I_{DQ}}$$

$$0.8 = 2\sqrt{\left(\frac{0.08}{2}\right)\left(\frac{W}{L}\right)_1 (0.01)} \Rightarrow \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 40$$

EX11.13

$$I_{e6} = (1 + \beta) I_{b6} = I_{b7}$$

$$I_{c7} = \beta I_{b7} = \beta (1 + \beta) I_{b6}$$

$$\frac{I_{c7}}{I_{b6}} = \beta (1 + \beta) = (100)(101) = 1.01 \times 10^4$$

EX11.14

$$I_{CQB} = 0.5 \text{ mA}, \quad I_{BQB} = \frac{0.5}{180} = 0.00278 \text{ mA} = I_{EQA}$$

$$I_{CQA} = 0.002747 \text{ mA}$$

$$r_{\pi A} = \frac{(90)(0.026)}{0.002747} = 851.8 \text{ k}\Omega$$

$$R_{oA} = \frac{r_{\pi A} + 300}{91} = \frac{851.8 + 300}{91} = 12.66 \text{ k}\Omega$$

$$r_{\pi B} = \frac{(180)(0.026)}{0.5} = 9.36 \text{ k}\Omega$$

$$R_o = 10 \parallel \frac{r_{\pi B} + R_{oA}}{1 + \beta_B} = 10 \parallel \left(\frac{9.36 + 12.66}{181} \right) = 10 \parallel 0.1217$$

or

$$R_o = 120 \Omega$$

EX11.15

$$I_1 = \frac{10 - 0.7 - (-10)}{R_1} = 0.6 \Rightarrow \underline{R_1 = 32.2 \text{ K}}$$

$$I_Q R_2 = V_T \ln \left(\frac{I_1}{I_Q} \right)$$

$$(0.2) R_2 = (0.026) \ln \left(\frac{0.6}{0.2} \right) \Rightarrow \underline{R_2 = 143 \Omega}$$

$$I_{R6} = I_1 \Rightarrow \underline{R_3 = 0}$$

$$10 = I_{C1} R_C + V_{CE1} - 0.7$$

$$10.7 = (0.1) R_C + 4 \Rightarrow \underline{R_C = 67 \text{ K}}$$

$$v_{o2} = -0.7 + 4 = 3.3 \text{ V}$$

$$v_{E4} = 3.3 - 1.4 = 1.9 \text{ V}$$

$$I_{R4} = \frac{v_{E4}}{R_4} \Rightarrow R_4 = \frac{1.9}{0.6} \Rightarrow \underline{R_4 = 3.17 \text{ K}}$$

$$v_{C3} = v_{O2} - 1.4 + v_{CE4} \\ = 3.3 - 1.4 + 3 = 4.9 \text{ V}$$

$$I_{R5} = I_{R4} = 0.6 = \frac{10 - 4.9}{R_5} \Rightarrow \underline{R_5 = 8.5 \text{ K}}$$

$$v_{E5} = 4.9 - 0.7 = 4.2 \text{ V} \Rightarrow R_6 = \frac{4.2 - 0.7}{0.6} = \underline{5.83 \text{ K}}$$

$$R_7 = \frac{0 - (-10)}{5} \Rightarrow \underline{R_7 = 2 \text{ K}}$$

EX11.16

$$R_{i2} = r_{\pi3} + (1 + \beta)r_{\pi4}$$

$$r_{\pi4} = \frac{(100)(0.026)}{0.6} = 4.333 \text{ K}$$

$$r_{\pi3} \approx \frac{\beta^2 V_T}{I_{R4}} = \frac{(100)^2 (0.026)}{0.6} = 433.3 \text{ K}$$

$$R_{i2} = 433.3 + (101)(4.333) \Rightarrow \underline{R_{i2} = 871 \text{ K}}$$

$$R_{i3} = r_{\pi5} + (1 + \beta)[R_6 + r_{\pi6} + (1 + \beta)R_7]$$

$$r_{\pi5} = \frac{(100)(0.026)}{0.6} = 4.333 \text{ K}$$

$$r_{\pi6} = \frac{(100)(0.026)}{5} = 0.52 \text{ K}$$

$$R_{i3} = 4.333 + (101)[5.83 + 0.52 + (101)(2)]$$

$$\underline{R_{i3} = 21.0 \text{ M}\Omega}$$

$$A_{d1} = \frac{g_m}{2}(R_C \parallel R_{i2})g_m = \frac{0.1}{0.026} = 3.846 \text{ mA/V}$$

$$A_{d1} = \frac{3.846}{2}(67 \parallel 871) = 119.64$$

$$A_2 = \left(\frac{I_{R4}}{2V_T} \right) (R_5 \parallel R_{i3}) = \frac{0.6}{2(0.026)} (8.5 \parallel 21000) = 98.037$$

$$A = A_{d1} \cdot A_2 = (119.64)(98.037) = 11,729$$

EX11.17

$$f_z = \frac{1}{2\pi R_o C_o} = \frac{1}{2\pi (10 \times 10^6)(0.2 \times 10^{-12})}$$

$$f_z = 79.6 \text{ kHz}$$

$$f_p = \frac{1}{2\pi R_{eq} C_o}$$

From Example 11.17, $R_{eq} = 51.98 \text{ }\Omega$

$$f_p = \frac{1}{2\pi (51.98)(0.2 \times 10^{-12})}$$

$$f_p = 15.3 \text{ GHz}$$

Test Your Understanding Solutions

TYU11.1

(a) $v_d = v_1 - v_2 = 2.10 - 2.12 = -0.02 \text{ V}$

$$v_{cm} = \frac{v_1 + v_2}{2} = \frac{2.10 + 2.12}{2} = 2.110 \text{ V}$$

 (b) $v_d = v_1 - v_2 = 0.25 - 0.002 \sin \omega t - [0.5 + 0.002 \sin \omega t]$

$$= -0.25 - 0.004 \sin \omega t \text{ (V)}$$

$$v_{cm} = \frac{0.25 - 0.002 \sin \omega t + 0.5 + 0.002 \sin \omega t}{2} = 0.375 \text{ V}$$

TYU11.2

(a) Need $v_{C1} = v_{C2} = 3 = 5 - (0.2)R_C \Rightarrow R_C = 10 \text{ k}\Omega$
 (b) $g_m = \frac{0.2}{0.026} = 7.692 \text{ mA/V}$
 $A_d = g_m R_C = (7.692)(10) = 76.9$

TYU11.3

(a) $v_o = A_d v_d + A_{cm} v_{cm}$
 $v_d = v_1 - v_2 = 0.995 \sin \omega t - 1.005 \sin \omega t = -0.01 \sin \omega t \text{ (V)}$

$$v_{cm} = \frac{v_1 + v_2}{2} = \frac{0.995 \sin \omega t + 1.005 \sin \omega t}{2} = 1.0 \sin \omega t \text{ (V)}$$

 Then

$$v_o = (80)(-0.01 \sin \omega t) + (-0.20)(1.0 \sin \omega t) = -1.0 \sin \omega t \text{ (V)}$$

 (b) $v_d = v_1 - v_2 = -0.01 \sin \omega t \text{ (V)}$
 $v_{cm} = 2 \text{ V}$
 Then

$$v_o = (80)(-0.01 \sin \omega t) + (-0.20)(2) = -0.4 - 0.8 \sin \omega t \text{ (V)}$$

TYU11.4

From Equation (11.41)

$$CMRR = \frac{g_m R_o}{\left(\frac{\Delta R_C}{R_C} \right)}$$

For $CMRR|_{dB} = 75 \text{ dB} \Rightarrow CMRR = 5623.4$

$$5623.4 = \frac{(3.86)(100)}{\Delta R_C} \cdot (10)$$

Then

Or $\Delta R_C = 0.686 \text{ K}$

TYU11.5

From Equation (11.49)

$$CMRR = \frac{1 + 2R_O g_m}{2 \left(\frac{\Delta g_m}{g_m} \right)}$$

For $CMRR|_{dB} = 90 \text{ dB} \Rightarrow CMRR = 31622.8$

Then $31622.8 = \left(\frac{1 + 2(100)(3.86)}{2\Delta g_m} \right) (3.86)$

Or $\Delta g_m = 0.0472 \text{ mA/V}$ or $\frac{\Delta g_m}{g_m} = \frac{0.0472}{3.86} = 0.0122 \Rightarrow 1.22\%$

TYU11.6

(a) $I_{EQ} = 0.2 \text{ mA}$, $I_{B1} = I_{B2} = \frac{0.2}{151} \Rightarrow 1.32 \mu\text{A}$

(b) $R_{id} = 2r_\pi = \frac{2(150)(0.026)}{0.2} = 39 \text{ k}\Omega$

$I_b = \frac{10 \sin \omega t (mV)}{39 \text{ k}\Omega} \Rightarrow I_b = 0.256 \sin \omega t (\mu\text{A})$

(c) $R_{icm} = \frac{1}{2} [r_\pi + (1 + \beta)(2R_o)]$; $r_\pi = \frac{39}{2} = 19.5 \text{ k}\Omega$

$R_{icm} = \frac{1}{2} [19.5 + (151)(2)(100)] \Rightarrow 15.11 \text{ M}\Omega$

$i_{cm} = \frac{1}{2} \cdot \frac{v_{cm}}{R_{icm}} = \frac{1}{2} \cdot \frac{3 \sin \omega t}{15.11} \Rightarrow 0.0993 \sin \omega t (\mu\text{A})$

TYU11.7

(a) $A_d = \sqrt{\left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right) \left(\frac{I_Q}{2} \right)} \cdot R_D$

$12 = \sqrt{\left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right) \left(\frac{0.4}{2} \right)} \cdot (7.5) \Rightarrow \left(\frac{W}{L} \right) = 256$

(b) $i_{D2} = \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right) (v_{GS2} - V_{TN})^2$

$0.2 = \left(\frac{0.1}{2} \right) (256) (v_{GS2} - 0.5)^2 \Rightarrow v_{GS2} = 0.625 \text{ V}$

$v_{DS2}(sat) = 0.625 - 0.5 = 0.125 \text{ V}$

$v_O = V^+ - i_{D2} R_D = 3 - (0.2)(7.5) = 1.5 \text{ V}$

$v_{CM}(\max) = v_O - v_{DS2}(sat) + v_{GS2} = 1.5 - 0.125 + 0.625$

or

$v_{CM}(\max) = 2 \text{ V}$

TYU11.8

From Example 11-8, $I_Q = 0.587 \text{ mA}$

$$A_d = \sqrt{\frac{K_n I_Q}{2}} \cdot R_D = \sqrt{\frac{(0.1)(0.587)}{2}} \cdot (16) \Rightarrow A_d = 2.74$$

$$\text{For } M_4, R_0 = \frac{1}{\lambda_4 I_Q} = \frac{1}{(0.02)(0.587)} \Rightarrow R_0 = 85.2 \text{ k}\Omega$$

$$g_m = 2K_n (V_{GS2} - V_{TN}) = 2(0.1)(2.71 - 1) = 0.342 \text{ mA/V}$$

$$A_{cm} = \frac{-g_m R_D}{1 + 2g_m R_0} = \frac{-(0.342)(16)}{1 + 2(0.342)(85.2)} \Rightarrow A_{cm} = -0.0923$$

$$CMRR_{dB} = 20 \log_{10} \left(\frac{2.74}{0.0923} \right) \Rightarrow CMRR_{dB} = 29.4 \text{ dB}$$

TYU11.9

$$(a) \text{ CMRR} = \frac{1}{2} \left[1 + 2 \sqrt{2 \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right) (I_Q) \cdot R_o} \right] = \frac{1}{2} \left[1 + 2 \sqrt{2 \left(\frac{0.1}{2} \right) (10)(0.1) \cdot (1000)} \right]$$

Or

$$\text{CMRR} = 316.73 \Rightarrow \text{CMRR}_{dB} = 50 \text{ dB}$$

$$(b) \text{ CMRR}_{dB} = 80 \text{ dB} \Rightarrow \text{CMRR} = 10^4$$

Then

$$10^4 = \frac{1}{2} \left[1 + 2 \sqrt{2 \left(\frac{0.1}{2} \right) (10)(0.1) \cdot R_o} \right] \Rightarrow R_o = 31.6 \text{ M}\Omega$$

TYU11.10

$$R_o = r_{o4} + r_{o2} (1 + g_{m4} r_{o4})$$

Assume $I_{REF} = I_O = 100 \mu\text{A}$ and $\lambda = 0.01 \text{ V}^{-1}$

$$r_{o2} = r_{o4} = \frac{1}{\lambda I_D} = \frac{1}{(0.01)(0.1)} \Rightarrow 1 \text{ M}\Omega$$

Let $K_n (\text{all devices}) = 0.1 \text{ mA/V}^2$

$$\text{Then } g_{m4} = 2\sqrt{K_n I_D} = 2\sqrt{(0.1)(0.1)} = 0.2 \text{ mA/V}$$

$$R_o = 1000 + 1000(1 + (0.2)(1000)) \Rightarrow 202 \text{ M}\Omega$$

$$\text{Now } V_{GS1} = V_{GS2} = \sqrt{\frac{I_D}{K_n}} + V_{TN} = \sqrt{\frac{0.05}{0.1}} + 1 = 1.707 \text{ V}$$

$$V_{DS1} (\text{sat}) = V_{GS1} - V_{TN} = 1.707 - 1 = 0.707 \text{ V}$$

$$\text{So } v_{o1} (\text{min}) = +4 - V_{GS1} + V_{DS1} (\text{sat}) = 4 - 1.707 + 0.707$$

$$v_{o1} (\text{min}) = 3 \text{ V} = 10 - I_D R_D = 10 - (0.05) R_D \Rightarrow R_D = 140 \text{ k}\Omega$$

For a one-sided output, the differential gain is:

$$A_d = \frac{1}{2} g_{m1} R_D \text{ where } g_{m1} = 2\sqrt{K_n I_{D1}}$$

$$= 2\sqrt{(0.1)(0.05)} = 0.1414 \text{ mA/V}$$

$$A_d = \frac{1}{2} (0.1414)(140) \Rightarrow A_d = 9.90$$

The common-mode gain is:

$$A_{cm} = \frac{\sqrt{2K_n I_{Q1}} \cdot R_D}{1 + 2\sqrt{2K_n I_{Q1}} \cdot R_o} = \frac{\sqrt{2(0.1)(0.1)} \cdot (140)}{1 + 2\sqrt{2(0.1)(0.1)} \cdot (202000)} \Rightarrow A_{cm} = 0.0003465$$

$$CMRR_{dB} = 20 \log_{10} \left| \frac{A_d}{A_{cm}} \right| \Rightarrow CMRR_{dB} = 89.1 \text{ dB}$$

Then

TYU11.11

a. $I_{B5} = \frac{I_Q}{\beta(1+\beta)} = \frac{0.5}{(180)(181)} \Rightarrow 15.3 \text{ nA}$

So $I_0 = 15.3 \text{ nA}$

b. For a balanced condition

$$V_{EC4} = V_{EC3} = V_{EB3} + V_{EB5} \Rightarrow V_{EC4} = 1.4 \text{ V}$$

$$V_{CE2} = V_{C2} - V_{E2} = (10 - 1.4) - (-0.7) \Rightarrow V_{CE2} = 9.3 \text{ V}$$

TYU11.12

$$r_{o2} = \frac{V_{AN}}{I_{CQ}} = \frac{120}{0.05} = 2400 \text{ k}\Omega$$

$$r_{o4} = \frac{V_{AP}}{I_{CQ}} = \frac{80}{0.05} = 1600 \text{ k}\Omega$$

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.05}{0.026} = 1.923 \text{ mA/V}$$

$$A_d = g_m (r_{o2} \parallel r_{o4}) = (1.923)(2400 \parallel 1600) = 1846$$

TYU11.13

$$P = (I_Q + I_{REF})(5 - (-5))$$

$$10 = (0.1 + I_{REF})(10) \Rightarrow I_{REF} = 0.9 \text{ mA}$$

$$R_1 = \frac{5 - 0.7 - (-5)}{I_{REF}} = \frac{9.3}{0.9} \Rightarrow R_1 = 10.3 \text{ k}\Omega$$

$$I_Q R_E = V_T \ln \left(\frac{I_{REF}}{I_Q} \right)$$

$$R_E = \frac{0.026}{0.1} \ln \left(\frac{0.9}{0.1} \right) \Rightarrow R_E = 0.571 \text{ k}\Omega$$

$$r_{o2} = \frac{V_{A2}}{I_{C2}} = \frac{120}{0.05} \Rightarrow 2.4 \text{ M}\Omega$$

$$r_{o4} = \frac{V_{A4}}{I_{C4}} = \frac{80}{0.05} \Rightarrow 1.6 \text{ M}\Omega$$

$$g_m = \frac{0.05}{0.026} = 1.923 \text{ mA/V}$$

$$A_d = g_m (r_{o2} \parallel r_{o4} \parallel R_L) = (1.923)(2400 \parallel 1600 \parallel 90) \Rightarrow A_d = 158$$

TYU11.14

(a) $R_o = r_{o2} \parallel r_{o4}$

$$r_{o2} = \frac{V_{AN}}{I_{CQ}} = \frac{120}{0.05} \Rightarrow 2.4 \text{ M}\Omega$$

$$r_{o4} = \frac{V_{AP}}{I_{CQ}} = \frac{80}{0.05} \Rightarrow 1.6 \text{ M}\Omega$$

$$R_o = 2.4 \parallel 1.6 = 0.96 \text{ M}\Omega$$

(b) $R_L = R_o = 0.96 \text{ M}\Omega$

TYU11.15

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.18)(0.1)} = 0.2683 \text{ mA/V}$$

$$r_{o2} = \frac{1}{\lambda_n I_{DQ}} = \frac{1}{(0.015)(0.1)} = 666.7 \text{ k}\Omega$$

$$r_{o4} = \frac{1}{\lambda_p I_{DQ}} = \frac{1}{(0.025)(0.1)} = 400 \text{ k}\Omega$$

$$A_d = g_m (r_{o2} \parallel r_{o4}) = (0.2683)(666.7 \parallel 400) = 67.1$$

TYU11.16

$$I_{E2} = 75 \mu\text{A}, I_{B2} = 0.497 \mu\text{A}, I_{C2} = 74.50 \mu\text{A}$$

$$I_{D1} = 25 + 0.497 = 25.497 \mu\text{A}$$

$$g_{m1} = 2\sqrt{K_n I_{D1}} = 2\sqrt{(0.05)(0.025497)} \Rightarrow 71.4 (\mu\text{A/V})$$

$$g_{m2} = \frac{I_{C2}}{V_T} = \frac{0.0745}{0.026} = 2.865 \text{ mA/V}$$

$$r_{\pi2} = \frac{(150)(0.026)}{0.0745} = 52.3 \text{ k}\Omega$$

Then

$$g_m^C = \frac{g_{m1}(1 + g_{m2}r_{\pi2})}{1 + g_{m1}r_{\pi2}} = \frac{(0.0714)[1 + (2.865)(52.3)]}{1 + (0.0714)(52.3)} = 2.275 \text{ mA/V}$$

TYU11.17

From Figure 11.41

$$r_{o4} = \frac{80}{0.5} = 160 \text{ k}\Omega$$

$$R_0 \cong \beta r_{o4} = (150)(160) \text{ k}\Omega \Rightarrow R_0 = 24 \text{ M}\Omega$$

From Figure 11.42

$$r_{o6} = \frac{1}{\lambda I_D} = \frac{1}{(0.0125)(0.5)} \Rightarrow r_{o6} = 160 \text{ k}\Omega$$

$$0.5 = 0.5(V_{GS} - 1)^2 \Rightarrow V_{GS} = 2 \text{ V}$$

$$g_{m6} = 2K_n(V_{GS} - V_{TN}) = 2(0.5)(2 - 1) = 1 \text{ mA/V}$$

$$r_{o4} = 160 \text{ k}\Omega$$

$$R_0 = (g_{m6})(r_{o6})(\beta r_{o4}) = (1)(160)(150)(160) \Rightarrow R_0 = 3.840 \text{ M}\Omega$$

TYU11.18

From Equation (11.126)

$$R_i = \frac{2(1 + \beta)\beta V_T}{I_Q} = \frac{2(121)(120)(0.026)}{0.5} \Rightarrow R_i = 1.51 \text{ M}\Omega$$

$$r_{\pi 11} = \frac{\beta V_T}{I_Q} = \frac{(120)(0.026)}{0.5} = 6.24 \text{ k}\Omega$$

$$R'_E = r_{\pi 11} \parallel R_3 = 6.24 \parallel 0.1 = 0.0984 \text{ k}\Omega$$

$$g_{m11} = \frac{I_Q}{V_T} = \frac{0.5}{0.026} = 19.23 \text{ mA/V}$$

$$r_{o11} = \frac{V_A}{I_Q} = \frac{120}{0.5} = 240 \text{ k}\Omega$$

$$\begin{aligned} \text{Then } R_{C11} &= r_{o11}(1 + g_{m11}R'_E) \\ &= 240[1 + (19.23)(0.0984)] \\ &= 694 \text{ k}\Omega \end{aligned}$$

$$r_{\pi 8} = \frac{\beta V_T}{I_{C8}} = \frac{(120)(0.026)}{2} = 1.56 \text{ k}\Omega$$

$$R_{b8} = r_{\pi 8} + (1 + \beta)R_4 = 1.56 + (121)(5) = 607 \text{ k}\Omega$$

$$\text{Then } R_{L7} = R_{C11} \parallel R_{b8} = 694 \parallel 607 = 324 \text{ k}\Omega$$

$$\text{Then } A_v = \left(\frac{I_Q}{2V_T} \right) R_{L7} = \left[\frac{0.5}{2(0.026)} \right] (324) \Rightarrow A_v = 3115$$

$$R_o = R_4 \parallel \left(\frac{r_{\pi 8} + Z}{1 + \beta} \right)$$

$$Z = R_{C11} \parallel R_{C7}$$

$$R_{C7} = \frac{V_A}{I_Q} = \frac{120}{0.5} = 240 \text{ k}\Omega$$

$$Z = 694 \parallel 240 = 178 \text{ k}\Omega$$

$$R_o = 5 \parallel \left(\frac{1.56 + 178}{121} \right) = 5 \parallel 1.48 \Rightarrow R_o = 1.14 \text{ k}\Omega$$

TYU11.19

$$A_v = \left(\frac{I_Q}{2V_T} \right) R_{L7}$$

$$10^3 = \left(\frac{0.5}{2(0.026)} \right) R_{L7} \Rightarrow \underline{R_{L7} = 104 \text{ k}\Omega}$$

Chapter 12

Exercise Solutions

EX12.1

$$(a) (i) A_f = \frac{A}{1 + A\beta} \Rightarrow 50 = \frac{5 \times 10^4}{1 + (5 \times 10^4)\beta} \Rightarrow \beta = 0.01998$$

$$(ii) \frac{A_f}{1/\beta} = \frac{50}{1/0.01998} = 0.999$$

$$(b) (i) 20 = \frac{100}{1 + (100)\beta} \Rightarrow \beta = 0.04$$

$$(ii) \frac{A_f}{1/\beta} = \frac{20}{1/0.04} = 0.80$$

EX12.2

$$(a) A_f = \frac{A}{1 + A\beta} \Rightarrow 50 = \frac{5 \times 10^5}{1 + (5 \times 10^5)\beta} \Rightarrow \beta = 0.019998$$

$$A = (5 \times 10^5)(0.85) = 4.25 \times 10^5$$

Now

$$A_f = \frac{4.25 \times 10^5}{1 + (4.25 \times 10^5)(0.019998)} = 49.99912$$

$$\text{Percent change} = \frac{49.99912 - 50}{50} \times 100\% = -1.76 \times 10^{-3}\%$$

$$(b) 20 = \frac{100}{1 + (100)\beta} \Rightarrow \beta = 0.04$$

$$A = (100)(0.85) = 85$$

Now

$$A_f = \frac{85}{1 + (85)(0.04)} = 19.318$$

$$\text{Percent change} = \frac{19.318 - 20}{20} \times 100\% = -3.41\%$$

EX12.3

$$(a) (i) 80 = \frac{5 \times 10^4}{1 + (5 \times 10^4)\beta} \Rightarrow \beta = 0.01248$$

$$(ii) \omega_{\beta H} = \omega_H (1 + \beta A_o) = (2\pi)(5) [1 + (0.01248)(5 \times 10^4)] = (2\pi)(3.125 \times 10^3) \text{ rad/s}$$

$$(b) (i) A_f(0) = \frac{5 \times 10^4}{1 + (5 \times 10^4) \left(\frac{0.01248}{2} \right)} = 159.7$$

$$\text{Percent change} = 100\%$$

$$(ii) \omega_{fH} = (2\pi)(5) \left[1 + \left(\frac{0.01248}{2} \right) (5 \times 10^4) \right] = (2\pi)(1.565 \times 10^3) \text{ rad/s}$$

$$\text{Percent change} \cong -50\%$$

EX12.4

(a)

$$\begin{aligned} v_{OA} &= A_1 A_2 v_i + A_2 v_n \\ &= (100)(10)v_i + (10)v_n \\ \frac{S_o}{N_o} &= \frac{1000v_i}{10v_n} = 100 \frac{S_i}{N_i} \end{aligned}$$

(b)

$$\begin{aligned} v_{OC} &= \frac{A_1 A_2}{1 + \beta A_1 A_2} v_i + \frac{A_2}{1 + \beta A_1 A_2} v_n \\ &= \frac{10^5}{1 + (0.001)10^5} v_i + \frac{10}{1 + (0.001)(10^5)} v_n \\ \frac{S_o}{N_o} &= \frac{10^3 v_i}{0.1 v_n} = 10^4 \frac{S_i}{N_i} \end{aligned}$$

EX12.5

a. $V_\varepsilon = V_S - V_{fb} = 100 - 99 = 1 \text{ mV}$

$$V_0 = A_v V_\varepsilon \Rightarrow A_v = \frac{5}{0.001} \Rightarrow A_v = 5000 \text{ V/V}$$

$$V_{fb} = \beta V_0 \Rightarrow \beta = \frac{V_{fb}}{V_0} = \frac{0.099}{5} \Rightarrow \beta = 0.0198 \text{ V/V}$$

$$A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{5000}{1 + (0.0198)(5000)} \Rightarrow A_{vf} = 50 \text{ V/V}$$

$$R_{yf} = R_i (1 + \beta A_v) = (5) [1 + (0.0198)(5000)] \Rightarrow R_{yf} = 500 \text{ k}\Omega$$

b. $R_{0f} = \frac{R_0}{1 + \beta A_v} = \frac{4}{1 + (0.0198)(5000)} \Rightarrow R_{0f} \Rightarrow 40 \Omega$

EX12.6

a. $I_\varepsilon = I_S - I_{fb} = 100 - 99 = 1 \mu\text{A}$

$$A_i = \frac{I_0}{I_\varepsilon} = \frac{5}{0.001} \Rightarrow A_i = 5000 \text{ A/A}$$

$$\beta = \frac{I_{fb}}{I_0} = \frac{0.099}{5} \Rightarrow \beta = 0.0198 \text{ A/A}$$

$$A_{if} = \frac{A_i}{1 + \beta A_i} = \frac{5000}{1 + (5000)(0.0198)} \Rightarrow A_{if} = 50 \text{ A/A}$$

$$R_{yf} = \frac{R_i}{1 + \beta A_i} = \frac{5}{1 + (0.0198)(5000)} \Rightarrow R_{yf} \Rightarrow 50 \Omega$$

b. $R_{0f} = (1 + \beta A_i) R_0 = [1 + (0.0198)(5000)](4) \Rightarrow R_{0f} = 400 \text{ k}\Omega$

EX12.7

$$\begin{aligned} \text{(a)} \quad A_{v_f} &= \left(1 + \frac{R_2}{R_1}\right) = \left(1 + \frac{60}{15}\right) = 5 \\ V_o &= A_{v_f} \cdot V_i = (5)(0.1) = 0.5 \text{ V} \\ \text{(b)} \quad \text{(i)} \quad A_{v_f} &= \frac{A_v}{1 + \frac{A_v}{\left(1 + \frac{R_2}{R_1}\right)}} = \frac{5 \times 10^4}{1 + \frac{5 \times 10^4}{5}} = 4.9995 \\ V_o &= (4.9995)(0.1) = 0.49995 \text{ V} \\ \text{(ii)} \quad V_\epsilon &= \frac{V_o}{A_v} = \frac{0.49995}{5 \times 10^4} \Rightarrow V_\epsilon = 9.999 \mu\text{V} \\ \text{(c)} \quad \text{(i)} \quad A_{v_f} &= \frac{5 \times 10^5}{1 + \frac{5 \times 10^5}{5}} = 4.99995 \\ \text{(ii)} \quad V_\epsilon &= \frac{0.499995}{5 \times 10^5} \Rightarrow V_\epsilon = 0.99999 \mu\text{V} \end{aligned}$$

EX12.8

Use a non inverting op-amp.

$$1 + \frac{R_2}{R_1} = 15 \Rightarrow \frac{R_2}{R_1} = 14$$

$$R_2 = 140 \text{ K}$$

Let $R_1 = 10 \text{ K}$

$$\beta = \frac{1}{1 + \frac{R_2}{R_1}} = 0.066667$$

Input resistance.

$$R_{if} = 5(0.06667)(5 \times 10^3) \cong 1.67 \text{ M}\Omega$$

$$R_{of} = \frac{50}{(0.06667)(5 \times 10^3)} \cong 0.15 \Omega$$

EX12.9

$$i_o = \left(\frac{h_{FE}}{1 + h_{FE}} \right) \cdot \frac{R_E}{\left(R_E + \frac{r_\pi}{1 + h_{FE}} \right)} \cdot i_s$$

$$r_\pi = \frac{(80)(0.026)}{0.5} = 4.16 \text{ k}\Omega$$

Then

$$\frac{r_\pi}{1 + h_{FE}} = \frac{4.16}{81} = 0.0514 \text{ k}\Omega$$

Then we want

$$\frac{i_o}{i_i} = 0.95 = \left(\frac{80}{81} \right) \left(\frac{R_E}{R_E + 0.0514} \right)$$

or

$$\left(\frac{R_E}{R_E + 0.0514} \right) = 0.9619$$

which yields

$$R_E (\text{min}) = 1.30 \text{ k}\Omega$$

and

$$V^+ = I_E R_E + 0.7 = \left(\frac{81}{80} \right) (0.5)(1.3) + 0.7 \Rightarrow \underline{V^+ (\text{min}) = 1.36 \text{ V}}$$

EX12.10

Use the configuration shown in figure 12.20.

$$R_S = 500 \Omega, R_L = 200 \Omega$$

$$1 + \frac{R_F}{R_1} = 15$$

Let

$$R_1 = 2 \text{ K}$$

For example, let $R_F = 28 \text{ K}$

EX12.11

$$(a) (i) V_G = \left(\frac{20}{20 + 30} \right) (10) - 5 = -1 \text{ V}$$

$$V_G = V_{GS} + I_{DQ} R_S - 5 = V_{GS} + K_n R_S (V_{GS} - V_{TN})^2 - 5$$

$$4 = V_{GS} + (2)(0.4)(V_{GS}^2 - 4V_{GS} + 4)$$

$$0.8V_{GS}^2 - 2.2V_{GS} - 0.8 = 0 \Rightarrow V_{GS} = 3.075 \text{ V}$$

So

$$I_{DQ} = 2(3.075 - 2)^2 = 2.312 \text{ mA}$$

$$(ii) V_i = V_{gs} + g_m V_{gs} R_S \Rightarrow V_{gs} = \frac{V_i}{1 + g_m R_S}$$

$$I_o = - \left(\frac{R_D}{R_D + R_L} \right) g_m \cdot \frac{V_i}{1 + g_m R_S}$$

$$A_{gf} = \frac{I_o}{V_i} = - \left(\frac{g_m}{1 + g_m R_S} \right) \left(\frac{R_D}{R_D + R_L} \right)$$

Now

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(2)(2.312)} = 4.30 \text{ mA/V}$$

$$A_{gf} = - \left[\frac{4.30}{1 + (4.30)(0.4)} \right] \left(\frac{2}{2 + 2} \right) = -0.7904 \text{ mA/V}$$

$$\begin{aligned} \text{(b) (i)} \quad 4 &= V_{GS} + (1.8)(0.4)(V_{GS}^2 - 4V_{GS} + 4) \\ 0.72V_{GS}^2 - 1.88V_{GS} - 1.12 &= 0 \Rightarrow V_{GS} = 3.111 \text{ V} \\ \text{So} \\ I_{DQ} &= (1.8)(3.111 - 2)^2 = 2.222 \text{ mA} \\ \text{(ii)} \quad g_m &= 2\sqrt{(1.8)(2.222)} = 4.0 \text{ mA/V} \\ A_{gf} &= -\left[\frac{4.0}{1 + (4.0)(0.4)}\right]\left(\frac{2}{2 + 2}\right) = -0.7692 \text{ mA/V} \\ \text{Percent change} &= \frac{0.7692 - 0.7904}{0.7904} \times 100\% = -2.68\% \end{aligned}$$

EX12.12

Use the circuit with the configuration shown in Figure 12.27.

The LED replaces R_L .

$$A_{gf} = 10 \text{ mS} = 10 \times 10^{-3} = \frac{1}{R_E} \Rightarrow R_E = 100 \Omega$$

EX12.13

$$\begin{aligned} \text{(a) (i)} \quad V_{GS} &= \left(\frac{150}{150 + 350}\right)(5) = 1.50 \text{ V} \\ I_{DQ} &= K_n (V_{GS} - V_{TN})^2 = (1.5)(1.5 - 0.8)^2 = 0.735 \text{ mA} \\ g_m &= 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1.5)(0.735)} = 2.1 \text{ mA/V} \\ V_o &= -g_m V_{gs} R_D ; \quad V_{gs} = \left(\frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_S}\right) \cdot V_i = \left(\frac{105}{105 + 20}\right) \cdot V_i = 0.84V_i \\ A_v &= -(0.84)(2.1)(2) = -3.528 \\ \text{(ii)} \quad \frac{V_{gs} - V_o}{R_F} + \frac{V_{gs}}{R_1 \parallel R_2} + \frac{V_{gs} - V_i}{R_S} &= 0 \\ V_{gs} \left(\frac{1}{R_F} + \frac{1}{R_1 \parallel R_2} + \frac{1}{R_S} \right) &= \frac{V_o}{R_F} + \frac{V_i}{R_S} \\ V_{gs} \left(\frac{1}{47} + \frac{1}{105} + \frac{1}{20} \right) &= \frac{V_o}{47} + \frac{V_i}{20} \\ V_{gs} &= V_o(0.26337) + V_i(0.61881) \end{aligned}$$

Now

$$\begin{aligned} \frac{V_o}{R_D} + g_m V_{gs} + \frac{V_o - V_{gs}}{R_F} &= 0 \\ \frac{V_o}{2} + \frac{V_o}{47} + \left(g_m - \frac{1}{R_F} \right) [V_o(0.26337) + V_i(0.61881)] &= 0 \end{aligned}$$

or

$$V_o(1.0687) + V_i(1.2863) = 0$$

which yields

$$A_v = \frac{V_o}{V_i} = -1.204$$

(b) $K_n = 1.275 \text{ mA/V}^2$

(i) $I_{DQ} = (1.275)(1.5 - 0.8)^2 = 0.62475 \text{ mA}$
 $g_m = 2\sqrt{(1.275)(0.62475)} = 1.785 \text{ mA/V}$
 $A_v = -(0.84)(1.785)(2) = -2.9988$
 Percent change $= \frac{2.9988 - 3.528}{3.528} \times 100\% = -15\%$

(ii) $V_o(0.52128) + (1.7637)[V_o(0.26337) + V_i(0.61881)] = 0$
 $V_o(0.98579) + V_i(1.091395) = 0$
 $A_v = \frac{V_o}{V_i} = -1.107$
 Percent change $= \frac{1.107 - 1.204}{1.204} \times 100\% = -8.06\%$

EX12.14

From EX12.13; $I_{DQ} = 0.735 \text{ mA}$, $g_m = 2.1 \text{ mA/V}$

(a) $I_x = \frac{V_x}{R_D} + \frac{V_x}{r_o} + g_m V_{gs} + \frac{V_x}{R_F + R_1 \parallel R_2 \parallel R_S}$

We find

$$V_{gs} = \left(\frac{R_1 \parallel R_2 \parallel R_S}{R_1 \parallel R_2 \parallel R_S + R_F} \right) \cdot V_x$$

$$\frac{I_x}{V_x} = \frac{1}{R_{of}} = \frac{1}{R_D} + \frac{1}{r_o} + \frac{1 + g_m(R_1 \parallel R_2 \parallel R_S)}{R_F + R_1 \parallel R_2 \parallel R_S}$$

Now

$$R_1 \parallel R_2 \parallel R_S = 150 \parallel 350 \parallel 20 = 16.8 \text{ k}\Omega$$

For $\lambda = 0 \Rightarrow r_o = \infty$

Then

$$\frac{1}{R_{of}} = \frac{1}{2} + \frac{1 + (2.1)(16.8)}{47 + 16.8} = 0.5 + 0.56865$$

or

$$R_{of} = 0.9358 \text{ k}\Omega$$

(b) For $\lambda = 0.04 \Rightarrow r_o = \frac{1}{(0.04)(0.735)} = 34.01 \text{ k}\Omega$

Then

$$\frac{1}{R_{of}} = 0.5 + 0.0294 + 0.56865$$

or

$$R_{of} = 0.9107 \text{ k}\Omega$$

EX12.15

$$V_{TH} = \left(\frac{5.5}{5.5 + 51} \right) (10) = 0.9735 \text{ V}$$

$$R_{TH} = 5.5 \parallel 51 = 4.965 \text{ k}\Omega$$

$$I_{BQ} = \frac{0.973 - 0.7}{4.96 + (121)(1)} = 0.00217 \text{ mA}$$

$$I_{CQ} = 0.2605 \text{ mA}$$

$$r_{\pi} = 11.98 \text{ k}\Omega, g_m = 10.02 \text{ mA/V}$$

$$R_{eq} = R_S \parallel R_1 \parallel R_2 \parallel r_{\pi} = (10) \parallel 51 \parallel 5.5 \parallel 12 = 2.598 \text{ k}\Omega$$

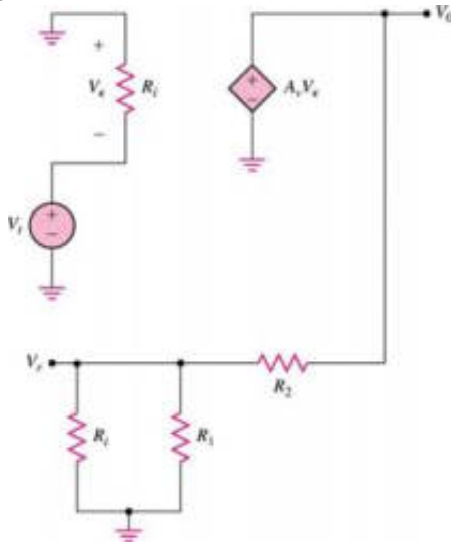
From Equation (12.99(b)):

$$T = (g_m R_C) \left(\frac{R_{eq}}{R_C + R_F + R_{eq}} \right)$$

$$= (10)(10) \left(\frac{2.598}{10 + 82 + 2.598} \right) \Rightarrow T = 2.75$$

EX12.16 Computer Analysis

EX12.17



$$V_e = -V_t, V_0 = A_v V_e = -A_v V_t$$

$$V_r = \left(\frac{R_1 \parallel R_i}{R_1 \parallel R_i + R_2} \right) V_0 = - \left(\frac{R_1 \parallel R_i}{R_1 \parallel R_i + R_2} \right) (A_v V_t)$$

$$T = - \frac{V_r}{V_t} = A_v \left(\frac{R_1 \parallel R_i}{R_1 \parallel R_i + R_2} \right)$$

or

$$T = \frac{A_v}{1 + \frac{R_2}{R_1 \parallel R_i}}$$

EX12.18

$$\phi = -\tan^{-1}\left(\frac{f}{10^3}\right) - 2 \tan^{-1}\left(\frac{f}{10^5}\right)$$

At $f_{180} \cong 10^5$ Hz, $\phi = -180^\circ$

$$|T(f_{180})| = 1 = \frac{\beta(3000)}{\sqrt{1 + \left(\frac{10^5}{10^3}\right)^2} \left[1 + \left(\frac{10^5}{10^5}\right)^2\right]}$$

or

$$1 = \frac{\beta(3000)}{(100)(2)} \Rightarrow \beta = 0.0667$$

EX12.19

$$A_f(0) = \frac{3000}{1 + \beta(3000)} = \frac{3000}{1 + (0.008)(3000)} = 120$$

$$|T| = 1 = \frac{(0.008)(3000)}{\sqrt{1 + \left(\frac{f}{10^3}\right)^2} \left[1 + \left(\frac{f}{10^5}\right)^2\right]}$$

By trial and error,

$$f \cong 2.28 \times 10^4 \text{ Hz}$$

$$\begin{aligned} \phi &= -\tan^{-1}\left(\frac{f}{10^3}\right) - 2 \tan^{-1}\left(\frac{f}{10^5}\right) \\ &= -\tan^{-1}\left(\frac{2.28 \times 10^4}{10^3}\right) - 2 \tan^{-1}\left(\frac{2.28 \times 10^4}{10^5}\right) = -87.49 - 2(12.84) = -113.18 \end{aligned}$$

Then

$$\text{Phase margin} = -113.18 - (-180) = 66.8^\circ$$

EX12.20

(a) $\phi = -180 = -\tan^{-1}\left(\frac{f_{180}}{10^3}\right) - 2 \tan^{-1}\left(\frac{f_{180}}{10^5}\right)$

$$f_{180} \cong 10^5 \text{ Hz}$$

$$|T(f_{180})| = \frac{250}{\sqrt{1 + \left(\frac{10^5}{10^3}\right)^2} \left[1 + \left(\frac{10^5}{10^5}\right)^2\right]} = \frac{250}{(100)(2)} = 1.25$$

$$|T| > 1 \text{ at } f_{180}$$

(b) $T = \frac{250}{\left(1 + j \frac{f}{f_{PD}}\right) \left(1 + j \frac{f}{10^3}\right) \left(1 + j \frac{f}{10^5}\right)^2}$

$$\phi = -120 = -\tan^{-1}\left(\frac{f}{f_{PD}}\right) - \tan^{-1}\left(\frac{f}{10^3}\right) - 2 \tan^{-1}\left(\frac{f}{10^5}\right)$$

Then

$$f \cong 0.577 \times 10^3 \text{ Hz}$$

Now

$$|T| = 1 = \frac{250}{\sqrt{1 + \left(\frac{0.577 \times 10^3}{f_{PD}}\right)^2} \sqrt{1 + \left(\frac{0.577 \times 10^3}{10^3}\right)^2}} \quad (1)$$

$$\left(\frac{0.577 \times 10^3}{f_{PD}}\right) = \frac{250}{1.155}$$

which yields

$$f_{PD} = 2.67 \text{ Hz}$$

EX12.21

$$A_f(0) = \frac{10^5}{1 + (0.025)(10^5)} = 40$$
$$f = (10) \left[1 + (0.025)(10^5) \right] \cong 25 \text{ kHz}$$

EX12.22

$$\phi = -135 = -\tan^{-1}\left(\frac{f_{135}}{f_{PD}}\right) - 2 \tan^{-1}\left(\frac{f_{135}}{10^5}\right) \Rightarrow f_{135} \cong 0.414 \times 10^5 \text{ Hz}$$

$$|T(f_{135})| = 1 = \frac{250}{\sqrt{1 + \left(\frac{0.414 \times 10^5}{f_{PD}}\right)^2} \left[1 + \left(\frac{0.414 \times 10^5}{10^5}\right)^2 \right]}$$

$$\frac{0.414 \times 10^5}{f_{PD}} = \frac{250}{1.171} \Rightarrow f_{PD} = 194 \text{ Hz}$$

Test Your Understanding Solutions

TYU12.1

$$(a) \quad A_f = \frac{A}{1 + A\beta}$$

$$50 = \frac{A}{1 + A(0.019)}$$

or

$$50 = A[1 - (50)(0.019)] \Rightarrow A = 10^3$$

$$(b) \quad A_f = \frac{5 \times 10^5}{1 + (5 \times 10^5)(0.019)} = 52.63$$

TYU12.2

$$\frac{dA_f}{A_f} = \frac{1}{(1 + \beta A)} \cdot \frac{dA}{A} = \left(\frac{A_f}{A} \right) \cdot \frac{dA}{A}$$

$$\frac{dA}{A} = \frac{dA_f}{A_f} \cdot \left(\frac{A}{A_f} \right) = (0.001) \left(\frac{5 \times 10^5}{100} \right) \Rightarrow \frac{dA}{A} = \pm 5\%$$

TYU12.3

$$(a) \quad (5 \times 10^5)(6) = (200 \times 10^3)A_f(0)$$

$$A_f(0) = 15$$

$$(b) \quad (5 \times 10^5)(6) = (100 \times 10^3)A_f(0)$$

$$A_f(0) = 30$$

TYU12.4

$$V_\varepsilon = V_S - V_{fb} = 100 - 99 = 1 \text{ mV}$$

$$A_g = \frac{I_0}{V_\varepsilon} = \frac{5 \text{ mA}}{1 \text{ mV}} \Rightarrow A_g = 5 \text{ A/V}$$

$$\beta = \frac{V_{fb}}{I_0} = \frac{99 \text{ mV}}{5 \text{ mA}} \Rightarrow \beta = 19.8 \text{ V/A}$$

$$A_{gf} = \frac{A_g}{1 + \beta A_g} = \frac{5}{1 + (19.8)(5)} \Rightarrow A_{gf} = 0.05 \text{ A/V} = 50 \text{ mA/V}$$

TYU12.5

$$I_\varepsilon = I_S - I_{fb} = 100 - 99 = 1 \text{ } \mu\text{A}$$

$$A_z = \frac{V_0}{I_\varepsilon} = \frac{5 \text{ V}}{1 \text{ } \mu\text{A}} \Rightarrow A_z = 5 \times 10^6 \text{ V/A}$$

$$\beta = \frac{I_{fb}}{V_0} = \frac{99 \text{ } \mu\text{A}}{5 \text{ V}} \Rightarrow \beta = 1.98 \times 10^{-5} \text{ A/V}$$

$$A_{zf} = \frac{A_z}{1 + \beta A_z} = \frac{5 \times 10^6}{1 + (1.98 \times 10^{-5})(5 \times 10^6)} \Rightarrow A_{zf} = 5 \times 10^4 \text{ V/A} = 50 \text{ V/mA}$$

TYU12.6

$$(a) \quad r_{\pi} = \frac{h_{FE} V_T}{I_{CQ}} = \frac{(120)(0.026)}{1.2} = 2.6 \text{ k}\Omega$$

$$A_{vf} = \frac{\left(\frac{1}{r_{\pi}} + g_m \right) R_E}{1 + \left(\frac{1}{r_{\pi}} + g_m \right) R_E} = \frac{(1 + h_{FE}) R_E}{r_{\pi} + (1 + h_{FE}) R_E} = \frac{(121)(1.5)}{2.6 + (121)(1.5)} = 0.985877$$

$$R_{if} = r_{\pi} + (1 + h_{FE}) R_E = 2.6 + (121)(1.5) = 184.1 \text{ k}\Omega$$

$$R_{of} = \frac{R_E}{1 + \left(\frac{1}{r_{\pi}} + g_m \right) R_E} = \frac{(R_E)(r_{\pi})}{r_{\pi} + (1 + h_{FE}) R_E} = \frac{(1.5)(2.6)}{2.6 + (121)(1.5)} \Rightarrow R_{of} = 21.18 \Omega$$

$$(b) \quad r_{\pi} = \frac{(180)(0.026)}{1.2} = 3.9 \text{ k}\Omega$$

$$A_{vf} = \frac{(181)(1.5)}{3.9 + (181)(1.5)} = 0.985839$$

$$\frac{\Delta A_{vf}}{A_{vf}} \times 100\% = -0.00385\%$$

$$R_{if} = 3.9 + (181)(1.5) = 275.4 \text{ k}\Omega$$

$$\frac{\Delta R_{if}}{R_{if}} \times 100\% = +49.6\%$$

$$R_{of} = \frac{(1.5)(3.9)}{3.9 + (181)(1.5)} \Rightarrow R_{of} = 21.24 \Omega$$

$$\frac{\Delta R_{of}}{R_{of}} \times 100\% = +0.283\%$$

TYU12.7

$$(a) \quad A_{vf} = \frac{g_m R_S}{1 + g_m R_S}, \quad g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.5)(0.25)} = 0.7071 \text{ mA/V}$$

$$A_{vf} = \frac{(0.7071)(3)}{1 + (0.7071)(3)} = 0.67962$$

$$R_{of} = \frac{R_S}{1 + g_m R_S} = \frac{3}{1 + (0.7071)(3)} = 0.96114 \text{ k}\Omega$$

$$(b) \quad g_m = 2\sqrt{(0.5)(1)} = 1.414 \text{ mA/V}$$

$$A_{vf} = \frac{(1.414)(3)}{1 + (1.414)(3)} = 0.80923$$

$$\frac{\Delta A_{vf}}{A_{vf}} \times 100\% = +19.1\%$$

$$R_{of} = \frac{3}{1 + (1.414)(3)} = 0.5723 \text{ k}\Omega$$

$$\frac{\Delta R_{of}}{R_{of}} \times 100\% = -40.5\%$$

TYU12.8 Computer Analysis

TYU12.9 Computer Analysis

TYU12.10

$$\begin{aligned} \text{(a) (i)} \quad A_{gf} &= \frac{(h_{FE} A_g)}{1 + (h_{FE} A_g) R_E} = \frac{(180)(10^2)}{1 + (180)(10^2)(1)} = 0.9999444 \text{ mA/V} \\ I_o &= A_{gf} \cdot V_i = (0.9999444)(1.5) = 1.4999166 \text{ mA} \\ \text{(ii)} \quad V_e &= V_i - I_o R_E = 1.5 - (1.4999166)(1) \Rightarrow V_e = 83.4 \mu\text{V} \\ \text{(b) (i)} \quad A_{gf} &= \frac{(144)(10^2)}{1 + (144)(10^2)(1)} = 0.9999306 \text{ mA/V} \\ I_o &= A_{gf} \cdot V_i = (0.9999306)(1.5) = 1.4998959 \text{ mA} \\ \frac{\Delta A_{gf}}{A_{gf}} \times 100\% &= \frac{0.9999306 - 0.9999444}{0.9999444} \times 100\% = -0.00138\% \\ \frac{\Delta I_o}{I_o} \times 100\% &= \frac{1.4998959 - 1.4999166}{1.4999166} \times 100\% = -0.00138\% \\ \text{(ii)} \quad V_e &= 1.5 - (1.4998959)(1) \Rightarrow V_e = 104 \mu\text{V} \end{aligned}$$

TYU12.11

$$\begin{aligned} \text{(a)} \quad R_{TH} &= R_1 \parallel R_2 = 51 \parallel 5.5 = 4.965 \text{ k}\Omega \\ V_{TH} &= \left(\frac{5.5}{5.5 + 51} \right) (10) = 0.9735 \text{ V} \\ I_{BQ} &= \frac{0.9735 - 0.7}{4.965 + (181)(0.5)} = 0.002865 \text{ mA} \\ I_{CQ} &= 0.5157 \text{ mA} \\ \text{Then} \quad g_m &= \frac{0.5157}{0.026} = 19.83 \text{ mA/V}, \quad r_\pi = \frac{(180)(0.026)}{0.5157} = 9.075 \text{ k}\Omega \\ \text{(i)} \quad V_o &= -g_m V_\pi R_C \\ V_\pi &= \left(\frac{R_1 \parallel R_2 \parallel r_\pi}{R_1 \parallel R_2 \parallel r_\pi + R_S} \right) \cdot V_i = \left[\frac{4.965 \parallel 9.075}{(4.965 \parallel 9.075) + 10} \right] \cdot V_i = 0.243 V_i \\ A_v &= \frac{V_o}{V_i} = -(0.243)(19.83)(10) = -48.187 \\ \text{(ii)} \quad (1) \quad &\frac{V_o}{R_C} + g_m V_\pi + \frac{V_o - V_\pi}{R_F} = 0 \\ (2) \quad &\frac{V_\pi - V_i}{R_S} + \frac{V_\pi}{R_{TH} \parallel r_\pi} + \frac{V_\pi - V_o}{R_F} = 0 \\ &V_\pi \left(\frac{1}{R_S} + \frac{1}{R_{TH} \parallel r_\pi} + \frac{1}{R_F} \right) = \frac{V_o}{R_F} + \frac{V_i}{R_S} \end{aligned}$$

$$V_{\pi} \left(\frac{1}{10} + \frac{1}{4.965 \parallel 9.075} + \frac{1}{60} \right) = \frac{V_o}{60} + \frac{V_i}{10}$$

$$V_{\pi} = V_o(0.038913) + V_i(0.233487)$$

Now

$$(1) \quad V_o \left(\frac{1}{R_C} + \frac{1}{R_F} \right) + V_{\pi} \left(g_m - \frac{1}{R_F} \right) = 0$$

$$V_o \left(\frac{1}{10} + \frac{1}{60} \right) + \left(19.83 - \frac{1}{60} \right) [V_o(0.038913) + V_i(0.233487)] = 0$$

$$V_o(0.887662) + V_i(4.62616) = 0$$

Then

$$A_v = \frac{V_o}{V_i} = -5.2116$$

$$(b) \quad (i) \quad I_{BQ} = \frac{0.9735 - 0.7}{4.965 + (121)(0.5)} = 0.004178 \text{ mA}$$

$$I_{CQ} = 0.5013 \text{ mA}$$

$$g_m = \frac{0.5013}{0.026} = 19.28 \text{ mA/V}, \quad r_{\pi} = 6.224 \text{ k}\Omega$$

$$R_{TH} \parallel r_{\pi} = 4.965 \parallel 6.224 = 2.7618 \text{ k}\Omega$$

$$V_{\pi} = \left(\frac{2.7618}{2.7618 + 10} \right) \cdot V_i = 0.2164 V_i$$

$$A_v = -g_m V_{\pi} R_C = -(19.28)(0.2164)(10) = -41.722$$

$$(ii) \quad V_{\pi} \left(\frac{1}{10} + \frac{1}{2.7618} + \frac{1}{60} \right) = \frac{V_o}{60} + \frac{V_i}{10}$$

$$V_{\pi} = V_o(0.034811) + V_i(0.208877)$$

Then

$$V_o(0.116667) + (19.26334)[V_o(0.034811) + V_i(0.208877)] = 0$$

$$V_o(0.787242) + V_i(4.023669) = 0$$

Then

$$A_v = \frac{V_o}{V_i} = -5.1111$$

$$(c) \quad (i) \quad \frac{41.722 - 48.187}{48.187} \times 100\% = -13.4\%$$

$$(ii) \quad \frac{5.1111 - 5.2116}{5.2116} \times 100\% = -1.93\%$$

TYU12.12

$$(a) \quad I_{CQ} = 0.5157 \text{ mA}, \quad g_m = 19.83 \text{ mA/V}, \quad r_{\pi} = 9.075 \text{ k}\Omega$$

$$(i) \quad R_o = R_C = 10 \text{ k}\Omega$$

$$(ii) \quad R_S \parallel R_{TH} \parallel r_{\pi} = 10 \parallel 4.965 \parallel 9.075 = 2.43 \text{ k}\Omega$$

$$I_x = \frac{V_x}{R_C} + g_m V_{\pi} + \frac{V_x}{R_F + 2.43}$$

$$V_{\pi} = \left(\frac{2.43}{2.43 + 60} \right) \cdot V_x = (0.03892) V_x$$

$$\frac{I_x}{V_x} = \frac{1}{R_{of}} = \frac{1}{10} + (19.83)(0.03892) + \frac{1}{62.43}$$

or

$$R_{of} = 1.126 \text{ k}\Omega$$

(b) $I_{CQ} = 0.5013 \text{ mA}$, $g_m = 19.28 \text{ mA/V}$, $r_\pi = 6.224 \text{ k}\Omega$

$$R_S \| R_{TH} \| r_\pi = 10 \| 4.965 \| 6.224 = 2.164 \text{ k}\Omega$$

(i) $R_o = 10 \text{ k}\Omega$

(ii) $\frac{I_x}{V_x} = \frac{1}{10} + (19.28) \left(\frac{2.164}{2.164 + 60} \right) + \frac{1}{62.164} = \frac{1}{R_{of}}$

or

$$R_{of} = 1.270 \text{ k}\Omega$$

TYU12.13

From Example 12.15, for $h_{FE} = 100$, $T = 4.10$.

Now for $h_{FE} = 150$,

$$R_{TH} = 4.965 \text{ k}\Omega, \quad V_{TH} = 0.9735 \text{ V}$$

$$I_{BQ} = \frac{0.9735 - 0.7}{4.965 + (151)(0.5)} = 0.003399 \text{ mA}$$

$$I_{CQ} = 0.5098 \text{ mA}, \quad g_m = 19.61 \text{ mA/V}, \quad r_\pi = 7.650 \text{ k}\Omega$$

$$R_{eq} = R_S \| R_{TH} \| r_\pi = 10 \| 4.965 \| 7.65 = 2.314 \text{ k}\Omega$$

We find

$$T = (g_m R_C) \left(\frac{R_{eq}}{R_{eq} + R_F + R_C} \right) = (19.61)(10) \left(\frac{2.314}{2.314 + 82 + 10} \right)$$

or

$$T = 4.811$$

$$\text{Percent change} = \frac{4.811 - 4.10}{4.10} \times 100\% = +17.3\%$$

TYU12.14

$$V_i = -V_\epsilon, \quad V_o = -A_v V_i$$

$$V_r = \left(\frac{R_1 \| R_i}{R_1 \| R_i + R_2} \right) \cdot V_o$$

$$T = -\frac{V_r}{V_i} = A_v \cdot \frac{1}{1 + \frac{R_2}{R_1 \| R_i}} = (10^4) \cdot \frac{1}{1 + \frac{20}{5 \| 50}} = 1.85 \times 10^3$$

TYU12.15

$$-135 = -\tan^{-1}\left(\frac{f_{135}}{10^3}\right) - 2\tan^{-1}\left(\frac{f_{135}}{10^5}\right)$$

$$f_{135} \cong 4.25 \times 10^4 \text{ Hz}$$

$$|T(f_{135})| = 1 = \frac{\beta(3000)}{\sqrt{1 + \left(\frac{f_{135}}{10^3}\right)^2} \left[1 + \left(\frac{f_{135}}{10^5}\right)^2\right]}$$

$$1 = \frac{\beta(3000)}{\sqrt{1 + \left(\frac{4.25 \times 10^4}{10^3}\right)^2} \left[1 + \left(\frac{4.25 \times 10^4}{10^5}\right)^2\right]} = \frac{\beta(3000)}{(42.51)(1.1806)}$$

Then

$$\beta = 0.0167$$

TYU12.16

$$T = A_i \beta = \frac{A_{i0} \beta}{\left(1 + j \cdot \frac{f}{f_1}\right) \left(1 + j \cdot \frac{f}{f_2}\right)}$$

$$\text{Phase} = -\left[\tan^{-1}\left(\frac{f}{f_1}\right) + \tan^{-1}\left(\frac{f}{f_2}\right)\right]$$

$$\text{Phase Margin} = 60^\circ \Rightarrow \text{Phase} = -120^\circ$$

$$-120^\circ = -\left[\tan^{-1}\left(\frac{f}{10^4}\right) + \tan^{-1}\left(\frac{f}{10^5}\right)\right]$$

$$\text{At } f' = 7.66 \times 10^4 \text{ Hz,}$$

$$\begin{aligned} \text{Phase} &= -\left[\tan^{-1}(7.66) + \tan^{-1}(0.766)\right] \\ &= -[82.56 + 37.45] \\ &= -120^\circ \end{aligned}$$

$$|T(f')| = 1 = \frac{(10^5) \beta}{\sqrt{1 + (7.66)^2} \times \sqrt{1 + (0.766)^2}}$$

$$1 = \frac{(10^5) \beta}{(7.725)(1.26)} \Rightarrow \beta = 9.73 \times 10^{-5}$$

TYU12.17

$$\text{Phase Margin} = 60^\circ \Rightarrow \text{Phase} = -120^\circ$$

$$\text{Phase} = -120^\circ = -3 \tan^{-1} \left(\frac{f'}{10^5} \right)$$

$$\tan^{-1} \left(\frac{f'}{10^5} \right) = 40^\circ \Rightarrow f' = 0.839 \times 10^5 \text{ Hz}$$

$$\begin{aligned} |T(f')| = 1 &= \frac{\beta(100)}{\left[\sqrt{1 + \left(\frac{f'}{10^5} \right)^2} \right]^3} \\ &= \frac{\beta(100)}{\left[\sqrt{1 + (0.839)^2} \right]^3} \Rightarrow \underline{\beta = 0.0222} \end{aligned}$$

Chapter 13

Exercise Solutions

EX13.1

$$I_{REF} = \frac{5 - 0.6 - 0.6 - (-5)}{25} = 0.352 \text{ mA}$$

$$I_{C10}(5) = (0.026) \ln \left(\frac{0.352}{I_{C10}} \right)$$

By trial and error

$$I_{C10} \cong 16 \mu\text{A}$$

Then

$$I_{C1} = I_{C2} = \frac{I_{C10}}{2} = 8 \mu\text{A}$$

EX13.2

$$I_{REF} = \frac{5 - 0.6 - 0.6 - (-5)}{40} = 0.22 \text{ mA}$$

$$I_{C17} = I_{C13B} = 0.75 I_{REF} = (0.75)(0.22) = 0.165 \text{ mA}$$

$$I_{C16} = \frac{0.165}{200} + \frac{(0.165)(0.1) + 0.6}{50} \\ = 0.000825 + 0.01233$$

$$I_{C16} = 13.2 \mu\text{A}$$

EX13.3

$$I_{C13A} = (0.25)(0.5) = 0.125 \text{ mA}$$

$$I_{R10} = \frac{0.6}{50} = 0.012 \text{ mA}$$

$$I_{C19} \cong I_{C13A} - I_{R10} = 0.125 - 0.012 = 0.113 \text{ mA}$$

$$I_{B19} = \frac{I_{C19}}{\beta} = \frac{0.113}{200} \Rightarrow 0.565 \mu\text{A}$$

$$I_{C18} = I_{R10} + I_{B19} = 12 + 0.565 = 12.565 \mu\text{A}$$

$$V_{BE19} = (0.026) \ln \left(\frac{0.113 \times 10^{-3}}{10^{-14}} \right) = 0.60185 \text{ V}$$

$$V_{BE18} = (0.026) \ln \left(\frac{12.565 \times 10^{-6}}{10^{-14}} \right) = 0.54474 \text{ V}$$

$$V_{BB} = 0.60185 + 0.54474 = 1.1466 \text{ V}$$

$$I_{C14} = (3 \times 10^{-14}) \exp \left(\frac{1.1466}{0.026} \right) \Rightarrow 0.113 \text{ mA}$$

EX13.4

$$r_{o6} = \frac{100}{0.0095} \Rightarrow 10.5 \text{ M}\Omega$$

Then, using results from Example 13.4

$$R_{act1} = r_{o6} [1 + g_{m6} (R_2 \parallel r_{\pi6})] = 10.5 [1 + (0.365)(1 \parallel 547)] \\ = 14.3 \text{ M}\Omega$$

$$r_{o4} = \frac{V_A}{I_{C4}} = \frac{100}{0.0095} \Rightarrow 10.5 \text{ M}\Omega$$

$$A_d = -\left(\frac{9.5}{0.026}\right)(10.5 \parallel 14.3 \parallel 4.07) = -889$$

EX13.5

$$R_{19} = r_{o13A} = \frac{100}{0.18} = 556 \text{ K}$$

$$R_{13} = r_{\pi22} + (1 + \beta_p)(R_{19} \parallel R_{20}) \\ = 7.22 + (51)(556 \parallel 111) \Rightarrow 4.73 \text{ M}\Omega$$

$$R_{act2} = \frac{100}{0.54} = 185 \text{ K}$$

$$R_{o17} = \frac{100}{0.54} = 185 \text{ K}$$

$$A_{v2} = \frac{-(200)(201)(50)(185 \parallel 4730 \parallel 185)}{4070[50 + [9.63 + (201)(0.1)]]} \\ = \frac{-182358786.9}{32450.1} \\ A_{v2} = -562$$

EX13.6

$$r_{o17} = \frac{100}{0.54} = 185 \text{ K} \quad r_{o13B} = \frac{100}{0.54} = 185 \text{ K}$$

$$R_{C17} = 185 [1 + (20.8)(0.1 \parallel 9.63)]$$

$$R_{C17} = 566 \text{ K}$$

$$R_{e22} = \frac{7.22 + 566 \parallel 185}{51} = 2.88 \text{ K}$$

$$R_{C19} = \frac{100}{0.18} = 556 \text{ K}$$

$$R_{e20} = \frac{0.65 + 2.88 \parallel 556}{51} = 0.0689 \\ = 68.9 \Omega$$

$$R_O = 22 + 68.9 = \underline{90.9 \Omega}$$

EX13.7

$$C_i = C_1 (1 + |A_2|) = 30(1 + 562) = 16890 \text{ pF}$$

$$R_{i2} = 4.07 \text{ M}\Omega$$

$$R_{o1} = R_{ac1} \parallel r_{o4} = 14.3 \parallel 10.5 = 6.05 \text{ M}\Omega$$

$$\text{Then } R_{eq} = R_{o1} \parallel R_{i2} = 6.05 \parallel 4.07 = 2.43 \text{ M}\Omega$$

Then

$$f_{PD} = \frac{1}{2\pi R_{eq} C_i} = \frac{1}{2\pi (2.43 \times 10^6)(16890 \times 10^{-12})} \\ = 3.88 \text{ Hz}$$

EX13.8

$$K_{p5} = \left(\frac{k_p'}{2} \right) \left(\frac{W}{L} \right)_5 = \left(\frac{0.04}{2} \right) (20) = 0.40 \text{ mA/V}^2$$

$$K_{p5} (V_{SG5} + V_{TP})^2 = \frac{5 - V_{SG5} - (-5)}{150}$$

$$60(V_{SG5}^2 - V_{SG5} + 0.25) = 10 - V_{SG5}$$

$$60V_{SG5}^2 - 59V_{SG5} + 5 = 0 \Rightarrow V_{SG5} = 0.8897 \text{ V}$$

$$I_{REF} = I_Q = \frac{10 - 0.8897}{150} \Rightarrow 60.74 \mu\text{A}$$

$$I_{D7} = I_{D8} = 60.74 \mu\text{A}$$

$$I_{D1} = I_{D2} = I_{D3} = I_{D4} = 30.37 \mu\text{A}$$

EX13.9

$$K_{p1} = K_{p2} = \left(\frac{0.04}{2} \right) (20) = 0.40 \text{ mA/V}^2$$

$$r_{o2} = r_{o4} = \frac{1}{\lambda I_{D2}} = \frac{1}{(0.02)(0.03037)} \Rightarrow 1.646 \text{ M}\Omega$$

$$A_d = \sqrt{2K_{p1}I_Q} (r_{o2} \parallel r_{o4}) = \sqrt{2(0.40)(0.06074)} (1.646 \parallel 1.646)$$

or

$$A_d = 181.4$$

Now

$$g_{m7} = 2 \sqrt{\left(\frac{k_n'}{2} \right) \left(\frac{W}{L} \right)_7 I_{D7}} = 2 \sqrt{\left(\frac{0.1}{2} \right) (20)(0.06074)} = 0.4929 \text{ mA/V}$$

$$r_{o7} = r_{o8} = \frac{1}{\lambda I_{D7}} = \frac{1}{(0.02)(0.06074)} = 823.2 \text{ k}\Omega$$

$$A_{v2} = g_{m7} (r_{o7} \parallel r_{o8}) = (0.4929)(823.2 \parallel 823.2)$$

or

$$A_{v2} = 202.9$$

Then

$$A_v = A_d A_{v2} = (181.4)(202.9) = 36,806 \Rightarrow 91.3 \text{ dB}$$

EX13.10

$$(a) \quad g_{m1} = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_1\left(\frac{I_{Q1}}{2}\right)} = 2\sqrt{\left(\frac{0.08}{2}\right)(22.5)\left(\frac{0.2}{2}\right)} = 0.60 \text{ mA/V}$$

$$r_{o2} = \frac{1}{(0.01)(0.1)} = 1000 \text{ k}\Omega$$

$$r_{o4} = \frac{1}{(0.015)(0.1)} = 666.7 \text{ k}\Omega$$

$$A_{d1} = g_{m1}(r_{o2} \parallel r_{o4}) = (0.6)(1000 \parallel 666.7) = 240$$

$$g_{m5} = 2\sqrt{\left(\frac{k'_p}{2}\right)\left(\frac{W}{L}\right)_5 I_{D5}} = 2\sqrt{\left(\frac{0.04}{2}\right)(80)(0.2)} = 1.131 \text{ mA/V}$$

$$r_{o5} = \frac{1}{(0.015)(0.2)} = 333.3 \text{ k}\Omega$$

$$r_{o9} = \frac{1}{(0.01)(0.2)} = 500 \text{ k}\Omega$$

$$A_2 = -g_{m5}(r_{o5} \parallel r_{o9}) = -(1.131)(333.3 \parallel 500) = -226.2$$

Then

$$A_v = A_{d1} A_2 = (240)(-226.2) = -54,288$$

(b)

$$K_{n6} = \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_6 = \left(\frac{0.08}{2}\right)(25) = 1.0 \text{ mA/V}^2$$

$$I_{D6} = K_{n6}(V_{GS6} - V_{TN})^2$$

$$0.04 = (1)(V_{GS6} - 0.7)^2 \Rightarrow V_{GS6} = 0.90 \text{ V} = V_{SG7}$$

Then

$$V_{GS8} = 2(0.9) = 1.8 \text{ V}$$

$$I_{D8} = 0.2 = \left(\frac{0.08}{2}\right)\left(\frac{W}{L}\right)_8 (1.8 - 0.7)^2 \Rightarrow \left(\frac{W}{L}\right)_8 = 4.13$$

EX13.11

$$I_{D1} = I_{D2} = 25 \mu\text{A}$$

$$g_{m1} = g_{m8} = 2\sqrt{\left(\frac{k'_p}{2}\right)\left(\frac{W}{L}\right) I_{DQ}} = 2\sqrt{\left(\frac{40}{2}\right)(25)(25)} \Rightarrow g_{m1} = g_{m8} = 224 \mu\text{A/V}$$

$$g_{m6} = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right) I_{DQ}} = 2\sqrt{\left(\frac{80}{2}\right)(25)(25)} \Rightarrow g_{m6} = 316 \mu\text{A/V}$$

$$r_{o1} = r_{o6} = r_{o8} = r_{o10} = \frac{1}{\lambda I_D} = \frac{1}{(0.02)(25)} = 2 \text{ M}\Omega$$

$$r_{o4} = \frac{1}{\lambda I_{D4}} = \frac{1}{(0.02)(50)} = 1 \text{ M}\Omega$$

$$R_{o8} = g_{m8}(r_{o8} \parallel r_{o10}) = (224)(2)(2) = 896 \text{ M}\Omega$$

$$R_{o6} = g_{m6}(r_{o6} \parallel r_{o1}) = 316(2)(1 \parallel 2) = 421 \text{ M}\Omega$$

Then

$$A_d = g_{m1}(R_{o6} \parallel R_{o8}) = 224(421 \parallel 896) \Rightarrow A_d = 64,158$$

EX13.12

$$(a) \quad K_p (V_{SG} - |V_{TP}|)^2 = \frac{V_{SG} - V_{BE}}{R_1}$$

$$(0.15)(V_{SG} - 0.8)^2 = \frac{V_{SG} - 0.6}{10}$$

We obtain

$$1.5V_{SG}^2 - 3.4V_{SG} + 1.56 = 0 \Rightarrow V_{SG} = 1.628 \text{ V}$$

Now

$$I_1 = I_2 = \frac{1.628 - 0.6}{10} = 0.1028 \text{ mA}$$

$$V_{C7} = V^+ - V_{EB} - V_{BE} = 5 - 0.6 - 0.6 = 3.8 \text{ V}$$

$$V_{C6} = V^- + V_{SG} = -5 + 1.628 = -3.372 \text{ V}$$

$$V_{CB7} = V_{BC6} = V_{C7} - V_{C6} = 3.8 - (-3.37) = 7.17 \text{ V}$$

(b)

$$\text{Set } V_{CB7} = V_{BC6} = 0 \Rightarrow V_{C7} = V_{C6}$$

$$V^+ - 1.2 = V^- + 1.628, \quad \text{Let } V^- = -V^+$$

$$2V^+ = 2.828, \quad \text{So that } V^+ = -V^- = 1.414 \text{ V}$$

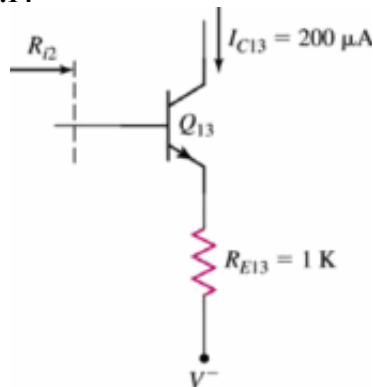
EX13.13

$$A_d = \sqrt{2K_p I_{Q5} (R_{12})}$$

$$= \sqrt{2(1)(0.2)(26)}$$

$$A_d = 16.4$$

EX13.14



$$\begin{aligned}
 r_{\pi 13} &= \frac{\beta V_T}{I_{C13}} = \frac{(200)(0.026)}{0.20} \\
 &= 26 \text{ k}\Omega \\
 R_{i2} &= r_{\pi 13} + (1 + \beta)R_{E13} = 26 + 201(1) \\
 &= 227 \text{ k}\Omega \\
 r_{o10} &= \frac{1}{\lambda I_{D10}} = \frac{1}{(0.02)(0.1)} = 500 \text{ k}\Omega \\
 r_{o12} &= \frac{V_A}{I_{C12}} = \frac{50}{0.1} = 500 \text{ k}\Omega \\
 g_{m12} &= \frac{I_{C12}}{V_T} = \frac{0.1}{0.026} = 3.85 \text{ mA/V} \\
 r_{\pi 12} &= \frac{\beta V_T}{I_{C12}} = \frac{(200)(0.026)}{0.1} = 52 \text{ k}\Omega \\
 R_{acr1} &= r_{o12} [1 + g_{m12}(r_{\pi 12} \parallel R_5)] \\
 &= 500 [1 + (3.85)(52 \parallel 0.5)] = 1453 \text{ k}\Omega \\
 A_d &= \sqrt{2K_n I_{Q5}} \cdot (r_{o10} \parallel R_{acr1} \parallel R_{i2}) \\
 &= \sqrt{2(0.6)(0.2)} \cdot (500 \parallel 1453 \parallel 227) \\
 &= (0.490)(141) \Rightarrow \underline{A_d = 69.1}
 \end{aligned}$$

EX13.15

From Example 13.15, $f_{PD} = 265 \text{ Hz}$

$$\begin{aligned}
 A_v &= A_{di} \cdot A_2 = (16.4)(1923) = 31,537 \\
 f_T &= f_{PD} \cdot A_v = (265)(31,537) = \underline{8.36 \text{ MHz}}
 \end{aligned}$$

Test Your Understanding Solutions

TYU13.1 Computer Analysis

TYU13.2 Computer Analysis

TYU13.3

$$I_{B1} = I_{B2} = \frac{9.5}{200} \mu\text{A} \quad I_{B1} = I_{B2} = 47.5 \text{ nA}$$

TYU13.4

$$\begin{aligned}
 V_{iN}(\text{max}) &= V^+ - V_{EB}(\text{on}) = 15 - 0.6 = 14.4 \text{ V} \\
 V_{iN}(\text{min}) &\cong 4V_{BE}(\text{on}) + V^+ \\
 &= 4(0.6) - 15 = -12.6 \text{ V} \\
 &\underline{-12.6 \leq V_{iN}(\text{cm}) \leq 14.4 \text{ V}}
 \end{aligned}$$

TYU13.5

- a. $V_0(\text{max}) \cong V^+ - 2V_{BE}(\text{on}) = 15 - 2(0.6)$
 $V_0(\text{max}) = 13.8 \text{ V}$
 $V_0(\text{min}) = 3V_{BE}(\text{on}) + V^- = 3(0.6) - 15$
 $V_0(\text{min}) \cong -13.2 \text{ V}$
 $-13.2 \leq V_0 \leq 13.8 \text{ V}$
- b. $V_0(\text{max}) = 5 - 1.2 = 3.8 \text{ V}$
 $V_0(\text{min}) \cong 3V_{BE} + V^- = 3(0.6) - 5 = -3.2 \text{ V}$
 $-3.2 \leq V_0 \leq 3.8 \text{ V}$

TYU13.6

$$I_{REF} = \frac{5 - V_{EB12} - V_{BE11} - (-5)}{40}$$

$$V_{EB12} = V_{BE11} = (0.026) \ln \left(\frac{I_{REF}}{5 \times 10^{-15}} \right)$$

Then by trial and error, $I_{REF} \cong 0.218 \text{ mA}$, and $V_{BE11} = V_{EB12} \cong 0.637 \text{ V}$

$$I_{C10}(5) = (0.026) \ln \left(\frac{0.218}{I_{C10}} \right)$$

By trial and error, $I_{C10} \cong 14.2 \mu\text{A}$

$$V_{BE10} = V_{BE11} - I_{C10}R_4 = 0.637 - (0.0142)(5) = 0.566 \text{ V}$$

$$I_{C6} = \frac{I_{C10}}{2} = 7.1 \mu\text{A}$$

$$V_{BE6} = (0.026) \ln \left(\frac{7.1 \times 10^{-6}}{5 \times 10^{-15}} \right) = 0.548 \text{ V}$$

TYU13.7

$$I_{REF} = \frac{10 - 0.6 - 0.6 - (-10)}{40} \Rightarrow I_{REF} = 0.47 \text{ mA}$$

$$I_{C10}R_4 = V_T \ln \left(\frac{I_{REF}}{I_{C10}} \right)$$

$$I_{C10}(5) = (0.026) \ln \left(\frac{0.47}{I_{C10}} \right)$$

By trial and error:

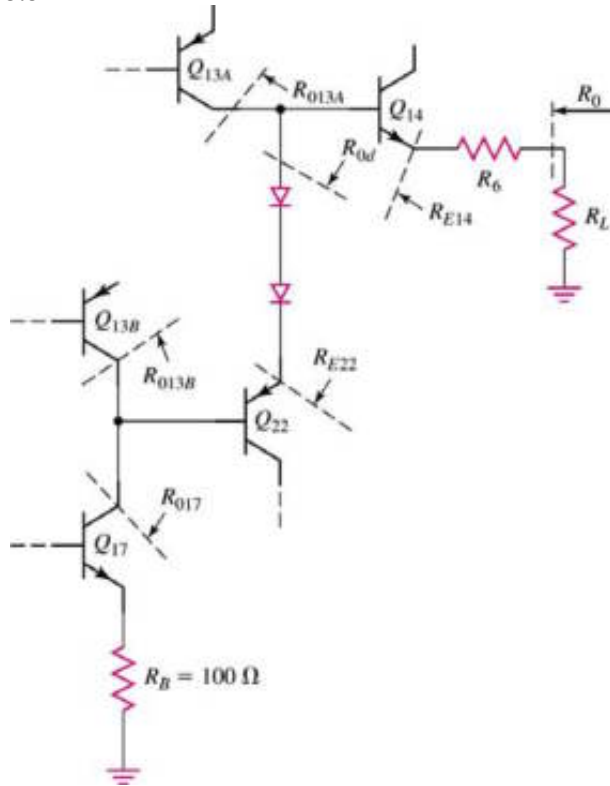
$$\Rightarrow I_{C10} \cong 17.2 \mu\text{A}$$

$$I_{C6} \cong \frac{I_{C10}}{2} \Rightarrow I_{C6} = 8.6 \mu\text{A}$$

$$I_{C13B} = (0.75)I_{REF} \Rightarrow I_{C13B} = 0.353 \text{ mA}$$

$$I_{C13A} = (0.25)I_{REF} \Rightarrow I_{C13A} = 0.118 \text{ mA}$$

TYU13.8



$$R_0 = R_6 + R_{E14}$$

$$R_{E14} = \frac{r_{\pi 14} + R_{0d} \parallel R_{013A}}{1 + \beta_n}$$

The diode resistance can be found as

$$I_D = I_S \exp\left(\frac{V_D}{V_T}\right)$$

$$\frac{1}{r_d} = \frac{\partial I_D}{\partial V_D} = I_S \left(\frac{1}{V_T}\right) \cdot \exp\left(\frac{V_D}{V_T}\right) = \frac{I_D}{V_T}$$

or

$$r_d = \frac{V_T}{I_D} = \frac{V_T}{I_{C13A}} = \frac{0.026}{0.18} \Rightarrow 144 \, \Omega$$

$$R_{E22} = \frac{r_{\pi 22} + R_{017} \parallel R_{013B}}{1 + \beta_p}$$

$$R_{013B} = r_{013B} = 92.6 \, \text{k}\Omega$$

$$R_{017} = r_{017} [1 + g_{m17} (R_8 \parallel r_{\pi 17})] = 283 \, \text{k}\Omega$$

From previous calculations

$$R_{E22} = 1.51 \, \text{k}\Omega$$

$$R_{0d} = 2r_d + R_{E22} = 2(0.144) + 1.51 = 1.80 \, \text{k}\Omega$$

$$R_{013A} = r_{013A} = 278 \, \text{k}\Omega$$

$$r_{\pi 14} = \frac{\beta_n V_T}{I_{C14}} = \frac{(200)(0.026)}{5} = 1.04 \, \text{k}\Omega$$

$$R_{E14} = \frac{1.04 + 1.8 \parallel 278}{201} \Rightarrow 14.1 \, \Omega$$

$$R_0 = R_6 + R_{E14} = 27 + 14.1 \Rightarrow R_0 \cong 41 \, \Omega$$

TYU13.9

$$I_{D1} = 20.2 = \left(\frac{40}{2}\right)(12.5)(V_{SG1} + V_{TP})^2 = 250(V_{SG1} - 0.5)^2 \Rightarrow V_{SG1} = 0.7843 \text{ V}$$

$$V_{SG5} = 0.9022 \text{ V}, \Rightarrow V_{SD6}(\text{sat}) = 0.9022 - 0.5 = 0.4022 \text{ V}$$

Then

$$v_{CM}(+) = 5 - 0.7843 - 0.4022 = 3.81 \text{ V}$$

$$I_{D3} = 20.2 = \left(\frac{100}{2}\right)(6.25)(V_{GS3} - 0.5)^2 \Rightarrow V_{GS3} = 0.7542 \text{ V}$$

$$V_{SD1}(\text{sat}) = 0.7843 - 0.5 = 0.2843 \text{ V}$$

Now

$$v_{CM}(-) = V^- + V_{GS3} + V_{SD1}(\text{sat}) - V_{SG1} = -5 + 0.7542 + 0.2843 - 0.7843 = -4.75 \text{ V}$$

$$\text{Then } -4.75 \leq v_{CM} \leq 3.81 \text{ V}$$

TYU13.10

$$V_{SG8} = V_{SG5} = 0.9022 \text{ V}$$

$$V_{SD8}(\text{sat}) = 0.9022 - 0.5 = 0.4022 \text{ V}$$

$$v_O(\text{max}) = 5 - 0.4022 = 4.6 \text{ V}$$

$$I_{D7} = \left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_7 (V_{GS7} - V_{TN})^2$$

$$40.4 = \left(\frac{100}{2}\right)(12.5)(V_{GS7} - 0.5)^2 \Rightarrow V_{GS7} = 0.7542 \text{ V}$$

$$V_{DS7}(\text{sat}) = 0.7542 - 0.5 = 0.2542 \text{ V}$$

$$v_O(\text{min}) = -5 + 0.2542 = -4.75 \text{ V}$$

$$\text{Then } -4.75 \leq v_O \leq 4.6 \text{ V}$$

TYU13.11

$$(a) \quad 0.25(V_{SG5} - 0.5)^2 = \frac{10 - V_{SG5}}{100}$$

$$25V_{SG5}^2 - 24V_{SG5} - 3.75 = 0 \Rightarrow V_{SG5} = 1.097 \text{ V}$$

$$I_{set} = I_Q = \frac{10 - 1.097}{100} \Rightarrow 89.03 \mu\text{A} = I_{D7} = I_{D8}$$

$$\text{Then } I_{D1} - I_{D4} = 44.52 \mu\text{A}$$

$$(b) \quad K_{p1} = K_{p2} = 0.25 \text{ mA/V}^2$$

$$r_{o2} = r_{o4} = \frac{1}{(0.02)(0.04452)} = 1123 \text{ k}\Omega$$

$$A_d = \sqrt{2K_{p1}I_Q(r_{o2} \parallel r_{o4})} = \sqrt{2(0.25)(0.08903)}(1123 \parallel 1123) = 118.5$$

$$g_{m7} = 2\sqrt{\left(\frac{0.1}{2}\right)(12.5)(0.08903)} = 0.4718 \text{ mA/V}$$

$$r_{o8} = r_{o7} = \frac{1}{(0.02)(0.08903)} = 561.6 \text{ k}\Omega$$

$$A_{v2} = (0.4718)(561.6 \parallel 561.6) = 132.5$$

$$\text{Then } A_v = (118.5)(132.5) = 15,701$$

TYU13.12

(a) $g_{m1} = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_1 I_{DQ}} = 2\sqrt{\left(\frac{0.1}{2}\right)(40)(0.1)} = 0.8944 \text{ mA/V}$

$$r_{o6} = r_{o8} = \frac{1}{(0.02)(0.3)} = 166.7 \text{ k}\Omega$$

$$A_d = Bg_{m1}(r_{o6} \parallel r_{o8}) = 3(0.8944)(166.7 \parallel 166.7) = 223.6$$

(b) $R_o = r_{o6} \parallel r_{o8} = 166.7 \parallel 166.7 = 83.33 \text{ k}\Omega$

$$f_{PD} = \frac{1}{2\pi R_o (C_L + C_P)} = \frac{1}{2\pi(83.33 \times 10^3)(2 \times 10^{-12})} \Rightarrow f_{PD} = 955 \text{ kHz}$$

$$GBW = (223.6)(955 \times 10^3) \Rightarrow 213.5 \text{ MHz}$$

TYU13.13

(a) $g_{m1} = 2\sqrt{\left(\frac{0.1}{2}\right)(40)(0.1)} = 0.8944 \text{ mA/V}$

$$r_{o6} = r_{o8} = r_{o10} = r_{o12} = \frac{1}{(0.02)(0.3)} = 166.7 \text{ k}\Omega$$

$$g_{m12} = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_{12} I_{D12}} = 2\sqrt{\left(\frac{0.1}{2}\right)(40)(0.3)} = 1.549 \text{ mA/V}$$

$$g_{m10} = 2\sqrt{\left(\frac{0.04}{2}\right)(40)(0.3)} = 0.9798 \text{ mA/V}$$

$$R_{o10} = g_{m10}(r_{o10} \parallel r_{o6}) = (0.9798)(166.7 \parallel 166.7) = 27,228 \text{ k}\Omega$$

$$R_{o12} = g_{m12}(r_{o12} \parallel r_{o8}) = (1.549)(166.7 \parallel 166.7) = 43,045 \text{ k}\Omega$$

$$A_d = (3)(0.8944)(27,228 \parallel 43,045) = 44,751$$

(b)

$$R_o = R_{o10} \parallel R_{o12} = 16,678 \text{ k}\Omega$$

$$f_{PD} = \frac{1}{2\pi(16,678 \times 10^3)(2 \times 10^{-12})} \Rightarrow 4.77 \text{ kHz}$$

$$GBW = (44,751)(4.77 \times 10^3) \Rightarrow 213.5 \text{ MHz}$$

TYU13.14

(a) $A_d = g_{m1}(R_{o6} \parallel R_{o8})$

From Example 13.11,

$$g_{m1} = 316 \mu A/V, R_{o8} = 316 M\Omega$$

Now

$$R_{o6} = g_{m6}(r_{o6})(r_{o4} \parallel r_{o1})$$

$$r_{o1} = 1 M\Omega, r_{o4} = 0.5 M\Omega$$

$$g_{m6} = \frac{I_{C6}}{V_T} = \frac{50}{0.026} \Rightarrow 1.923 mA/V$$

$$r_{o6} = \frac{V_{A6}}{I_{C6}} = \frac{80}{50} = 1.6 M\Omega$$

Then

$$R_{o6} = (1.923)(1600)(0.5 \parallel 1) = 1026 M\Omega$$

$$A_d = (316)(1026 \parallel 316) \Rightarrow A_d = 76,343$$

$$f_{PD} = \frac{1}{2\pi(316 \parallel 1026) \times 10^6 \times 2 \times 10^{-12}} \Rightarrow f_{PD} = 329 Hz$$

(b) $f_{PD} \cdot A_d = (329)(76,343) \Rightarrow 25.1 MHz$

TYU13.15

For Q_7 and R_1

$$V_{SG} = V_{BE7} + I_1 R_1 = 0.6 + I_1(5)$$

For M_8 :

$$I_2 = K_p(V_{SG} + V_{TP})^2$$

$$I_2 = 0.3(V_{SG} - 1.4)^2$$

By trial and error:

$$V_{SG} = 2.54 V$$

$$I_1 = I_2 = 0.388 mA$$

TYU13.16

For J_6 biased in the saturation region

$$\Rightarrow I_{C3} = I_{DSS} = 300 \mu A$$

Q_1, Q_2, Q_3 are matched

$$\Rightarrow I_{C1} = I_{C2} = I_{C3} = 300 \mu A$$

Chapter 14

Exercise Solutions

EX14.1

$$(a) \quad A_{CL} = \frac{-40}{1 + \frac{1}{A_{OL}}(41)} = \frac{-40}{1 + \frac{41}{2 \times 10^5}} = -39.9918$$

$$(b) \quad A_{CL} = \frac{-40}{1 + \frac{41}{5 \times 10^4}} = -39.9672$$

$$(c) \quad \text{Percent change} = \frac{39.9672 - 39.9918}{39.9918} \times 100\% = -0.0615\%$$

EX14.2

$$(a) \quad \frac{1}{R_{if}} = \frac{1}{40} + \frac{1}{80} \cdot \frac{(1 + 5 \times 10^4)}{(1)} = 0.025 + 625.0$$

$$\text{so that } R_{if} = 1.6 \Omega$$

$$(b) \quad \frac{1}{R_{if}} = \frac{1}{40} + \frac{1}{80} \cdot \frac{\left(1 + 5 \times 10^4 + \frac{1}{10}\right)}{\left(1 + \frac{1}{10} + \frac{1}{80}\right)} = 0.025 + \frac{1}{80} \cdot \frac{(5.00011 \times 10^4)}{(1.1125)}$$

$$\text{so that } R_{if} = 1.78 \Omega$$

EX14.3

$$\begin{aligned} R_{if} &= \frac{40(1 + 10^4) + 99\left(1 + \frac{40}{1}\right)}{1 + \frac{99}{1}} \\ &\cong \frac{4 \times 10^5 + 4.059 \times 10^3}{100} \\ R_{if} &= 4.04 \times 10^3 \text{ k}\Omega \Rightarrow \underline{R_{if} = 4.04 \text{ M}\Omega} \end{aligned}$$

EX14.4

$$1 + \frac{R_2}{R_1} = 100$$

$$a. \quad \frac{1}{R_{of}} = \frac{1}{100} \left[\frac{10^5}{100} \right] = 10 \Rightarrow \underline{R_{of} = 0.1 \Omega}$$

$$b. \quad \frac{1}{R_{of}} = \frac{1}{10} \left[\frac{10^5}{100} \right] = 10^2$$

$$\underline{R_{of} = 10^{-2} \text{ k}\Omega \Rightarrow R_{of} = 10 \Omega}$$

EX14.5

(a) $f_T = (2 \times 10^5)(5) = (30)f_{3-dB} \Rightarrow f_{3-dB} = 33.3 \text{ kHz}$

(b) $v_O = A_{CLO} \cdot v_I = (30)(100 \sin(2\pi f t)) \mu\text{V}$

or

$$v_O = 3 \sin(2\pi f t) \text{ mV}$$

(c) (i) $v_{O,peak} \cong 3 \text{ mV}$

(ii) $v_O = \frac{3}{\sqrt{1 + \left(\frac{50}{33.3}\right)^2}} = 1.663 \text{ mV}$

(iii) $v_O = \frac{3}{\sqrt{1 + \left(\frac{200}{33.3}\right)^2}} = 0.493 \text{ mV}$

EX14.6

(a) $v_O = (SR) \cdot t$

(i) $v_O = (1.25)(2) = 2.5 \text{ V}$

(ii) $v_O = (1.25)(4) = 5 \Rightarrow v_O = 4 \text{ V}$

(iii) $v_O = 4 \text{ V}$

(b) $v_O = 4 = (1.25)(t) \Rightarrow t = 3.2 \mu\text{s}$

EX14.7

(a) $f_{\max} = \frac{SR}{2\pi V_{PO}} = \frac{0.63 \times 10^6}{2\pi(0.25)} \Rightarrow 401 \text{ kHz}$

(b) $f_{\max} = \frac{0.63 \times 10^6}{2\pi(2)} \Rightarrow 50.1 \text{ kHz}$

(c) $f_{\max} = \frac{0.63 \times 10^6}{2\pi(8)} \Rightarrow 12.5 \text{ kHz}$

EX14.8

$$V_{OS} = V_T \ln\left(\frac{I_{S2}}{I_{S1}}\right)$$

$$\frac{2}{26} = \ln\left(\frac{I_{S2}}{I_{S1}}\right)$$

Then

$$I_{S2} = I_{S1} \exp\left(\frac{2}{26}\right) = 2.16 \times 10^{-15} \text{ A}$$

$$\text{Percent change} = \frac{2.16 - 2}{2} \times 100\% = 8\%$$

EX14.9

We need

$$i_{C1} = i_{C2}, v_{EC3} = v_{EC4} = 0.6 \text{ V, and } v_{CE1} = v_{CE2} = 10 \text{ V}$$

By Equation (14.60(a))

$$\begin{aligned} i_{C1} &= I_{S1} \left[\exp \left(\frac{v_{BE1}}{V_T} \right) \right] \left(1 + \frac{10}{50} \right) \\ &= I_{S3} \left[\exp \left(\frac{v_{EB3}}{V_T} \right) \right] \left(1 + \frac{0.6}{50} \right) \end{aligned}$$

By Equation (14.60(b))

$$\begin{aligned} i_{C2} &= I_{S2} \left[\exp \left(\frac{v_{BE2}}{V_T} \right) \right] \left(1 + \frac{10}{50} \right) \\ &= I_{S4} \left[\exp \left(\frac{v_{EB4}}{V_T} \right) \right] \left(1 + \frac{0.6}{50} \right) \end{aligned}$$

$I_{S1} = I_{S2}$, take the ratio:

$$\exp \left(\frac{v_{BE1} - v_{BE2}}{V_T} \right) = \frac{I_{S3}}{I_{S4}}$$

$$\begin{aligned} v_{BE1} - v_{BE2} = V_{OS} &= V_T \ln \left(\frac{I_{S3}}{I_{S4}} \right) \\ &= 0.026 \cdot \ln(1.05) \\ \Rightarrow V_{OS} &= \underline{1.27 \text{ mV}} \end{aligned}$$

EX14.10

$$\begin{aligned} V_{OS} &= \frac{1}{2} \cdot \sqrt{\frac{I_Q}{2K_n}} \cdot \left(\frac{\Delta K_n}{K_n} \right) \\ 0.020 &= \frac{1}{2} \cdot \sqrt{\frac{150}{2(50)}} \cdot \left(\frac{\Delta K_n}{50} \right) \\ \Rightarrow \Delta K_n &= \underline{1.63 \mu A/V^2} \\ \Rightarrow \frac{\Delta K_n}{K_n} &= \frac{1.63}{50} \Rightarrow \underline{3.26\%} \end{aligned}$$

EX14.11

$$\begin{aligned} \text{Want } \left(\frac{R_5}{R_5 + R_4} \right) V^+ &= 5 \text{ mV} \\ R_5 \parallel R_4 \text{ so } \frac{R_5}{R_4} \cdot V^+ &= 0.005 \\ R_5 &= \frac{(0.005)(100)}{10} = 0.05 \text{ k}\Omega \\ \Rightarrow R_5 &= \underline{50 \Omega} \end{aligned}$$

EX14.12

$$R'_1 = 25 \parallel 1 = 0.9615 \text{ k}\Omega$$

$$R'_2 = 75 \parallel 1 = 0.9868 \text{ k}\Omega$$

$$\text{For } I_Q = 100 \mu\text{A} \Rightarrow i_{C1} = i_{C2} = 50 \mu\text{A}$$

From Equation (14.75)

$$(0.026) \ln \left(\frac{50 \times 10^{-6}}{10^{-14}} \right) + (0.050)(0.9615)$$

$$= (0.026) \ln \left(\frac{i_{C2}}{I_{S4}} \right) + (0.050)(0.9868)$$

$$0.58065 + 0.048075$$

$$= (0.026) \ln \left(\frac{i_{C2}}{I_{S4}} \right) + 0.04934$$

$$\ln \left(\frac{i_{C2}}{I_{S4}} \right) = 22.284$$

$$\frac{50 \times 10^{-6}}{I_{S4}} = 4.7625 \times 10^9$$

$$I_{S4} \cong 1.05 \times 10^{-14} \text{ A}$$

EX14.13

(a) (i) Yes

$$(ii) R_3 = R_1 \parallel R_2 = 20 \parallel 120 = 17.14 \text{ k}\Omega$$

(b) (i) Yes

$$(ii) v_O = 0 = I_{B1} R_2 - I_{B2} R_3 \left(1 + \frac{R_2}{R_1} \right)$$

$$(0.75)(120) = (0.85)R_3 \left(1 + \frac{120}{20} \right) \Rightarrow R_3 = 15.13 \text{ k}\Omega$$

Test Your Understanding Solutions

TYU14.1

$$v_{1CM}(\max) = V^+ - V_{SD1}(\text{sat}) - V_{SG1}$$

$$v_{1CM}(\min) = V^- + V_{DS4}(\text{sat}) + V_{SD1}(\text{sat}) - V_{SG1}$$

We have:

$$I_{REF} = 100 \mu A, k'_n = 80 \mu A/V^2, k'_p = 40 \mu A/V^2,$$

$$\left(\frac{W}{L}\right) = 25$$

For M_1 :

$$I_D = 50 = \left(\frac{40}{2}\right)(25)(V_{SG1} + V_{TP})^2$$

$$\text{So } 50 = 500(V_{SG1} - 0.5)^2 \Rightarrow V_{SG1} = 0.816 \text{ V}$$

$$V_{SD1}(\text{sat}) = 0.816 - 0.5 = 0.316 \text{ V}$$

Then

$$v_{CM}(\max) = V^+ - 0.316 - 0.816 = V^+ - 1.13 \text{ V}$$

For M_4 :

$$I_D = 100 = \left(\frac{80}{2}\right)(25)(V_{GS4} - V_{TN})^2$$

$$\text{So } 100 = 1000(V_{GS4} - 0.5)^2 \Rightarrow V_{GS4} = 0.816 \text{ V}$$

$$V_{DS4}(\text{sat}) = 0.816 - 0.5 = 0.316 \text{ V}$$

$$V_{CM}(\min) = V^- + 0.316 + 0.316 - 0.816 = V^- - 0.184$$

So

$$\underline{V^- - 0.184 \leq v_{CM} \leq V^+ - 1.13 \text{ V}}$$

TYU14.2

$$v_o(\max) = V^+ - V_{SD8}(\text{sat}) - V_{SG10}$$

$$v_o(\min) = V^- + V_{DS4}(\text{sat}) + V_{DS6}(\text{sat})$$

Now

$$V_{SG8} = V_{SG10} = \sqrt{\frac{50}{(40/2)(25)}} + 0.5 = 0.816 \text{ V}$$

$$V_{SD8}(\text{sat}) = V_{SD10}(\text{sat}) = 0.316 \text{ V}$$

$$\text{So } v_o(\max) = V^+ - 0.316 - 0.816 = V^+ - 1.13$$

Also

$$V_{GS6} = \sqrt{\frac{50}{(80/2)(25)}} + 0.5 = 0.724 \text{ V}$$

$$V_{GS4} = \sqrt{\frac{100}{(80/2)(25)}} + 0.5 = 0.816 \text{ V}$$

$$V_{DS6}(\text{sat}) = 0.724 - 0.5 = 0.224 \text{ V}$$

$$V_{DS4}(\text{sat}) = 0.816 - 0.5 = 0.316 \text{ V}$$

So

$$v_o(\min) = V^- + 0.316 + 0.224 = V^- + 0.54$$

Then

$$\underline{V^- + 0.54 \leq v_o \leq V^+ - 1.13 \text{ V}}$$

TYU14.3

$$(a) \quad -\frac{R_2}{R_1} = -\frac{250}{25} = -10.0$$

$$A_{CL} = -(1 - 0.001)(10.0) = -9.99$$

Now

$$-9.99 = \frac{-10}{1 + \frac{11}{A_{OL}}} \Rightarrow A_{OL} = 10,989$$

$$(b) \quad A_{CL} = -(1 - 0.0005)(10) = -9.995$$

We find

$$-9.995 = \frac{-10}{1 + \frac{11}{A_{OL}}} \Rightarrow A_{OL} = 21,989$$

TYU14.4

$$A_{CL} = \frac{A_{CL}(\infty)}{1 + \left[\frac{A_{CL}(\infty)}{A_{OL}} \right]}$$

a.

$$A_{CL}(\infty) = 1 + \frac{R_2}{R_1} = 1 + \frac{495}{5} = 100$$

$$A_{CL} = \frac{100}{1 + \frac{100}{10^5}} \Rightarrow \underline{A_{CL} = 99.90}$$

$$\underline{A_{CL}(\infty) = 100}$$

$$b. \quad \frac{dA_{CL}}{A_{CL}} = 10 \times \frac{100}{10^5} = \underline{0.01\%}$$

$$A_{CL} = 99.90 - (0.0001)(99.90) \\ \Rightarrow \underline{A_{CL} = 99.89}$$

TYU14.5

$$(a) \quad \frac{A_{CL}}{A_{CL}(\infty)} = \frac{1}{1 + \frac{A_{CL}(\infty)}{A_{OL}}}$$

$$0.999 = \frac{1}{1 + \frac{A_{CL}(\infty)}{2 \times 10^4}} \Rightarrow A_{CL}(\infty) = 20.02$$

$$(b) \quad 0.9995 = \frac{1}{1 + \frac{A_{CL}(\infty)}{2 \times 10^4}} \Rightarrow A_{CL}(\infty) = 10.005$$

TYU14.6

$$\frac{i_i}{I_1} = \left(\frac{R_f}{R_i} \right)$$

a. $\frac{i_i}{I_1} = \frac{0.1}{10^4} = 1 \times 10^{-5}$

b. $\frac{i_i}{I_1} = \frac{10}{10^4} = 1 \times 10^{-3}$

TYU14.7

Voltage follower $R_2 = 0, R_1 = \infty$

$$R_f = R_i (1 + A_{0L}) = 10(1 + 5 \times 10^5) \\ \cong 5 \times 10^6 \text{ k}\Omega \Rightarrow R_f = 5000 \text{ M}\Omega$$

TYU14.8

(a) (i) $f_T = (5 \times 10^4)(15) = (25)f_{3-dB}$
or $f_{3-dB} = f_{\max} = 30 \text{ kHz}$

(ii) $V_{PO} = \frac{SR}{2\pi f_{\max}} = \frac{0.8 \times 10^6}{2\pi(30 \times 10^3)} = 4.24 \text{ V}$

(b) (i) $f_T = (5 \times 10^5)(10) = (25)f_{3-dB} \Rightarrow f_{3-dB} = 200 \text{ kHz}$

(ii) $V_{PO} = \frac{SR}{2\pi f_{\max}} = \frac{0.8 \times 10^6}{2\pi(200 \times 10^3)} = 0.637 \text{ V}$

TYU14.9

$$v_0 = I_{B1} R_3 = (10^{-6})(200 \times 10^3)$$

a. $\Rightarrow v_0 = 0.20 \text{ V}$

$$R_4 = R_1 \parallel R_2 \parallel R_3 = 100 \parallel 50 \parallel 200$$

b. $\Rightarrow R_4 = 28.6 \text{ k}\Omega$

Chapter 15

Exercise Solutions

EX15.1

$$f_{3-dB} = \frac{1}{2\pi RC} \Rightarrow RC = \frac{1}{2\pi f_{3-dB}} = \frac{1}{2\pi(25 \times 10^3)} = 6.366 \times 10^{-6}$$

$$\text{Set } C_3 = 50 \text{ pF} = 1.414C \Rightarrow C = 35.36 \text{ pF}$$

$$C_4 = (0.707)C = 25 \text{ pF}$$

$$\text{Then } R = \frac{6.366 \times 10^{-6}}{C} = \frac{6.366 \times 10^{-6}}{35.36 \times 10^{-12}} \Rightarrow R = 180 \text{ k}\Omega$$

EX15.2

$$(a) R_{eq} = \frac{1}{f_c C} = \frac{1}{(100 \times 10^3)(1.2 \times 10^{-12})} \Rightarrow R_{eq} = 8.33 \text{ M}\Omega$$

$$(b) C = \frac{1}{f_c R_{eq}} = \frac{1}{(50 \times 10^3)(50 \times 10^6)} \Rightarrow C = 0.4 \text{ pF}$$

EX15.3

$$\text{Low-frequency gain: } T = -\frac{C_1}{C_2} = -\frac{30}{5} = -6$$

$$f_{3dB} = \frac{f_c C_2}{2\pi C_F} = \frac{(100 \times 10^3)(5 \times 10^{-12})}{2\pi(12 \times 10^{-12})} \Rightarrow f_{3dB} = 6.63 \text{ kHz}$$

EX15.4

$$f_o = \frac{1}{2\pi\sqrt{3}RC} \Rightarrow RC = \frac{1}{2\pi\sqrt{3}(22.5 \times 10^3)} = 4.084 \times 10^{-6}$$

$$\text{Set } R = 10 \text{ k}\Omega$$

$$C = \frac{4.084 \times 10^{-6}}{10 \times 10^3} \Rightarrow C = 408 \text{ pF}$$

$$R_2 = 8R = 80 \text{ k}\Omega$$

EX15.5

$$f_0 = \frac{1}{2\pi RC} \Rightarrow C = \frac{1}{2\pi f_0 R}$$

$$C = \frac{1}{2\pi(800)(10^4)} \Rightarrow C \cong 0.02 \text{ }\mu\text{F}$$

$$R_2 = 2R_1 = 2(10) \Rightarrow R_2 = 20 \text{ k}\Omega$$

EX15.6

$$V_{TH} = \left(\frac{R_1}{R_1 + R_2} \right) \cdot V_H$$

Set $R_1 = 10 \text{ k}\Omega$

$$\text{Then } 0.5 = \left(\frac{10}{10 + R_2} \right) (9) \Rightarrow R_2 = 170 \text{ k}\Omega$$

EX15.7

$$V_{TH} - V_{TL} = \left(\frac{R_1}{R_1 + R_2} \right) (V_H - V_L)$$

$$0.10 = \left(\frac{R_1}{R_1 + R_2} \right) (10 - [-10])$$

$$1 + \frac{R_2}{R_1} = \frac{20}{0.10} = 200 \Rightarrow \underline{\underline{\frac{R_2}{R_1} = 199}}$$

$$V_S = \left(\frac{R_2}{R_1 + R_2} \right) V_{REF}$$

$$V_{REF} = \left(1 + \frac{R_1}{R_2} \right) V_S = \left(1 + \frac{1}{199} \right) (1) \Rightarrow \underline{\underline{V_{REF} = 1.005 \text{ V}}}$$

$$I = \frac{V_H - V_{BE}(\text{on}) - V_\gamma}{R + 0.1}$$

$$R + 0.1 = \frac{10 - 0.7 - 0.7}{0.2} = 43 \text{ k}\Omega$$

$$\underline{\underline{R = 42.9 \text{ k}\Omega}}$$

EX15.8

At $t = 0^-$ let $v_O = -5 \text{ V}$ so $v_X = -2.5 \text{ V}$: For $t > 0$

$$v_X = 10 + (-2.5 - 10) \exp\left(\frac{-t}{\tau_X}\right)$$

When $v_X = 5 \text{ V}$, output switches

$$5 = 10 - 12.5 \exp\left(\frac{-t_1}{\tau_X}\right)$$

$$\exp\left(\frac{-t_1}{\tau_X}\right) = \frac{10 - 5}{12.5} = \frac{5}{12.5}$$

$$\exp\left(\frac{+t_1}{\tau_X}\right) = \frac{12.5}{5} \Rightarrow t_1 = \tau_X \ln\left(\frac{12.5}{5}\right) \Rightarrow t_1 = \tau_X (0.916)$$

During the next part of the cycle

$$v_X = -5 + [5 - (-5)] \exp\left(\frac{-t}{\tau_X}\right)$$

When $v_X = -2.5 \text{ V}$, output switches

$$-2.5 = -5 + 10 \exp\left(\frac{-t_2}{\tau_X}\right)$$

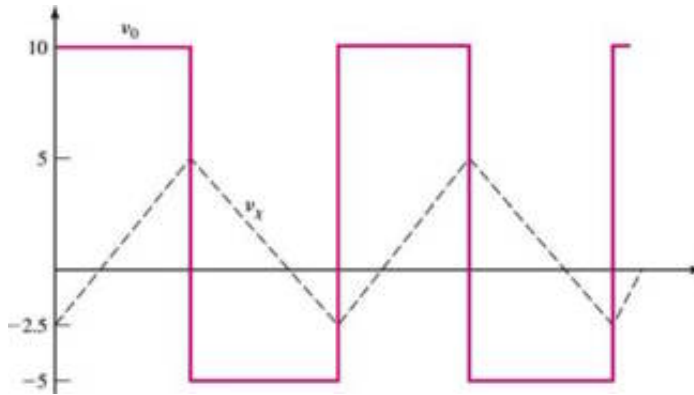
$$\exp\left(\frac{-t_2}{\tau_X}\right) = \frac{5 - 2.5}{10} = \frac{2.5}{10}$$

$$\exp\left(\frac{+t_2}{\tau_X}\right) = \frac{10}{2.5} \Rightarrow t_2 = \tau_X \ln\left(\frac{10}{2.5}\right) \Rightarrow t_2 = \tau_X (1.39)$$

$$\text{Period} = t_1 + t_2 = T = (0.916 + 1.39)\tau_X = 2.31\tau_X, \text{ Frequency} = f = \frac{1}{2.31\tau_X}$$

$$\tau_X = (50 \times 10^3)(0.01 \times 10^{-6}) = 5 \times 10^{-4} \text{ s} \Rightarrow f = 866 \text{ Hz}$$

$$\text{Duty cycle} = \frac{t_1}{t_1 + t_2} \times 100\% = \frac{0.916}{0.916 + 1.39} \times 100\% \Rightarrow \text{Duty cycle} = 39.7\%$$



EX15.9

a.

$$\tau_X = R_X C_X$$

$$v_Y = \left(\frac{R_1}{R_1 + R_2}\right) \cdot v_O = \left(\frac{10}{10 + 90}\right)(12) = 1.2 \text{ V}$$

$$\beta = \left(\frac{R_1}{R_1 + R_2}\right) = 0.10$$

$$T = \tau_X \ln\left(\frac{1 + V_Y/V_P}{1 - \beta}\right) = \tau_X \ln\left(\frac{1 + (0.7)/(12)}{1 - 0.10}\right)$$

$$T = 50 \times 10^{-6} = \tau_X \ln(1.18) = \tau_X (0.162)$$

$$R_X = \frac{50 \times 10^{-6}}{(0.1 \times 10^{-6})(0.162)} \Rightarrow R_X = 3.09 \text{ k}\Omega$$

b. Recovery time

$$v_X = V_P + (-1.2 - V_P) \exp\left(\frac{-t}{\tau_X}\right)$$

When $v_X = V_\gamma$ $t = t_2$

$$0.7 = 12 + (-1.2 - 12) \exp\left(\frac{-t_2}{\tau_X}\right)$$

$$\exp\left(\frac{-t_2}{\tau_X}\right) = \frac{12 - 0.7}{13.2} = 0.856$$

$$t_2 = \tau_X \ln\left(\frac{1}{0.856}\right) = \tau_X (0.155)$$

$$\tau_X = (3.09 \times 10^3)(0.1 \times 10^{-6}) = 3.09 \times 10^{-4} \text{ s}, \Rightarrow t_2 = 48.0 \mu\text{s}$$

EX15.10

(a) $T = 1.1RC = (1.1)(20 \times 10^3)(0.012 \times 10^{-6}) \Rightarrow T = 0.264 \text{ ms}$

(b) $RC = \frac{T}{1.1} = \frac{120 \times 10^{-6}}{1.1} = 1.09 \times 10^{-4}$

If $C = 0.01 \mu\text{F}$, then $R = \frac{1.09 \times 10^{-4}}{0.01 \times 10^{-6}} \Rightarrow R = 10.9 \text{ k}\Omega$

EX15.11

$$f = \frac{1}{0.693(R_A + 2R_B)C} = \frac{1}{(0.693)[20 + 2(80)] \times 10^3 \times (0.01 \times 10^{-6})} \Rightarrow f = 802 \text{ Hz}$$

$$\text{Duty cycle} = \frac{R_A + R_B}{R_A + 2R_B} \times 100\% = \frac{20 + 80}{20 + 2(80)} \times 100\% \Rightarrow \text{Duty cycle} = 55.6\%$$

EX15.12

$$\bar{P} = \frac{1}{2} \cdot \frac{V_P^2}{R_L}$$

$$V_P = \sqrt{2R_L \bar{P}} = \sqrt{2(8)(1)} \Rightarrow V_P = 4 \text{ V}$$

$$I_P = \frac{V_P}{R_L} = \frac{4}{8} \Rightarrow I_P = 0.5 \text{ A}$$

a.

$$V_{CE} = 12 - 4 = 8 \text{ V}$$

b.

$$I_C \approx 0.5 \text{ A}$$

So $P = I_C \cdot V_{CE} = (0.5)(8) \Rightarrow P = 4 \text{ W}$

EX15.13

(a) (i) $V_P = \sqrt{2R_L \overline{P}_L} = \sqrt{2(20)(5)} = 14.14 \text{ V}$

$$I_P = \frac{V_P}{R_L} = \frac{14.14}{20} = 0.707 \text{ A}$$

(ii) $V_S = \frac{\pi R_L P_S}{V_P}$, We have $P_S = 5 \text{ W}$

$$V_S = \frac{\pi(20)(5)}{14.14} = 22.2 \text{ V}$$

(b) (i) $V_P = \sqrt{2(8)(10)} = 12.65 \text{ V}$

$$I_P = \frac{12.65}{8} = 1.58 \text{ A}$$

(ii) $V_S = \frac{\pi(8)(10)}{12.65} = 19.9 \text{ V}$

EX15.14

Line
Now

$$\text{regulation} = \frac{dV_0}{dV^+} = \frac{dV_0}{dV_Z} \cdot \frac{dV_Z}{dV^+}$$

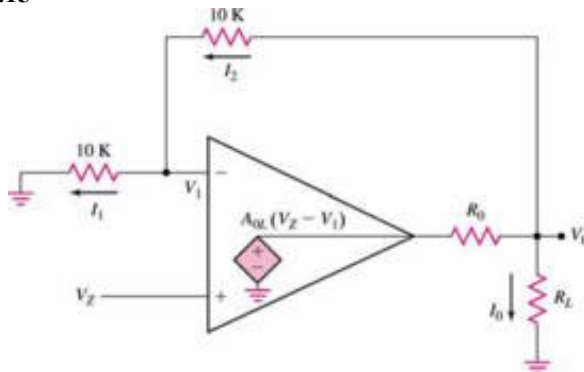
$$\frac{dV_0}{dV_Z} = \left(1 + \frac{10}{10}\right) = 2$$

$$\frac{dV_Z}{dV^+} = \left(\frac{r_z}{r_z + R_1}\right) = \frac{10}{10 + 4400} = 0.00227$$

So Line
0.454%

$$\text{regulation} = (2)(0.00227) = 0.00454$$

EX15.15



$$\frac{V_1}{10} = \frac{V_0 - V_1}{10} \Rightarrow V_1 \left(\frac{1}{10} + \frac{1}{10} \right) = \frac{V_0}{10}$$

$$V_1 \left(\frac{2}{10} \right) = \frac{V_0}{10} \Rightarrow V_0 = 2V_1 \Rightarrow V_1 = \frac{V_0}{2}$$

$$\frac{V_0 - V_1}{10} + \frac{V_0}{R_L} + \frac{V_0 - A_{0L}(V_Z - V_1)}{R_0} = 0$$

$$\begin{aligned} \frac{V_0}{10} + \frac{V_0}{R_L} + \frac{V_0}{R_0} - \frac{A_{0L}V_Z}{R_0} &= \frac{V_1}{10} - \frac{A_{0L}V_1}{R_0} \\ &= \frac{V_0}{2(10)} - \frac{A_{0L}V_0}{2R_0} \end{aligned}$$

$$\frac{V_0}{10} + I_0 + \frac{V_0}{0.5} - \frac{1000(6.3)}{0.5} = \frac{V_0}{20} - \frac{(1000)V_0}{2(0.5)}$$

$$V_0[0.10 + 2.0 - 0.05 + 1000] + I_0 = 12,600$$

$$V_0(1002.05) + I_0 = 12,600$$

For $I_0 = 1 \text{ mA} \Rightarrow V_0 = 12.5732$

For $I_0 = 100 \text{ mA} \Rightarrow V_0 = 12.4744$

$$\begin{aligned} \text{Load reg} &= \frac{V_0(\text{NL}) - V_0(\text{FL})}{V_0(\text{NL})} \times 100\% \\ &= \frac{12.5732 - 12.4744}{12.5732} \times 100\% \\ \text{Load reg} &= 0.786\% \end{aligned}$$

EX15.16

a.

$$I_{C3} = \frac{V_Z - 3V_{BE}(\text{on})}{R_1 + R_2 + R_3}$$

$$I_{C3} = \frac{5.6 - 3(0.6)}{3.9 + 3.4 + 0.576} = \frac{3.8}{7.88} \Rightarrow \underline{I_{C3} = 0.482 \text{ mA}}$$

$$I_{C4}R_4 = V_T \ln \left(\frac{I_{C3}}{I_{C4}} \right)$$

$$I_{C4}(0.1) = (0.026) \ln \left(\frac{0.482}{I_{C4}} \right)$$

By trial and error

$$\underline{I_{C4} = 0.213 \text{ mA}}$$

$$V_{B7} = 2(0.6) + (0.482)(3.9) \Rightarrow \underline{V_{B7} = 3.08 \text{ V}}$$

b.

$$\left(\frac{R_{13}}{R_{13} + R_{12}} \right) V_0 = V_{B8} = V_{B7}$$

$$\left(\frac{2.23}{2.23 + R_{12}} \right) (5) = 3.08$$

$$(2.23)(5) = (3.08)(2.23) + (3.08) R_{12}$$

$$11.15 = 6.868 = 3.08 R_{12} \Rightarrow \underline{R_{12} = 1.39 \text{ k}\Omega}$$

Test Your Understanding Solutions

TYU15.1

$$(a) \quad f_{3-dB} = \frac{1}{2\pi RC} \Rightarrow RC = \frac{1}{2\pi f_{3-dB}} = \frac{1}{2\pi(200)} = 7.958 \times 10^{-4}$$

For example, let $C = 0.01 \mu\text{F}$

$$\text{Then } R = \frac{7.958 \times 10^{-4}}{0.01 \times 10^{-6}} \Rightarrow R = 79.58 \text{ k}\Omega$$

$$R_1 = \frac{79.58}{3.546} = 22.44 \text{ k}\Omega$$

$$R_2 = \frac{79.58}{1.392} = 57.17 \text{ k}\Omega$$

$$R_3 = \frac{79.58}{0.2024} = 393.2 \text{ k}\Omega$$

$$(b) \quad (i) \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{200}{100} \right)^6}} = 0.124 \Rightarrow -18.1 \text{ dB}$$

$$(ii) \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{200}{300} \right)^6}} = 0.959 \Rightarrow -0.365 \text{ dB}$$

TYU15.2

$$(a) \quad RC = \frac{1}{2\pi f_{3-dB}} = \frac{1}{2\pi(30 \times 10^3)} = 5.305 \times 10^{-6}$$

For example, let $R = 100 \text{ k}\Omega$

$$\text{Then } C = \frac{5.305 \times 10^{-6}}{100 \times 10^3} \Rightarrow C = 53.05 \text{ pF}$$

$$C_1 = 1.082C = 57.4 \text{ pF}$$

$$C_2 = 0.9241C = 49.02 \text{ pF}$$

$$C_3 = 2.613C = 138.6 \text{ pF}$$

$$C_4 = 0.3825C = 20.29 \text{ pF}$$

$$(b) \quad (0.99) = \frac{1}{\sqrt{1 + \left(\frac{f}{30}\right)^8}} \Rightarrow \left(\frac{f}{30}\right)^8 = 0.020304$$

which yields $f = 18.43 \text{ kHz}$

TYU15.3

$$\text{1-pole} \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{12}{10}\right)^2}} \Rightarrow -3.87 \text{ dB}$$

$$\text{2-pole} \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{12}{10}\right)^4}} \Rightarrow -4.88 \text{ dB}$$

$$\text{3-pole} \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{12}{10}\right)^6}} \Rightarrow -6.0 \text{ dB}$$

$$\text{4-pole} \quad |T| = \frac{1}{\sqrt{1 + \left(\frac{12}{10}\right)^8}} \Rightarrow -7.24 \text{ dB}$$

TYU15.4

$$f_c C = \frac{1}{R_{eq}} = \frac{1}{25 \times 10^6} = 4 \times 10^{-8}$$

For example, let $f_c = 50 \text{ kHz}$,

$$C = \frac{4 \times 10^{-8}}{50 \times 10^3} \Rightarrow C = 0.8 \text{ pF}$$

TYU15.5

$$f_o = \frac{1}{2\pi\sqrt{6}RC} \Rightarrow C = \frac{1}{2\pi\sqrt{6}(15 \times 10^3)(20 \times 10^3)} \Rightarrow C = 217 \text{ pF}$$

$$R_2 = 29R = (29)(15) = 435 \text{ k}\Omega$$

TYU15.6

$$f_o = \frac{1}{2\pi\sqrt{L \cdot \left(\frac{C_1 C_2}{C_1 + C_2}\right)}} = \frac{1}{2\pi\sqrt{(10^{-6}) \left[\frac{(10^{-9})^2}{2 \times 10^{-9}}\right]}} \Rightarrow f_o = 7.12 \text{ MHz}$$

$$\frac{C_2}{C_1} = g_m R$$

$$g_m = \frac{C_2}{C_1} \cdot \frac{1}{R} = \frac{1}{4 \times 10^3} \Rightarrow g_m = 0.25 \text{ mA/V}$$

We have

$$g_m = 2 \left(\frac{k'}{2} \right) \left(\frac{W}{L} \right) (V_{GS} - V_{Th})$$

$$k' \cong 20 \mu\text{A/V}^2, V_{GS} - V_{Th} \cong 1 \text{ V}$$

$$\text{So } \frac{W}{L} = \frac{0.25 \times 10^{-3}}{(20 \times 10^{-6})(1)} = 12.5$$

and a value of $W/L = 12.5$ is certainly reasonable.

TYU15.7

$$V_{TH} = - \left(\frac{R_1}{R_2} \right) \cdot V_L$$

$$0.2 = - \left(\frac{R_1}{R_2} \right) (-12)$$

Set $R_2 = 200 \text{ k}\Omega$

$$\text{Then } \frac{R_1}{200} = \frac{0.2}{12} \Rightarrow R_1 = 3.33 \text{ k}\Omega$$

TYU15.8

a.

$$V_S = \left(\frac{R_2}{R_1 + R_2} \right) V_{REF} = \left(\frac{10}{1+10} \right) (2)$$

$$\underline{V_S = 1.82 \text{ V}}$$

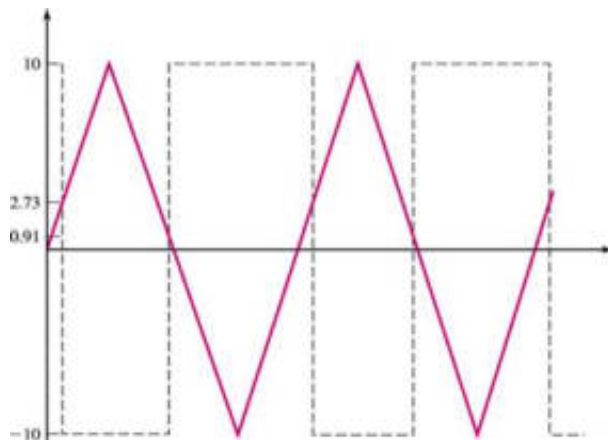
$$V_{TH} = V_S + \left(\frac{R_1}{R_1 + R_2} \right) V_H = 1.82 + \left(\frac{1}{1+10} \right) (10)$$

$$\underline{V_{TH} = 2.73 \text{ V}}$$

$$V_{TL} = V_S + \left(\frac{R_1}{R_1 + R_2} \right) V_L = 1.82 + \left(\frac{1}{1+10} \right) (-10)$$

$$\underline{V_{TL} = 0.91 \text{ V}}$$

b.



TYU15.9

$$V_{TH} - V_{TL} = \frac{R_1}{R_2}(V_H - V_L)$$

$$0.5 = \frac{R_1}{R_2}[9 - (-9)] = 18\left(\frac{R_1}{R_2}\right)$$

Set $R_1 = 10 \text{ k}\Omega$,

$$\text{Then } R_2 = \frac{(18)(10)}{0.5} = 360 \text{ k}\Omega$$

$$\text{Now } V_S = \left(1 + \frac{R_1}{R_2}\right) \cdot V_{REF}$$

$$-2 = \left(1 + \frac{10}{360}\right) \cdot V_{REF} \Rightarrow V_{REF} = -1.946 \text{ V}$$

TYU15.10

$$(a) f = \frac{1}{2.2R_X C_X} = \frac{1}{(2.2)(20 \times 10^3)(0.05 \times 10^{-6})} = 454.5 \text{ Hz}$$

50% duty cycle

$$(b) R_X = \frac{1}{(2.2)f C_X} = \frac{1}{(2.2)(1.2 \times 10^3)(0.05 \times 10^{-6})} \Rightarrow R_X = 7.576 \text{ k}\Omega$$

TYU15.11

$$\beta = \frac{R_1}{R_1 + R_2} = \frac{20}{20 + 40} = 0.333$$

$$\tau_X = R_X C_X = (10^4)(0.01 \times 10^{-6}) = 1 \times 10^{-4} \text{ s}$$

$$T = \tau_X \ln\left(\frac{1 + V_Y/V_P}{1 - \beta}\right) = (10^{-4}) \ln\left(\frac{1 + (0.7)/8}{1 - 0.333}\right) \Rightarrow T = 48.9 \mu\text{s}$$

Recovery time

$$V_Y = \left(\frac{R_1}{R_1 + R_2}\right) \cdot V_O = \left(\frac{20}{20 + 40}\right)(8) = 2.667 \text{ V}$$

$$0.7 = 8 + (-2.667 - 8) \exp\left(\frac{-t_2}{\tau_X}\right)$$

$$\exp\left(\frac{-t_2}{\tau_X}\right) = \frac{8 - 0.7}{10.66} = 0.6844$$

$$t_2 = \tau_X \ln\left(\frac{1}{0.6844}\right) \Rightarrow t_2 = 37.9 \mu\text{s}$$

TYU15.12

$$f = \frac{1}{(0.693)(R_A + 2R_B)C} \Rightarrow R_A + 2R_B = \frac{1}{(0.693)fC} = \frac{1}{(0.693)(10^3)(0.01 \times 10^{-6})}$$

$$R_A + 2R_B = 1.443 \times 10^5$$

$$\text{Duty cycle} = 55\% = \frac{(R_A + R_B)}{(R_A + 2R_B)} \times 100\%$$

$$0.55 = \frac{1.443 \times 10^5 - R_B}{1.443 \times 10^5}$$

$$R_B = (1.443 \times 10^5)(1 - 0.55) \Rightarrow R_B = 64.9 \text{ k}\Omega \text{ and } R_A = 14.43 \text{ k}\Omega$$

TYU15.13

(a) (i) For A_1 : $A_{v1} = \left(1 + \frac{R_2}{R_1}\right) = \left(1 + \frac{40}{20}\right) = 3$

For A_2 : $A_{v2} = -\frac{R_4}{R_3} = -\frac{60}{20} = -3$

(ii) $V_L(\text{peak}) = 12 - (-12) = 24 \text{ V}$

$$I_L(\text{peak}) = \frac{24}{0.5} = 48 \text{ mA}$$

$$P_L(\text{avg}) = \left(\frac{24}{\sqrt{2}}\right)\left(\frac{0.048}{\sqrt{2}}\right) = 0.576 \text{ W}$$

(iii) $v_{o1}(\text{peak}) = 12 \Rightarrow v_i(\text{peak}) = \frac{12}{3} = 4 \text{ V}$

(b) Need $A_{v1} = 6 = \left(1 + \frac{R_2}{20}\right) \Rightarrow R_2 = 100 \text{ k}\Omega$

$$A_{v2} = -6 = -\frac{R_4}{20} \Rightarrow R_4 = 120 \text{ k}\Omega$$

Chapter 16

Exercise Solutions

EX16.1

$$\begin{aligned} \text{(a)} \quad v_O &= V_{DD} - i_D R_D = V_{DD} - \left(\frac{k'_n}{2} \right) \left(\frac{W}{L} \right) R_D [2(v_I - V_{TN})v_O - v_O^2] \\ 0.1 &= 3 - \left(\frac{0.1}{2} \right) (4) R_D [2(3 - 0.5)(0.1) - (0.1)^2] \Rightarrow R_D = 29.6 \text{ k}\Omega \\ \text{(b)} \quad i_{D,\max} &= \frac{3 - 0.1}{29.6} = 0.0980 \text{ mA} \\ P_{D,\max} &= (0.098)(3) = 0.294 \text{ mW} \\ \text{(c)} \quad &\text{From Equation (16.9),} \\ &\left(\frac{0.1}{2} \right) (4) (29.6) V_{Ot}^2 + V_{Ot} - 3 = 0 \\ 5.92 V_{Ot}^2 + V_{Ot} - 3 &= 0 \Rightarrow V_{Ot} = 0.632 \text{ V} \\ V_{Ot} &= V_{It} - V_{TN} \Rightarrow V_{It} = 1.132 \text{ V} \end{aligned}$$

EX16.2

$$\begin{aligned} \text{(a)} \quad \text{(i)} \quad v_O &= V_{DD} - V_{TNL} = 3 - 0.4 = 2.6 \text{ V} \\ \text{(ii)} \quad &\text{From Equation (16.21),} \\ &\left(\frac{0.1}{2} \right) (16) [2(2.6 - 0.4)v_O - v_O^2] = \left(\frac{0.1}{2} \right) (2)(3 - v_O - 0.4)^2 \\ &\text{which yields} \\ 9v_O^2 - 40.4v_O + 6.76 &= 0 \Rightarrow v_O = 0.174 \text{ V} \\ \text{(b)} \quad i_{D,\max} &= \left(\frac{0.1}{2} \right) (2)(3 - 0.174 - 0.4)^2 = 0.589 \text{ mA} \\ P_{D,\max} &= (0.589)(3) = 1.766 \text{ mW} \\ \text{(c)} \quad V_{It} &= \frac{3 - 0.4 + 0.4(1 + \sqrt{8})}{1 + \sqrt{8}} = 1.08 \text{ V} \\ V_{Ot} &= 1.08 - 0.4 = 0.68 \text{ V} \end{aligned}$$

EX16.3

$$\begin{aligned} \text{(a)} \quad &\text{From Equation (16.27(b)),} \\ &\frac{6}{2} [2(3 - 0.4)v_O - v_O^2] = [-(-0.8)]^2 \\ \text{or} \quad 3v_O^2 - 15.6v_O + 0.64 &= 0 \Rightarrow v_O = 0.0414 \text{ V} \\ \text{(b)} \quad i_{D,\max} &= \left(\frac{0.1}{2} \right) (2)[-(-0.8)]^2 = 0.064 \text{ mA} \\ P_{D,\max} &= (0.064)(3) = 0.192 \text{ mW} \end{aligned}$$

$$(c) \sqrt{\frac{6}{2}}(V_{It} - 0.4) = -(-0.8) \Rightarrow V_{It} = 0.862 \text{ V}$$

$$\text{Driver: } V_{It} = 0.862 \text{ V, } V_{Ot} = 0.462 \text{ V}$$

$$\text{Load: } V_{It} = 0.862 \text{ V, } V_{Ot} = 3 - 0.8 = 2.2 \text{ V}$$

EX16.4

$$V_{OH} = 2.5 - \left\{ 0.5 + (0.3) \left[\sqrt{0.73 + V_{OH}} - \sqrt{0.73} \right] \right\}$$

$$V_{OH} - 2.2563 = -(0.3) \sqrt{0.73 + V_{OH}}$$

$$V_{OH}^2 - 4.5126V_{OH} + 5.09098 = (0.09)(0.73 + V_{OH})$$

$$V_{OH}^2 - 4.60264V_{OH} + 5.02528 = 0 \Rightarrow V_{OH} = 1.781 \text{ V}$$

EX16.5

$$(a) (i) \frac{K_D}{K_L} [2(v_I - V_{TND})v_O - v_O^2] = (-V_{TNL})^2$$

$$\left(\frac{5}{1} \right) [2(1.8 - 0.4)v_O - v_O^2] = [-(-0.6)]^2$$

$$5v_O^2 - 14v_O + 0.36 = 0 \Rightarrow v_O = 26 \text{ mV}$$

$$(ii) 2 \left(\frac{K_D}{K_L} \right) [2(v_I - V_{TND})v_O - v_O^2] = (-V_{TNL})^2$$

$$2 \left(\frac{5}{1} \right) [2(1.8 - 0.4)v_O - v_O^2] = [-(-0.6)]^2$$

$$10v_O^2 - 28v_O + 0.36 = 0 \Rightarrow v_O = 12.9 \text{ mV}$$

$$(b) i_{D,\max} = K_L (-V_{TNL})^2 = \left(\frac{100}{2} \right) (1) [-(-0.6)]^2 = 18 \mu\text{ A}$$

$$P = i_{D,\max} \cdot V_{DD} = (18)(1.8) = 32.4 \mu\text{ W}$$

EX16.6

$$(a) \left(\frac{K_D}{3K_L} \right) [2(v_I - V_{TND})v_O - v_O^2] = (-V_{TNL})^2$$

$$\frac{12}{(3)(1)} [2(2.5 - 0.4)v_O - v_O^2] = [-(-0.6)]^2$$

$$4v_O^2 - 16.8v_O + 0.36 = 0 \Rightarrow v_O = 21.5 \text{ mV}$$

$$(b) \frac{4}{(3)(1)} [2(2.5 - 0.4)v_O - v_O^2] = [-(-0.6)]^2$$

$$1.333v_O^2 - 5.6v_O + 0.36 = 0 \Rightarrow v_O = 65.3 \text{ mV}$$

EX16.7

$$V_{It} = \frac{V_{DD}}{2} = \frac{2.1}{2} = 1.05 \text{ V}$$

$$V_{OPt} = V_{It} - V_{TD} = 1.05 - (-0.4) = 1.45 \text{ V}$$

$$(a) \quad V_{ONt} = V_{It} - V_{TN} = 1.05 - 0.4 = 0.65 \text{ V}$$

$$V_{It} = \frac{2.1 + (-0.4) + \sqrt{0.5}(0.4)}{1 + \sqrt{0.5}} = 1.16 \text{ V}$$

$$V_{OPt} = 1.16 + 0.4 = 1.56 \text{ V}$$

$$(b) \quad V_{ONt} = 1.16 - 0.4 = 0.76 \text{ V}$$

$$V_{It} = \frac{2.1 + (-0.4) + \sqrt{2}(0.4)}{1 + \sqrt{2}} = 0.938 \text{ V}$$

$$V_{OPt} = 0.938 + 0.4 = 1.338 \text{ V}$$

$$(c) \quad V_{ONt} = 0.538 \text{ V}$$

EX16.8

$$P = f \cdot C_L \cdot V_{DD}^2$$

$$(0.10 \times 10^{-6}) = f (0.5 \times 10^{-12}) (3)^2$$

$$f = 2.22 \times 10^4 \text{ Hz} \Rightarrow \underline{f = 22.2 \text{ kHz}}$$

EX16.9

$$(a) \quad V_{It} = \frac{V_{DD} + V_{TP} + \sqrt{\frac{K_n}{K_p}} \cdot V_{TN}}{1 + \sqrt{\frac{K_n}{K_p}}} = \frac{1.8 - 0.4 + (0.4)\sqrt{\frac{200}{80}}}{1 + \sqrt{\frac{200}{80}}} \Rightarrow V_{It} = 0.7874 \text{ V}$$

$$V_{OPt} = V_{It} - V_{TP} = 0.7874 + 0.4 = 1.187 \text{ V}$$

$$V_{ONt} = V_{It} - V_{TN} = 0.7874 - 0.4 = 0.3874 \text{ V}$$

$$(b) \quad \frac{K_n}{K_p} = \frac{200}{80} = 2.5$$

$$V_{IL} = 0.4 + \frac{(1.8 - 0.4 - 0.4)}{(2.5 - 1)} \left[2\sqrt{\frac{2.5}{2.5 + 3}} - 1 \right] \Rightarrow V_{IL} = 0.6323 \text{ V}$$

$$V_{IH} = 0.4 + \frac{(1.8 - 0.4 - 0.4)}{(2.5 - 1)} \left[\frac{2(2.5)}{\sqrt{3(2.5) + 1}} - 1 \right] \Rightarrow V_{IH} = 0.8767 \text{ V}$$

$$V_{OHU} = \frac{1}{2} \{ (1 + 2.5)(0.6323) + 1.8 - (2.5)(0.4) + 0.4 \} \Rightarrow V_{OHU} = 1.7065 \text{ V}$$

$$V_{OLU} = \frac{(0.8767)(1 + 2.5) - 1.8 - (2.5)(0.4) + 0.4}{2(2.5)} \Rightarrow V_{OLU} = 0.1337 \text{ V}$$

$$(c) \quad NM_L = 0.6323 - 0.1337 = 0.4986 \text{ V}$$

$$NM_H = 1.7065 - 0.8767 = 0.8298 \text{ V}$$

EX16.10

3 PMOS in series and 3 NMOS in parallel.

Worst Case: Only one NMOS is ON in Pull-down mode $\Rightarrow$ same as the CMOS inverter $\Rightarrow W_n = W$.

All 3 PMOS are on during pull-up mode $\Rightarrow W_p = 3(2W) = 6W$.

EX16.11

NMOS: Worst Case, M_{NA} , M_{NB} on, $W_n = 2(W)$ or M_{NC} , M_{ND} or M_{NE} on $\Rightarrow W_n = 2(W)$.

PMOS: M_{PA} and M_{PC} on or M_{PB} and M_{PD} on $\Rightarrow W_p = 2(2W) = 4W$

If M_{PD} and M_{PE} on, need $W_p = 2(4W) = 8W$

EX16.12

(a) $v_O = \phi - V_{TN} = 2.5 - 0.4 = 2.1 \text{ V}$

(b) For $v_{DS} = 0$, $v_O = 1.8 \text{ V}$

(c) $v_O = \phi - V_{TN} = 2.5 - 0.4 = 2.1 \text{ V}$

(d) $v_O = \phi - V_{TN} = 1.5 - 0.4 = 1.1 \text{ V}$

EX16.13

(a) $v'_I = \phi - V_{TN} = 3.3 - 0.5 = 2.8 \text{ V}$

$$\frac{K_D}{K_L} [2(v'_I - V_{TN})v_O - v_O^2] = [V_{DD} - v_O - V_{TN}]^2$$

$$\frac{K_D}{K_L} [2(2.8 - 0.5)(0.1) - (0.1)^2] = [3.3 - 0.1 - 0.5]^2$$

which yields

$$\frac{K_D}{K_L} = 16.2$$

(b) $v'_I = \phi - V_{TN} = 2.8 - 0.5 = 2.3 \text{ V}$

$$\frac{K_D}{K_L} [2(2.3 - 0.5)(0.1) - (0.1)^2] = [3.3 - 0.1 - 0.5]^2$$

which yields

$$\frac{K_D}{K_L} = 20.8$$

EX16.14

$16 K \Rightarrow 16384 \text{ cells}$

Total Power = $125 \text{ mW} = (2.5)I_T \Rightarrow I_T = 50 \text{ mA}$

$$I = \frac{50 \text{ mA}}{16384} \Rightarrow I = 3.05 \mu\text{A}$$

Then, for each cell,

Now, $I \cong \frac{V_{DD}}{R}$ or $R = \frac{V_{DD}}{I} = \frac{2.5}{3.05} \Rightarrow R = 0.82 \text{ M}\Omega$

Test Your Understanding Solutions

TYU16.1

$$\begin{aligned}
 P_{D,\max} &= i_{D,\max} \cdot V_{DD} \\
 0.50 &= i_{D,\max} (1.8) \Rightarrow i_{D,\max} = 0.2778 \text{ mA} \\
 i_{D,\max} &= 0.2778 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right)_L (1.8 - 0.12 - 0.4)^2 \Rightarrow \left(\frac{W}{L} \right)_L = 3.39 \\
 \frac{K_D}{K_L} [2(1.4 - 0.4)(0.12) - (0.12)^2] &= (1.8 - 0.12 - 0.4)^2 \Rightarrow \frac{K_D}{K_L} = 7.26 \\
 \frac{K_D}{K_L} = 7.26 &= \frac{(W/L)_D}{(W/L)_L} = \frac{(W/L)_D}{3.39} \Rightarrow \left(\frac{W}{L} \right)_D = 24.6
 \end{aligned}$$

TYU16.2

$$\begin{aligned}
 P_{D,\max} &= i_{D,\max} \cdot V_{DD} \\
 0.2 &= i_{D,\max} (1.8) \Rightarrow i_{D,\max} = 0.111 \text{ mA} \\
 i_{D,\max} &= 0.111 = \left(\frac{0.1}{2} \right) \left(\frac{W}{L} \right)_L [-(-0.6)]^2 \Rightarrow \left(\frac{W}{L} \right)_L = 6.17 \\
 \frac{K_D}{K_L} &= \frac{[-(-0.6)]^2}{2(1.8 - 0.4)(0.08) - (0.08)^2} = 1.654 = \frac{(W/L)_D}{(W/L)_L} = \frac{(W/L)_D}{6.17} \\
 \text{so } \left(\frac{W}{L} \right)_D &= 10.2
 \end{aligned}$$

TYU16.3

$$\begin{aligned}
 \text{(a)} \quad I &= (0.098)(100,000) \text{ mA}, \Rightarrow I = 9.8 \text{ A} \\
 P &= (0.294)(100,000) \text{ mW}, \Rightarrow P = 29.4 \text{ W} \\
 \text{(b)} \quad I &= (0.589)(100,000) \text{ mA}, \Rightarrow I = 58.9 \text{ A} \\
 P &= (1.766)(100,000) \text{ mW}, \Rightarrow P = 176.6 \text{ W} \\
 \text{(c)} \quad I &= (0.064)(100,000) \text{ mA}, \Rightarrow I = 6.4 \text{ A} \\
 P &= (0.194)(100,000) \text{ mW}, \Rightarrow P = 19.2 \text{ W}
 \end{aligned}$$

TYU16.4

$$\begin{aligned}
 \text{(a)} \quad P &= i_D \cdot V_{DD} \\
 50 &= i_D (2.5) \Rightarrow i_D = 20 \mu\text{A} \\
 i_D &= 20 = \left(\frac{100}{2} \right) \left(\frac{W}{L} \right)_L [-(-0.6)]^2 \Rightarrow \left(\frac{W}{L} \right)_L = 1.11 \\
 \frac{K_D}{K_L} [2(2.5 - 0.4)(0.05) - (0.05)^2] &= [-(-0.6)]^2 \Rightarrow \frac{K_D}{K_L} = 1.735 \\
 \frac{K_D}{K_L} = 1.735 &= \frac{(W/L)_D}{(W/L)_L} = \frac{(W/L)_D}{1.11} \Rightarrow \left(\frac{W}{L} \right)_D = 1.93 \\
 \text{(b)} \quad 3(1.735)[2(2.5 - 0.4)V_{OL} - V_{OL}^2] &= [-(-0.6)]^2 \\
 5.205V_{OL}^2 - 21.861V_{OL} + 0.36 &= 0 \Rightarrow V_{OL} = 16.5 \text{ mV}
 \end{aligned}$$

TYU16.5

$$(a) \quad \frac{1}{2} \cdot \frac{(W/L)_D}{(W/L)_L} [2(2.1 - 0.4)(0.08) - (0.08)^2] = (2.5 - 0.08 - 0.4)^2 \Rightarrow \left(\frac{W}{L}\right)_D = 15.4$$

$$(b) \quad i_{D,\max} = \left(\frac{100}{2}\right)(0.5)(2.5 - 0.08 - 0.4)^2 = 102 \mu A$$

$$P = i_{D,\max} \cdot V_{DD} = (102)(2.5) = 255 \mu W$$

TYU16.6

$$(a) \quad \frac{1}{2} \frac{(W/L)_D}{(W/L)_L} [2(2.5 - 0.4)(0.08) - (0.08)^2] = [-(-0.6)]^2 \Rightarrow \left(\frac{W}{L}\right)_D = 1.09$$

$$(b) \quad i_{D,\max} = \left(\frac{100}{2}\right)(0.5)[-(-0.6)]^2 = 9 \mu A$$

$$P = i_{D,\max} \cdot V_{DD} = (9)(2.5) = 22.5 \mu W$$

TYU16.7

a.

$$K_n = K_p = 50 \mu A/V^2$$

$$V_{It} = 2.5 V$$

$$i_{D(\max)} = K_n (V_{It} - V_{TN})^2 = 50(2.5 - 0.8)^2 \Rightarrow \underline{i_{D(\max)} = 145 \mu A}$$

b.

$$K_n = K_p = 200 \mu A/V^2$$

$$V_{It} = 2.5 V$$

$$i_{D(\max)} = (200)(2.5 - 0.8)^2 \Rightarrow \underline{i_{D(\max)} = 578 \mu A}$$

TYU16.8

$$(a) \quad V_{It} = \frac{5 + (-2) + 0.8}{1 + 1} = 1.9 V$$

$$V_{OPt} = V_{It} - V_{TP} = 1.9 - (-2) = 3.9 V$$

$$V_{ONt} = V_{It} - V_{TN} = 1.9 - 0.8 = 1.1 V$$

$$(b) \quad V_{IL} = 0.8 + \frac{3}{8}[5 + (-2) - 0.8] = 1.625 V$$

$$V_{IH} = 0.8 + \frac{5}{8}[5 + (-2) - 0.8] = 2.175 V$$

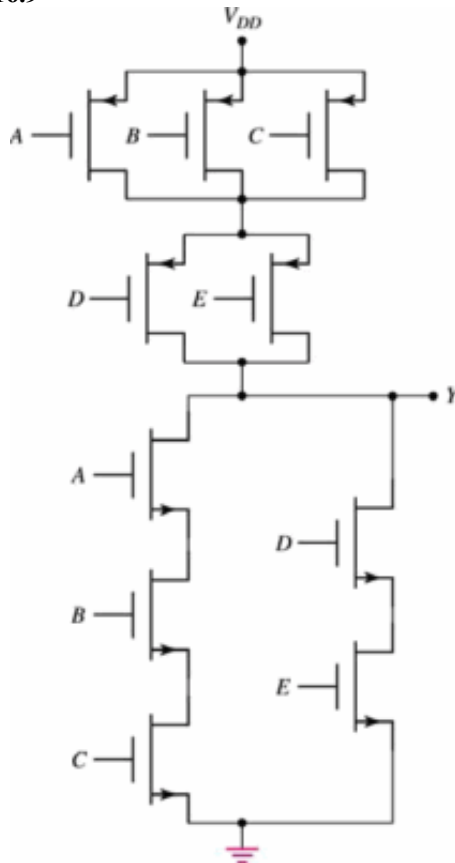
$$V_{OLU} = \frac{1}{2}[2(2.175) - 5 - 0.8 - (-2)] = 0.275 V$$

$$V_{OHU} = \frac{1}{2}[2(1.625) + 5 - 0.8 - (-2)] = 4.725 V$$

$$(c) \quad NM_L = 1.625 - 0.275 = 1.35 V$$

$$NM_H = 4.725 - 2.175 = 2.55 V$$

TYU16.9



TYU16.10

NMOS – 2 transistors in series

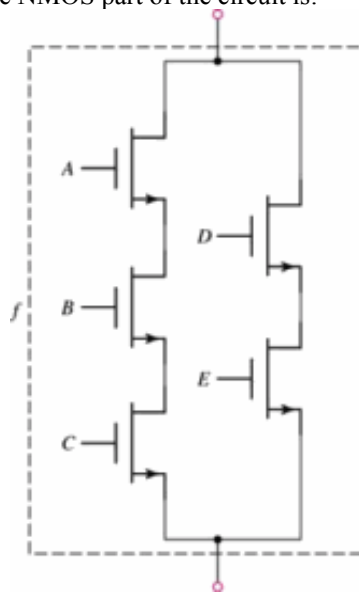
$$W_n = 2(W) = 2W$$

PMOS – 2 transistors in series

$$W_p = 2(2W) = 4W$$

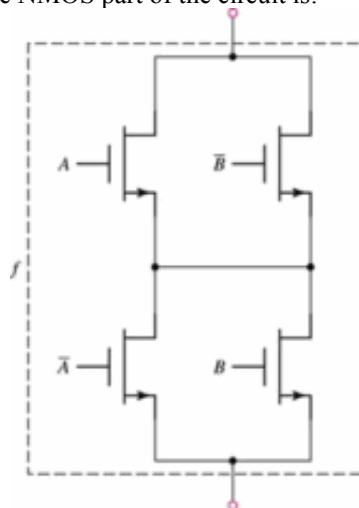
TYU16.11

The NMOS part of the circuit is:



TYU16.12

The NMOS part of the circuit is:



TYU16.13

Insert Figure X-TYU16.13

TYU16.14

For NMOS, $\phi - v_O \geq 0.4 \text{ V}$ or $\phi - v_I \geq 0.4 \text{ V}$

At $v_I = 2.1 \text{ V}$, $2.1 = 2.5 - 0.2t \Rightarrow t = 2 \text{ s}$

NMOS conducting for $2 \leq t \leq 12.5 \text{ s}$

For PMOS, $v_I - 0 \geq 0.4 \text{ V}$

At $v_I = 0.4 \text{ V}$, $0.4 = 2.5 - 0.2t \Rightarrow t = 10.5 \text{ s}$

PMOS conducting for $0 \leq t \leq 10.5 \text{ s}$

TYU16.15

(a) $1 \text{ K} \Rightarrow 32 \times 32$ array

Each row and column requires a 5-bit word $\Rightarrow$ 6 transistors per row and column, $\Rightarrow 32 \times 6 + 32 \times 6 = 384$ transistors plus buffer transistors.

(b) $4 \text{ K} \Rightarrow 64 \times 64$ array

Each row and column requires a 6-bit word $\Rightarrow$ 7 transistors per row and column $\Rightarrow 64 \times 7 + 64 \times 7 = 896$ transistors plus buffer transistors.

(c) $16 \text{ K} \Rightarrow 128 \times 128$ array

Each row and column requires a 7-bit word $\Rightarrow$ 8 transistors per row and column
 $\Rightarrow 128 \times 8 + 128 \times 8 = 2048$ transistors plus buffer transistors.

TYU16.16

From Equation (16.82)

$$\frac{(W/L)_{nA}}{(W/L)_{nI}} = \frac{2(V_{DD}V_{TN}) - 3V_{TN}^2}{(V_{DD} - 2V_{TN})^2} = \frac{2(2.5)(0.4) - 3(0.4)^2}{(2.5 - 2(0.4))^2} = 0.526$$

From Equation (16.84)

$$\frac{(W/L)_p}{(W/L)_{nB}} = \frac{k'_n}{k'_p} \cdot \frac{2(V_{DD}V_{TN}) - 3V_{TN}^2}{(V_{DD} + V_{TP})^2} = (2.5) \left[\frac{2(2.5)(0.4) - 3(0.4)^2}{(2.5 - 0.4)^2} \right] = 0.862$$

So $\left(\frac{W}{L}\right)$ of transmission gate device must be < 0.526 times the $\left(\frac{W}{L}\right)$ of the NMOS transistors in the inverter cell. The $\left(\frac{W}{L}\right)$ of the PMOS transistors must be < 0.862 times the $\left(\frac{W}{L}\right)$ of the transmission gate devices. Then the $\left(\frac{W}{L}\right)$ of the PMOS devices must be < 0.453 times $\left(\frac{W}{L}\right)$ of NMOS devices in cell.

TYU16.17

Initial voltage across the storage capacitor $= V_{DD} - V_{TN} = 3 - 0.5 = 2.5 \text{ V}$.

Now

$$-I = C \frac{dV}{dt} \quad \text{or} \quad V = -\frac{I}{C} \cdot t + K$$

where $K = 2.5 \text{ V}$, $t = 1.5 \text{ ms}$, $V = \frac{2.5}{2} = 1.25 \text{ V}$, and $C = 0.05 \text{ pF}$. Then

$$1.25 = 2.5 - \frac{I(1.5 \times 10^{-3})}{(0.05 \times 10^{-12})} \Rightarrow$$

$$I = 4.17 \times 10^{-11} \text{ A} \Rightarrow I = 41.7 \text{ pA}$$

Chapter 17

Exercise Solutions

EX17.1

$$(a) \quad i_E = \frac{-0.7 - (-1.8)}{R_E} = 0.11 \Rightarrow R_E = 10 \text{ k}\Omega$$

$$i_{C1} = i_{C2} = \frac{i_E}{2} = 0.055 \text{ mA}$$

$$R_C = \frac{1.8 - 1.45}{0.055} = 6.364 \text{ k}\Omega$$

$$(b) \quad (i) \quad v_1 = 0.5 \text{ V}, \quad v_E = 0.5 - 0.7 = -0.2 \text{ V}$$

$$i_E = \frac{-0.2 - (-1.8)}{10} = 0.16 \text{ mA}$$

$$v_{O1} = 1.8 - (0.16)(6.364) = 0.782 \text{ V}$$

$$v_{O2} = 1.8 \text{ V}$$

$$(ii) \quad v_1 = -0.5 \text{ V}, \quad i_E = 0.11 \text{ mA}$$

$$v_{O1} = 1.8 \text{ V}$$

$$v_{O2} = 1.8 - (0.11)(6.364) = 1.1 \text{ V}$$

$$(c) \quad (i) \quad i_E = 0.16 \text{ mA}, \quad P = (0.16)[1.8 - (-1.8)] = 0.576 \text{ mW}$$

$$(ii) \quad i_E = 0.11 \text{ mA}, \quad P = (0.11)[1.8 - (-1.8)] = 0.396 \text{ mW}$$

EX17.2

$$P(i_{CXY} + i_{CR} + i_3 + i_4)(5.2)$$

$$v_X = v_Y = \text{logic } 1 \Rightarrow i_{CXY} = 3.22 \text{ mA}$$

$$i_{CR} = 0$$

$$i_3 = \frac{-0.7 + 5.2}{1.5} = 3 \text{ mA}$$

$$i_4 = \frac{-1.4 + 5.2}{1.5} = 2.53 \text{ mA}$$

$$a. \quad P = (3.22 + 0 + 3 + 2.53)(5.2) \Rightarrow \underline{P = 45.5 \text{ mW}}$$

$$v_X = v_Y = \text{logic } 0 \Rightarrow i_{CXY} = 0$$

$$i_{CR} = 2.92 \text{ mA}$$

$$i_3 = 2.53 \text{ mA}$$

$$i_4 = 3 \text{ mA}$$

$$b. \quad P = (0 + 2.92 + 2.53 + 3)(5.2) \Rightarrow \underline{P = 43.9 \text{ mW}}$$

EX17.3

$$v_{B5} = -1 + 0.7 = -0.3 \text{ V}$$

$$R_1 = \frac{0.3}{0.5} = 0.6 \text{ k}\Omega$$

$$R_2 = \frac{-0.3 - 1.4 - (-3.3)}{0.5} = 3.2 \text{ k}\Omega$$

$$R_5 = \frac{-1 - (-3.3)}{0.5} = 4.6 \text{ k}\Omega$$

EX17.4

$$I_{MAX} = 1 = \frac{-0.7 - (-5.2)}{R_3} \Rightarrow R_3 = R_4 = 4.5 \text{ K}$$

$$i_{CXY} = \frac{-0.7 - 0.7 - (-5.2)}{1.18} = 3.22 \text{ mA}$$

$$i_5 = i_1 = 1.40 \text{ mA}$$

$$i_3 = 1.0 \text{ mA}$$

$$i_4 = \frac{-1.4 - (-5.2)}{4.5} = 0.844 \text{ mA}$$

$$P = (3.22 + 1.4 + 1.4 + 1.0 + 0.844)(5.2) = 40.8 \text{ mW}$$

EX17.5

$$i_L = \left(\frac{v_{oR} - 0.7 - (-5.2)}{(1.18)(51)} \right) (10)$$

$$i_3 = \frac{v_{oR} - (-5.2)}{1.5}$$

$$\left[\frac{0 - (V_{oR} + 0.7)}{0.24} \right] (51) = \frac{v_{oR} + 5.2}{1.5} + \left(\frac{v_{oR} + 4.5}{(1.18)(51)} \right) (10)$$

$$-v_{oR} \left[\left(\frac{51}{0.24} \right) + \frac{1}{1.5} + \frac{1 - (6)}{(1.18)(51)} \right] = \frac{(0.7)(51)}{0.24} + \frac{5.2}{1.5} + \frac{4.5(10)}{(1.18)(51)}$$

$$-v_{oR} [212.50 + 0.6667 + 0.166168] = 148.75 + 3.4666 + 0.747757$$

$$-v_{oR} [213.3328] = 152.9644$$

$$v_{oR} = -0.7170$$

EX17.6

$$r_{\pi 3} = \frac{(100)(0.026)}{1} = 2.6 \text{ K}$$

$$g_{m3} = \frac{1}{0.026} = 38.46 \text{ mA/V}$$

$$I_{b3} = \frac{V_n}{R_{C2} + r_{\pi 3} + (1 + \beta)R_3} = \frac{V_n}{0.24 + 2.6 + (101)(4.5)}$$

$$I_{b3} = \frac{V_n}{457.34}$$

$$V_O = -I_{b3}(R_{C2} + r_{\pi3}) = -\left(\frac{V_n}{457.34}\right)(0.24 + 2.6)$$

$$V_O = -0.00621 V_n$$

$$V'_O = (1 + \beta)I_{b3}R_3 = (101)\left(\frac{V_n}{457.34}\right)(4.5)$$

$$V'_O = 0.9938 V_n$$

EX17.7

$$P = I_Q \cdot V_{CC} \Rightarrow 0.2 = I_Q(1.7) \Rightarrow I_Q = 117.6 \mu\text{A}$$

$$Q_R \text{ on} \Rightarrow v_0 = 1.7 - I_Q R_C = 1.7 - 0.4 \Rightarrow R_C = \frac{0.4}{0.1176} \Rightarrow R_C = 3.40 \text{ k}\Omega$$

$$V_R = \frac{1.7 + 1.3}{2} \Rightarrow V_R = 1.5 \text{ V}$$

EX17.8

(a)

$$v_X = v_Y = 5 \text{ V}$$

$$v_1 = V_{BE}(\text{sat}) + 2V_Y = 0.8 + 2(0.7) = 2.2 \text{ V}$$

$$i_1 = \frac{5 - 2.2}{4} = 0.70 \text{ mA}$$

$$i_{RC} = \frac{V_{CC} - V_{CE}(\text{sat})}{R_C} = \frac{5 - 0.1}{4} = 1.225 \text{ mA}$$

$$P = (i_1 + i_{RC})V_{CC} = (0.70 + 1.225)(5)$$

$$\text{or } P = 9.625 \text{ mW}$$

$$v_X = v_Y = 0 \Rightarrow v_1 = 0.70 \text{ V}$$

$$i_1 = \frac{V_{CC} - v_1}{R_1} = \frac{5 - 0.70}{4} = 1.075 \text{ mA}$$

$$(b) \quad P = i_1 \cdot V_{CC} = (1.075)(5) \Rightarrow P = 5.375 \text{ mW}$$

EX17.9

$$(a) \quad v_X = v_Y = 0.1 \text{ V}$$

$$v_{B1} = 0.1 + 0.8 = 0.9 \text{ V}$$

$$i_1 = i_{B1} = \frac{5 - 0.9}{12} = 0.342 \text{ mA}$$

$$i_{C1} \cong 0, \quad i_{B2} = i_{C2} = 0, \quad i_{Bo} = i_{Co} = 0$$

(b) $v_X = v_Y = 5 \text{ V}$
 $v_{B1} = 0.8 + 0.8 + 0.7 = 2.3 \text{ V}$
 $i_1 = i_{B1} = \frac{5 - 2.3}{12} = 0.225 \text{ mA}$
 $i_{B2} = |i_{C1}| = (1 + 0.2)i_{B1} = 0.27 \text{ mA}$
 $v_{C2} = 0.8 + 0.1 = 0.9 \text{ V}$
 $i_2 = i_{C2} = \frac{5 - 0.9}{4} = 1.025 \text{ mA}$
 $i_{E2} = i_{B2} + i_{C2} = 0.27 + 1.025 = 1.295 \text{ mA}$
 $i_{Bo} = i_{E2} - \frac{0.8}{R_B} = 1.295 - \frac{0.8}{2} = 0.895 \text{ mA}$
 $i_3 = i_{Co} = \frac{5 - 0.1}{6} = 0.8167 \text{ mA}$

EX17.10

(a) $v_X = v_Y = 3.6 \text{ V}$
 $i_1 = \frac{5 - (0.8 + 0.8 + 0.7)}{12} = 0.225 \text{ mA}$
 $i_{B2} = (1 + 0.2)(0.225) = 0.27 \text{ mA}$
 $i_{C2} = \frac{5 - (0.8 + 0.1)}{4} = 1.025 \text{ mA}$
 $i_{Bo} = 0.27 + 1.025 - \frac{0.8}{2} = 0.895 \text{ mA}$
 $i'_L = \frac{5 - (0.1 + 0.8)}{12} = 0.3417 \text{ mA}$
 $i_L(\text{max}) = \beta i_{Bo} = N i'_L$
 $(25)(0.895) = N(0.3417) \Rightarrow N = 65$
(b) $i_L(\text{max}) = 12 = N i'_L = N(0.3417) \Rightarrow N = 35$

EX17.11

(a) $i_C = \frac{5 - 0.4}{2.25} = 2.044 \text{ mA}$
 $i'_C = \frac{2 + 2.044}{1 + \frac{1}{15}} = 3.791 \text{ mA}$
 $i'_B = \frac{i'_C}{\beta} = \frac{3.791}{15} = 0.253 \text{ mA}$
 $i_D = i_B - i'_B = 2 - 0.253 = 1.747 \text{ mA}$

(b) $i_C = 2.044 + 10 = 12.044 \text{ mA}$

$$i'_C = \frac{2 + 12.044}{1 + \frac{1}{15}} = 13.166 \text{ mA}$$

$$i'_B = \frac{13.166}{15} = 0.878 \text{ mA}$$

$$i_D = 2 - 0.878 = 1.122 \text{ mA}$$

(c) $i_D = 0$, $i'_B = 2 \text{ mA}$, $i'_C = (2)(15) = 30 \text{ mA}$
 $i_L = 30 - 2.044 \approx 28 \text{ mA}$

EX17.12

(a) $v_1 = 0.4 + 0.3 = 0.7$, $i_1 = \frac{5 - 0.7}{40} = 0.1075 \text{ mA}$

All transistor currents are zero.

$$P = (0.1075)(5 - 0.4) \Rightarrow 495 \mu\text{W}$$

(b) $v_1 = 1.4 \text{ V}$, $i_1 = i_{B2} = \frac{5 - 1.4}{40} = 0.090 \text{ mA}$

$$v_{C2} = 0.7 + 0.4 = 1.1 \text{ V}, \quad i_2 = i_{C2} = \frac{5 - 1.1}{12} = 0.325 \text{ mA}$$

$$i_{B0} \approx i_{B2} + i_{C2} = 0.09 + 0.325 = 0.415 \text{ mA}$$

$$i_{C0} \approx 0$$

$$P = (i_1 + i_2)(5) = (0.09 + 0.325)(5) = 2.08 \text{ mW}$$

Test Your Understanding Solutions

TYU17.1

$$(a) \quad i_E = 0.8 = \frac{0.75 - 0.7 - (-1.8)}{R_E} \Rightarrow R_E = 2.31 \text{ k}\Omega$$

$$R_{C2} = \frac{1.8 - 1.1}{0.8} = 0.875 \text{ k}\Omega$$

$$(b) \quad i_E = \frac{1.1 - 0.7 - (-1.8)}{2.3125} = 0.951 \text{ mA}$$

$$R_{C1} = \frac{1.8 - 1.1}{0.95135} = 0.736 \text{ k}\Omega$$

TYU17.2

$$\text{logic } 1 = -0.7 \text{ V}$$

$$Q_1 \text{ and } Q_2 \text{ on when } v_X = v_Y = -0.7 \text{ V}$$

$$i_E = \frac{-0.7 - 0.7 - (-5.2)}{R_E} = 2.5 \Rightarrow R_E = 1.52 \text{ k}\Omega$$

$$v_{\text{NOR}} = -1.5 \Rightarrow R_{C1} = \frac{0 - (-1.5 + 0.7)}{2.5} \Rightarrow R_{C1} = 320 \Omega$$

$$V_R = \frac{-1.5 - 0.7}{2} \Rightarrow V_R = -1.1 \text{ V}$$

$$Q_R \text{ on} \Rightarrow i_E = \frac{-1.1 - 0.7 - (-5.2)}{1.52} = 2.237 \text{ mA}$$

$$R_{C2} = \frac{0 - (-1.5 + 0.7)}{2.237} \Rightarrow R_{C2} = 358 \Omega$$

$$R_3 = R_4 = \frac{-0.7 - (-5.2)}{2.5} \Rightarrow R_3 = R_4 = 1.8 \text{ k}\Omega$$

TYU17.3

State	A	B	C	Q_{01}	Q_{02}	Q_{03}	Q_1	Q_2	Q_R	v_0
1	0	0	0	off	off	off	off	on	on	0
2	1	0	0	"on"	off	off	off	on	on	0
3	0	1	0	off	on	off	off	on	"off"	1
4	0	0	1	off	off	on	on	off	on	0
5	1	1	0	on	on	off	off	on	off	1
6	1	0	1	on	off	on	on	off	"off"	1
7	0	1	1	off	on	on	on	off	on	0
8	1	1	1	on	on	on	on	off	off	1

$(A \text{ AND } C) \text{ OR } (B \text{ AND } \bar{C})$
true for true for
 states 6 and 8 states 3 and 5
Output goes high for these 4 states

TYU17.4

A	B	C	v_o
0	0	0	0
1	0	0	1
0	1	0	1
0	0	1	1
1	1	0	0
1	0	1	0
0	1	1	0
1	1	1	1

$$\Rightarrow (A \oplus B) \oplus C$$

TYU17.5

(a) $i_1 = \frac{5 - (0.1 + 0.7)}{15} = 0.28 \text{ mA}$
 $i_2 = i_R = i_B = i_{RC} = 0$
 $v_O = 5 \text{ V}$

(b) Same as part (a).

(c) $i_2 = i_1 = \frac{5 - (0.8 + 0.7 + 0.7)}{15} = 0.1867 \text{ mA}$
 $i_R = \frac{0.8}{15} = 0.0533 \text{ mA}$
 $i_B = 0.1867 - 0.0533 = 0.1334 \text{ mA}$
 $i_{RC} = \frac{5 - 0.1}{6} = 0.8167 \text{ mA}$
 $v_O = 0.1 \text{ V}$

TYU17.6

(a) $i_B = 0.1134 \text{ mA}$, $i_{RC} = 0.8167 \text{ mA}$
 $i'_L = \frac{5 - (0.1 + 0.7)}{15} = 0.28 \text{ mA}$
 $i_{RC} + N i'_L = \beta i_B$
 $0.8167 + N(0.28) = (30)(0.1134) \Rightarrow N = 9$

(b) $i_{C,\max} = \beta i_B = (30)(0.1134) = 3.4 < 12 \text{ mA}$
 $\Rightarrow N = 9$

TYU17.7

From EX17.9, $i_{Bo} = 0.895 \text{ mA}$, $i_3 = 0.8167 \text{ mA}$
 $v_O = 0.1 \text{ V}$, so $v'_{B1} = 0.1 + 0.8 = 0.9 \text{ V}$
 $i'_L = \frac{5 - 0.9}{12} = 0.3417 \text{ mA}$
 $i_{Co,\max} = \beta i_{Bo} = i_3 + N i'_L$
 $(25)(0.895) = 0.8167 + N(0.3417)$
 $N = 63.1 \Rightarrow N = 63$

TYU17.8

Q_1 in saturation

$$i_{B1} = \frac{5 - 0.9}{6} \Rightarrow i_{B1} = 0.683 \text{ mA}$$

$$|i_{C1}| = i_{B2} = i_{C2} = 0$$

$$i_{B0} = i_{C0} = 0$$

$$v_{B4} = 0.1 + 0.7 = 0.8 \text{ V}$$

$$i_{B4} = \frac{0.1}{(21)(4)} \Rightarrow i_{B4} = 1.19 \text{ } \mu\text{A}$$

$$i_{C4} = 23.8 \text{ } \mu\text{A}$$

$$i_{B3} = i_{C3} = 0$$

TYU17.9

(a) $v_X = v_Y = 0.4$, $v_{B1} = 0.4 + 0.7 = 1.1 \text{ V}$

$$i_1 = \frac{5 - 1.1}{2.8} = 1.393 \text{ mA}$$

$$P = i_1(5 - 0.4) = (1.393)(5 - 0.4) = 6.41 \text{ mW}$$

(b) $v_X = v_Y = 3.6 \text{ V}$

$$v_{B1} = 2.1, i_1 = \frac{5 - 2.1}{2.8} = 1.036 \text{ mA}$$

$$v_{C2} = 0.7 + 0.4 = 1.1 \text{ V}, i_2 = \frac{5 - 1.1}{0.76} = 5.132 \text{ mA}$$

$$v_{E4} = 1.1 - 0.7 = 0.4 \text{ V}, i_{R4} = \frac{0.4}{3.5} = 0.1143 \text{ mA}$$

$$i_{R3} = \left(\frac{\beta}{1 + \beta} \right) i_{R4} = \left(\frac{25}{26} \right) (0.1143) = 0.1099 \approx 0.11 \text{ mA}$$

$$P = (i_1 + i_2 + i_{R3})(5) = (1.036 + 5.132 + 0.11)(5)$$

or

$$P = 31.4 \text{ mW}$$

TYU17.10

(a) $v_X = 0.4 \text{ V}$, $v_{E1} = 0.4 + 0.7 = 1.1 \text{ V}$

$$i_{R1} = \frac{5 - 1.1}{40} \Rightarrow 97.5 \text{ } \mu\text{A}$$

$$Q_2 \text{ cutoff, } i_{R2} = 0$$

(b) $v_X = 3.6 \text{ V}$, $v_{E1} = 3(0.7) = 2.1 \text{ V}$

$$i_{R1} = \frac{5 - 2.1}{40} \Rightarrow 72.5 \text{ } \mu\text{A}$$

$$v_{C2} = 2(0.7) + 0.4 = 1.8 \text{ V}$$

$$i_{R2} = \frac{5 - 1.8}{50} \Rightarrow 64 \text{ } \mu\text{A}$$

TYU17.11

(a) $v_X = 0.4 \text{ V}$, $v_{E1} = 0.4 + 0.7 = 1.1 \text{ V}$

$$i_{R1} = \frac{3.5 - 1.1}{40} \Rightarrow 60 \mu\text{A}$$

$$i_{R2} = 0$$

(b) $v_X = 2.1 \text{ V}$, $v_{E1} = 2.1 \text{ V}$

$$i_{R1} = \frac{3.5 - 2.1}{40} \Rightarrow 35 \mu\text{A}$$

$$v_{C2} = 1.8 \text{ V}$$

$$i_{R2} = \frac{3.5 - 1.8}{50} \Rightarrow 34 \mu\text{A}$$